

时频冲突检测与消解

摘 要

本文研究有限带宽下大量用频计划的时频冲突检测与消解。把每个用频计划展开为它在“频段 $\times$ 时隙”栅格上占用的时频单元集合，则“冲突”等价于两集合相交、“无冲突”等价于诸集合两两不交。这一刻画把四个问题统一为同一语言，并使每个结果都能被一个与求解器代码完全分离的校验器复核。

问题 1 由半开区间判据给出精确检测：成对枚举（双指针归并）与按频段分桶的扫描线两种算法独立实现，输出完全一致。150 个计划中检出 297 对冲突，涉及 148 个计划、431 对相交的使用次；冲突图只有 1 个连通分量（规模 148）却很稀疏（平均度 3.96、密度 0.0266）；B-C 类冲突占六成以上。时频单元占用率仅 22.9%，表明**冲突源于局部拥塞而非资源总量不足**。

问题 2 建立 0-1 规划：每个计划恰选一个动作（保持 / 频移 / 时移 / 撤销），以“覆盖同一时频单元的变量至多取一”作为无冲突的充要编码，并按题目优先级构造七级词典序目标（撤销 A/B/C $\rightarrow$ 调整 A/B/C $\rightarrow$ 归一化幅度）。求解采用现任解保持的逐级搜索：每级加入由可行现任解导出的上界割，该割不删去任何最优解，却保证报告值永不劣于任何已知可行解。结果为保留 14、调整 124、撤销 12，且**撤销的全部是 C 类，A、B 类一个未撤**。撤销三级 0/0/12 经 CP-SAT 闭合对偶间隙，为**已证明的全局最优**；由此直接得到推论：在不撤销 A、B 类的前提下“零撤销”不可行。调整各级未闭合间隙，本文只报告“值 / 证明下界 / 状态”，不作最优性声明。

问题 3 归结为二维集合装填。指出题干“不限制平移幅度”存在实质歧义：既有计划固定（解释 A）与既有计划亦可重排（解释 B）。以 A 为主解释（与结果模板一一对应、可独立复核），得最多再安排 109 个 C 类装备，且 CP-SAT 对偶界、HiGHS 线性松弛界与 MILP 界三者同时等于 109，间隙为 0，利用率由 24.6% 升至 36.8%；对解释 B 则由面积守恒给上界、由首次适配重排给构造下界，得区间 [557, 673]。

问题 4 为 C 类增加“改间隔”动作后重解同一模型：撤销数由 12 降至 9（降幅四分之一），代价是多调整 3 个计划、且时间范围延长 $75\Delta t$ ——本文明确指出并讨论了这一代价。

检验：四个独立校验器全部通过（零剩余冲突）；24 个随机微型实例与穷举最优 24/24 一致；两个求解器、两种标量化交叉验证。灵敏度分析给出一条有实用价值的结论：把最大频段平移幅度由 $10\Delta f$ 放宽到 $15\Delta f$ ，撤销数即由 12 降为 0，且频域放宽的效果约为时域的两倍。

关键词：时频冲突检测；冲突图；词典序整数规划；约束规划（CP-SAT）；二维集合装填；最优性界

1 问题重述

1.1 背景与资源模型

在同一空域、同一时段内工作的用频装备越来越多，若两部装备在重叠的频段上同时发射，就会互相干扰。为在计划阶段（而非事后）消除干扰，需要把“谁在什么时间用哪一段频率”的全部申请汇总起来，先检测其中潜在的时频冲突，再通过调整少量计划参数或撤销个别计划把冲突消解掉。

题目将时频资源离散化：时域以 Δt 为最小不可拆分单位，频域把有限带宽等分为 100 个宽度为 Δf 的频段，编号 $0, 1, \dots, 99$ 。于是“时频资源”就是一张以频段编号为行、以时刻为列的二维栅格，每个栅格单元 (b, t) 表示“第 b 个频段在第 t 个时隙”这一份最小资源。冲突即两个用频计划占用了同一个单元。

1.2 用频计划的参数与冲突的判定

附件 1 给出 150 个用频装备提报的用频计划，按编号首字母分为 A、B、C 三类（各类装备数与参数见表 2）。每个计划由四个参数确定：

- (i) **频段区间**：连续占用的频段 $[f, f + w)$ ，占用的频段数 w 在任何调整下保持不变；
- (ii) **首次使用的时间区间**： $[s, s + d)$ ， d 为每次使用的时长；
- (iii) **间隔时长** g ：相邻两次使用之间的空闲时长；
- (iv) **使用次数** n ：按上述规则重复使用 n 次。

题目以装备 A001 为例规定了推进规则：首次使用 $[35, 40)$ 、间隔 60，则第二次使用为 $[40 + 60, 45 + 60) = [100, 105)$ 。即第 k 次 ($k = 0, 1, \dots, n - 1$) 使用的时间区间为 $[s + k(d + g), s + d + k(d + g))$ ，所有区间均为半开区间。**时频冲突**的判定是：两个用频计划的频段区间存在交叠，且它们的某一次使用在时间上存在交叠。

1.3 消解规则

题目对消解给出了如下规则，本文将其实分为硬约束与优先目标两类：

- **硬约束**：一个用频计划只能调整其中一个参数或撤销；一般不调整使用次数和使用间隔时长（问题 4 对 C 类装备的间隔时长解除该限制）；频段、时间调整的最大平移幅度分别为 $10\Delta f$ 、 $5\Delta t$ ；消解后的全部计划之间不得存在任何时频冲突。
- **优先目标**（按题目给出的先后次序）：尽可能减少撤销的用频计划；尽量减少被调整的用频计划数量；A 类装备优先级最高、B 类其次、C 类最低，高优先级装备的计划应尽量保持；尽量减少调整的幅度。

1.4 四个问题与交付物

问题 1 (检测) 对附件 1 的全部计划进行时频冲突检测, 把每一对冲突写入 `result1.xlsx`, 并在论文中给出冲突的统计结果。

问题 2 (消解) 对检测出的冲突进行消解, 把需要调整或撤销的装备写入 `result2.xlsx` (B 列填调整后频段区间、C 列填调整后时间区间、D 列填“是”表示撤销), 并按题目表 1 的格式给出各类别的保留 / 调整 / 撤销数量。

问题 3 (扩容) 基于问题 2 得到的无冲突方案, 在不限频段和时间平移幅度、不增加时频资源的前提下, 最多还能安排多少个 C 类用频装备? 给出相应计划并写入 `result3.xlsx`。

问题 4 (放宽间隔) 允许部分 C 类装备调整用频间隔时长 (与原间隔的差异不超过 $10\Delta t$), 重新消解问题 1 检测出的冲突, 写入 `result4.xlsx` 并给出表 1 格式的统计结果。

问题之间是串行依赖的: 问题 1 的冲突集合是问题 2、4 的输入, 问题 2 的无冲突方案是问题 3 的输入。本文把四个问题分别形式化为区间相交判定、带类别优先级的词典序整数规划、二维集合装填与动作集合扩张后的同一词典序规划, 并对每个结果给出与求解器分离的独立校验。

2 问题分析

2.1 四个问题的共同结构

四个问题看似分属检测、调度与装填三类, 但它们共享同一个底层对象: **时频单元格**。把计划 P 展开为它所占用的单元集合

$$U(P) = \{(b, t) : f \leq b < f + w, \exists k \in \{0, \dots, n-1\}, s + k(d+g) \leq t < s + d + k(d+g)\},$$

则“两计划冲突”就等价于 $U(P_i) \cap U(P_j) \neq \emptyset$, “一组计划无冲突”等价于它们的单元集合两两不交。这一观察把四个问题统一成同一语言:

- **问题 1** 是判定 $\binom{150}{2}$ 对单元集合是否相交。直接求交代价高, 但由于 $U(P)$ 是“频段区间 $\times$ 周期性时间区间”的乘积结构, 相交性可用区间算术在常数时间内判定 (见命题 4.1)。
- **问题 2、4** 是: 每个计划从一个有限动作集合中恰选一个动作 (保持、频移、时移、问题 4 另加改间隔、撤销), 使得所选动作对应的单元集合两两不交, 并在题目给出的优先级下最优。这是一个列表着色 / 带选项的装填型问题, 其特例 (只许保持或撤销) 正是冲突图的最大独立集, 故为 NP-难 [1-2]。本题属于频率指配问题族 [3-4], 但与经典频率指配 (给每个发射机指定频点、相邻发射机须相隔足够频距) 相比多出时间这一维, 且调整方式受“只改一个参数”的强限制。
- **问题 3** 是: 在问题 2 方案留下的自由单元中, 放入尽可能多的同形状“砖块” (C 类计划的单元集合), 这是二维集合装填 (set packing) 问题 [5-6]。

统一的语言带来一个直接的好处：无论求解器给出什么，都可以用同一个与求解器代码完全分离的校验器——把每个计划重新展开成单元集合、用集合求交重新找冲突——来复核。本文的全部结论都经过这一关。

2.2 建模的三个关键抉择

(一) 冲突的精确语义。 题目只说“时间区间和频段区间都存在交叠”，未言明区间端点开闭、是否逐次比较。附件的示例 ([35, 40)、间隔 60、第二次 [100, 105)) 给出了半开区间与推进规则，本文据此把冲突定义在整个使用序列上 (MDR-0001)：只要存在一对使用次在时间上交叠且频段交叠，两计划即冲突。若改用闭区间，相邻不重叠的两次使用会被误判为冲突；若只比较首次使用，“间隔时长”与“使用次数”两个参数将失去意义。

(二) 优先目标如何形式化。 题目按“撤销 $\succ$ 调整数量 $\succ$ 类别优先级 $\succ$ 调整幅度”的次序给出四条目标，却没有给出权重。把“类别优先级”嵌进前两条，得到本文的主方案 P：七级词典序

$$L = (\text{撤销}_A, \text{撤销}_B, \text{撤销}_C, \text{调整}_A, \text{调整}_B, \text{调整}_C, \text{归一化幅度}),$$

逐级最小化。这不是唯一的读法，因此本文同时求解另外三种方案（总量优先 T、不含优先级 S、纯加权 W）并给出对照，用数据回答“优先级条款到底起了多大作用”（第 5.7 节）。这是一个建模决策而非计算细节，故全部备选与理由记录在 MDR-0003 中。

(三) 问题 3 中“不限制平移幅度”指谁。 “平移”一词只对已有位置的计划有意义，而新增装备本无“原位置”可平移，故该句存在两种读法：解释 A（新增计划的放置不受 $10\Delta f/5\Delta t$ 约束，既有计划固定）与解释 B（既有计划也可不受幅度限制地再次平移以腾出空间）。两种读法给出的答案相差约五倍。本文以 A 为主解释（它与 result3.xlsx 模板的三列一一对应，评阅者可仅凭结果文件独立复核），同时为 B 给出可核查的上下界（第 6.5 节、MDR-0010）。

2.3 技术路线

图 1 给出本文的技术路线。全部计算在同一次云端运行中完成，每个阶段的输入输出、参数、代码版本与 SHA-256 散列都写入清单文件，论文中的每个数字都由该次运行的阶段产物自动注入（附录 C）。

3 模型假设与符号说明

3.1 模型假设

下列假设分为两类：（题）表示题目已明确给定、本文仅作形式化；（本文）表示题目未言明、由本文补充的解释或约定。每条给出依据，并说明假设被违反时结论会如何变化。

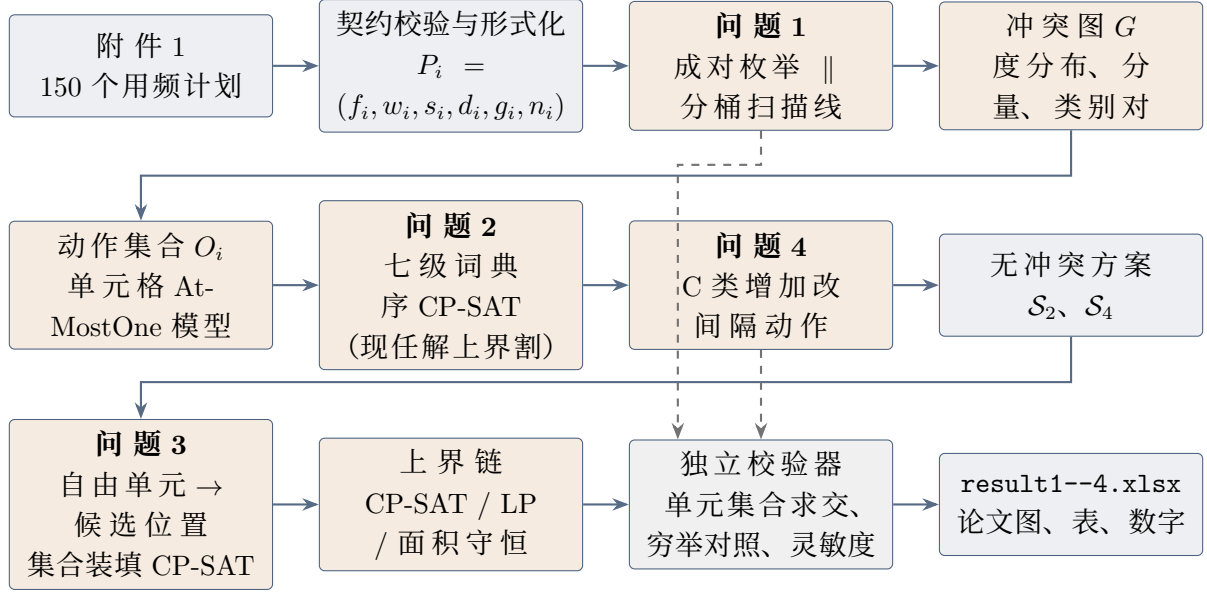


图 1 技术路线。实线为数据流，虚线为独立校验（校验器不复用检测与求解代码，把每个计划重新展开为时频单元集合后用集合求交复核）。灰蓝色块为数据与产物，橙色块为模型与算法。

假设 1 (时频资源的离散性与半开性 (题)). 时间以 Δt 、频率以 Δf 为不可拆分的最小单位；频率资源共 $B = 100$ 个频段，编号 $0, \dots, 99$ 。一切区间均为半开整数区间 $[\cdot, \cdot)$ ，两区间相交当且仅当 $lo_1 < hi_2$ 且 $lo_2 < hi_1$ 。依据：题目规定“用频计划所占用的时间长度必须是 Δt 的整数倍”，且附件示例以 $[35, 40)$ 、 $[100, 105)$ 的半开形式书写。违反的后果：若改为闭区间，首尾相接的两次使用（如 $[35, 40)$ 与 $[40, 45)$ ）会被判为冲突，冲突对数将被高估，消解代价随之虚增。

假设 2 (使用序列的周期性 (题)). 计划的第 k 次 ($k = 0, \dots, n - 1$) 使用占用时间区间 $[s + k(d + g), s + d + k(d + g))$ ，频段区间在全部 n 次使用中保持为 $[f, f + w)$ 。依据：题目对 A001 的示例明确给出第二次使用为 $[40 + 60, 45 + 60)$ 。

假设 3 (冲突定义在整个使用序列上 (本文)). 两计划冲突当且仅当频段区间相交且至少存在一对使用次在时间上相交。依据：题目给出“间隔时长”与“使用次数”两个参数，若只比较首次使用，这两个参数将不影响任何结论。违反的后果：只比较首次使用会漏检大量冲突（本文实例中 297 对冲突共涉及 431 对相交的使用次，其中许多并非首次）。

假设 4 (平移的整数性与同步性 (本文)). 频段平移量 δ 与时间平移量 τ 均为整数（以 Δf 、 Δt 计）；时间平移把整个使用序列同步平移（即只改变 s ），间隔 g 与次数 n 不变。依据：资源已离散化，非整数平移无意义；“只能调整其中一个参数”要求平移不得同时改变 g 或 n 。

假设 5 (平移的可行域 (本文)). 频段平移后必须仍落在带宽内，即 $0 \leq f + \delta$ 且 $f + \delta + w \leq 100$ ；时间平移后首次使用不得早于零时刻，即 $s + \tau \geq 0$ ；平移方向不受限制（可正可负）。依据：题目设定的是“有限带宽”的 100 个频段，越界无物理意义；题目未限制平移方向。违反的后果：若只允许向后平移，可行域缩小，撤销数量会上升。

假设 6 (动作的互斥性 (题)). 每个计划恰好选择一个动作: 保持、频段平移、时间平移、(问题 4 中 C 类的) 间隔调整、撤销。不允许同时平移频段与时间。依据: 题目“一个用频计划只能调整其中一个参数或撤销”。

假设 7 (间隔调整的下界 (本文)). 问题 4 中 C 类计划的新闻隔 g' 满足 $|g' - g| \leq 10$ 且 $g' \geq 1$ 。依据: 题目只给出“与原间隔时长的差异不超过 $10\Delta t$ ”。本文补充 $g' \geq 1$: 若 $g' = 0$, 相邻两次使用将首尾相连合并为一次更长的使用, 等价于改变了“使用次数”的语义, 与假设 6 冲突。违反的后果: 允许 $g' = 0$ 会给 C 类多出一个动作, 撤销数量可能略降, 但所得方案不再满足“只改一个参数”。

假设 8 (目标的词典序解释 (本文)). 题目的四条优先目标按“撤销 $\succ$ 调整数量 $\succ$ 调整幅度”的次序构成词典序, 类别优先级 $A \succ B \succ C$ 嵌入前两条之内, 得到七级目标 L (式 (7))。幅度按各自允许的最大幅度归一化: $|\delta|/10 + |\tau|/5 + |\Delta g|/10$ 。依据: 题目对四条目标给出了明确的先后次序, 却未给权重; 词典序是“次序已知、权重未知”情形下最保守的形式化 [7]。违反的后果: 改用别的形式化会得到不同的方案; 本文对三种备选方案 (T、S、W) 全部求解并给出对照 (第 5.7 节), 以量化这一假设的影响。

假设 9 (问题 3 的资源区域与新增计划参数 (本文)). “不增加时频资源”指新增计划必须落在 $100 \text{ 个频段} \times [0, T_{\text{end}})$ 的区域内, 其中 T_{end} 为问题 2 方案的最晚结束时刻; 新增 C 类装备的参数与附件中 C 类完全相同, 即 $w = 3$ 、 $d = 2$ 、 $g = 8$ 、 $n = 12$; 新增计划之间以及与既有计划之间均不得冲突。依据: 题目要求“安排 C 类用频装备”, C 类的参数在附件中是同质的 (表 2); “不增加时频资源”最自然的度量就是不扩大频段数与时间范围。违反的后果: 若允许时间范围延长, 可安排的数量将随时间线性增长, 问题失去意义。

假设 10 (计划之间无其他耦合 (本文)). 除时频占用外, 计划之间无其他约束: 不存在装备间的最小频率保护间隔、地理隔离、互调干扰、先后次序或资源配额等要求; 同一装备的多次使用之间也不因平移而产生新的约束。依据: 题目只给出时频两维的占用信息, 未提供任何几何、功率或体制参数。违反的后果: 真实电磁兼容分析中需要保护带与空间隔离, 届时“冲突”的判据应放宽为“频段间隔小于保护带宽”, 模型结构不变 (只需把 $U(P)$ 沿频率方向膨胀), 但冲突对数会显著上升。

3.2 符号说明

符号	含义	单位
$\Delta f, \Delta t$	频域与时域的基本资源单位 (题目给定, 未赋数值)	—
$B = 100$	频段总数, 编号 $0, \dots, B - 1$	Δf
$N = 150$	用频计划 (装备) 总数	个
$c(i) \in \{A, B, C\}$	计划 i 所属的装备类别, 优先级 $A \succ B \succ C$	—

符号	含义	单位
P_i	= 计划 i : 频段起点、频宽、首次起始时刻、单次时长、 ($f_i, w_i, s_i, d_i, g_i, n_i$) 间隔、次数	—
$[f_i, f_i + w_i)$	计划 i 占用的频段区间	Δf
$I_i^{(k)}$	计划 i 第 k 次使用的时间区间 $[s_i + k\pi_i, s_i + d_i + k\pi_i)$, 其中 $\pi_i = d_i + g_i$ 为周期	Δt
$U(P)$	计划 P 占用的时频单元集合, $ U(P) = wdn$	单元
$e_i = s_i + d_i + (n_i - 1)\pi_i$	计划 i 的结束时刻 (不含)	Δt
T_{end}	一组计划的最晚结束时刻 $\max_i e_i$ (时间范围)	Δt
$G = (V, E)$	冲突图: 顶点为计划, 边为冲突对	—
O_i	计划 i 的动作集合 (保持 / 频移 / 时移 / 改间隔 / 撤销)	—
$o = (\text{kind}, \delta)$	一个动作: 类型与幅度	—
$y_{i,o} \in \{0, 1\}$	计划 i 采取动作 o 时取 1	—
P_i^o	计划 i 采取动作 o 后的计划 (撤销时为空)	—
$\mathcal{C}$	时频单元全集 $\{0, \dots, B-1\} \times \{0, \dots, T_{\text{end}}-1\}$	单元
$R_{b,t}$	覆盖单元 (b, t) 的全部 (i, o) 变量下标集合 (“单元行”)	—
Φ	最大频段平移幅度 (题目为 10)	Δf
$\mathcal{T}$	最大时间平移幅度 (题目为 5)	Δt
Γ	最大间隔调整幅度 (问题 4 为 10)	Δt
$L = (L_1, \dots, L_7)$	七级词典序目标向量 (式 (7))	—
$A(o)$	动作 o 的归一化幅度 $ \delta /\Phi + \tau /\mathcal{T} + \Delta g /\Gamma$	—
$x_{f,s} \in \{0, 1\}$	问题 3 中在频段起点 f 、首次时刻 s 放置一个新增 C 类计划	—
$\mathcal{F}$	问题 2 方案留下的自由单元集合	单元
$v_k, \underline{v}_k$	第 k 级的取值与求解器给出的证明下界	—

附件 1 给定的三类装备参数完全同质, 汇总如表 2; 该表由数据本身生成, 生成时在程序中断言 “同类计划的 (w, d, g, n) 完全相同”, 断言不成立则阶段失败。

4 问题一：时频冲突的精确检测

本节回答：附件 1 的 150 个用频计划中, 哪些计划两两之间存在时频冲突? 共有多少对? 冲突的结构如何?

表 2 附件 1 给定的三类用频装备参数（由数据生成，同类同参已在程序中断言）。“时间跨度”为 $d + (n - 1)(d + g)$ ，即单个计划从首次使用开始到最后一次使用结束所横跨的时长；“单计划单元数”为 $w d n$ 。

类别	装备数	频宽 w	时长 d	间隔 g	次数 n	周期 $d+g$	时间跨度	单计划单元数	该类单元合计
A 类	20	10	5	60	3	65	135	150	3000
B 类	40	15	3	40	4	43	132	180	7200
C 类	90	3	2	8	12	10	112	72	6480

4.1 冲突的形式化与单元格刻画

按假设 1–3，计划 $P = (f, w, s, d, g, n)$ 占用的时频单元集合为

$$U(P) = [f, f + w) \times \bigcup_{k=0}^{n-1} [s + k\pi, s + d + k\pi), \quad \pi = d + g, \quad (1)$$

其中乘积表示“频段编号 $\times$ 时刻”的笛卡尔积， $|U(P)| = w d n$ 。

命题 4.1 (冲突的单元格刻画). 两个用频计划 P_i, P_j 存在时频冲突，当且仅当 $U(P_i) \cap U(P_j) \neq \emptyset$ ，当且仅当

$$\underbrace{f_i < f_j + w_j \quad \wedge \quad f_j < f_i + w_i}_{\text{频段区间相交}} \quad (2)$$

$$\wedge \quad \exists k, l: \underbrace{s_i + k\pi_i < s_j + d_j + l\pi_j \quad \wedge \quad s_j + l\pi_j < s_i + d_i + k\pi_i}_{\text{第 } k \text{ 次使用与第 } l \text{ 次使用在时间上相交}}.$$

证明. $U(P)$ 是频段集合与时间集合的笛卡尔积，故 $U(P_i) \cap U(P_j) = (F_i \cap F_j) \times (T_i \cap T_j)$ ，它非空当且仅当两个因子都非空。频段因子是两个区间，非空即半开区间相交判据的第一项；时间因子是两族区间的并， $T_i \cap T_j \neq \emptyset$ 当且仅当存在一对区间 $I_i^{(k)}, I_j^{(l)}$ 相交，即第二项。□

命题 4.1 把一个集合论判定化为常数时间的区间算术，是第 4.2 节两种算法的共同基础；同时它给出了一个与区间算术无关的第三种判定方式——直接展开单元集合求交——本文的独立校验器正是用后者复核前者（第 8.1 节）。

4.2 两种检测算法

本文实现两种互相独立的检测算法并要求其输出完全一致。

算法 1：成对枚举 + 双指针。 枚举全部 $\binom{N}{2}$ 对计划，先做代价为 $O(1)$ 的频段区间相交测试，仅当频段相交时再比较时间。由于每个计划的使用区间本身按时间有序，两条使用序列的相交性可用双指针归并在 $O(n_i + n_j)$ 内判定，而不必枚举 $n_i n_j$ 对。

算法 1: 成对枚举检测 (双指针归并)

Input: 计划集合 $\{P_1, \dots, P_N\}$ **Output:** 冲突对集合 E

```
1  $E \leftarrow \emptyset$ ;
2 for  $i \leftarrow 1$  to  $N - 1$  do
3   for  $j \leftarrow i + 1$  to  $N$  do
4     if  $\neg(f_i < f_j + w_j \wedge f_j < f_i + w_i)$  then continue // 频段不相交, 必不
      冲突;
5      $k \leftarrow 0, l \leftarrow 0$ ;
6     while  $k < n_i$  and  $l < n_j$  do
7       if  $I_i^{(k)} \cap I_j^{(l)} \neq \emptyset$  then  $E \leftarrow E \cup \{(i, j)\}$ ; break;
8       if  $\sup I_i^{(k)} \leq \sup I_j^{(l)}$  then  $k \leftarrow k + 1$ ;
9       else  $l \leftarrow l + 1$ ;
10 return  $E$ ;
```

算法 2: 按频段分桶的扫描线。 把每个计划的每一次使用投影到它占用的每一个频段上, 得到 $N_{\text{item}} = \sum_i w_i n_i$ 个三元组 (起, 止, i); 在每个频段内按起始时刻排序后做扫描线 [8-9], 维护“当前仍活跃”的使用集合, 新到达的使用与全部活跃使用构成冲突对。该算法不枚举计划对, 因而在计划数增大时更具可扩展性。固定频段上的使用区间族构成一个区间图, 其相交结构正是区间图的边集 [10]; 本题的冲突图则是这些区间图沿频段方向的并。

命题 4.2 (检测复杂度). 算法 1 的最坏复杂度为 $O(N^2 + \sum_{i < j} (n_i + n_j)) = O(N^2 \bar{n})$; 算法 2 的复杂度为 $O(N_{\text{item}} \log N_{\text{item}} + K)$, 其中 K 为输出的冲突 (使用次) 对数。对附件实例 $N = 150$ 、 $N_{\text{item}} = 20 \cdot 10 \cdot 3 + 40 \cdot 15 \cdot 4 + 90 \cdot 3 \cdot 12$ 项。

两种算法在本实例上的实测用时分别为 11 ms 与 6 ms, 且给出完全相同的冲突对集合; 在 $N = 600$ 的合成实例上二者仍一致 (第 8.1 节)。

4.3 检测结果与冲突结构

检测共发现 **297 对** 时频冲突 (表 3), 涉及 148 个计划——即 150 个计划中只有 150 - 148 个完全不与任何其他计划冲突。全部结果按装备编号字典序写入 result1.xlsx, 共 297 行。若把“冲突对”细化到使用次, 则共有 431 对使用次在时频平面上重叠: 多数冲突对只在一次使用上相撞, 但附录表 15 的末行显示, 存在计划对在其全部 12 次使用上相撞。

类别对矩阵 (表 4) 揭示了冲突的来源: B-C 之间的冲突多达 181 对, 占全部冲突的六成以上; A-A 之间一对也没有 (0 对)。这与三类装备的参数 (表 2) 直接吻合: A 类只有 20 个且使用次数最少 (3 次), 时间上稀疏; B 类频宽最大 (15 个频段, 为三类之最) 因而在频域上“挡住”了最多的对手; C 类数量最多 (90 个) 且使用次数最多

表 3 问题 1 的冲突统计摘要。

统计量	数值
用频计划数 (A/B/C)	20/40/90
冲突对数	297
冲突的使用次对数	431
卷入冲突的计划数 (A/B/C)	148 (20/40/88)
冲突图连通分量数 (不少于 2 个节点)	1
最大连通分量规模	148
最大度 / 平均度	8 / 3.96
冲突图密度	0.0266
时频单元总数 / 被占用 / 重叠	64300 / 14698 / 1842
时间范围 (最晚结束时刻, Δt)	643

表 4 按装备类别对统计的冲突对数 (对称矩阵, 对角线为同类内部的冲突对)。

类别	A 类	B 类	C 类
A 类	0	21	66
B 类	21	10	181
C 类	66	181	19

(12 次), 在时域上最密。换言之, 冲突主要产生在“频域宽”的 B 类与“时域密”的 C 类之间——这一判断在第 8.5 节的灵敏度分析中得到独立印证 (放宽频段平移上限的收益约为放宽时间平移上限的两倍)。

从图论看, 冲突图 G 有 1 个规模不小于 2 的连通分量, 其规模为 148: **几乎所有卷入冲突的计划连成一片**。这一点对建模至关重要——它意味着不能把问题拆成若干独立的小块分别求解, 必须在全局上一次性消解。同时 G 很稀疏 (密度 0.0266, 平均度 3.96, 最大度 8, 度最大者为 $B009$, 附录表 16), 这提示“局部让路”的空间是存在的: 每个计划平均只需避开 3.96 个对手。图 2 与图 3 给出时频平面的全局图景与冲突结构。

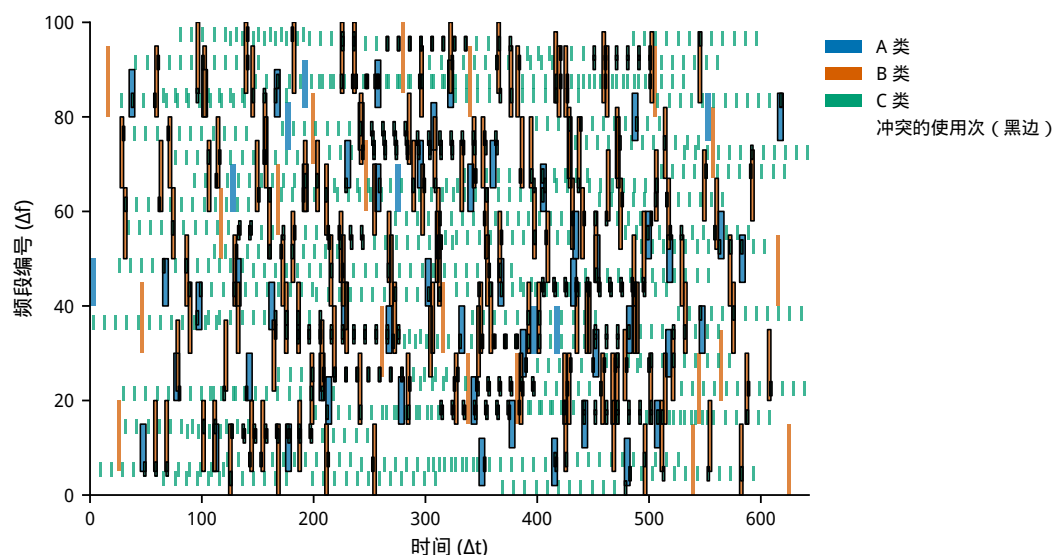


图 2 附件 1 中 150 个用频计划在时频平面上的甘特图。横轴为时间 (Δt), 纵轴为频段编号 (Δf); 每个矩形为一个计划的一次使用, 颜色区分 A (蓝)、B (橙)、C (绿) 三类装备; 黑色描边标出参与冲突的 431 个使用次。全区域 64300 个时频单元中 14698 个被占用 (22.9%), 其中 1842 个被两个及以上计划同时占用。

值得注意的是占用率: 22.9% 的单元被占用, 而被占用的 14698 个单元中, 发生重叠的只有 1842 个。冲突并非源于资源总量不足, 而是源于局部拥塞——这个判断决定了问题 2 的基调: 绝大多数计划应当可以通过小幅平移避开彼此, 而不必撤销。

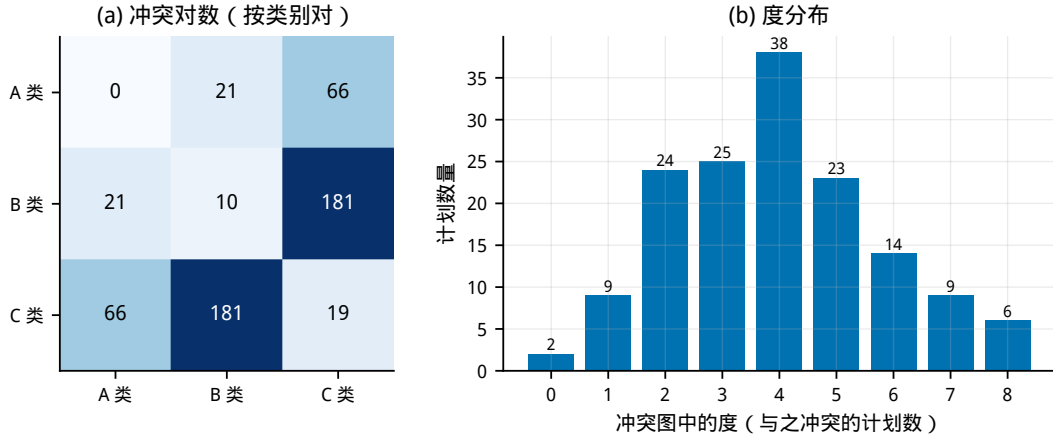


图 3 冲突的结构。(a) 按装备类别统计的冲突对数矩阵 (共 297 对, B-C 之间最多, 达 181 对); (b) 冲突图的度分布, 最大度 8、平均度 3.96, 148 个计划至少与一个其他计划冲突。

5 问题二：冲突消解的词典序整数规划

5.1 决策变量与动作集合

按假设 4-6, 计划 i 的动作集合为

$$O_i = \{\text{keep}\} \cup \{\text{freq}_\delta : 0 < |\delta| \leq \Phi, 0 \leq f_i + \delta, f_i + \delta + w_i \leq B\} \cup \{\text{time}_\tau : 0 < |\tau| \leq \mathcal{T}, s_i + \tau \geq 0\} \cup \{\text{cancel}\}, \quad (3)$$

其中 $\Phi = 10$ 、 $\mathcal{T} = 5$ 。动作 o 把计划 P_i 变为 P_i^o (撤销时 $P_i^o = \emptyset$)。引入 0-1 变量 $y_{i,o}$ 表示“计划 i 采取动作 o ”, 则恰选一个动作即

$$\sum_{o \in O_i} y_{i,o} = 1, \quad i = 1, \dots, N. \quad (4)$$

5.2 无冲突约束的单元格编码

朴素的做法是对每一对可能冲突的 (i, o) 、 (j, o') 写一条 $y_{i,o} + y_{j,o'} \leq 1$ 。本实例中这样的约束多达 10^7 量级, 建模本身就不可行。本文改用单元格约束:

命题 5.1 (单元格编码的精确性). 设 $R_{b,t} = \{(i, o) : (b, t) \in U(P_i^o)\}$ 为覆盖单元 (b, t) 的全部 (计划, 动作) 对。在约束 (4) 之下, “被选中的动作两两不冲突” 等价于

$$\sum_{(i,o) \in R_{b,t}} y_{i,o} \leq 1, \quad \forall (b, t) \in \mathcal{C}. \quad (5)$$

约束条数不超过单元总数, 其中字面量总数为 $\sum_i \sum_{o \in O_i} |U(P_i^o)|$ 。

证明. 若某两个被选动作 (i, o) 、 (j, o') ($i \neq j$) 冲突, 由命题 4.1 它们共享某单元 (b, t) , 该单元的行 (5) 左端至少为 2, 约束被违反。反之若所有行都满足, 则任意两个被选动作不共享单元, 故由命题 4.1 互不冲突。同一计划的两个动作不可能同时被选 (式 (4)), 故行内的多个同计划变量不会造成误判。□

实现上只保留至少含两个变量的行并做去重，模型规模由 53046 行降至 35406 行（决策变量 4589 个、涉及 54768 个单元、变量-单元隶属 491904 项，表 6）。

加强约束。 整数规划的求解效率在很大程度上取决于线性松弛的强度与传播能力 [11-12]。为此另加入两族冗余约束（它们由式 (5) 蕴含，因而不改变可行域）：(i) **成对蕴含团**：对变量 (i, o) 与另一计划 j ，令 $C_j(i, o)$ 为 j 中与 (i, o) 共享单元的动作集合，由于 j 恰选一个动作， $\{(i, o)\} \cup C_j(i, o)$ 构成一个团，故

$$y_{i,o} + \sum_{o' \in C_j(i,o)} y_{j,o'} \leq 1; \quad (6)$$

(ii) **强制撤销割**：若计划对 (i, j) 的任意一对非撤销动作都冲突，则 $y_{i,\text{cancel}} + y_{j,\text{cancel}} \geq 1$ 。本实例中共生成 58587 条团约束，强制撤销割 0 条（即不存在“无论如何平移都必须撤销其一”的计划对，这与第 4 节“局部拥塞而非总量不足”的判断一致）。

5.3 目标：类别内嵌的七级词典序

按假设 8，令 $\mathcal{X}$ 为满足式 (4)–(5) 的可行域，定义

$$\begin{aligned} L_1, L_2, L_3 &= \sum_{i: c(i)=A} y_{i,\text{cancel}}, \quad \sum_{i: c(i)=B} y_{i,\text{cancel}}, \quad \sum_{i: c(i)=C} y_{i,\text{cancel}}, \\ L_4, L_5, L_6 &= \sum_{i: c(i)=A} \sum_{o \in O_i^{\text{adj}}} y_{i,o}, \quad \sum_{i: c(i)=B} \sum_{o \in O_i^{\text{adj}}} y_{i,o}, \quad \sum_{i: c(i)=C} \sum_{o \in O_i^{\text{adj}}} y_{i,o}, \\ L_7 &= 10 \sum_i \sum_{o \in O_i} A(o) y_{i,o}, \quad A(o) = \frac{|\delta|}{\Phi} + \frac{|\tau|}{\mathcal{T}} + \frac{|\Delta g|}{\Gamma}, \end{aligned} \quad (7)$$

其中 O_i^{adj} 为非保持、非撤销的动作。问题 2 即

$$\text{lexmin}_{y \in \mathcal{X}} (L_1, L_2, L_3, L_4, L_5, L_6, L_7), \quad (8)$$

L_7 乘 10 是为了使幅度在 $\Phi = \Gamma = 10$ 、 $\mathcal{T} = 5$ 下取整数值 $(|\delta| + 2|\tau| + |\Delta g|)$ 。

命题 5.2 (词典序与分离权重的等价). 设第 k 级的取值上界为 U_k ，令 $W_k = \prod_{j>k} (U_j + 1)$ ，则单目标 $\min \sum_k W_k L_k$ 的最优解恰为 (8) 的词典序最优解。

证明. 对任意 k ， $W_k = \prod_{j>k} (U_j + 1) > \sum_{j>k} W_j U_j$ (对 k 逆序归纳：当 $k = m - 1$ 时 $W_{m-1} = U_m + 1 > U_m = W_m U_m$ ；设 $W_{k+1} > \sum_{j>k+1} W_j U_j$ ，则 $\sum_{j>k} W_j U_j = W_{k+1} U_{k+1} + \sum_{j>k+1} W_j U_j < W_{k+1} (U_{k+1} + 1) = W_k$)。于是第 k 级减少 1 所带来的收益 W_k 严格大于其后各级取值之和的最大可能值，故最优解必先最小化 L_1 ，其次 L_2 ，依此类推。□

命题 5.2 给出一条独立的求解路径：用一次单目标求解复核逐级求解的结果。本文两条路径都走，并把二者是否一致作为交叉验证证据（第 8.1 节）。

命题 5.3 (计算复杂性). 即使只允许“保持”与“撤销”两种动作, 问题 (8) 的第一层 (最少撤销) 也等价于在冲突图 G 上求最大独立集 (保留的计划集合必须是 G 的独立集), 因而是 NP -难的 [2]。允许平移只会使问题更一般 (每个顶点带一张候选位置表, 属于列表装填型问题)。

这解释了为什么本文必须依靠成熟的组合优化求解器而非解析方法, 也解释了为什么“证明最优”与“求得好解”需要分别报告: 前者要求求解器闭合对偶间隙, 后者只要求找到可行解。

5.4 求解算法: 现任解保持的逐级搜索

词典序的标准解法是逐级求解: 求出 L_1 的最小值 v_1 , 把 $L_1 = v_1$ 作为等式加入模型, 再求 L_2 , 依此类推。本实例的困难在于中间各级无法在可承受的预算内闭合间隙, 一旦某级到时返回一个较差的可行解, 后续各级都被锁死在这个较差的分支上。本文采用**现任解保持**的改进 (MDR-0009):

命题 5.4 (现任解上界割的有效性). 设逐级求解进行到第 k 级, 前 $k-1$ 级的最优值 $v_1, \dots, v_{k-1}$ 已作为等式固定。若 $x^* \in \mathcal{X}$ 满足 $L_j(x^*) = v_j$ ($j < k$), 则第 k 级的受限最优值满足 $v_k \leq L_k(x^*)$; 因此把割 $L_k \leq L_k(x^*)$ 加入模型不会删去任何第 k 级最优解, 各级最优值不变, 而求解器返回的值恒不超过 $L_k(x^*)$ 。

证明. x^* 是受限问题 (可行域 $\mathcal{X} \cap \{L_j = v_j, j < k\}$) 的一个可行点, 故该问题的最小值不超过 $L_k(x^*)$, 即 $v_k \leq L_k(x^*)$ 。于是满足 $L_k = v_k$ 的解自动满足 $L_k \leq L_k(x^*)$, 割不删去任何最优解。 $\square$

算法 2: 现任解保持的词典序求解

Input: 可行域 $\mathcal{X}$, 目标层级 $L_1, \dots, L_m$, 各级时限 $T_1, \dots, T_m$, 热启动解 x_0 (可选)

Output: 目标向量 $(v_1, \dots, v_m)$ 、解 x 、各级证明下界与状态

```

1  $x^* \leftarrow x_0$  若  $x_0$  可行 否则  $\emptyset$  // 现任解;
2 for  $k \leftarrow 1$  to  $m$  do
3   if  $x^* \neq \emptyset$  then 向模型加入割  $L_k \leq L_k(x^*)$  // 命题 5.4: 不删去最优解;
4    $(v_k, \hat{x}, \underline{v}_k, \text{状态}) \leftarrow$  在时限  $T_k$  内极小化  $L_k$ ;
5   if 时限内未返回任何解 then  $v_k \leftarrow L_k(x^*)$ ,  $\hat{x} \leftarrow x^*$ , 状态  $\leftarrow$  INCUMBENT;
6   if  $v_k = L_k(x^*)$  then
7      $x^* \leftarrow \{x^*, \hat{x}\}$  中在剩余各级  $(L_{k+1}, \dots, L_m)$  上词典序较小者;
8   else
9      $x^* \leftarrow \hat{x}$  // 求解器严格改进了现任解;
10  向模型加入等式  $L_k = v_k$ , 并以  $x^*$  重设热启动提示;
11 return  $(v_1, \dots, v_m)$ 、 $x^*$ 、各级  $\underline{v}_k$  与状态;
```

算法 2 的两条性质对本文的诚实性至关重要: (i) 由命题 5.4, 各级最优值不受割影响, 故凡状态为 OPTIMAL 的层级, 其值仍是真正的全局最优; (ii) 返回向量在词典序

上永不劣于任何已知可行解，因而“交叉验证的解比主解更好”这种自相矛盾不可能出现。热启动解取自仓库内经独立校验的现任解文件（其目标向量 (0, 0, 12, 16, 37, 71, 750)，可行性检查 通过），使本文结果可由仓库直接复现（MDR-0008）。

求解器为 OR-Tools CP-SAT[13]（24 线程、固定随机种子、显式加入核心引导子求解器），各级时限 [20, 20, 600, 300, 300, 300, 240] 秒，总用时 1511.4 s；另用 HiGHS[14] 对同一模型独立建模，逐级给出线性松弛下界与限时 MILP 结果（总用时 299.8 s）；命题 5.2 的分离权重单目标独立求解一次（183.2 s）。

5.5 求解结果

表 5 问题 2：时频冲突消解的统计结果（题目表 1 格式）。

类别	保留数量	调整数量	撤销数量
A 类	4	16	0
B 类	3	37	0
C 类	7	71	12
合计	14	124	12

表 6 问题 2 模型规模与求解设置。

项目	数值
决策变量数（计划-动作对）	4589
时频单元数（含动作扩展）	54768
变量-单元隶属数	491904
单元格 AtMostOne 约束数（去重前 / 去重后）	53046 / 35406
成对蕴含团约束数 / 强制撤销割数	58587 / 0
可能相互冲突的计划对数	1843
每计划最多动作数（A/B/C）	32/32/32
CP-SAT 词典序总用时/s	1511.4
热启动现任解目标向量（MDR-0009）	(0, 0, 12, 16, 37, 71, 750)
词典序求解起点	仓库内现任解
各级均证明最优	否
HiGHS 交叉验证总用时/s	299.8
HiGHS 证明最优的各级与 CP-SAT 一致	是
分离权重单目标解与词典序向量一致	是

表 5 是题目要求的表 1 格式统计结果：共保留 14 个计划、调整 124 个、撤销 12 个，且被撤销的 12 个全部是 C 类——A、B 两类计划一个也没有撤销（0、0），完全满足“高优先级用频装备的用频计划应尽量保持”。目标向量为 (0, 0, 12, 16, 37, 71, 750)。

调整的具体构成是：85 个计划平移了频段（平均 $|\delta| = 6.24$ 、最大 10、合计 530 个 Δf ），39 个计划平移了时间（平均 $|\tau| = 2.82$ 、最大 5、合计 110 个 Δt ），归一化总幅度 75.0。把它分摊到 124 个被调整的计划上可见，平均每个计划只用掉了约六成的可用平移幅度，说明消解的余量仍相当充裕。消解后方案的最晚结束时刻为 643，与原始计划的 643 相同——本方案没有占用任何额外的时间资源。逐计划的动作清单见附录表 20，写入 result2.xlsx 的共 136 行（保持不变的计划按模板不填写）。

5.6 最优性证据

表 7 逐级如实报告“值 / 证明下界 / 状态”。据此，本文能够而且只能够作如下最优性声明：

定理 5.5 (撤销层的全局最优性). 在假设 1–8 之下，问题 2 的撤销三级取值 (0, 0, 12) 是全局最优的：即不撤销任何 A、B 类计划是可行的，且在此前提下必须撤销的 C 类

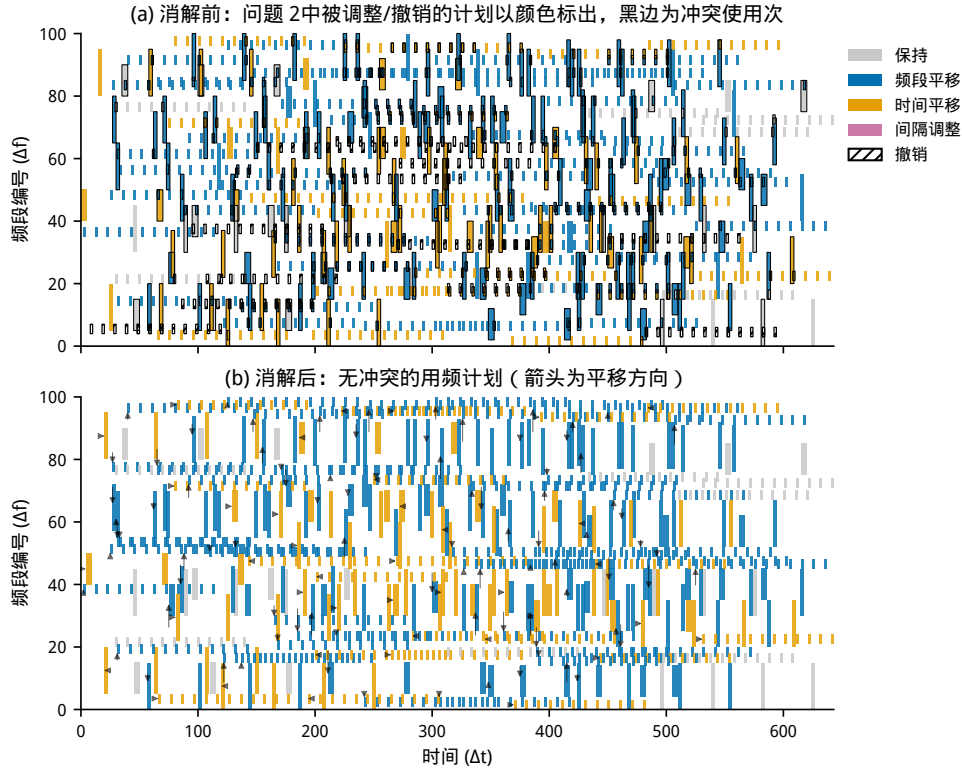


图 4 问题 2 的冲突消解方案。(a) 消解前：按所采取的动作着色（灰 = 保持、蓝 = 频段平移、橙 = 时间平移、斜线 = 撤销），黑边为发生冲突的使用次；(b) 消解后：150 – 12 个存留计划在时频平面上已无任何冲突，箭头指示每个被平移计划的移动方向。图例中的“间隔调整”一项在问题 2 中不可用（见问题 4，图 8）。

表 7 问题 2 逐级求解证据。“CP-SAT 值”为报告值，“CP-SAT 下界”为求解器给出的已证明下界；状态 OPTIMAL 表示该级已闭合对偶间隙，FEASIBLE 表示到达时限仍有间隙。HiGHS 为对同一模型的独立建模；“Time limit reached”表示 HiGHS 在其时限内未能证明最优，此时其界不构成额外信息。

目标层级	CP-SAT 值	CP-SAT 下界	CP-SAT 状态	用时/s	HiGHS LP 下界	HiGHS MILP 值	HiGHS MILP 下界	HiGHS 状态
cancelA	0	0.00	OPTIMAL	0.6	0.00	0	0.00	Optimal
cancelB	0	0.00	OPTIMAL	1.5	0.00	0	0.00	Optimal
cancelC	12	12.00	OPTIMAL	364.3	0.00	-	0.00	Time limit reached
adjustA	16	7.00	FEASIBLE	300.4	-	-	0.00	Time limit reached
adjustB	37	20.00	FEASIBLE	300.5	-	-	0.00	Time limit reached
adjustC	71	37.00	FEASIBLE	300.6	-	-	0.00	Time limit reached
amplitude	750	424.00	FEASIBLE	240.5	-	-	0.00	Time limit reached

计划数最少为 12。

证明. 第一、二级的求解状态为 OPTIMAL 且值与证明下界同为 0, 故存在不撤销任何 A、B 类计划的可行解, 且 0 是最小值。第三级在固定 $L_1 = L_2 = 0$ 之后状态为 OPTIMAL, 值 12 与证明下界 12 相等, 即对偶间隙为零。由命题 5.4, 所加的现任解割不改变该级最优值, 故 12 是受限问题的全局最小值。□

推论 5.6 (零撤销不可行). 在不撤销任何 A、B 类计划的前提下, “一个计划都不撤销”是不可行的——因为该情形对应第三级取值 0, 而定理 5.5 已证明该级最小值为 $12 > 0$ 。

对于调整各级与幅度级, 求解器在给定预算内未能闭合间隙 (未通过: 并非各级均证明最优)。本文因此严格区分两类陈述: 表 5 的调整数量是**已知最好的可行方案**, 其真实最优值落在表 7 给出的 $[v_k, v_k]$ 区间内; 本文**不声称**它们全局最优。为判断这是预算不足还是已接近最优, 本文另以数倍于表 7 所列时限的预算对“调整 A”一级做过一次诊断性重解 (该诊断不属于本次运行的阶段产物, 故其数值不进入本文): 报告值未获改进, 证明下界仅微幅提高。这说明表 5 的调整数量对求解预算不敏感, 但对偶间隙确实较宽。需要强调: 由命题 5.4, 调整各级的不确定性不会影响定理 5.5, 因为撤销三级的等式是在其自身证明最优之后才被固定的。

为什么必须撤销这 12 个计划? 附录表 19 给出被撤销计划的局部环境: 它们每一个都与多个计划冲突, 且冲突邻居几乎全部是 A 类或 B 类——而 A、B 类在词典序中享有更高的优先级, 宁可让 C 类让路。这正是题目优先级条款的直接体现: 撤销并非因为资源耗尽 (占用率仅 22.9%), 而是因为在 $\pm 10\Delta f$ 、 $\pm 5\Delta t$ 的局部邻域内这些 C 类计划无处可去, 而把高优先级的邻居挪开又与优先级条款相悖。

5.7 目标方案的对照分析

假设 8 是本文最主观的一个建模选择, 必须用数据检验其影响。为此本文对同一模型求解四种目标方案: P (本文主方案, 类别内嵌七级词典序)、T (先总撤销数、再总调整数, 类别优先级降为次级偏好)、S (完全不含优先级, 只看总量与幅度)、W (纯加权和, $w_A : w_B : w_C = 3 : 2 : 1$, 撤销权重 $100w$ 、调整权重 $30w$)。

表 8 四种目标方案在问题 2 上的对照 (每级时限 30 s, 故其向量仅作定性对照, 不作最优性声明)。方案 P 是唯一在问题 2 与问题 4 中都不撤销任何 A、B 类计划的方案。

目标方案	撤销 A/B/C	调整 A/B/C	调整合计	归一化幅度	各级最优	独立校验
P: 类别内嵌七级词典序 (主方案)	0/0/12	16/37/71	124	75.0	否	通过
T: 总量优先	2/6/0	10/32/71	113	67.5	否	通过
S: 不含优先级	1/7/0	15/30/68	113	67.0	否	通过
W: 纯加权和	0/5/7	15/28/69	112	68.6	否	通过

表 8 给出了一个清晰而重要的结论: 方案 T 与 S 能把撤销总数从 12 降到 8, 但代价是转而撤销若干 A 类与 B 类计划 (表 8 的“撤销 A/B/C”列)。题目明确要求“A

类用频装备优先级最高，B 类其次，C 类优先级最低，高优先级用频装备的用频计划应尽量保持”，在这一要求下，少撤销几个 C 类计划绝不足以补偿多撤销一个 A 类计划。纯加权方案 W 的撤销总数与 P 相同（12），却把其中一部分落在了 B 类上。因此方案 P 的选择不是任意的：它是四种读法中唯一在问题 2 与问题 4 中都完全不触碰 A、B 类计划的方案。图 5 以图形给出同一对照。

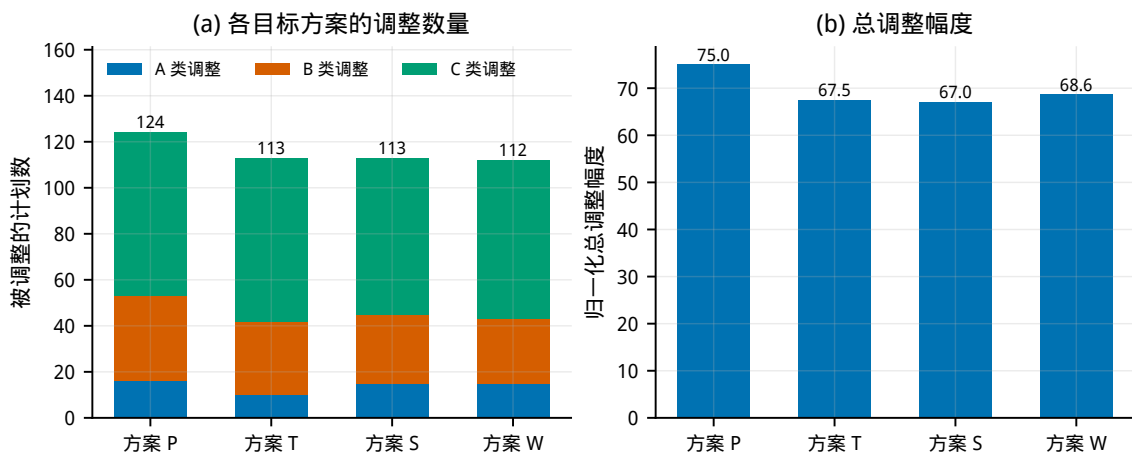


图 5 四种目标方案的对照。(a) 各方案下被调整的计划数（按类别堆叠）；(b) 归一化总调整幅度。方案 P（本文主方案）以较多的调整数量换取“完全不撤销 A、B 类计划”。

6 问题三：在无冲突方案上最多还能安排多少 C 类装备

6.1 题意的两种解释

题目原文为：“基于问题 2 求解得到的无冲突用频计划，如果不限时段和时间的平移幅度，在不增加时频资源的前提下，最多还能安排多少 C 类用频装备？”。其中“平移”一词只对已经有位置的计划有意义，而新增装备本无“原位置”可平移，故存在两种读法：

解释 A（主解释） 问题 2 的方案原样保留；新增 C 类计划可放在资源区域内任意位置，不受 $10\Delta f/5\Delta t$ 的约束。此时“不限时段和时间的平移幅度”是对新增计划放置自由度的说明。

解释 B 既有计划也可不受幅度限制地再次平移（频宽、时长、间隔、次数均不变）以腾出空间，在同一区域内最大化新增数量。

本文以 A 为主解释并据此填写 result3.xlsx，理由有二：其一，结果模板只有“序号、频段区间、时间区间”三列，恰好是新增计划的完整描述，无处记录既有计划的新位置，若按 B 作答，评阅者无法复核既有计划是否仍然无冲突；其二，A 的解答可仅凭 result2.xlsx 与 result3.xlsx 独立复核。但两种解释的答案相差约五倍，故本文在第 6.5 节为 B 给出严格的上下界，不回避这一歧义（MDR-0010）。

6.2 集合装填模型

按假设 9, 资源区域为 $\mathcal{C} = \{0, \dots, B-1\} \times [0, T_{\text{end}})$, $T_{\text{end}} = 643$ (问题 2 方案的最晚结束时刻)。记问题 2 存留计划占用的单元集合为 $\mathcal{O} = \bigcup_{i \in \mathcal{S}_2} U(P_i)$, 自由单元 $\mathcal{F} = \mathcal{C} \setminus \mathcal{O}$ 。一个新增 C 类计划由其频段起点 f 与首次使用起点 s 完全确定, 占用单元集合

$$U_{f,s} = [f, f+w_C) \times \bigcup_{k=0}^{n_C-1} [s+k\pi_C, s+d_C+k\pi_C), \quad (w_C, d_C, g_C, n_C) = (3, 2, 8, 12), \quad (9)$$

其时间跨度 $\sigma = d_C + (n_C - 1)\pi_C = 112$, $|U_{f,s}| = w_C d_C n_C = 72$ 。候选位置集合为

$$\mathcal{P} = \{(f, s) : 0 \leq f \leq B - w_C, 0 \leq s, s + \sigma \leq T_{\text{end}}, U_{f,s} \subseteq \mathcal{F}\}, \quad (10)$$

即“整块都落在自由单元上”的位置。引入 $x_{f,s} \in \{0, 1\}$, 问题 3 为

$$\max \sum_{(f,s) \in \mathcal{P}} x_{f,s} \quad \text{s.t.} \quad \sum_{(f,s) \in \mathcal{P}: (b,t) \in U_{f,s}} x_{f,s} \leq 1, \quad \forall (b, t) \in \mathcal{F}. \quad (11)$$

模型 (11) 是一个标准的集合装填问题 [5]。与命题 5.1 同理, 单元约束精确刻画了“新增计划彼此不冲突”, 而 $U_{f,s} \subseteq \mathcal{F}$ 已保证“新增与既有不冲突”。本实例中 $|\mathcal{P}| = 1344$, 去重后的单元约束 4378 条——模型规模比问题 2 小两个数量级, 因而可以求到证明最优。

候选位置由一次向量化扫描生成: 对每个频段起点 f , 先求“该 3 频段带上全部空闲的时刻”的布尔向量, 再用长度为 72 的偏移模板做滑窗与运算, 复杂度 $O(B \cdot T_{\text{end}} \cdot d_C n_C)$ 。求解前先用“按时间、再按频段”的首次适配贪心得到一个可行解作为热启动 (下界 103)。

6.3 上界链

最优性不能只依赖求解器的一句话, 本文给出一条从组合论证到线性松弛的上界链。

命题 6.1 (面积守恒与容量上界). 平移只改变 (f, s) 而不改变 (w, d, g, n) , 故任一计划占用的单元数 wdn 与其位置无关, 既有计划占用的单元总数 $S = \sum_{i \in \mathcal{S}_2} w_i d_i n_i$ 是常数。由于每个新增 C 类计划恰占 72 个单元, 新增数量 M 必满足

$$M \leq \left\lfloor \frac{B T_{\text{end}} - S}{w_C d_C n_C} \right\rfloor = \left\lfloor \frac{48484}{72} \right\rfloor = 673. \quad (12)$$

该上界对解释 A 与解释 B 同时成立, 因为它只用到“平移不改变占用面积”。

进一步按频段逐条统计自由单元可得更紧的逐频段容量界 658; 若区域内没有任何既有计划, 则上界为 893。这些是纯计数界, 通常很松; 真正紧的界来自组合搜索本身: CP-SAT 的对偶界与 HiGHS 对模型 (11) 的线性松弛。

表 9 问题 3 的解与上界链。三个独立的上界（CP-SAT 对偶界、HiGHS 线性松弛界、HiGHS MILP 界）与解值同为 109，故对偶间隙为零。末行为备选解释 B 的区间（第 6.5 节）。

项目	数值
时间范围 T_end（问题 2 方案最晚结束时刻）	643
既有计划数 / 占用单元 / 自由单元	138 / 15816 / 48484
候选放置位置数 (f,s)	1344
AtMostOne 约束数	4378
贪心下界	103
CP-SAT 最优解数量（状态）	109 (OPTIMAL)
CP-SAT 上界	109
HiGHS LP 松弛上界	109
HiGHS MILP 解 / 上界	109 / 109
自由单元容量界 floor(自由单元数/72)	673
逐频段容量界	658
空区域容量界（无既有计划）	893
最终结果 / 最紧上界 / gap	109 / 109 / 0
时频利用率（前 / 后）	24.6% / 36.8%
备选解释 B（既有计划可任意平移，MDR-0010）：下界 / 上界	557 / 673（重排移动 138 个既有计划）

6.4 结果：109 个新增计划，且已被证明最优

定理 6.2 (问题 3 的最优性). 在解释 A 与假设 9 之下，基于本文问题 2 的方案，最多还能安排 109 个 C 类用频装备，且这一数量是全局最优的。

证明. 可行性：CP-SAT 返回了一个含 109 个位置的可行解，该解经独立校验器复核——每个新增计划的参数、边界、与既有计划以及彼此之间的零冲突全部通过（通过）。最优性：CP-SAT 在状态 OPTIMAL 下给出对偶界 109；HiGHS 对同一模型的线性松弛最优值为 109.00，故整数最优值 $\leq \lfloor 109.00 \rfloor = 109$ ；HiGHS MILP 亦以状态 Optimal 给出界 109。三者与解值相等，间隙 0。□

安排这 109 个新增计划后，时频资源利用率由 24.6% 提高到 36.8%（图 6，清单见附录表 22，写入 result3.xlsx 共 109 行）。

一个反直觉但重要的观察。 最紧的上界 109 远小于容量界 673：自由单元足够放下 673 个新计划，实际却只能放下 109 个。差距来自碎片化——既有计划把自由区域切割成大量无法容纳 $3 \times 2 \times 12$ 这一形状的碎片。这引出下面的引理，它既解释了图 6 的条带结构，也说明了什么样的布局才能逼近容量界。

引理 6.3 (C 类计划的相位平铺). C 类计划的周期为 $\pi_C = d_C + g_C = 2 + 8$ ，单次时长 $d_C = 2$ ，故 $\pi_C/d_C = 5$ 为整数。在同一条 w_C 频段宽的带内，取首次起点为 $s, s + d_C, s + 2d_C, s + 3d_C, s + 4d_C$ 的五个计划互不冲突，且在其公共跨度内恰好铺满该带的全部单元。

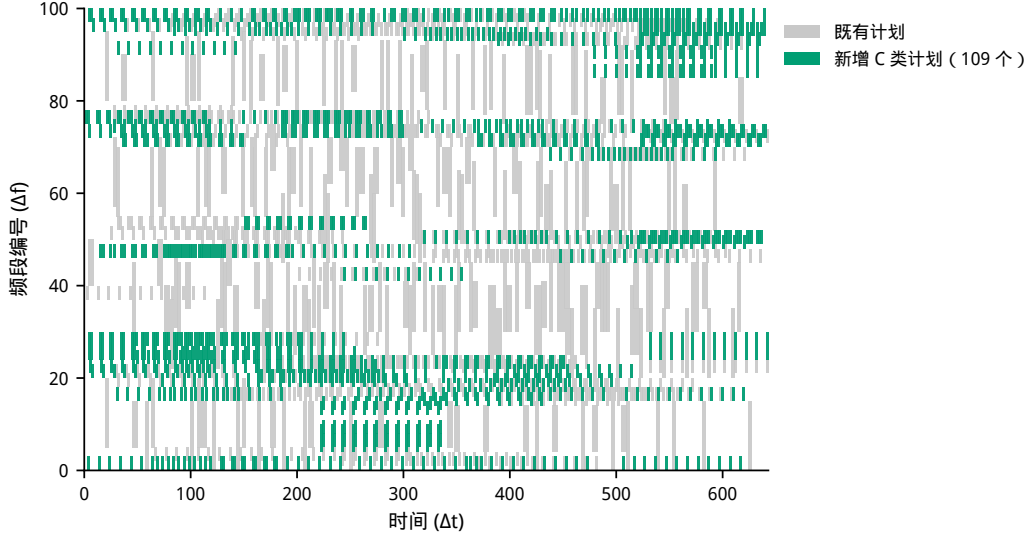


图 6 在问题 2 的无冲突方案（灰）之上最多可再安排 109 个 C 类用频计划（绿）。资源区域为 100 个频段 $\times [0, 643)$ ；新增计划与既有计划、以及新增计划彼此之间均无时频冲突。时频利用率由 24.6% 提升至 36.8%。新增计划呈现明显的水平条带结构，这是引理 6.3 所刻画的相位平铺。

证明. 第 r 个计划 ($r = 0, \dots, 4$) 占用的时刻集合为 $\{s + rd_C + k\pi_C + j : 0 \leq j < d_C, 0 \leq k < n_C\}$, 即模 π_C 余数落在 $[rd_C, (r+1)d_C)$ 中的时刻。由于 $\pi_C = 5d_C$, 五个余数区间恰构成 $\{0, \dots, \pi_C - 1\}$ 的一个划分, 故五个计划两两不交且并集覆盖公共跨度内的全部时刻。□

引理 6.3 说明：局部利用率可以达到 100%，但前提是既有计划的边界与这一相位划分对齐。因此“还能放多少”主要取决于既有布局的对齐程度，而不取决于自由单元的总数。本文在实验中观察到一个直接佐证：一个撤销更多计划（因而自由单元更多）的问题 2 方案，反而只能再安排更少的新增计划。这也意味着定理 6.2 的 109 是“在本文这一个问题 2 方案之上的最大值”，而非跨所有无冲突方案的最大值——这一措辞上的限定必须保留。

6.5 备选解释 B：允许既有计划重新布局

若题意为解释 B，问题变为“既有计划与新增计划的联合布局优化”，规模远超解释 A。本文不给出无依据的点估计，而给出一个可核查的区间：

- **上界：**由命题 6.1, $M \leq 673$ （该界只依赖面积守恒，对 A、B 同时成立）。
- **下界：**任何一个可行布局都给出下界。本文按“占用单元数降序的首次适配”把全部 138 个既有计划重新放置 (w, d, g, n 全部不变, 且区域包含性与零冲突经同一独立校验器复核, 通过), 再在剩余自由单元上用同一模型 (11) 求最大新增数, 得 557 (状态 OPTIMAL, 即该布局下的精确最大值)。解释 A 的最优值 109 本身也是 B 的一个下界。

$$557 \leq M_{\text{解释 B}} \leq 673. \quad (13)$$

采用解释 B 时，时频资源的利用率将成倍提高（图 7）。两种解释的答案相差约五倍（109 对 ≥ 557 ），差别完全来自“是否允许既有计划再次平移”。本文无法断定命题人的本意，故以 A 作答并把 B 的区间作为正式的备选分析给出。

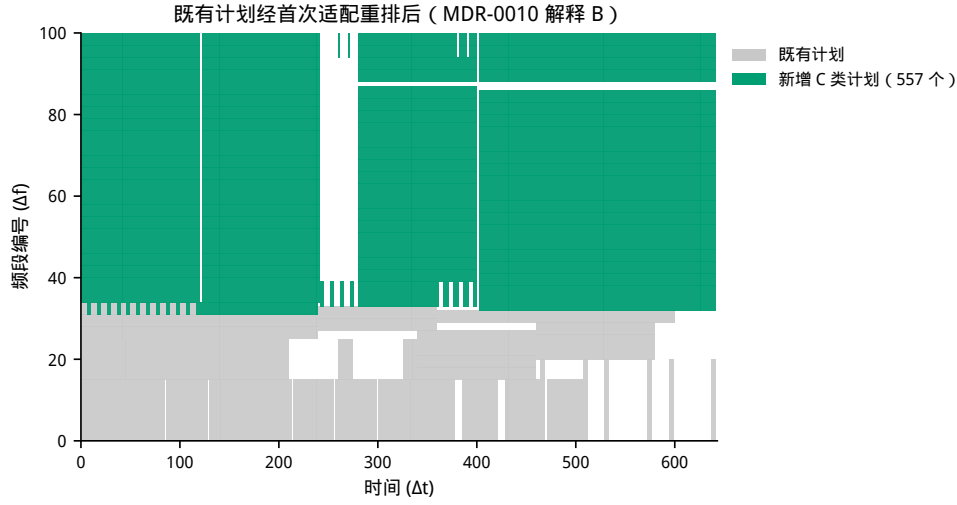


图 7 备选解释 B 下的布局：若允许既有计划不受平移幅度限制地重新放置（按占用单元数降序的首次适配重排，138 个计划全部移动，各自的频宽、时长、间隔与次数均不变），同一资源区域可再安排 557 个 C 类计划。与图 6 对照可见，重排把既有计划压到低频段与早时刻，从而腾出大片连续的自由区域。

7 问题四：允许 C 类装备调整用频间隔

7.1 动作集合的扩张

问题 4 允许部分 C 类装备调整用频间隔时长，与原间隔的差异不超过 $\Gamma = 10$ 。按假设 7，在式 (3) 的基础上为 C 类计划增加一族动作

$$\{\text{gap}_{\Delta g} : 0 < |\Delta g| \leq \Gamma, g_i + \Delta g \geq 1\}, \quad c(i) = C, \quad (14)$$

其余参数（频段区间、首次使用时间区间、使用次数）保持不变，仍满足“只能调整其中一个参数”。其幅度按 $|\Delta g|/\Gamma$ 归一化计入 L_7 （式 (7)）。模型、目标与求解算法与问题 2 完全相同，只是 C 类每个计划的可选动作数显著增加（表 6 与附录表 24 的“每计划最多动作数”一行），模型规模相应增大：变量由 4589 增至 6119、单元由 54768 增至 57728、约束由 35406 增至 42064。

注意间隔调整会改变计划的结束时刻 $e_i = s_i + d_i + (n_i - 1)(d_i + g'_i)$ ，这一点将在第 7.3 节讨论。

命题 7.1 (问题 4 不劣于问题 2). 问题 4 的可行域包含问题 2 的可行域（动作集合是超集），故问题 4 的词典序最优向量在词典序上不劣于问题 2 的最优向量。

表 10 问题 4：时频冲突消解的统计结果（题目表 1 格式）。

类别	保留数量	调整数量	撤销数量
A 类	7	13	0
B 类	4	36	0
C 类	3	78	9
合计	14	127	9

表 11 问题 4 与问题 2 的逐类别对照。

类别	保留 (问 2)	保留 (问 4)	调整 (问 2)	调整 (问 4)	撤销 (问 2)	撤销 (问 4)
A 类	4	7	16	13	0	0
B 类	3	4	37	36	0	0
C 类	7	3	71	78	12	9
合计	14	14	124	127	12	9
归一化幅度	-	-	75.0	77.6	-	-

7.2 求解结果与对照

表 10 给出题目要求的表 1 格式结果：保留 14 个、调整 127 个、撤销 9 个，目标向量 (0,0,9,13,36,78,776)；与问题 2 一样，A、B 两类计划仍然一个也没有撤销。写入 result4.xlsx 共 136 行。

核心结论由表 11 给出：允许 C 类调整间隔，可把必须撤销的计划从 12 个减少到 9 个（-3，降幅四分之一），代价是多调整 3 个计划、归一化总幅度增加 2.6。词典序不劣性检查通过（通过），与命题 7.1 一致。在 127 个被调整的计划中，31 个使用了新增的间隔调整动作（平均 $|\Delta g| = 4.90$ 、最大 10），其余仍为频段平移（58 个）或时间平移（38 个）。

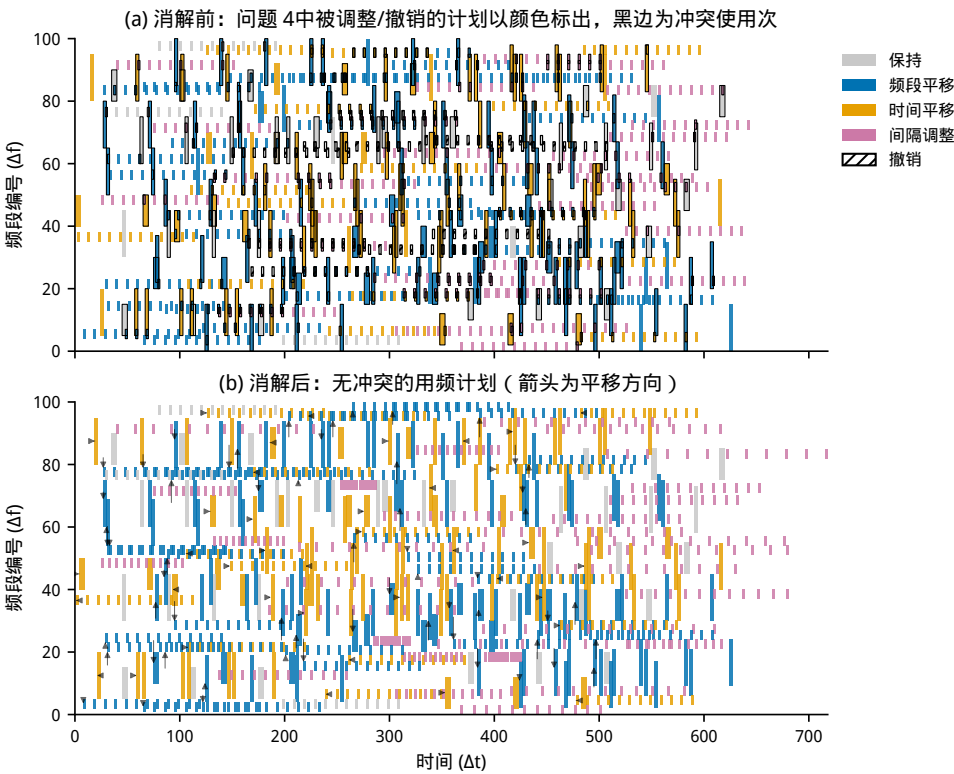


图 8 问题 4 的冲突消解方案，图例与图 4 相同，另增“间隔调整”一类（31 个 C 类计划改变了使用间隔）。

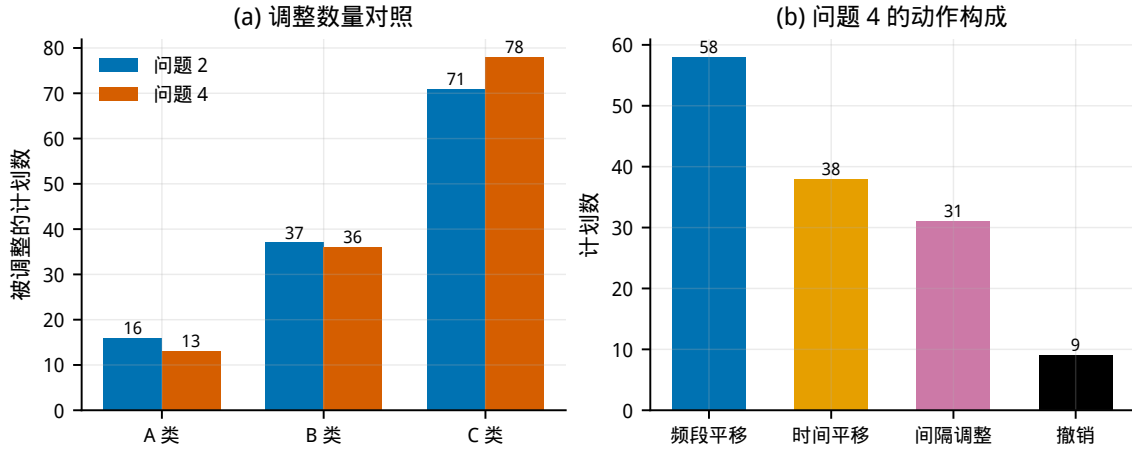


图 9 问题 4 与问题 2 的对照。(a) 各类被调整的计划数；(b) 问题 4 所采取动作的构成。允许 C 类调整间隔后，必须撤销的计划由 12 个降为 9 个。

间隔调整为什么有效？ 频段平移与时间平移都是把整个使用序列刚性地搬动，而改变间隔是把序列在时间轴上拉伸或压缩：它同时改变了第 $1, 2, \dots, n-1$ 次使用的位置，且各次的位移量 $k\Delta g$ 互不相同。对于 C 类这种“次数多、周期短”的计划 ($n = 12$ 、 $\pi_C = 2 + 8$)，一次 $|\Delta g| = 1$ 的调整就能让最后一次使用移动 $n_C - 1 = 12 - 1$ 个 Δt ，远超时间平移上限 $\mathcal{T} = 5$ 。这正是它能解开那些“平移无解”的死结的原因。

7.3 最优性证据与时间范围的代价

表 12 问题 4 逐级求解证据（列的含义同表 7）。与问题 2 不同，本题的“撤销 C”一级在时限内未能闭合间隙，故 9 是已知最好的可行值而非已证明的最优值。

目标层级	CP-SAT 值	CP-SAT 下界	CP-SAT 状态	用时/s	HiGHS LP 下界	HiGHS MILP 值	HiGHS MILP 下界	HiGHS 状态
cancelA	0	0.00	OPTIMAL	0.6	0.00	0	0.00	Optimal
cancelB	0	0.00	OPTIMAL	1.4	0.00	0	0.00	Optimal
cancelC	9	3.00	FEASIBLE	600.6	0.00	-	-	Time limit reached
adjustA	13	5.00	FEASIBLE	300.7	-	-	-	Time limit reached
adjustB	36	17.00	FEASIBLE	300.8	-	-	-	Time limit reached
adjustC	78	40.00	FEASIBLE	300.7	-	-	-	Time limit reached
amplitude	776	354.00	FEASIBLE	240.4	-	-	-	Time limit reached

表 12 显示：撤销 A、B 两级为 0 且状态 OPTIMAL（故“不撤销任何 A、B 类计划”在问题 4 中同样被证明可行且最优）；撤销 C 一级的值 9 与证明下界 3 之间仍有间隙，状态 FEASIBLE。因此本文对问题 4 的表述是：**9 是已知最好的可行方案，且严格优于问题 2 的 12**；其真实最优值落在 $[3, 9]$ 之间。这里体现了现任解保持算法的价值：热启动提示的向量为 $(0, 0, 10, 12, 35, 75, 752)$ ，其第三个分量即提示解的撤销 C 数量，而词典序搜索在保持前两级不变的前提下把它进一步改进到 9。

一个必须指出的代价。 间隔变长会把最后一次使用推后，故问题 4 方案的最晚结束时刻为 718，比问题 2 的 643 延长了 75 个 Δt 。题目只在问题 3 中提出“不增加时频资源”，对问题 4 并未设此约束，故本文不对结束时刻设限；但若评阅者按“资源总量不

变”理解问题 4，则本方案实际上多占用了时间资源。这是一个真实的取舍：用 $75\Delta t$ 的时间跨度，换回若干本来必须撤销的用频计划（撤销数 $12 \rightarrow 9$ ）。若不接受这一代价，只需在模型中加入硬约束 $e_i \leq 643$ 重新求解；该变体本文未做，列为局限（第 9 节）。

交叉验证的诚实说明。 问题 4 的分离权重单目标在其时限内只回到一个更差的向量，未能复现词典序解（未通过）。这不构成矛盾——词典序解严格更优，且由命题 5.4 其值永不劣于任何已知可行解——但它说明：在该规模上，单次加权求解弱于逐级求解。HiGHS 在其证明最优的各级与 CP-SAT 一致（通过）。

8 模型检验

组合优化的结果不能只靠“求解器说它是最优的”来支撑。本节按五条独立的证据线检验本文的模型与结果：（一）与求解器代码分离的可行性校验；（二）算法之间的交叉验证；（三）最优性的证明与界；（四）穷举对照与确定性；（五）灵敏度与可扩展性。每一条都给出可核查的产物，并明确区分“已证明”与“已知最好”。

8.1 与求解器分离的独立校验

本文实现了一套**独立校验器**，它刻意不复用检测与求解的任何代码：对给定的一组计划，它按定义 (1) 把每个计划显式展开为时频单元的集合，再用集合求交逐对判断冲突。这是命题 4.1 的“另一侧”——区间算术与集合展开是同一事实的两种算法实现，二者一致才说明实现正确，而不是同一段代码自证。

表 13 四个问题的独立校验结论。校验器与检测/求解代码分离，按定义展开时频单元集合并用集合求交复核。

对象	独立校验方法	校验内容	结论
问题 1 冲突检测	cell-set intersection	真值 297 对，报告 297 对，缺失 0，多余 0	通过
问题 2 消解方案	independent per-plan legality checks + cell-set conflict scan	150 个决策，存留 138，剩余冲突 0	通过
问题 3 新增计划	parameter/bound checks + cell-set conflict scan over existing and new plans	新增 109，剩余冲突 0	通过
问题 4 消解方案	independent per-plan legality checks + cell-set conflict scan	150 个决策，存留 141，剩余冲突 0	通过

校验内容逐题不同：问题 1 复核冲突对集合（真值 297 对，缺失 0、多余 0）；问题 2、4 逐计划检查“恰有一个动作”、“调整后与原计划恰有一个参数不同”、 $|\delta| \leq \Phi$ 、 $|\tau| \leq \mathcal{T}$ 、 $|\Delta g| \leq \Gamma$ 、 $g' \geq 1$ 与边界条件，再对存留计划做零冲突扫描，并重新计算表 5、10 与目标向量与求解器报告比对；问题 3 检查新增计划的参数、区域包含性，以及“既有-既有”、“既有-新增”、“新增-新增”三类零冲突。四个校验全部通过（表 13：通过、通过、通过、通过），备选解释 B 的重排布局亦通过同一校验（通过，另加“ w, d, g, n 不变”的参数不变性断言）。**结果工作簿在四个校验全部通过之前不会被写出——这是流水线中的硬门禁。**

8.2 算法之间的交叉验证

两种检测算法。 成对枚举（算法 1）与按频段分桶的扫描线给出完全相同的 297 对冲突；在 $n = 150, \dots, 600$ 的合成实例上二者也始终一致（通过，附录表 31）。加上第 8.1 节的单元集合校验器，问题 1 的答案由三种互不相同的算法共同确认。

两个求解器。 本文用 HiGHS[14] 对问题 2、4 的模型独立建模（0–1 线性规划形式），逐级给出线性松弛下界与限时 MILP 结果。在 HiGHS 自身证明最优的每一级上，其值与 CP-SAT 完全一致（通过、通过）。问题 3 更进一步：CP-SAT 对偶界、HiGHS 线性松弛界与 HiGHS MILP 界三者同时等于 109（表 9）。

两种标量化。 命题 5.2 给出词典序的等价单目标形式。本文以此做第二次独立搜索（不以词典序解热启动，而以仓库内的现任解热启动）：问题 2 返回了与逐级求解相同的目标向量（通过）。问题 4 则未能复现（未通过）——单次加权求解只回到一个更差的向量。本文如实报告这一点：它不是矛盾（词典序解严格更优，且由命题 5.4 永不劣于任何已知可行解），而是说明在该规模上逐级求解强于单次加权求解。

诚实的界定。 需要指出：两种标量化都以同一个现任解为起点，因此问题 2 的“一致”主要说明两条独立的搜索路径都没有找到更好的解，而不等于独立地证明了最优性。本文的最优性证据只有一个来源——表 7、12、9 中的“证明下界”一列。

8.3 最优性：已证明的与未证明的

本文对最优性的声明严格限定在求解器闭合了对偶间隙的层级上：

- **已证明最优：**问题 2 的撤销三级 $(0, 0, 12)$ （定理 5.5，三级状态均为 OPTIMAL，值与证明下界相等）；问题 4 的撤销 A、B 两级（均为 0，状态 OPTIMAL）；问题 3 的 109（定理 6.2，三个独立上界同时等于解值，间隙 0）。
- **已知最好但未证明最优：**问题 2 的调整三级与幅度级（未通过）；问题 4 的撤销 C 级与其后各级（未通过）。它们的真实最优值落在表 7、12 给出的 $[\underline{v}_k, v_k]$ 区间内。

由命题 5.4，现任解上界割不改变任何一级的最优值，因此上述“已证明”的层级确实是全局最优；同时该割保证报告向量在词典序上永不劣于任何已知可行解，故未证明的层级至少不会比先前任何一次搜索更差。线性松弛下界在本模型上很弱（表 7 中撤销级的 LP 下界为 0），这是装填型约束的典型现象：分数解可以把每个计划“摊薄”到多个动作上而不违反任何单元约束。因此紧的界只能来自 CP-SAT 的组合推理。

8.4 穷举对照与确定性

最优性证据之外，还需检验求解器被正确使用——目标是否真的按词典序最小化了、约束是否真的编码了“无冲突”。为此本文构造 24 个随机微型实例（计划数 ≤ 6 、动作

集合缩小), 对每个实例枚举全部动作组合 (最多 262144 种) 直接求出词典序最优向量, 再与本文求解器的结果比对: 24/24 **全部一致** (通过, 附录表 29)。同样的实例用相同随机种子重复求解, 24/24 给出完全相同的解 (通过)。

局限: 确定性只在微型实例上检验过。150 个计划的完整模型单次求解需数十分钟, 本文未做重复求解, 故 150 规模上的确定性记为“未检验”。由于 CP-SAT 的多线程搜索在固定种子下仍可能受线程调度影响, 大规模模型的位对位可复现性不应被假定; 但本文的结论不依赖于此——撤销层的最优性由证明下界保证, 与是否得到同一个最优解无关。

8.5 灵敏度分析

表 14 消解方案对最大平移幅度 $(\Phi, \mathcal{T})$ 的灵敏度。第四行 (10, 5) 为题目给定值, 即正文的问题 2 主结果。每个情形的解都通过了独立校验 (通过)。

最大频段平移	最大时间平移	撤销 A/B/C	调整 A/B/C	调整合计	归一化幅度	各级最优	用时/s
5	2	0/0/59	19/36/26	81	31.0	是	5.0
5	5	0/0/42	16/35/44	95	45.6	否	250.7
10	2	0/0/24	19/36/59	114	64.4	否	262.1
10	5	0/0/12	16/37/71	124	75.0	否	1511.4
15	5	0/0/0	12/34/79	125	98.4	否	241.1
20	10	0/0/0	5/37/75	117	120.2	否	238.5

题目把最大平移幅度定为 $(\Phi, \mathcal{T}) = (10, 5)$, 但未说明这一取值的来源, 故它是最值得做灵敏度分析的参数。本文在 6 组取值上重解同一模型 (表 14、图 10), 得到三条结论:

1. **撤销数量对频段平移上限极其敏感。**把 Φ 由 10 放宽到 15, 撤销数由 12 直接降到 0——一个计划都不必撤销; 反之收紧到 $\Phi = 5$, 撤销数升至 59。这再次印证第 4 节的判断: **撤销源于局部拥塞与幅度上限, 而非资源总量不足。**
2. **频域比时域更“贵”。**把两个上限分别放宽, 可以定量比较二者的作用: Φ 由 5 增至 10 ($\mathcal{T}$ 固定) 使撤销数减少约三十个; 而 $\mathcal{T}$ 由 2 增至 5 (Φ 固定) 只减少约十余个。**频域放宽的效果约为时域的两倍——**这与问题 1 中“B-C 类冲突占六成以上、而 B 类频宽最大”的统计一致。
3. **一条有实际价值的管理建议。**若能把最大频段平移幅度由 $10\Delta f$ 放宽到 $15\Delta f$, 代价仅是多调整一个计划、归一化总幅度由 75.0 升至表 14 第五行所示的水平, 却可以完全避免撤销。对用频管理部门而言, 这意味着“适度放宽频段调整权限”比“增加带宽”更具性价比。

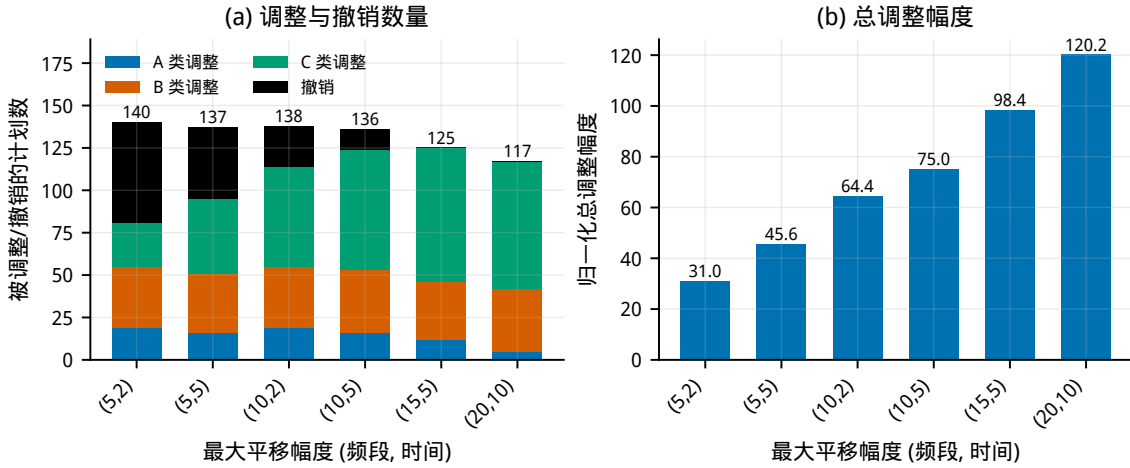


图 10 消解方案对最大平移幅度的灵敏度。(a) 被调整（按类别堆叠）与被撤销（黑）的计划数；(b) 归一化总调整幅度。频段平移上限由 $10\Delta f$ 放宽到 $15\Delta f$ 时撤销数降为 0，收紧到 $5\Delta f$ 时升至 59，说明冲突主要源于频域拥塞。

单调性检查及其有效范围（诚实说明）。 动作集合更大的情形其最优向量应当不劣于基准，更小的情形应当不优于基准；该检查通过（通过）。但必须指出：自现任解保持算法引入后，动作集合更大的情形 $((15, 5), (20, 10))$ 中基准解本身可行，会成为现任解并给出上界割，因而其“不劣于基准”是由构造保证的，不再是独立检验；只有动作集合更小的情形 $((5, 2), (5, 5), (10, 2))$ 中基准解一般不可行、不产生割，其“不优于基准”仍是真正的检验。

其他不确定性来源。 目标函数的形式化（假设 8）由四方案对照量化（表 8、附录表 26）；问题 3 的题意歧义由解释 A 与 B 的区间式 (13) 量化； T_{end} 的两种取法（问题 2 方案的最晚结束时刻与原始计划的最晚结束时刻）在本实例中相同（均为 643），故不产生差异。

8.6 可扩展性

附录表 31 与图 11 给出 $n = 150, \dots, 600$ 的合成实例基准。两种检测算法均为亚秒级（最大 0.15 s 与 0.02 s），说明**检测不是瓶颈**；消解模型的变量数与约束数随 n 近似线性增长。**局限**：在固定 120 s 时限下，全部合成实例的求解状态均为 FEASIBLE（未通过），即本文没有证据表明该方法在更大规模上仍能证明最优。这与问题的 NP-难性（命题 5.3）一致：可扩展的是建模与求可行解，不是证明最优性。

9 模型评价与推广

9.1 模型的优点

1. **统一的单元格语言。**四个问题——检测、消解、增容、放宽间隔——被统一到“时频单元集合是否相交”这一个判据上（命题 4.1、5.1）。这既使模型精确（单元约束是无冲突的充要编码，不是近似或松弛），又使校验独立（同一判据的另一种算法实现

即构成校验器)。

2. **可证明的部分被明确证明。**问题 2 撤销层的全局最优性 (定理 5.5) 与问题 3 的最优性 (定理 6.2) 都给出了闭合的对偶间隙, 后者更由三个独立上界同时确认。推论 5.6 直接回答了“能否一个都不撤销”这一自然追问。
3. **未证明的部分被诚实标注。**调整各级只报告“值 / 证明下界 / 状态” (表 7、12), 不作最优性声明。现任解上界割 (命题 5.4) 在数学上保证报告值不劣于任何已知可行解, 从而消除了“交叉验证击败主解”这类自相矛盾。
4. **主观选择被量化。**目标函数的形式化是本文最主观的一步, 四方案对照 (表 8) 把“优先级条款值多少”变成了可以看见的数字: 坚持不撤销 A、B 类计划的代价, 是撤销总数从 8 上升到 12。
5. **题意歧义被正面处理。**问题 3 的两种读法给出相差约五倍的答案, 本文不回避, 而是以模板一致性为准选定主解释, 并为另一解释给出可核查的区间式 (13)。
6. **结果可复现。**全部计算在同一次云端运行中完成, 每个数字都由阶段产物自动注入论文, 每个阶段记录代码版本、参数与输入输出的 SHA-256 (附录 C)。

9.2 模型的局限

1. **调整层的最优性未闭合。**问题 2 的调整三级与幅度级、问题 4 的撤销 C 级均未证明最优 (未通过、未通过)。其间隙较宽, 主要原因是线性松弛在装填型约束上很弱 (撤销级的 LP 下界为 0)。改进方向是加入更强的有效不等式 (如基于冲突图团的覆盖不等式) 或列生成。
2. **问题 3 的答案依赖于具体的问题 2 方案。**定理 6.2 的 109 是“在本文这一个无冲突方案之上”的最大值。由引理 6.3, 可装填数量主要取决于既有布局与相位划分的对齐程度, 而非自由单元的总数——因此不同的问题 2 最优解会给出不同的答案。若要回答“跨所有无冲突方案的最大值”, 需把问题 2 与问题 3 合并为一个联合优化, 规模将显著增大。
3. **问题 4 的方案延长了时间范围。**间隔变长把最后一次使用推后, 最晚结束时刻由 643 增至 718 (第 7.3 节)。题目未对问题 4 提出“不增加时频资源”的要求, 但若按此理解, 应加入硬约束 $e_i \leq 643$ 重解; 该变体本文未做。
4. **离散化与动作集合是题目给定的。** Δf 、 Δt 的粒度、“只改一个参数”的限制、平移幅度上限均为外生。连续化模型、多参数联合调整、功率或方向性维度都未建模。
5. **物理层机制未建模 (假设 10)。**真实电磁兼容需要保护带、空间隔离、互调与杂散抑制等; 本文的“冲突”是纯几何的。好在模型对此是可扩展的: 只需把 $U(P)$ 沿频率 (或时间) 方向膨胀一个保护宽度, 全部命题与算法原样成立, 只是冲突对数会上升。
6. **大规模的可证明最优性未验证。**合成实例在固定时限下只达到可行解 (未通过)。

9.3 模型的推广

(一) **带保护带的冲突判据。** 把频段方向的占用由 $[f, f + w)$ 膨胀为 $[f - \rho, f + w + \rho)$ (ρ 为保护频段数), 时间方向同理膨胀为收发转换时间, 命题 4.1–5.1 与全部算法不变。这使模型可直接用于真实的频谱指配。

(二) **在线与滚动消解。** 实际用频计划是陆续提报的。把已批准的计划视为固定占用、新提报的计划视为待放置对象, 问题即退化为问题 3 的装填模型; 若允许有限次回溯调整, 则是问题 2 模型在小邻域上的重解。由于检测是亚秒级的 (表 31), 在线判定完全可以实时完成。

(三) **鲁棒性与不确定性。** 若计划的起始时刻带有抖动 $\pm\epsilon$, 可把每次使用的时间区间膨胀 ϵ 后重解, 得到“对抖动免疫”的方案; 方案的代价随 ϵ 的增长曲线正是第 8.5 节灵敏度分析的变体。

(四) **同类问题。** 本文的“每个对象恰选一个位置、位置之间不得共享资源单元、按优先级词典序最优”的结构, 同样适用于会议室与航班时刻分配、光网络波长指配、数据中心作业的时间–机架放置等问题; 单元格 AtMostOne 编码与现任解保持的词典序搜索可原样迁移。

9.4 给用频管理部门的建议

基于本文的定量结果, 提出三条建议: **其一**, 适度放宽频段调整权限比增加带宽更有效——把最大频段平移幅度由 $10\Delta f$ 提高到 $15\Delta f$, 即可把必须撤销的计划数从 12 降为 0 (第 8.5 节); **其二**, 允许 C 类装备微调用频间隔是低成本的松弛——它把撤销数从 12 降到 9 (表 11), 但需注意其代价是时间范围延长 $75\Delta t$; **其三**, 若提报时能引导 C 类装备按引理 6.3 的相位错开 (首次时刻相差 d_C 的整数倍), 同一频带的利用率可接近 100%, 从源头上减少冲突。

AI 工具使用声明

本研究在问题分析、方案比较、程序编写与调试、云端实验组织、数值核验、图表绘制以及论文文字组织等环节使用了生成式 AI 工具 (Anthropic Claude Code, 模型 Claude Fable 5.1); 所有数据处理、计算与排版在 Modal 云平台执行。使用过程与人工核验方式详见支撑材料中的《AI 工具使用详情.pdf》。

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A 补充表格

本附录收录正文中引用但因篇幅置于此处的完整表格。全部表格由 tables 阶段生成（.tex 与同名 .csv 并存），数值与结果工作簿一致。

A.1 问题 1：冲突强度、度分布与全部冲突对

表 15 冲突强度分布：横跨 k 对使用次的冲突对数量。多数冲突对只在一次使用上相撞。

冲突使用次对数 k	冲突对数量
1	236
2	45
3	3
4	2
6	2
7	4
8	2
10	2
12	1

表 16 冲突图中度最大的 10 个用频装备。

装备编号	冲突计划数（度）
B009	8
B016	8
B024	8
B027	8
B030	8
C076	8
B008	7
B011	7
B013	7
B018	7

表 17 问题 1 检测出的全部 297 对时频冲突（按装备编号字典序，与 result1.xlsx 一致）。“冲突使用次对数”为该计划对中出现时间重叠的使用次对数，“频段重叠宽度”以 Δf 计。

序号	装备 1	装备 2	类别对	冲突使用次对数	频段重叠宽度
1	A001	B001	AB	1	10
2	A001	C002	AC	1	3
3	A001	C012	AC	1	3
4	A001	C022	AC	1	3
5	A002	B010	AB	1	10
6	A002	C009	AC	1	2
7	A002	C010	AC	1	2
8	A002	C019	AC	1	3
9	A002	C029	AC	1	3
10	A003	B009	AB	2	8
11	A003	C008	AC	2	3
12	A003	C018	AC	1	3
13	A003	C028	AC	1	3
14	A004	B009	AB	1	2
15	A004	B014	AB	1	10
16	A004	B018	AB	1	5
17	A004	C016	AC	1	3
18	A004	C017	AC	1	3

序号	装备 1	装备 2	类别对	冲突使用次对数	频段重叠宽度
19	A004	C027	AC	1	1
20	A005	B007	AB	1	10
21	A005	B012	AB	1	10
22	A005	C014	AC	1	3
23	A005	C024	AC	1	3
24	A005	C054	AC	1	3
25	A006	B004	AB	1	5
26	A006	B008	AB	1	10
27	A006	C006	AC	1	3
28	A007	B011	AB	1	10
29	A007	B022	AB	1	7
30	A007	C023	AC	1	3
31	A007	C053	AC	1	1
32	A008	C028	AC	1	1
33	A008	C038	AC	1	3
34	A008	C039	AC	1	3
35	A008	C049	AC	1	3
36	A008	C058	AC	1	1
37	A008	C059	AC	1	3
38	A009	B012	AB	1	5
39	A009	C033	AC	2	1
40	A009	C043	AC	1	3
41	A009	C053	AC	1	3
42	A009	C054	AC	1	2
43	A010	C027	AC	1	3
44	A010	C037	AC	1	3
45	A010	C057	AC	1	3
46	A011	B027	AB	1	5
47	A011	C034	AC	1	3
48	A011	C044	AC	1	3
49	A011	C074	AC	1	3
50	A012	B024	AB	1	5
51	A012	B028	AB	1	5
52	A012	C036	AC	1	3
53	A012	C046	AC	1	3
54	A012	C056	AC	1	3
55	A012	C076	AC	1	3
56	A013	B016	AB	1	7
57	A013	C022	AC	1	3
58	A013	C032	AC	1	3
59	A013	C041	AC	1	2
60	A013	C052	AC	1	3
61	A014	C040	AC	1	3
62	A014	C050	AC	1	3
63	A014	C070	AC	1	3
64	A014	C080	AC	1	1

序号	装备 1	装备 2	类别对	冲突使用次对数	频段重叠宽度
65	A014	C090	AC	1	3
66	A015	B030	AB	2	5
67	A015	C067	AC	1	2
68	A015	C077	AC	1	3
69	A015	C078	AC	2	3
70	A016	C067	AC	1	3
71	A016	C087	AC	1	3
72	A017	B033	AB	1	8
73	A017	C065	AC	1	3
74	A017	C075	AC	2	3
75	A017	C085	AC	1	3
76	A018	B028	AB	1	10
77	A018	B033	AB	1	3
78	A018	C065	AC	1	1
79	A018	C066	AC	1	3
80	A018	C076	AC	1	1
81	A018	C085	AC	1	3
82	A019	C073	AC	1	3
83	A019	C082	AC	1	3
84	A020	C039	AC	1	3
85	A020	C069	AC	1	3
86	A020	C079	AC	1	3
87	A020	C089	AC	1	3
88	B001	C001	BC	2	3
89	B001	C002	BC	1	3
90	B001	C012	BC	1	3
91	B002	B011	BB	1	10
92	B002	C003	BC	2	3
93	B002	C013	BC	2	3
94	B002	C014	BC	1	3
95	B003	C004	BC	1	3
96	B003	C005	BC	1	3
97	B003	C015	BC	1	3
98	B003	C024	BC	1	3
99	B004	C016	BC	1	3
100	B004	C017	BC	1	3
101	B004	C027	BC	1	3
102	B005	C009	BC	1	3
103	B005	C010	BC	1	2
104	B005	C019	BC	1	3
105	B005	C029	BC	1	3
106	B005	C030	BC	1	3
107	B006	C001	BC	1	3
108	B006	C011	BC	2	3
109	B006	C012	BC	1	1
110	B006	C021	BC	1	3

序号	装备 1	装备 2	类别对	冲突使用次对数	频段重叠宽度
111	B006	C051	BC	1	3
112	B006	C052	BC	1	3
113	B007	C004	BC	1	3
114	B007	C013	BC	1	3
115	B007	C014	BC	1	3
116	B007	C024	BC	2	3
117	B008	C006	BC	1	3
118	B008	C015	BC	2	3
119	B008	C016	BC	1	3
120	B008	C025	BC	1	1
121	B008	C026	BC	1	3
122	B008	C055	BC	1	3
123	B009	C008	BC	2	1
124	B009	C017	BC	2	1
125	B009	C018	BC	1	1
126	B009	C027	BC	2	3
127	B009	C028	BC	1	3
128	B009	C057	BC	1	3
129	B010	C009	BC	1	3
130	B010	C010	BC	1	2
131	B010	C030	BC	1	3
132	B011	C012	BC	1	2
133	B011	C013	BC	1	3
134	B011	C023	BC	1	3
135	B011	C043	BC	1	3
136	B011	C053	BC	2	3
137	B012	C024	BC	1	3
138	B012	C025	BC	1	2
139	B012	C054	BC	1	3
140	B013	B018	BB	3	10
141	B013	B023	BB	1	10
142	B013	C015	BC	1	3
143	B013	C025	BC	2	3
144	B013	C026	BC	1	3
145	B013	C036	BC	1	3
146	B013	C055	BC	1	3
147	B014	C017	BC	1	3
148	B014	C027	BC	1	3
149	B014	C037	BC	1	3
150	B014	C056	BC	1	3
151	B014	C057	BC	1	3
152	B015	C018	BC	1	3
153	B015	C028	BC	1	3
154	B015	C038	BC	2	3
155	B015	C039	BC	1	3
156	B015	C058	BC	1	3

序号	装备 1	装备 2	类别对	冲突使用次对数	频段重叠宽度
157	B015	C059	BC	1	3
158	B016	C021	BC	1	3
159	B016	C022	BC	1	3
160	B016	C031	BC	1	3
161	B016	C041	BC	2	3
162	B016	C042	BC	1	1
163	B016	C051	BC	1	3
164	B016	C052	BC	1	3
165	B017	C014	BC	1	3
166	B017	C033	BC	1	1
167	B017	C043	BC	1	3
168	B017	C054	BC	1	3
169	B018	B023	BB	2	5
170	B018	C026	BC	1	3
171	B018	C036	BC	1	3
172	B018	C055	BC	1	3
173	B018	C056	BC	1	3
174	B019	B024	BB	2	10
175	B019	C028	BC	1	2
176	B019	C058	BC	1	2
177	B020	C019	BC	2	3
178	B020	C020	BC	1	3
179	B020	C029	BC	1	3
180	B020	C030	BC	1	3
181	B020	C040	BC	1	3
182	B021	C031	BC	1	1
183	B021	C032	BC	1	3
184	B021	C041	BC	1	3
185	B021	C042	BC	1	3
186	B021	C051	BC	1	1
187	B021	C052	BC	1	3
188	B021	C072	BC	1	3
189	B022	C033	BC	1	3
190	B022	C044	BC	1	3
191	B022	C054	BC	1	2
192	B022	C073	BC	1	3
193	B022	C074	BC	1	2
194	B023	C034	BC	1	3
195	B023	C035	BC	2	3
196	B023	C045	BC	1	3
197	B023	C074	BC	1	1
198	B023	C075	BC	1	3
199	B024	C037	BC	1	3
200	B024	C046	BC	1	3
201	B024	C047	BC	1	3
202	B024	C076	BC	1	2

序号	装备 1	装备 2	类别对	冲突使用次对数	频段重叠宽度
203	B024	C077	BC	1	3
204	B024	C086	BC	1	3
205	B025	B040	BB	2	5
206	B025	C049	BC	1	3
207	B026	C031	BC	1	1
208	B026	C042	BC	1	3
209	B026	C061	BC	1	3
210	B026	C062	BC	2	3
211	B026	C071	BC	2	3
212	B027	B032	BB	2	13
213	B027	B033	BB	2	2
214	B027	C033	BC	1	3
215	B027	C043	BC	1	3
216	B027	C044	BC	2	3
217	B027	C063	BC	1	3
218	B027	C073	BC	1	3
219	B028	C036	BC	1	3
220	B028	C045	BC	2	3
221	B028	C066	BC	1	3
222	B028	C075	BC	1	3
223	B028	C076	BC	1	1
224	B029	B034	BB	2	15
225	B029	B038	BB	1	5
226	B029	C046	BC	1	3
227	B029	C047	BC	1	3
228	B029	C076	BC	1	2
229	B029	C077	BC	1	3
230	B029	C086	BC	2	3
231	B030	C038	BC	1	3
232	B030	C048	BC	2	3
233	B030	C049	BC	1	3
234	B030	C068	BC	1	3
235	B030	C069	BC	1	3
236	B030	C078	BC	2	3
237	B030	C079	BC	1	3
238	B031	C042	BC	1	3
239	B031	C061	BC	2	3
240	B031	C062	BC	2	3
241	B031	C071	BC	2	3
242	B031	C072	BC	1	3
243	B031	C081	BC	1	3
244	B032	C063	BC	1	3
245	B032	C073	BC	1	3
246	B033	C064	BC	2	3
247	B033	C065	BC	2	3
248	B033	C075	BC	1	3

序号	装备 1	装备 2	类别对	冲突使用次对数	频段重叠宽度
249	B034	C067	BC	1	3
250	B034	C076	BC	1	2
251	B034	C086	BC	1	3
252	B034	C087	BC	1	3
253	B035	C048	BC	1	3
254	B035	C068	BC	1	3
255	B035	C069	BC	1	3
256	B035	C078	BC	2	3
257	B035	C079	BC	1	3
258	B035	C088	BC	1	3
259	B035	C089	BC	1	3
260	B036	C070	BC	1	3
261	B036	C090	BC	1	3
262	B037	C064	BC	1	3
263	B037	C074	BC	1	3
264	B037	C075	BC	1	2
265	B037	C083	BC	1	2
266	B037	C084	BC	2	3
267	B038	C065	BC	1	1
268	B038	C076	BC	1	3
269	B038	C085	BC	1	3
270	B038	C086	BC	1	3
271	B039	C067	BC	1	2
272	B039	C068	BC	1	3
273	B039	C078	BC	1	3
274	B039	C088	BC	1	3
275	B040	C049	BC	1	3
276	B040	C069	BC	1	3
277	B040	C070	BC	1	3
278	B040	C090	BC	2	1
279	C005	C025	CC	2	2
280	C019	C029	CC	7	2
281	C021	C051	CC	2	2
282	C022	C032	CC	2	3
283	C023	C033	CC	4	1
284	C025	C055	CC	4	1
285	C027	C057	CC	10	2
286	C028	C058	CC	7	3
287	C031	C051	CC	8	3
288	C033	C043	CC	12	1
289	C037	C077	CC	3	3
290	C038	C048	CC	6	2
291	C039	C049	CC	7	2
292	C043	C053	CC	8	2
293	C050	C070	CC	1	2
294	C061	C071	CC	6	1

序号	装备 1	装备 2	类别对	冲突使用次对数	频段重叠宽度
295	C067	C077	CC	3	1
296	C069	C079	CC	7	1
297	C076	C086	CC	10	2

A.2 问题 2：被撤销计划与逐计划动作

表 19 问题 2 中被撤销计划的局部环境。“冲突邻居数”为该计划在冲突图中的度，“其中 A/B 类”为这些邻居中高优先级装备的个数，“可用平移动作数”为式 (3) 中该计划的合法平移动作数。每个被撤销的计划都被高优先级邻居包围，这正是它们不得不让路的原因。

装备编号	频段	首次时间	冲突邻居数	其中 A/B 类	可用平移动作数
C010	[4,7)	[8,10)	3	3	24
C017	[36,39)	[95,97)	4	4	30
C018	[20,23)	[107,109)	3	3	30
C019	[12,15)	[86,88)	4	3	30
C024	[62,65)	[150,152)	4	4	30
C034	[62,65)	[294,296)	2	2	30
C035	[57,60)	[272,274)	1	1	30
C037	[31,34)	[270,272)	4	3	30
C054	[64,67)	[218,220)	5	5	30
C055	[52,55)	[213,215)	4	3	30
C067	[33,36)	[451,453)	5	4	30
C090	[3,6)	[482,484)	3	3	23

表 20 问题 2 的逐计划动作清单（只列出被调整或被撤销的计划，与 result2.xlsx 一致）。

装备编号	动作	原频段	新频段	原首次时间	新首次时间
A003	频段平移	[20,30)	[30,40)	[75,80)	[75,80)
A005	时间平移	[60,70)	[60,70)	[125,130)	[129,134)
A006	时间平移	[40,50)	[40,50)	[0,5)	[4,9)
A007	频段平移	[73,83)	[65,75)	[175,180)	[175,180)
A008	频段平移	[15,25)	[5,15)	[211,216)	[211,216)
A009	频段平移	[65,75)	[62,72)	[228,233)	[228,233)
A010	时间平移	[30,40)	[30,40)	[265,270)	[267,272)
A011	时间平移	[60,70)	[60,70)	[273,278)	[272,277)
A012	频段平移	[40,50)	[31,41)	[300,305)	[300,305)
A013	时间平移	[82,92)	[82,92)	[190,195)	[186,191)

装备编号	动作	原频段	新频段	原首次时间	新首次时间
A014	频段平移	[2,12)	[4,14)	[348,353)	[348,353)
A015	时间平移	[25,35)	[25,35)	[385,390)	[387,392)
A016	频段平移	[30,40)	[35,45)	[415,420)	[415,420)
A017	频段平移	[50,60)	[52,62)	[432,437)	[432,437)
A018	频段平移	[45,55)	[35,45)	[451,456)	[451,456)
A020	频段平移	[10,20)	[4,14)	[375,380)	[375,380)
B001	时间平移	[80,95)	[80,95)	[15,18)	[20,23)
B002	频段平移	[65,80)	[57,72)	[27,30)	[27,30)
B003	频段平移	[50,65)	[55,70)	[30,33)	[30,33)
B005	频段平移	[5,20)	[0,15)	[57,60)	[57,60)
B006	频段平移	[85,100)	[79,94)	[95,98)	[95,98)
B007	频段平移	[60,75)	[55,70)	[62,65)	[62,65)
B008	频段平移	[40,55)	[31,46)	[85,88)	[85,88)
B009	时间平移	[22,37)	[22,37)	[77,80)	[81,84)
B010	时间平移	[5,20)	[5,20)	[24,27)	[20,23)
B011	频段平移	[70,85)	[78,93)	[155,158)	[155,158)
B012	时间平移	[55,70)	[55,70)	[167,170)	[169,172)
B013	时间平移	[45,60)	[45,60)	[180,183)	[185,188)
B014	时间平移	[30,45)	[30,45)	[185,188)	[190,193)
B015	频段平移	[15,30)	[25,40)	[197,200)	[197,200)
B016	频段平移	[85,100)	[78,93)	[235,238)	[235,238)
B017	频段平移	[60,75)	[57,72)	[202,205)	[202,205)
B018	频段平移	[40,55)	[49,64)	[225,228)	[225,228)
B019	时间平移	[25,40)	[25,40)	[217,220)	[219,222)
B020	时间平移	[0,15)	[0,15)	[124,127)	[120,123)
B021	频段平移	[80,95)	[78,93)	[295,298)	[295,298)
B022	频段平移	[65,80)	[75,90)	[307,310)	[307,310)
B023	时间平移	[50,65)	[50,65)	[310,313)	[308,311)
B024	时间平移	[30,45)	[30,45)	[305,308)	[308,311)
B025	频段平移	[15,30)	[25,40)	[337,340)	[337,340)
B026	频段平移	[80,95)	[77,92)	[375,378)	[375,378)
B027	频段平移	[65,80)	[55,70)	[342,345)	[342,345)
B028	频段平移	[45,60)	[52,67)	[365,368)	[365,368)
B029	时间平移	[30,45)	[30,45)	[357,360)	[362,365)
B030	频段平移	[15,30)	[25,40)	[384,387)	[384,387)
B031	频段平移	[83,98)	[77,92)	[415,418)	[415,418)
B032	频段平移	[67,82)	[76,91)	[427,430)	[427,430)
B033	时间平移	[52,67)	[52,67)	[430,433)	[425,428)
B035	频段平移	[15,30)	[20,35)	[457,460)	[457,460)
B037	频段平移	[58,73)	[52,67)	[462,465)	[462,465)
B038	频段平移	[40,55)	[30,45)	[485,488)	[485,488)
B039	时间平移	[20,35)	[20,35)	[477,480)	[479,482)
B040	频段平移	[5,20)	[0,15)	[424,427)	[424,427)
C001	频段平移	[90,93)	[95,98)	[40,42)	[40,42)
C002	频段平移	[82,85)	[76,79)	[27,29)	[27,29)
C004	频段平移	[60,63)	[52,55)	[31,33)	[31,33)

装备编号	动作	原频段	新频段	原首次时间	新首次时间
C005	频段平移	[55,58)	[52,55)	[33,35)	[33,35)
C006	频段平移	[47,50)	[50,53)	[25,27)	[25,27)
C007	频段平移	[35,38)	[37,40)	[2,4)	[2,4)
C009	频段平移	[13,16)	[18,21)	[31,33)	[31,33)
C010	撤销	[4,7)	-	[8,10)	-
C011	时间平移	[96,99)	[96,99)	[80,82)	[82,84)
C012	频段平移	[83,86)	[75,78)	[65,67)	[65,67)
C013	时间平移	[70,73)	[70,73)	[75,77)	[80,82)
C014	频段平移	[65,68)	[72,75)	[92,94)	[92,94)
C015	频段平移	[50,53)	[49,52)	[110,112)	[110,112)
C016	频段平移	[42,45)	[50,53)	[88,90)	[88,90)
C017	撤销	[36,39)	-	[95,97)	-
C018	撤销	[20,23)	-	[107,109)	-
C019	撤销	[12,15)	-	[86,88)	-
C020	时间平移	[2,5)	[2,5)	[65,67)	[66,68)
C021	频段平移	[95,98)	[97,100)	[125,127)	[125,127)
C022	频段平移	[86,89)	[93,96)	[147,149)	[147,149)
C023	频段平移	[76,79)	[75,78)	[171,173)	[171,173)
C024	撤销	[62,65)	-	[150,152)	-
C025	频段平移	[54,57)	[49,52)	[132,134)	[132,134)
C026	时间平移	[46,49)	[46,49)	[145,147)	[143,145)
C027	频段平移	[33,36)	[27,30)	[165,167)	[165,167)
C028	频段平移	[24,27)	[19,22)	[168,170)	[168,170)
C029	频段平移	[11,14)	[15,18)	[137,139)	[137,139)
C030	频段平移	[6,9)	[15,18)	[122,124)	[122,124)
C031	时间平移	[94,97)	[94,97)	[265,267)	[268,270)
C032	频段平移	[86,89)	[96,99)	[246,248)	[246,248)
C033	频段平移	[74,77)	[72,75)	[252,254)	[252,254)
C034	撤销	[62,65)	-	[294,296)	-
C035	撤销	[57,60)	-	[272,274)	-
C036	时间平移	[46,49)	[46,49)	[266,268)	[269,271)
C037	撤销	[31,34)	-	[270,272)	-
C038	时间平移	[22,25)	[22,25)	[285,287)	[283,285)
C039	时间平移	[16,19)	[16,19)	[264,266)	[265,267)
C040	频段平移	[5,8)	[1,4)	[242,244)	[242,244)
C041	频段平移	[90,93)	[97,100)	[303,305)	[303,305)
C042	频段平移	[83,86)	[93,96)	[326,328)	[326,328)
C043	时间平移	[72,75)	[72,75)	[253,255)	[250,252)
C044	频段平移	[66,69)	[70,73)	[334,336)	[334,336)
C045	频段平移	[53,56)	[49,52)	[317,319)	[317,319)
C046	频段平移	[42,45)	[45,48)	[327,329)	[327,329)
C047	频段平移	[36,39)	[45,48)	[341,343)	[341,343)
C048	时间平移	[21,24)	[21,24)	[346,348)	[345,347)
C050	频段平移	[5,8)	[1,4)	[306,308)	[306,308)
C051	时间平移	[94,97)	[94,97)	[225,227)	[222,224)
C052	频段平移	[86,89)	[94,97)	[204,206)	[204,206)

装备编号	动作	原频段	新频段	原首次时间	新首次时间
C053	频段平移	[71,74)	[75,78)	[213,215)	[213,215)
C054	撤销	[64,67)	-	[218,220)	-
C055	撤销	[52,55)	-	[213,215)	-
C056	时间平移	[41,44)	[41,44)	[202,204)	[201,203)
C057	频段平移	[32,35)	[22,25)	[185,187)	[185,187)
C058	频段平移	[24,27)	[22,25)	[218,220)	[218,220)
C059	时间平移	[16,19)	[16,19)	[201,203)	[200,202)
C060	时间平移	[2,5)	[2,5)	[198,200)	[194,196)
C061	频段平移	[90,93)	[95,98)	[450,452)	[450,452)
C062	频段平移	[86,89)	[92,95)	[420,422)	[420,422)
C064	频段平移	[61,64)	[67,70)	[455,457)	[455,457)
C065	频段平移	[54,57)	[49,52)	[478,480)	[478,480)
C066	时间平移	[45,48)	[45,48)	[442,444)	[439,441)
C067	撤销	[33,36)	-	[451,453)	-
C068	频段平移	[27,30)	[17,20)	[461,463)	[461,463)
C069	时间平移	[15,18)	[15,18)	[441,443)	[443,445)
C070	频段平移	[6,9)	[15,18)	[415,417)	[415,417)
C071	时间平移	[92,95)	[92,95)	[390,392)	[392,394)
C072	频段平移	[86,89)	[96,99)	[386,388)	[386,388)
C073	频段平移	[77,80)	[72,75)	[398,400)	[398,400)
C074	频段平移	[64,67)	[70,73)	[396,398)	[396,398)
C075	频段平移	[57,60)	[49,52)	[391,393)	[391,393)
C076	频段平移	[43,46)	[45,48)	[384,386)	[384,386)
C077	频段平移	[31,34)	[22,25)	[361,363)	[361,363)
C078	频段平移	[26,29)	[17,20)	[389,391)	[389,391)
C079	频段平移	[17,20)	[14,17)	[391,393)	[391,393)
C080	时间平移	[0,3)	[0,3)	[368,370)	[370,372)
C081	时间平移	[95,98)	[95,98)	[485,487)	[484,486)
C082	频段平移	[82,85)	[91,94)	[507,509)	[507,509)
C085	频段平移	[51,54)	[46,49)	[492,494)	[492,494)
C086	频段平移	[42,45)	[48,51)	[405,407)	[405,407)
C087	频段平移	[37,40)	[45,48)	[525,527)	[525,527)
C088	时间平移	[21,24)	[21,24)	[528,530)	[531,533)
C090	撤销	[3,6)	-	[482,484)	-

A.3 问题 3：新增计划清单

序号	频段区间	首次时间区间	间隔时长	使用次数
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表 22 问题 3 在解释 A 下新增的 109 个 C 类用频计划（与 result3.xlsx 一致；间隔时长与使用次数按假设 9 与附件 C 类相同）。

序号	频段区间	首次时间区间	间隔时长	使用次数
1	[75,78)	[1,3)	8	12
2	[0,3)	[3,5)	8	12
3	[75,78)	[3,5)	8	12
4	[24,27)	[4,6)	8	12
5	[27,30)	[4,6)	8	12
6	[72,75)	[4,6)	8	12
7	[97,100)	[4,6)	8	12
8	[21,24)	[5,7)	8	12
9	[24,27)	[6,8)	8	12
10	[27,30)	[6,8)	8	12
11	[97,100)	[6,8)	8	12
12	[20,23)	[7,9)	8	12
13	[95,98)	[8,10)	8	12
14	[46,49)	[14,16)	8	12
15	[46,49)	[17,19)	8	12
16	[73,76)	[27,29)	8	12
17	[15,18)	[30,32)	8	12
18	[90,93)	[31,33)	8	12
19	[70,73)	[36,38)	8	12
20	[70,73)	[38,40)	8	12
21	[27,30)	[48,50)	8	12
22	[23,26)	[49,51)	8	12
23	[0,3)	[68,70)	8	12
24	[18,21)	[73,75)	8	12
25	[15,18)	[74,76)	8	12
26	[46,49)	[79,81)	8	12
27	[46,49)	[81,83)	8	12
28	[94,97)	[84,86)	8	12
29	[27,30)	[90,92)	8	12
30	[21,24)	[91,93)	8	12
31	[24,27)	[91,93)	8	12
32	[97,100)	[127,129)	8	12
33	[0,3)	[130,132)	8	12
34	[27,30)	[133,135)	8	12
35	[24,27)	[143,145)	8	12
36	[75,78)	[149,151)	8	12
37	[52,55)	[151,153)	8	12
38	[52,55)	[153,155)	8	12
39	[97,100)	[159,161)	8	12
40	[94,97)	[160,162)	8	12
41	[21,24)	[163,165)	8	12

序号	频段区间	首次时间区间	间隔时长	使用次数
42	[19,22)	[166,168)	8	12
43	[72,75)	[185,187)	8	12
44	[75,78)	[185,187)	8	12
45	[75,78)	[187,189)	8	12
46	[72,75)	[188,190)	8	12
47	[18,21)	[192,194)	8	12
48	[46,49)	[195,197)	8	12
49	[97,100)	[201,203)	8	12
50	[19,22)	[210,212)	8	12
51	[22,25)	[220,222)	8	12
52	[4,7)	[222,224)	8	12
53	[7,10)	[222,224)	8	12
54	[12,15)	[222,224)	8	12
55	[5,8)	[224,226)	8	12
56	[8,11)	[224,226)	8	12
57	[13,16)	[224,226)	8	12
58	[41,44)	[244,246)	8	12
59	[15,18)	[257,259)	8	12
60	[0,3)	[258,260)	8	12
61	[97,100)	[288,290)	8	12
62	[93,96)	[300,302)	8	12
63	[73,76)	[316,318)	8	12
64	[14,17)	[319,321)	8	12
65	[49,52)	[319,321)	8	12
66	[93,96)	[324,326)	8	12
67	[97,100)	[331,333)	8	12
68	[19,22)	[337,339)	8	12
69	[22,25)	[337,339)	8	12
70	[22,25)	[339,341)	8	12
71	[17,20)	[341,343)	8	12
72	[0,3)	[362,364)	8	12
73	[70,73)	[370,372)	8	12
74	[73,76)	[370,372)	8	12
75	[70,73)	[372,374)	8	12
76	[15,18)	[377,379)	8	12
77	[18,21)	[385,387)	8	12
78	[92,95)	[388,390)	8	12
79	[20,23)	[403,405)	8	12
80	[49,52)	[403,405)	8	12
81	[97,100)	[408,410)	8	12
82	[67,70)	[437,439)	8	12
83	[45,48)	[447,449)	8	12
84	[97,100)	[452,454)	8	12
85	[85,88)	[479,481)	8	12
86	[89,92)	[479,481)	8	12
87	[67,70)	[482,484)	8	12

序号	频段区间	首次时间区间	间隔时长	使用次数
88	[73,76)	[495,497)	8	12
89	[0,3)	[506,508)	8	12
90	[97,100)	[506,508)	8	12
91	[15,18)	[509,511)	8	12
92	[48,51)	[510,512)	8	12
93	[49,52)	[522,524)	8	12
94	[85,88)	[522,524)	8	12
95	[88,91)	[522,524)	8	12
96	[91,94)	[522,524)	8	12
97	[94,97)	[522,524)	8	12
98	[70,73)	[523,525)	8	12
99	[49,52)	[524,526)	8	12
100	[92,95)	[524,526)	8	12
101	[70,73)	[525,527)	8	12
102	[49,52)	[526,528)	8	12
103	[94,97)	[526,528)	8	12
104	[72,75)	[527,529)	8	12
105	[94,97)	[528,530)	8	12
106	[97,100)	[528,530)	8	12
107	[72,75)	[529,531)	8	12
108	[24,27)	[531,533)	8	12
109	[27,30)	[531,533)	8	12

A.4 问题 4：模型规模、目标方案与逐计划动作

表 24 问题 4 的模型规模与求解设置。

项目	数值
决策变量数（计划-动作对）	6119
时频单元数（含动作扩展）	57728
变量-单元隶属数	602064
单元格 AtMostOne 约束数（去重前 / 去重后）	54347 / 42064
成对蕴含团约束数 / 强制撤销割数	84855 / 0
可能相互冲突的计划对数	2267
每计划最多动作数（A/B/C）	32/32/49
CP-SAT 词典序总用时/s	1748.2
热启动现任解目标向量（MDR-0009）	(0, 0, 10, 12, 35, 75, 752)
词典序求解起点	仓库内现任解
各级均证明最优	否
HiGHS 交叉验证总用时/s	309.2
HiGHS 证明最优的各级与 CP-SAT 一致	是
分离权重单目标解与词典序向量一致	否

表 25 问题 4 中被撤销计划的局部环境(列的含义同表 19)。

装备编号	频段	首次时间	冲突邻居数	其中 A/B 类	可用调整动作数
C024	[62,65)	[150,152)	4	4	47
C027	[33,36)	[165,167)	6	5	47
C028	[24,27)	[168,170)	6	5	47
C033	[74,77)	[252,254)	6	4	47
C037	[31,34)	[270,272)	4	3	47
C044	[66,69)	[334,336)	3	3	47
C047	[36,39)	[341,343)	2	2	47
C054	[64,67)	[218,220)	5	5	47
C074	[64,67)	[396,398)	4	4	47

表 26 四种目标方案在问题 4 上的对照（列的含义同表 8，另给出使用间隔调整动作的计划数）。

目标方案	撤销 A/B/C	调整 A/B/C	调整合计	归一化幅度	间隔调整数	独立校验
P	0/0/9	13/36/78	127	77.6	31	通过
T	2/2/5	12/32/75	119	68.6	25	通过
S	3/6/0	12/29/67	108	63.2	11	通过
W	0/0/12	12/33/70	115	71.5	31	通过

表 27 问题 4 的逐计划动作清单（只列出被调整或被撤销的计划，与 result4.xlsx 一致）。

装备编号	动作	原频段	新频段	原首次时间	新首次时间	原间隔	新间隔
A003	频段平移	[20,30)	[10,20)	[75,80)	[75,80)	60	60
A004	时间平移	[35,45)	[35,45)	[95,100)	[93,98)	60	60
A005	时间平移	[60,70)	[60,70)	[125,130)	[129,134)	60	60
A006	时间平移	[40,50)	[40,50)	[0,5)	[4,9)	60	60
A007	频段平移	[73,83)	[65,75)	[175,180)	[175,180)	60	60
A008	频段平移	[15,25)	[22,32)	[211,216)	[211,216)	60	60
A010	频段平移	[30,40)	[20,30)	[265,270)	[265,270)	60	60
A011	时间平移	[60,70)	[60,70)	[273,278)	[276,281)	60	60
A012	频段平移	[40,50)	[32,42)	[300,305)	[300,305)	60	60
A013	时间平移	[82,92)	[82,92)	[190,195)	[186,191)	60	60
A014	时间平移	[2,12)	[2,12)	[348,353)	[353,358)	60	60
A015	频段平移	[25,35)	[30,40)	[385,390)	[385,390)	60	60
A017	时间平移	[50,60)	[50,60)	[432,437)	[433,438)	60	60
B001	时间平移	[80,95)	[80,95)	[15,18)	[19,22)	40	40
B002	频段平移	[65,80)	[60,75)	[27,30)	[27,30)	40	40
B003	频段平移	[50,65)	[54,69)	[30,33)	[30,33)	40	40
B005	时间平移	[5,20)	[5,20)	[57,60)	[59,62)	40	40
B006	频段平移	[85,100)	[79,94)	[95,98)	[95,98)	40	40
B008	频段平移	[40,55)	[35,50)	[85,88)	[85,88)	40	40
B009	频段平移	[22,37)	[30,45)	[77,80)	[77,80)	40	40
B010	时间平移	[5,20)	[5,20)	[24,27)	[21,24)	40	40
B011	频段平移	[70,85)	[79,94)	[155,158)	[155,158)	40	40
B012	时间平移	[55,70)	[55,70)	[167,170)	[170,173)	40	40
B013	时间平移	[45,60)	[45,60)	[180,183)	[182,185)	40	40
B014	时间平移	[30,45)	[30,45)	[185,188)	[187,190)	40	40
B015	频段平移	[15,30)	[25,40)	[197,200)	[197,200)	40	40
B016	频段平移	[85,100)	[79,94)	[235,238)	[235,238)	40	40
B018	时间平移	[40,55)	[40,55)	[225,228)	[220,223)	40	40
B019	时间平移	[25,40)	[25,40)	[217,220)	[219,222)	40	40
B020	频段平移	[0,15)	[4,19)	[124,127)	[124,127)	40	40
B021	时间平移	[80,95)	[80,95)	[295,298)	[299,302)	40	40
B022	频段平移	[65,80)	[75,90)	[307,310)	[307,310)	40	40
B023	频段平移	[50,65)	[60,75)	[310,313)	[310,313)	40	40
B024	时间平移	[30,45)	[30,45)	[305,308)	[310,313)	40	40

装备编号	动作	原频段	新频段	原首次时间	新首次时间	原间隔	新间隔
B025	频段平移	[15,30)	[24,39)	[337,340)	[337,340)	40	40
B026	时间平移	[80,95)	[80,95)	[375,378)	[370,373)	40	40
B027	时间平移	[65,80)	[65,80)	[342,345)	[338,341)	40	40
B028	时间平移	[45,60)	[45,60)	[365,368)	[360,363)	40	40
B029	频段平移	[30,45)	[25,40)	[357,360)	[357,360)	40	40
B030	频段平移	[15,30)	[6,21)	[384,387)	[384,387)	40	40
B031	时间平移	[83,98)	[83,98)	[415,418)	[418,421)	40	40
B032	频段平移	[67,82)	[62,77)	[427,430)	[427,430)	40	40
B033	频段平移	[52,67)	[60,75)	[430,433)	[430,433)	40	40
B034	时间平移	[30,45)	[30,45)	[445,448)	[446,449)	40	40
B035	频段平移	[15,30)	[6,21)	[457,460)	[457,460)	40	40
B036	频段平移	[0,15)	[9,24)	[495,498)	[495,498)	40	40
B038	时间平移	[40,55)	[40,55)	[485,488)	[486,489)	40	40
B039	频段平移	[20,35)	[30,45)	[477,480)	[477,480)	40	40
B040	频段平移	[5,20)	[2,17)	[424,427)	[424,427)	40	40
C001	间隔调整	[90,93)	[90,93)	[40,42)	[40,42)	8	15
C002	频段平移	[82,85)	[76,79)	[27,29)	[27,29)	8	8
C004	频段平移	[60,63)	[51,54)	[31,33)	[31,33)	8	8
C005	频段平移	[55,58)	[51,54)	[33,35)	[33,35)	8	8
C006	间隔调整	[47,50)	[47,50)	[25,27)	[25,27)	8	5
C007	时间平移	[35,38)	[35,38)	[2,4)	[1,3)	8	8
C008	频段平移	[20,23)	[23,26)	[29,31)	[29,31)	8	8
C009	频段平移	[13,16)	[20,23)	[31,33)	[31,33)	8	8
C010	频段平移	[4,7)	[2,5)	[8,10)	[8,10)	8	8
C012	频段平移	[83,86)	[76,79)	[65,67)	[65,67)	8	8
C013	间隔调整	[70,73)	[70,73)	[75,77)	[75,77)	8	5
C014	频段平移	[65,68)	[75,78)	[92,94)	[92,94)	8	8
C015	时间平移	[50,53)	[50,53)	[110,112)	[106,108)	8	8
C016	频段平移	[42,45)	[50,53)	[88,90)	[88,90)	8	8
C017	频段平移	[36,39)	[27,30)	[95,97)	[95,97)	8	8
C018	时间平移	[20,23)	[20,23)	[107,109)	[108,110)	8	8
C019	频段平移	[12,15)	[20,23)	[86,88)	[86,88)	8	8
C020	频段平移	[2,5)	[1,4)	[65,67)	[65,67)	8	8
C021	时间平移	[95,98)	[95,98)	[125,127)	[126,128)	8	8
C022	频段平移	[86,89)	[76,79)	[147,149)	[147,149)	8	8
C023	时间平移	[76,79)	[76,79)	[171,173)	[170,172)	8	8
C024	撤销	[62,65)	-	[150,152)	-	8	-
C025	间隔调整	[54,57)	[54,57)	[132,134)	[132,134)	8	4
C026	时间平移	[46,49)	[46,49)	[145,147)	[148,150)	8	8
C027	撤销	[33,36)	-	[165,167)	-	8	-
C028	撤销	[24,27)	-	[168,170)	-	8	-
C029	间隔调整	[11,14)	[11,14)	[137,139)	[137,139)	8	9
C030	频段平移	[6,9)	[1,4)	[122,124)	[122,124)	8	8
C031	频段平移	[94,97)	[97,100)	[265,267)	[265,267)	8	8
C032	频段平移	[86,89)	[94,97)	[246,248)	[246,248)	8	8
C033	撤销	[74,77)	-	[252,254)	-	8	-

装备编号	动作	原频段	新频段	原首次时间	新首次时间	原间隔	新间隔
C034	间隔调整	[62,65)	[62,65)	[294,296)	[294,296)	8	9
C035	时间平移	[57,60)	[57,60)	[272,274)	[274,276)	8	8
C036	频段平移	[46,49)	[55,58)	[266,268)	[266,268)	8	8
C037	撤销	[31,34)	-	[270,272)	-	8	-
C038	间隔调整	[22,25)	[22,25)	[285,287)	[285,287)	8	1
C039	时间平移	[16,19)	[16,19)	[264,266)	[261,263)	8	8
C040	时间平移	[5,8)	[5,8)	[242,244)	[239,241)	8	8
C041	频段平移	[90,93)	[97,100)	[303,305)	[303,305)	8	8
C042	间隔调整	[83,86)	[83,86)	[326,328)	[326,328)	8	5
C043	间隔调整	[72,75)	[72,75)	[253,255)	[253,255)	8	1
C044	撤销	[66,69)	-	[334,336)	-	8	-
C045	频段平移	[53,56)	[49,52)	[317,319)	[317,319)	8	8
C046	频段平移	[42,45)	[45,48)	[327,329)	[327,329)	8	8
C047	撤销	[36,39)	-	[341,343)	-	8	-
C048	间隔调整	[21,24)	[21,24)	[346,348)	[346,348)	8	18
C049	间隔调整	[17,20)	[17,20)	[313,315)	[313,315)	8	3
C050	间隔调整	[5,8)	[5,8)	[306,308)	[306,308)	8	7
C051	时间平移	[94,97)	[94,97)	[225,227)	[222,224)	8	8
C052	频段平移	[86,89)	[94,97)	[204,206)	[204,206)	8	8
C053	频段平移	[71,74)	[75,78)	[213,215)	[213,215)	8	8
C054	撤销	[64,67)	-	[218,220)	-	8	-
C055	间隔调整	[52,55)	[52,55)	[213,215)	[213,215)	8	14
C056	间隔调整	[41,44)	[41,44)	[202,204)	[202,204)	8	14
C057	间隔调整	[32,35)	[32,35)	[185,187)	[185,187)	8	14
C058	频段平移	[24,27)	[14,17)	[218,220)	[218,220)	8	8
C059	频段平移	[16,19)	[19,22)	[201,203)	[201,203)	8	8
C061	间隔调整	[90,93)	[90,93)	[450,452)	[450,452)	8	13
C062	频段平移	[86,89)	[77,80)	[420,422)	[420,422)	8	8
C063	频段平移	[73,76)	[80,83)	[433,435)	[433,435)	8	8
C064	间隔调整	[61,64)	[61,64)	[455,457)	[455,457)	8	9
C065	间隔调整	[54,57)	[54,57)	[478,480)	[478,480)	8	16
C066	间隔调整	[45,48)	[45,48)	[442,444)	[442,444)	8	15
C067	频段平移	[33,36)	[27,30)	[451,453)	[451,453)	8	8
C068	时间平移	[27,30)	[27,30)	[461,463)	[463,465)	8	8
C069	频段平移	[15,18)	[24,27)	[441,443)	[441,443)	8	8
C070	间隔调整	[6,9)	[6,9)	[415,417)	[415,417)	8	13
C071	间隔调整	[92,95)	[92,95)	[390,392)	[390,392)	8	14
C072	频段平移	[86,89)	[95,98)	[386,388)	[386,388)	8	8
C073	时间平移	[77,80)	[77,80)	[398,400)	[402,404)	8	8
C074	撤销	[64,67)	-	[396,398)	-	8	-
C075	间隔调整	[57,60)	[57,60)	[391,393)	[391,393)	8	15
C076	频段平移	[43,46)	[42,45)	[384,386)	[384,386)	8	8
C077	频段平移	[31,34)	[21,24)	[361,363)	[361,363)	8	8
C078	间隔调整	[26,29)	[26,29)	[389,391)	[389,391)	8	18
C079	间隔调整	[17,20)	[17,20)	[391,393)	[391,393)	8	1
C080	间隔调整	[0,3)	[0,3)	[368,370)	[368,370)	8	10

装备编号	动作	原频段	新频段	原首次时间	新首次时间	原间隔	新间隔
C081	时间平移	[95,98)	[95,98)	[485,487)	[482,484)	8	8
C082	间隔调整	[82,85)	[82,85)	[507,509)	[507,509)	8	17
C083	间隔调整	[71,74)	[71,74)	[531,533)	[531,533)	8	9
C084	间隔调整	[67,70)	[67,70)	[510,512)	[510,512)	8	9
C085	间隔调整	[51,54)	[51,54)	[492,494)	[492,494)	8	15
C086	时间平移	[42,45)	[42,45)	[405,407)	[402,404)	8	8
C087	间隔调整	[37,40)	[37,40)	[525,527)	[525,527)	8	12
C088	间隔调整	[21,24)	[21,24)	[528,530)	[528,530)	8	6
C089	频段平移	[15,18)	[24,27)	[497,499)	[497,499)	8	8
C090	时间平移	[3,6)	[3,6)	[482,484)	[478,480)	8	8

A.5 模型检验：微型实例穷举对照与规模基准

表 29 24 个随机微型实例上的穷举对照。对每个实例枚举全部动作组合，求出词典序最优向量，与本文的 CP-SAT 求解器结果逐一比对；最大枚举组合数为 262144。

实例	计划数	冲突对	动作组合数	穷举最优向量	CP-SAT 向量	一致	穷举用时/s
1	4	1	3584	(0, 0, 0, 0, 0, 1, 2)	(0, 0, 0, 0, 0, 1, 2)	是	0.01
2	5	2	21952	(0, 0, 0, 0, 0, 2, 3)	(0, 0, 0, 0, 0, 2, 3)	是	0.03
3	6	0	196608	(0, 0, 0, 0, 0, 0, 0)	(0, 0, 0, 0, 0, 0, 0)	是	0.54
4	4	2	4096	(0, 0, 0, 0, 1, 1, 4)	(0, 0, 0, 0, 1, 1, 4)	是	0.00
5	5	2	24576	(0, 0, 0, 1, 0, 2, 5)	(0, 0, 0, 1, 0, 2, 5)	是	0.02
6	6	2	200704	(0, 0, 0, 0, 1, 1, 4)	(0, 0, 0, 0, 1, 1, 4)	是	0.15
7	4	1	2688	(0, 0, 0, 0, 0, 1, 2)	(0, 0, 0, 0, 0, 1, 2)	是	0.01
8	5	2	28672	(0, 0, 0, 1, 0, 1, 2)	(0, 0, 0, 1, 0, 1, 2)	是	0.04
9	6	6	262144	(0, 0, 0, 1, 1, 2, 7)	(0, 0, 0, 1, 1, 2, 7)	是	0.08
10	4	4	3584	(0, 0, 0, 1, 1, 1, 6)	(0, 0, 0, 1, 1, 1, 6)	是	0.00
11	5	2	13440	(0, 0, 0, 2, 0, 1, 5)	(0, 0, 0, 2, 0, 1, 5)	是	0.01
12	6	5	200704	(0, 0, 0, 1, 1, 1, 5)	(0, 0, 0, 1, 1, 1, 5)	是	0.13
13	4	2	2688	(0, 0, 0, 0, 1, 0, 2)	(0, 0, 0, 0, 1, 0, 2)	是	0.01
14	5	1	28672	(0, 0, 0, 1, 0, 0, 1)	(0, 0, 0, 1, 0, 0, 1)	是	0.05
15	6	4	229376	(0, 0, 0, 0, 2, 1, 5)	(0, 0, 0, 0, 2, 1, 5)	是	0.10
16	4	0	3584	(0, 0, 0, 0, 0, 0, 0)	(0, 0, 0, 0, 0, 0, 0)	是	0.01
17	5	3	28672	(0, 0, 0, 0, 1, 1, 4)	(0, 0, 0, 0, 1, 1, 4)	是	0.03
18	6	5	129024	(0, 0, 1, 1, 1, 0, 3)	(0, 0, 1, 1, 1, 0, 3)	是	0.03
19	4	2	2304	(0, 0, 0, 1, 0, 1, 4)	(0, 0, 0, 1, 0, 1, 4)	是	0.00
20	5	3	28672	(0, 0, 0, 0, 2, 1, 5)	(0, 0, 0, 0, 2, 1, 5)	是	0.03
21	6	3	107520	(0, 0, 0, 1, 1, 1, 4)	(0, 0, 0, 1, 1, 1, 4)	是	0.09
22	4	2	4096	(0, 0, 0, 0, 1, 0, 2)	(0, 0, 0, 0, 1, 0, 2)	是	0.01
23	5	3	24576	(0, 0, 0, 1, 1, 1, 5)	(0, 0, 0, 1, 1, 1, 5)	是	0.02
24	6	3	172032	(0, 0, 0, 0, 0, 3, 4)	(0, 0, 0, 0, 0, 3, 4)	是	0.16

表 31 合成实例上的规模基准 ($n = 150, \dots, 600$, 时间范围随 n 等比放大以保持密度)。CP-SAT 一行采用固定 120 s 时限, 故其状态只说明 “在该预算内找到了可行解”, 不代表可证明最优。

n	时间范围	冲突对	成对检测/s	扫描线/s	变量数	约束数	建模/s	CP-SAT 状态	求解/s	撤销合计	调整合计
150	651	329	0.01	0.004	4582	32154	0.3	FEASIBLE	121.1	20	115
300	1191	717	0.04	0.102	9276	64600	0.9	FEASIBLE	122.5	73	190
600	2252	1423	0.15	0.016	18485	132121	1.6	FEASIBLE	124.2	162	381

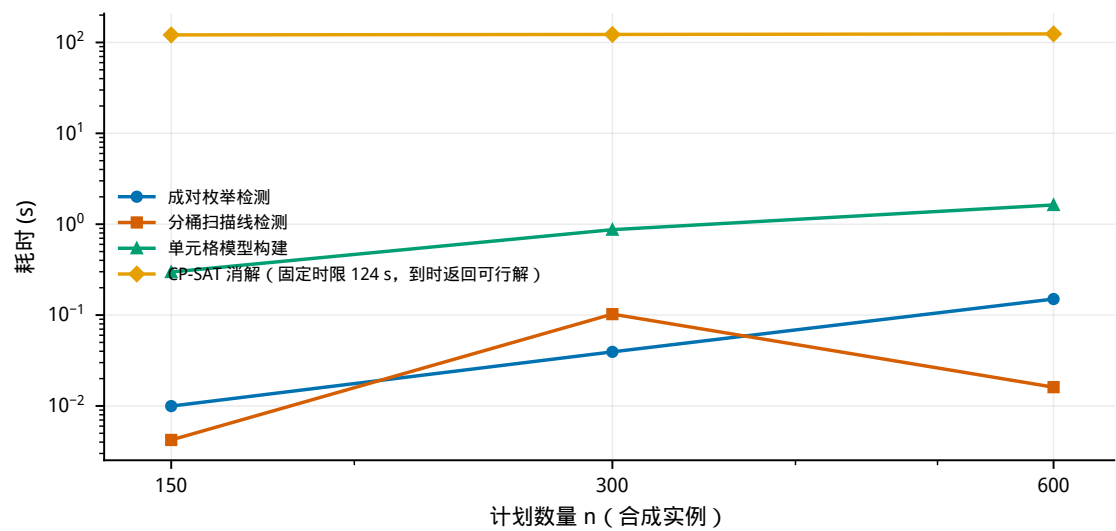


图 11 算法的规模基准。合成实例 $n = 150, \dots, 600$ 下两种检测算法与模型构建的用时, 以及 CP-SAT 在固定 120 s 时限下的求解时间 (纵轴为对数刻度)。检测算法始终为亚秒级, 求解时间则受时限截断。

B 支撑材料文件清单

下表列出支撑材料压缩包内的全部文件 (相对路径) 及其用途。压缩包内另含 MANIFEST.json, 记录每个文件的 SHA-256。

文件	说明
AI_USAGE.md	AI 工具使用记录 (源文件)
AI 工具使用详情.pdf	AI 工具使用详情
MANIFEST.json	支撑材料成员清单与 SHA-256
Makefile	本地快捷命令 (只提交云端任务)
README.md	仓库说明与复现入口
REPRODUCE.md	复现说明
UNLICENSE	公有领域声明 (Unlicense)

文件	说明
docs/ARCHITECTURE.md	工程规范、建模标准、决策记录
docs/CONTRIBUTING.md	工程规范、建模标准、决策记录
docs/DATA_NOTES.md	工程规范、建模标准、决策记录
docs/ENGINEERING_STANDARD.md	工程规范、建模标准、决策记录
docs/MODELING_STANDARD.md	工程规范、建模标准、决策记录
docs/RESULTS.md	工程规范、建模标准、决策记录
docs/RUNBOOK.md	工程规范、建模标准、决策记录
docs/adr/0001-cloud-only-execution.md	工程规范、建模标准、决策记录
docs/adr/0002-stage-run-manifest-model.md	工程规范、建模标准、决策记录
docs/adr/0003-independent-repos-vendored-runtime.md	工程规范、建模标准、决策记录
docs/mdr/0000-template.md	工程规范、建模标准、决策记录
docs/mdr/0001-conflict-semantics.md	工程规范、建模标准、决策记录
docs/mdr/0002-resolution-options.md	工程规范、建模标准、决策记录
docs/mdr/0003-lexicographic-objective.md	工程规范、建模标准、决策记录
docs/mdr/0004-solver-and-formulation.md	工程规范、建模标准、决策记录
docs/mdr/0005-q3-packing-interpretation.md	工程规范、建模标准、决策记录
docs/mdr/0006-q4-interval-adjustment.md	工程规范、建模标准、决策记录
docs/mdr/0007-verification-design.md	工程规范、建模标准、决策记录

文件	说明
docs/mdr/0008-warm-starts-and-time-limits.md	工程规范、建模标准、决策记录
docs/mdr/0009-incumbent-preserving-lexicographic-search.md	工程规范、建模标准、决策记录
docs/mdr/0010-q3-unlimited-shift-interpretation.md	工程规范、建模标准、决策记录
docs/mdr/README.md	工程规范、建模标准、决策记录
docs/problem_brief.md	工程规范、建模标准、决策记录
forge/__init__.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/config.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/context.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/contracts.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/events.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/hashing.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/manifest.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/modal_app.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/packaging.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/pdfqa.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/plotting.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/runner.py	云端运行时（阶段执行、清单、契约、排版、打包）
forge/tex.py	云端运行时（阶段执行、清单、契约、排版、打包）

文件	说明
forge/xlsx.py	云端运行时（阶段执行、清单、契约、排版、打包）
manuscript/ai_usage.tex	论文 LaTeX 源码
manuscript/main.tex	论文 LaTeX 源码
manuscript/references.bib	论文 LaTeX 源码
manuscript/sections/00_abstract.tex	论文 LaTeX 源码
manuscript/sections/01_restatement.tex	论文 LaTeX 源码
manuscript/sections/02_analysis.tex	论文 LaTeX 源码
manuscript/sections/03_assumptions.tex	论文 LaTeX 源码
manuscript/sections/04_models.tex	论文 LaTeX 源码
manuscript/sections/08_validation.tex	论文 LaTeX 源码
manuscript/sections/09_evaluation.tex	论文 LaTeX 源码
manuscript/sections/10_ai_statement.tex	论文 LaTeX 源码
manuscript/sections/ai_usage_body.tex	论文 LaTeX 源码
manuscript/sections/appendix_repro.tex	论文 LaTeX 源码
manuscript/sections/appendix_tables.tex	论文 LaTeX 源码
pipelines/__init__.py	本题建模与求解程序
pipelines/common/__init__.py	通用阶段：数据接入、门禁、论文、质量检查、打包发布
pipelines/common/stages.py	通用阶段：数据接入、门禁、论文、质量检查、打包发布
pipelines/common/validation.py	通用阶段：数据接入、门禁、论文、质量检查、打包发布
pipelines/d/__init__.py	本题建模与求解程序
pipelines/d/contracts.py	本题建模与求解程序
pipelines/d/detect.py	本题建模与求解程序
pipelines/d/highs_util.py	本题建模与求解程序
pipelines/d/pack.py	本题建模与求解程序
pipelines/d/plans.py	本题建模与求解程序
pipelines/d/report.py	本题建模与求解程序
pipelines/d/resolve.py	本题建模与求解程序
pipelines/d/stages.py	本题建模与求解程序
pipelines/d/synth.py	本题建模与求解程序
pipelines/d/validators.py	本题建模与求解程序
pyproject.toml	项目元数据与静态检查配置

文件	说明
runs/20260912-102234-ea48b61/bench/events.json	结构化运行日志
l	
runs/20260912-102234-ea48b61/bench/manifest.json	阶段清单(输入输出散列、环境、参数)
runs/20260912-102234-ea48b61/bench/numbers.json	
runs/20260912-102234-ea48b61/bench/scaling.json	
runs/20260912-102234-ea48b61/bench/tiny.json	
runs/20260912-102234-ea48b61/compare/comparison.json	
runs/20260912-102234-ea48b61/compare/events.jsonl	结构化运行日志
runs/20260912-102234-ea48b61/compare/manifest.json	阶段清单(输入输出散列、环境、参数)
runs/20260912-102234-ea48b61/compare/numbers.json	
runs/20260912-102234-ea48b61/detect/conflicts.json	
runs/20260912-102234-ea48b61/detect/events.jsonl	结构化运行日志
runs/20260912-102234-ea48b61/detect/manifest.json	阶段清单(输入输出散列、环境、参数)
runs/20260912-102234-ea48b61/detect/numbers.json	
runs/20260912-102234-ea48b61/detect/plans.json	
runs/20260912-102234-ea48b61/detect/stats.json	
runs/20260912-102234-ea48b61/detect/validation_report.json	
runs/20260912-102234-ea48b61/figures/events.jsonl	论文图(矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/fig_bench_scaling.pdf	论文图(矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/fig_bench_scaling.png	论文图(矢量 PDF 与预览 PNG)

文件	说明
runs/20260912-102234-ea48b61/figures/figures/f ig_q1_gantt.pdf	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q1_gantt.png	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q1_structure.pdf	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q1_structure.png	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q2_resolution.pdf	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q2_resolution.png	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q2_schemes.pdf	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q2_schemes.png	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q3_layout.pdf	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q3_layout.png	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q3_repack.pdf	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q3_repack.png	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q4_comparison.pdf	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q4_comparison.png	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q4_resolution.pdf	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_q4_resolution.png	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_sensitivity.pdf	论文图 (矢量 PDF 与预览 PNG)
runs/20260912-102234-ea48b61/figures/figures/f ig_sensitivity.png	论文图 (矢量 PDF 与预览 PNG)

文件	说明
runs/20260912-102234-ea48b61/figures/manifest.json	论文图（矢量 PDF 与预览 PNG）
runs/20260912-102234-ea48b61/figures/numbers.json	论文图（矢量 PDF 与预览 PNG）
runs/20260912-102234-ea48b61/ingest/data/附件 1_Sheet1.parquet	原始数据接入结果（散列、题目文本、规范化数据）
runs/20260912-102234-ea48b61/ingest/events.jsonl	原始数据接入结果（散列、题目文本、规范化数据）
runs/20260912-102234-ea48b61/ingest/inventory.json	原始数据接入结果（散列、题目文本、规范化数据）
runs/20260912-102234-ea48b61/ingest/manifest.json	原始数据接入结果（散列、题目文本、规范化数据）
runs/20260912-102234-ea48b61/ingest/problem_text.txt	原始数据接入结果（散列、题目文本、规范化数据）
runs/20260912-102234-ea48b61/lint/events.jsonl	静态检查报告
runs/20260912-102234-ea48b61/lint/lint_report.json	静态检查报告
runs/20260912-102234-ea48b61/lint/manifest.json	静态检查报告
runs/20260912-102234-ea48b61/lint/mypy.txt	静态检查报告
runs/20260912-102234-ea48b61/lint/ruff_check.txt	静态检查报告
runs/20260912-102234-ea48b61/lint/ruff_format.txt	静态检查报告
runs/20260912-102234-ea48b61/pack/events.jsonl	结构化运行日志
runs/20260912-102234-ea48b61/pack/manifest.json	阶段清单（输入输出散列、环境、参数）
runs/20260912-102234-ea48b61/pack/new_plans.json	
runs/20260912-102234-ea48b61/pack/numbers.json	
runs/20260912-102234-ea48b61/pack/pack_report.json	
runs/20260912-102234-ea48b61/pack/repack_plans.json	
runs/20260912-102234-ea48b61/pack/validation_report.json	

文件	说明
runs/20260912-102234-ea48b61/qa/events.jsonl	论文与结果文件质量门禁报告
runs/20260912-102234-ea48b61/qa/manifest.json	论文与结果文件质量门禁报告
runs/20260912-102234-ea48b61/qa/qa_report.json	论文与结果文件质量门禁报告
runs/20260912-102234-ea48b61/resolve/alternatives.json	
runs/20260912-102234-ea48b61/resolve/decisions.json	
runs/20260912-102234-ea48b61/resolve/events.jsonl	结构化运行日志
runs/20260912-102234-ea48b61/resolve/manifest.json	阶段清单(输入输出散列、环境、参数)
runs/20260912-102234-ea48b61/resolve/numbers.json	
runs/20260912-102234-ea48b61/resolve/resolved_plans.json	
runs/20260912-102234-ea48b61/resolve/solver_report.json	
runs/20260912-102234-ea48b61/resolve/validation_report.json	
runs/20260912-102234-ea48b61/resolve_interval/alternatives.json	
runs/20260912-102234-ea48b61/resolve_interval/decisions.json	
runs/20260912-102234-ea48b61/resolve_interval/events.jsonl	结构化运行日志
runs/20260912-102234-ea48b61/resolve_interval/manifest.json	阶段清单(输入输出散列、环境、参数)
runs/20260912-102234-ea48b61/resolve_interval/numbers.json	
runs/20260912-102234-ea48b61/resolve_interval/resolved_plans.json	
runs/20260912-102234-ea48b61/resolve_interval/solver_report.json	

文件	说明
runs/20260912-102234-ea48b61/resolve_interval/validation_report.json	
runs/20260912-102234-ea48b61/results/events.jsonl	结果工作簿与求解输出
runs/20260912-102234-ea48b61/results/manifest.json	结果工作簿与求解输出
runs/20260912-102234-ea48b61/results/numbers.json	结果工作簿与求解输出
runs/20260912-102234-ea48b61/results/result1.xlsx	结果工作簿与求解输出
runs/20260912-102234-ea48b61/results/result2.xlsx	结果工作簿与求解输出
runs/20260912-102234-ea48b61/results/result3.xlsx	结果工作簿与求解输出
runs/20260912-102234-ea48b61/results/result4.xlsx	结果工作簿与求解输出
runs/20260912-102234-ea48b61/results/results_report.json	结果工作簿与求解输出
runs/20260912-102234-ea48b61/sensitivity/cases.json	
runs/20260912-102234-ea48b61/sensitivity/events.jsonl	结构化运行日志
runs/20260912-102234-ea48b61/sensitivity/manifest.json	阶段清单(输入输出散列、环境、参数)
runs/20260912-102234-ea48b61/sensitivity/numbers.json	
runs/20260912-102234-ea48b61/tables/events.jsonl	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/manifest.json	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/numbers.json	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_bench_scaling.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_bench_scaling.tex	论文表(LaTeX 片段与 CSV)

文件	说明
runs/20260912-102234-ea48b61/tables/tables/tab_bench_tiny.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_bench_tiny.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_data.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_data.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q1_category.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q1_category.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q1_conflicts.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q1_conflicts.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q1_multiplicity.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q1_multiplicity.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q1_summary.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q1_summary.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q1_top_degree.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q1_top_degree.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_adjustments.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_adjustments.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_cancelled.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_cancelled.tex	论文表(LaTeX 片段与 CSV)

文件	说明
runs/20260912-102234-ea48b61/tables/tables/tab_q2_model.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_model.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_schemes.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_schemes.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_solver.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_solver.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_table1.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q2_table1.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q3_plans.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q3_plans.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q3_summary.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q3_summary.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_adjustments.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_adjustments.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_cancelled.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_cancelled.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_compare.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_compare.tex	论文表(LaTeX 片段与 CSV)

文件	说明
runs/20260912-102234-ea48b61/tables/tables/tab_q4_model.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_model.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_schemes.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_schemes.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_solver.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_solver.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_table1.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_q4_table1.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_sensitivity.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_sensitivity.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_validation.csv	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/tables/tables/tab_validation.tex	论文表(LaTeX 片段与 CSV)
runs/20260912-102234-ea48b61/test/coverage.json	测试报告与覆盖率
runs/20260912-102234-ea48b61/test/events.jsonl	测试报告与覆盖率
runs/20260912-102234-ea48b61/test/junit.xml	测试报告与覆盖率
runs/20260912-102234-ea48b61/test/manifest.json	测试报告与覆盖率
runs/20260912-102234-ea48b61/test/pytest.txt	测试报告与覆盖率
runs/20260912-102234-ea48b61/test/test_report.json	测试报告与覆盖率
runs/20260912-102234-ea48b61/validate/data_profile.json	数据契约校验报告与数据画像

文件	说明
runs/20260912-102234-ea48b61/validate/events.jsonl	数据契约校验报告与数据画像
runs/20260912-102234-ea48b61/validate/manifest.json	数据契约校验报告与数据画像
runs/20260912-102234-ea48b61/validate/validation_report.json	数据契约校验报告与数据画像
tests/__init__.py	单元 / 性质 / 集成测试
tests/conftest.py	单元 / 性质 / 集成测试
tests/integration/__init__.py	单元 / 性质 / 集成测试
tests/integration/test_ingest_validate.py	单元 / 性质 / 集成测试
tests/unit/__init__.py	单元 / 性质 / 集成测试
tests/unit/test_config.py	单元 / 性质 / 集成测试
tests/unit/test_contracts.py	单元 / 性质 / 集成测试
tests/unit/test_d_hints.py	单元 / 性质 / 集成测试
tests/unit/test_d_pack.py	单元 / 性质 / 集成测试
tests/unit/test_d_plans_detect.py	单元 / 性质 / 集成测试
tests/unit/test_d_report.py	单元 / 性质 / 集成测试
tests/unit/test_d_resolve.py	单元 / 性质 / 集成测试
tests/unit/test_hashing.py	单元 / 性质 / 集成测试
tests/unit/test_manifest.py	单元 / 性质 / 集成测试
tests/unit/test_packaging_text.py	单元 / 性质 / 集成测试
tests/unit/test_runner.py	单元 / 性质 / 集成测试
tests/unit/test_xlsx.py	单元 / 性质 / 集成测试
tools/__init__.py	本地控制面：提交云端任务、下载、校验散列
tools/cli.py	本地控制面：提交云端任务、下载、校验散列

C 复现记录

论文中的全部数值、表格与图形均由同一次云端运行（run 20260912-102234-ea48b61, 代码版本 073408459d426ad3354178c73b523a0708d57937, 构建日期 2026-09-12）生成。各阶段状态与输出摘要如下。

阶段	状态	用时/s	输出摘要 (SHA-256 前 12 位)
bench	completed	375.0	45603a6ec806
compare	completed	0.0	16ef61ffedc3
detect	completed	1.0	b64918146437
figures	completed	14.0	4d7ea02b97ce
fmt	completed	0.0	2b76b8e10758
ingest	completed	0.0	1c6f1c5a038c
lint	completed	14.0	cd5e1ca31d0d
pack	completed	129.0	a2d2e3130377
package	completed	17.0	2a060d975f8e
paper	running	0.0	
qa	completed	1.0	5fb541f84015
release	completed	0.0	4c85432a40b5
resolve	completed	2286.0	1ed01f40c94d
resolve_interval	completed	2546.0	ab59b5991863
results	completed	1.0	e2f110745065
sensitivity	completed	1003.0	1ee2b3594487
tables	completed	4.0	5c4704b3ec4a
test	completed	11.0	d964bbe98b59
validate	completed	0.0	4dad00dd8173

C.1 复现步骤

本仓库的设计原则是“唯一计算面”：一切数据解析、数值计算、优化求解、测试、绘图、L^AT_EX 排版与打包都在云端 (Modal) 执行，本地只负责编辑文本、提交任务、下载产物并核对 SHA-256。上表中 paper 一行显示为 running，是因为该表由 paper 阶段自身在编译前生成，其自身状态此刻尚未写入；其余阶段均已 completed。复现步骤如下：

1. 安装 Modal 客户端并完成鉴权后，在仓库根目录执行 `python3 tools/cli.py provision` 验证云端工具链 (T_EX Live、中文字体、CP-SAT、HiGHS)。
2. 数据接入与契约校验: `python3 tools/cli.py run ingest,validate,detect --new-run --size small`。
3. 两个重阶段分离运行 (各约 40 分钟): `run resolve --size large --spawn` 与 `run resolve_interval --size large --spawn`，随后 `wait <ID>` 或轮询 `status`。
4. 其余科学阶段: `run pack,bench,sensitivity --size large`，再 `run compare,results,figures,tables`。

5. 门禁与论文: `run lint,test` 与 `run paper,qa,package,release`。
6. 下载与核对: `download --stage <name>` 会逐文件核对 SHA-256; `release --version <tag>` 生成交付目录与发布清单。

各阶段的确切参数（求解器线程数、逐级时限）记录在仓库的运行手册中，也记录在每个阶段的 `manifest.json` 内。

C.2 可复现性的保证

- **固定输入**：题目 PDF 与附件在仓库中原样保存，`ingest` 阶段记录其 SHA-256，任何代码不得改写。
- **固定环境**：容器镜像的依赖版本固定；每个阶段的 `manifest.json` 记录 Python 版本、平台、CPU 数与全部包版本。
- **固定随机源**：所有随机性经统一的种子接口派生，求解器的随机种子固定。
- **固定热启动**：词典序求解的现任解提示保存在仓库的 `configs/hints/` 中（附带来源 `run`、代码版本与目标向量）。由命题 5.4，带提示重跑所得向量在词典序上不会劣于提示，因此本文结果可由仓库直接复现。提示只影响搜索起点，不改变可行域，也不改变任何一级的最优值。
- **数字的单一来源**：论文正文不允许手工键入结果数字；每个数值或来自阶段登记的宏，或来自 `tables` 阶段生成的表格文件。若两个阶段登记了同名而不同值的数字，论文构建会报告冲突并由质检门禁拦截。

C.3 已知的不确定性

150 个计划的完整模型采用多线程求解，单次求解耗时数十分钟，本文未做重复求解以验证位对位的确定性（记为“未检验”）；微型实例上的确定性检验为 24/24 通过。需要强调：本文已证明最优的结论（定理 5.5、6.2）由求解器的证明下界保证，与是否复现出同一个最优解无关；未证明最优的层级则本就以区间形式报告。

D 完整程序

以下按文件收录建模、求解、检验、绘图、排版与打包的全部源程序，文件名与附录 B 一致。运行方式见附录 C。

Makefile

```
1 # Thin wrappers over tools/cli.py. Nothing here computes locally.
2 PY ?= python3
3 CLI := $(PY) tools/cli.py
4 STAGE ?= ingest
5 SIZE ?= small
6 RUN ?=
7 RUNFLAG := $(if $(RUN),--run-id $(RUN),)
8 FORCEFLAG := $(if $(FORCE),--force,)
9
10 .PHONY: fmt help provision run pipeline exec status logs runs download release lint test paper qa package smoke
11
12 help:
13     @echo "make provision          # verify cloud toolchain"
14     @echo "make run STAGE=q1 SIZE=medium # run one stage (RUN=<id> to continue a run, FORCE=1 to rerun)"
15     @echo "make pipeline                # ingest,validate,lint,test in one container"
16     @echo "make status | logs STAGE=q1 | runs | download STAGE=results"
17     @echo "make paper | qa | package | release VERSION=v1.0.0"
18
19 provision:
20     $(CLI) provision
21
22 run:
23     $(CLI) run $(STAGE) $(RUNFLAG) --size $(SIZE) $(FORCEFLAG)
24
25 pipeline:
26     $(CLI) run ingest,validate,lint,test $(RUNFLAG) --size $(SIZE) $(FORCEFLAG)
27
28 exec:
29     $(CLI) exec --code '$(CODE)' $(RUNFLAG)
30
31 status:
32     $(CLI) status $(RUNFLAG)
33
34 logs:
35     $(CLI) logs $(STAGE) $(RUNFLAG)
36
37 runs:
38     $(CLI) runs
39
40 download:
41     $(CLI) download $(RUNFLAG) $(if $(STAGE),--stage $(STAGE),)
42
43 lint:
44     $(CLI) run lint $(RUNFLAG) $(FORCEFLAG)
45
46 test:
47     $(CLI) run test $(RUNFLAG) $(FORCEFLAG)
48
49 paper:
50     $(CLI) run paper $(RUNFLAG) --size $(SIZE) $(FORCEFLAG)
51
52 qa:
53     $(CLI) run qa $(RUNFLAG) $(FORCEFLAG)
54
55 package:
56     $(CLI) run package $(RUNFLAG) $(FORCEFLAG)
57
58 release:
59     $(CLI) run release $(RUNFLAG) $(FORCEFLAG) && $(CLI) release --version $(VERSION) $(RUNFLAG) --force
60
61 smoke:
62     $(CLI) run ingest,validate,lint,test,paper,qa,package,release --new-run --param require_results=false --param
63     ↪ allow_missing_results=true
```

```

64 fmt:
65     $(CLI) fmt $(RUNFLAG)

```

configs/default.toml

```

1 # Forge 运行时配置（默认档）。Stage 代码通过 ctx.cfg("a.b.c") 读取；
2 # `--profile <name>` 会叠加 configs/<name>.toml；`--param config_overrides='{...}'` 可临时覆盖。
3
4 [run]
5 seed = 20260912
6
7 [gates]
8 lint_required = ["ruff_check", "ruff_format", "mypy"]
9 mypy_paths = ["forge"]
10
11 [paper]
12 main = "main.tex"
13 # 需在同一 run 中已完成的 stage；其 tables/*.tex 与 figures/*.pdf/png 会被并入排版
14 sources = ["figures", "tables"]
15 max_body_pages = 30
16 max_overfull_pt = 20.0
17 code_globs = [
18     "forge/**/*.py", "pipelines/**/*.py", "tools/**/*.py", "tests/**/*.py",
19     "configs/*.toml", "configs/*.json", "Makefile", "pyproject.toml",
20 ]
21
22 [qa]
23 required_stages = ["lint", "test"]
24 results_stage = "results"
25 require_results = true
26 identity_denylist = [] # 硬门禁（正则）：出现即失败
27 identity_watchlist = ["大学", "学院", "赛区", "队号", "指导教师", "[\\w.-]+@[\\w.-]+\\.\\w+"] # 软提示：写入报告供复核
28
29 [package]
30 stages = ["ingest", "validate", "detect", "resolve", "pack", "resolve_interval", "compare", "bench", "sensitivity", "
    ↳ results", "figures", "tables", "lint", "test", "qa"]
31 code_globs = [
32     "forge/**/*.py", "pipelines/**/*.py", "tools/**/*.py", "tests/**/*.py", "configs/**",
33     "manuscript/**/*.tex", "manuscript/**/*.bib", "docs/**/*.md", "Makefile", "pyproject.toml",
34     "README.md", "UNLICENSE", "AI_USAGE.md",
35 ]
36 max_bytes = 20971520
37
38 [release]
39 paper_name = "论文.pdf"

```

configs/project.json

```

1 {
2   "letter": "D",
3   "package": "d",
4   "slug": "spectrum-conflict",
5   "title": "时频冲突检测与消解",
6   "app": "cumcm2026-d-spectrum-conflict",
7   "volume": "cumcm2026-d-artifacts",
8   "problem_pdf": "D题.pdf",
9   "attachments_dir": "附件",
10  "templates_dir": "附件/附件2",
11  "result_files": [
12    "result1.xlsx",

```

```

13     "result2.xlsx",
14     "result3.xlsx",
15     "result4.xlsx"
16 ],
17 "github_repo": "cumcm2026-d-spectrum-conflict"
18 }

```

forge/__init__.py

```

1 """Forge — cloud-only research engineering runtime (CUMCM 2026).
2
3 Every computation (data parsing, modelling, solving, tests, figures, LaTeX, PDF QA,
4 packaging) runs inside Modal. The local machine only edits text, drives git,
5 submits jobs, downloads artifacts and verifies transport hashes.
6 """
7
8 __version__ = "1.0.0"

```

forge/config.py

```

1 """TOML configuration with profile overlays and dotted access."""
2
3 from __future__ import annotations
4
5 import json
6 import tomllib
7 from pathlib import Path
8 from typing import Any
9
10 from .hashing import sha256_json
11
12 _MISSING = object()
13
14
15 def deep_merge(base: dict[str, Any], override: dict[str, Any]) -> dict[str, Any]:
16     out = dict(base)
17     for key, value in override.items():
18         if isinstance(value, dict) and isinstance(out.get(key), dict):
19             out[key] = deep_merge(out[key], value)
20         else:
21             out[key] = value
22     return out
23
24
25 class Config:
26     """Read-only dotted-access view over a merged configuration mapping."""
27
28     def __init__(self, data: dict[str, Any]) -> None:
29         self._data = json.loads(json.dumps(data))
30
31     def get(self, key: str, default: Any = None) -> Any:
32         cursor: Any = self._data
33         for part in key.split("."):
34             if not isinstance(cursor, dict) or part not in cursor:
35                 return default
36             cursor = cursor[part]
37         return cursor
38
39     def __getitem__(self, key: str) -> Any:
40         value = self.get(key, _MISSING)

```

```

41         if value is _MISSING:
42             raise KeyError(key)
43         return value
44
45     def section(self, key: str) -> dict[str, Any]:
46         value = self.get(key, {})
47         return dict(value) if isinstance(value, dict) else {}
48
49     def as_dict(self) -> dict[str, Any]:
50         return json.loads(json.dumps(self._data))
51
52     @property
53     def digest(self) -> str:
54         return sha256_json(self._data)
55
56
57 def load_project(repo: Path) -> dict[str, Any]:
58     return json.loads((repo / "configs" / "project.json").read_text(encoding="utf-8"))
59
60
61 def load_config(repo: Path, profile: str | None = None, overrides: dict[str, Any] | None = None) -> Config:
62     data = tomlib.loads((repo / "configs" / "default.toml").read_text(encoding="utf-8"))
63     if profile:
64         data = deep_merge(data, tomlib.loads((repo / "configs" / f"{profile}.toml").read_text(encoding="utf-8")))
65     if overrides:
66         data = deep_merge(data, overrides)
67     return Config(data)

```

forge/context.py

```

1  """Stage execution context: paths, config, params, logging, seeds and paper numbers."""
2
3  from __future__ import annotations
4
5  import json
6  import random
7  import re
8  import zlib
9  from pathlib import Path
10 from typing import Any
11
12 from .config import Config, load_config, load_project
13 from .events import EventLogger
14
15 _NUMBER_KEY = re.compile(r"^[A-Za-z]+$")
16
17
18 class StageContext:
19     """Everything a stage needs. Stages must write only under ``stage_dir``."""
20
21     def __init__(self, run_id: str, stage: str, params: dict[str, Any], repo: Path, vol: Path) -> None:
22         self.run_id = run_id
23         self.stage = stage
24         self.params = dict(params)
25         self.repo = repo
26         self.vol = vol
27         self.project = load_project(repo)
28         overrides = self.params.get("config_overrides") or None
29         self.config: Config = load_config(repo, self.params.get("profile"), overrides)
30         self.run_dir = vol / "runs" / run_id
31         self.stage_dir = self.run_dir / stage
32         self.stage_dir.mkdir(parents=True, exist_ok=True)
33         self.log = EventLogger(self.stage_dir / "events.jsonl", run_id, stage)

```

```

34     self._numbers: dict[str, str] = {}
35
36     # ---- raw inputs (immutable, at repo root, exactly as delivered by the organisers) ----
37     @property
38     def problem_pdf(self) -> Path:
39         return self.repo / self.project["problem_pdf"]
40
41     @property
42     def attachments(self) -> Path:
43         return self.repo / self.project["attachments_dir"]
44
45     @property
46     def templates(self) -> Path:
47         return self.repo / self.project["templates_dir"]
48
49     # ---- outputs ----
50     def out(self, *parts: str) -> Path:
51         path = self.stage_dir.joinpath(*parts)
52         path.parent.mkdir(parents=True, exist_ok=True)
53         return path
54
55     def dep(self, stage: str) -> Path:
56         """Directory of a completed dependency stage in the same run (raises otherwise)."""
57         directory = self.run_dir / stage
58         manifest = directory / "manifest.json"
59         if not manifest.exists():
60             raise RuntimeError(f"dependency stage '{stage}' has not run in run {self.run_id}")
61         status = json.loads(manifest.read_text(encoding="utf-8")).get("status")
62         if status != "completed":
63             raise RuntimeError(f"dependency stage '{stage}' has status {status!r} in run {self.run_id}")
64         return directory
65
66     def has_stage(self, stage: str) -> bool:
67         manifest = self.run_dir / stage / "manifest.json"
68         return manifest.exists() and json.loads(manifest.read_text(encoding="utf-8")).get("status") == "completed"
69
70     def dep_data(self, stage: str, name: str) -> Any:
71         return json.loads((self.dep(stage) / name).read_text(encoding="utf-8"))
72
73     # ---- parameters / configuration ----
74     def param(self, key: str, default: Any = None) -> Any:
75         return self.params.get(key, default)
76
77     def cfg(self, key: str, default: Any = None) -> Any:
78         return self.config.get(key, default)
79
80     # ---- determinism ----
81     def seed(self, salt: str = "") -> int:
82         base = int(self.cfg("run.seed", 0))
83         return (base + zlib.crc32(salt.encode("utf-8"))) % (2**31 - 1)
84
85     def seed_everything(self, salt: str = "") -> int:
86         value = self.seed(salt)
87         random.seed(value)
88         try:
89             import numpy as np
90
91             np.random.seed(value) # noqa: NPY002 - legacy global RNG used by third-party libraries
92         except ImportError: # pragma: no cover
93             pass
94         self.log.info("seed", salt=salt, value=value)
95         return value
96
97     # ---- numbers that flow into the paper as \val<Key> macros ----
98     def number(self, key: str, value: Any, fmt: str | None = None) -> str:
99         if not _NUMBER_KEY.match(key):

```

```

100         raise ValueError(f"paper number keys must be letters only (TeX macro names); got {key!r}")
101     if fmt is not None:
102         text = format(value, fmt)
103     elif isinstance(value, bool):
104         text = str(value)
105     elif isinstance(value, int):
106         text = str(value)
107     elif isinstance(value, float):
108         text = f"{value:.4f}"
109     else:
110         text = str(value)
111     self._numbers[key] = text
112     return text
113
114 def flush_numbers(self) -> None:
115     if not self._numbers:
116         return
117     path = self.stage_dir / "numbers.json"
118     merged: dict[str, str] = {}
119     if path.exists():
120         merged.update(json.loads(path.read_text(encoding="utf-8")))
121     merged.update(self._numbers)
122     path.write_text(json.dumps(merged, ensure_ascii=False, indent=2, sort_keys=True), encoding="utf-8")
123
124 # ---- json helpers ----
125 def write_json(self, name: str, obj: Any) -> Path:
126     path = self.out(name)
127     path.write_text(json.dumps(obj, ensure_ascii=False, indent=2, default=str), encoding="utf-8")
128     return path
129
130 def read_json(self, path: Path) -> Any:
131     return json.loads(path.read_text(encoding="utf-8"))

```

forge/contracts.py

```

1  """Lightweight DataFrame contracts (schema + range + monotonicity + custom checks)."""
2
3  from __future__ import annotations
4
5  import re
6  from collections.abc import Callable
7  from dataclasses import dataclass
8  from typing import Any
9
10 import numpy as np
11 import pandas as pd
12
13
14 @dataclass(frozen=True)
15 class Column:
16     name: str
17     kind: str = "any" # int | float | number | str | datetime | any
18     nullable: bool = False
19     min: float | None = None
20     max: float | None = None
21     regex: str | None = None
22     unique: bool = False
23     monotonic: str | None = None # increasing | non_decreasing | decreasing | non_increasing
24     allowed: tuple[Any, ...] | None = None
25     step: float | None = None # constant difference between consecutive values
26
27
28 @dataclass(frozen=True)

```

```

29 class FrameContract:
30     name: str
31     columns: tuple[Column, ...]
32     min_rows: int = 1
33     max_rows: int | None = None
34     extra_columns: str = "forbid" # forbid | allow
35     checks: tuple[Callable[[pd.DataFrame], list[str]], ...] = ()
36
37
38 def normalise_columns(df: pd.DataFrame) -> pd.DataFrame:
39     out = df.copy()
40     out.columns = pd.Index([str(c).strip() for c in out.columns])
41     return out
42
43
44 def validate_frame(df: pd.DataFrame, contract: FrameContract) -> dict[str, Any]:
45     df = normalise_columns(df)
46     errors: list[str] = []
47     warnings: list[str] = []
48     expected = [c.name for c in contract.columns]
49     missing = [c for c in expected if c not in df.columns]
50     extra = [c for c in df.columns if c not in expected]
51     if missing:
52         errors.append(f"missing columns: {missing}")
53     if extra and contract.extra_columns == "forbid":
54         errors.append(f"unexpected columns: {extra}")
55     n = len(df)
56     if n < contract.min_rows:
57         errors.append(f"rows {n} < min_rows {contract.min_rows}")
58     if contract.max_rows is not None and n > contract.max_rows:
59         errors.append(f"rows {n} > max_rows {contract.max_rows}")
60
61     for col in contract.columns:
62         if col.name not in df.columns:
63             continue
64         series = df[col.name]
65         nulls = int(series.isna().sum())
66         if nulls and not col.nullable:
67             errors.append(f"{col.name}: {nulls} null values")
68         values = series.dropna()
69         if col.kind in {"int", "float", "number"}:
70             numeric = pd.to_numeric(values, errors="coerce")
71             bad = int(numeric.isna().sum())
72             if bad:
73                 errors.append(f"{col.name}: {bad} non-numeric values")
74             numeric = numeric.dropna().astype(float)
75             if col.kind == "int" and len(numeric) and not np.all(np.mod(numeric.to_numpy(), 1) == 0):
76                 errors.append(f"{col.name}: non-integer values present")
77             if col.min is not None and len(numeric) and float(numeric.min()) < col.min:
78                 errors.append(f"{col.name}: min {numeric.min()} < {col.min}")
79             if col.max is not None and len(numeric) and float(numeric.max()) > col.max:
80                 errors.append(f"{col.name}: max {numeric.max()} > {col.max}")
81             if col.monotonic and len(numeric) > 1:
82                 diff = np.diff(numeric.to_numpy())
83                 ok = {
84                     "increasing": bool((diff > 0).all()),
85                     "non_decreasing": bool((diff >= 0).all()),
86                     "decreasing": bool((diff < 0).all()),
87                     "non_increasing": bool((diff <= 0).all()),
88                 }[col.monotonic]
89                 if not ok:
90                     errors.append(f"{col.name}: not {col.monotonic}")
91             if col.step is not None and len(numeric) > 1:
92                 diff = np.diff(numeric.to_numpy())
93                 if not np.allclose(diff, col.step):
94                     errors.append(f"{col.name}: step is not constant {col.step} (min {diff.min()}, max {diff.max()})")

```

```

95     elif col.kind == "str":
96         if col.regex:
97             pattern = re.compile(col.regex)
98             bad_values = [str(v) for v in values.astype(str) if not pattern.fullmatch(str(v))]
99             if bad_values:
100                 errors.append(f"{col.name}: {len(bad_values)} values violate /{col.regex}/, e.g. {bad_values[:3]}")
101     elif col.kind == "datetime":
102         parsed = pd.to_datetime(values, errors="coerce")
103         bad = int(parsed.isna().sum())
104         if bad:
105             errors.append(f"{col.name}: {bad} unparsable datetimes")
106     if col.unique and values.duplicated().any():
107         errors.append(f"{col.name}: {int(values.duplicated().sum())} duplicated values")
108     if col.allowed is not None:
109         unexpected = sorted(set(values.unique()) - set(col.allowed), key=str)
110         if unexpected:
111             errors.append(f"{col.name}: values outside allowed set: {unexpected[:5]}")
112
113     for check in contract.checks:
114         try:
115             errors.extend(check(df))
116         except Exception as exc:
117             errors.append(f"check {getattr(check, '__name__', check)!r} raised {exc!r}")
118
119     return {
120         "contract": contract.name,
121         "ok": not errors,
122         "rows": n,
123         "columns": list(df.columns),
124         "errors": errors,
125         "warnings": warnings,
126     }

```

forge/events.py

```

1  """Structured JSONL event logging with a human-readable mirror on stdout."""
2
3  from __future__ import annotations
4
5  import json
6  import time
7  from datetime import UTC, datetime
8  from pathlib import Path
9  from typing import Any
10
11
12  def _short(value: Any, limit: int = 200) -> str:
13      text = value if isinstance(value, str) else json.dumps(value, ensure_ascii=False, default=str)
14      return text if len(text) <= limit else text[: limit - 1] + "..."
15
16
17  class EventLogger:
18      """Append-only event stream: one JSON object per line, mirrored to stdout."""
19
20      def __init__(self, path: Path, run_id: str, stage: str) -> None:
21          path.parent.mkdir(parents=True, exist_ok=True)
22          self._fh = path.open("a", encoding="utf-8")
23          self.path = path
24          self.run_id = run_id
25          self.stage = stage
26          self._t0 = time.monotonic()
27
28      def emit(self, event: str, level: str = "info", **fields: Any) -> None:

```

```

29     record = {
30         "ts": datetime.now(UTC).isoformat(timespec="milliseconds"),
31         "elapsed_s": round(time.monotonic() - self._t0, 3),
32         "run_id": self.run_id,
33         "stage": self.stage,
34         "level": level,
35         "event": event,
36         **fields,
37     }
38     self._fh.write(json.dumps(record, ensure_ascii=False, default=str) + "\n")
39     self._fh.flush()
40     extra = " ".join(f"{k}={_short(v)}" for k, v in fields.items())
41     print(f"[{record['elapsed_s']:9.3f}s] {level.upper():5s} {self.stage}:{event} {extra}", flush=True)
42
43     def info(self, event: str, **fields: Any) -> None:
44         self.emit(event, "info", **fields)
45
46     def warn(self, event: str, **fields: Any) -> None:
47         self.emit(event, "warn", **fields)
48
49     def error(self, event: str, **fields: Any) -> None:
50         self.emit(event, "error", **fields)
51
52     def metric(self, name: str, value: Any, **fields: Any) -> None:
53         self.emit("metric", "info", name=name, value=value, **fields)
54
55     def close(self) -> None:
56         if not self._fh.closed:
57             self._fh.close()

```

forge/hashing.py

```

1  """Content hashing helpers (SHA-256 everywhere; JSON hashed in canonical form)."""
2
3  from __future__ import annotations
4
5  import hashlib
6  import json
7  from collections.abc import Iterable
8  from pathlib import Path
9  from typing import Any
10
11  CHUNK = 1 << 20
12
13
14  def sha256_file(path: Path) -> str:
15     h = hashlib.sha256()
16     with path.open("rb") as fh:
17         for chunk in iter(lambda: fh.read(CHUNK), b""):
18             h.update(chunk)
19     return h.hexdigest()
20
21
22  def sha256_bytes(data: bytes) -> str:
23     return hashlib.sha256(data).hexdigest()
24
25
26  def sha256_json(obj: Any) -> str:
27     payload = json.dumps(obj, sort_keys=True, ensure_ascii=False, separators=(",", ":"), default=str)
28     return sha256_bytes(payload.encode("utf-8"))
29
30
31  def hash_tree(root: Path, ignore: Iterable[str] = ()) -> dict[str, dict[str, Any]]:

```

```

32     """Return ``{relative_posix_path: {"sha256": ..., "bytes": ...}}`` for every file under ``root``.
33
34     Any path component contained in ``ignore`` excludes the file (directory names or file names).
35     The result is sorted by path so that its canonical JSON digest is stable.
36     """
37     ignored = set(ignore)
38     out: dict[str, dict[str, Any]] = {}
39     if not root.exists():
40         return out
41     for path in sorted(root.rglob("*")):
42         if not path.is_file():
43             continue
44         rel = path.relative_to(root)
45         if ignored & set(rel.parts):
46             continue
47         out[rel.as_posix()] = {"sha256": sha256_file(path), "bytes": path.stat().st_size}
48     return out
49
50
51 def tree_digest(tree: dict[str, dict[str, Any]]) -> str:
52     return sha256_json({k: v["sha256"] for k, v in sorted(tree.items())})

```

forge/manifest.py

```

1  """Run/stage manifests: the unit of lineage, reproducibility and audit."""
2
3  from __future__ import annotations
4
5  import importlib.metadata
6  import json
7  import os
8  import platform
9  import sys
10 import traceback
11 from dataclasses import asdict, dataclass, field, fields
12 from datetime import UTC, datetime
13 from pathlib import Path
14 from typing import Any
15
16 from .hashing import hash_tree, tree_digest
17
18 TRACKED_PACKAGES = (
19     "numpy",
20     "scipy",
21     "pandas",
22     "pyarrow",
23     "openpyxl",
24     "matplotlib",
25     "scikit-learn",
26     "statsmodels",
27     "ortools",
28     "highspy",
29     "numba",
30     "cvxpy",
31     "networkx",
32     "pymupdf",
33     "pytest",
34     "hypothesis",
35     "ruff",
36     "mypy",
37 )
38 RECORD_FILES = ("manifest.json", "events.jsonl") # run records, excluded from the outputs digest
39

```

```

40
41 def utc_now() -> str:
42     return datetime.now(UTC).isoformat(timespec="seconds")
43
44
45 def runtime_info() -> dict[str, Any]:
46     packages: dict[str, str] = {}
47     for name in TRACKED_PACKAGES:
48         try:
49             packages[name] = importlib.metadata.version(name)
50         except importlib.metadata.PackageNotFoundError:
51             continue
52     return {
53         "python": sys.version.split()[0],
54         "platform": platform.platform(),
55         "cpu_count": os.cpu_count(),
56         "modal_task_id": os.getenv("MODAL_TASK_ID"),
57         "modal_is_remote": os.getenv("MODAL_IS_REMOTE") == "1" or os.getenv("MODAL_TASK_ID") is not None,
58         "packages": packages,
59         "env": {k: os.environ.get(k) for k in ("OMP_NUM_THREADS", "OPENBLAS_NUM_THREADS", "PYTHONHASHSEED")},
60     }
61
62
63 @dataclass
64 class Manifest:
65     schema: str = "forge.manifest/1"
66     run_id: str = ""
67     stage: str = ""
68     status: str = "running" # running | completed | failed
69     started_at: str = field(default_factory=utc_now)
70     finished_at: str | None = None
71     duration_s: float | None = None
72     code_ref: dict[str, Any] = field(default_factory=dict)
73     params: dict[str, Any] = field(default_factory=dict)
74     config_digest: str = ""
75     deps: dict[str, str] = field(default_factory=dict)
76     inputs: dict[str, dict[str, Any]] = field(default_factory=dict)
77     outputs: dict[str, dict[str, Any]] = field(default_factory=dict)
78     outputs_digest: str = ""
79     metrics: dict[str, Any] = field(default_factory=dict)
80     runtime: dict[str, Any] = field(default_factory=dict)
81     error: dict[str, Any] | None = None
82
83 @classmethod
84 def start(
85     cls,
86     run_id: str,
87     stage: str,
88     params: dict[str, Any],
89     code_ref: dict[str, Any],
90     config_digest: str,
91     deps: dict[str, str],
92     inputs: dict[str, dict[str, Any]],
93 ) -> Manifest:
94     manifest = cls(
95         run_id=run_id,
96         stage=stage,
97         params=params,
98         code_ref=code_ref,
99         config_digest=config_digest,
100         deps=deps,
101         inputs=inputs,
102     )
103     manifest.runtime = runtime_info()
104     return manifest
105

```

```

106     def complete(self, stage_dir: Path, metrics: dict[str, Any]) -> None:
107         self.outputs = hash_tree(stage_dir, ignore=RECORD_FILES)
108         self.outputs_digest = tree_digest(self.outputs)
109         self.metrics = json.loads(json.dumps(metrics, default=str))
110         self.status = "completed"
111         self._finish()
112
113     def fail(self, exc: BaseException) -> None:
114         self.error = {"type": type(exc).__name__, "message": str(exc), "traceback": traceback.format_exc()}
115         self.status = "failed"
116         self._finish()
117
118     def _finish(self) -> None:
119         self.finished_at = utc_now()
120         started = datetime.fromisoformat(self.started_at)
121         self.duration_s = round((datetime.fromisoformat(self.finished_at) - started).total_seconds(), 3)
122
123     def write(self, path: Path) -> None:
124         path.parent.mkdir(parents=True, exist_ok=True)
125         path.write_text(json.dumps(asdict(self), ensure_ascii=False, indent=2, default=str), encoding="utf-8")
126
127     @classmethod
128     def read(cls, path: Path) -> Manifest:
129         raw = json.loads(path.read_text(encoding="utf-8"))
130         known = {f.name for f in fields(cls)}
131         return cls(**{k: v for k, v in raw.items() if k in known})
132
133     def verify_outputs(self, stage_dir: Path) -> list[str]:
134         """Return a list of mismatches between the manifest and the files on disk (empty == verified)."""
135         problems: list[str] = []
136         current = hash_tree(stage_dir, ignore=RECORD_FILES)
137         for rel, meta in self.outputs.items():
138             if rel not in current:
139                 problems.append(f"missing: {rel}")
140             elif current[rel]["sha256"] != meta["sha256"]:
141                 problems.append(f"sha256 mismatch: {rel}")
142         for rel in current:
143             if rel not in self.outputs:
144                 problems.append(f"unexpected file: {rel}")
145         return problems

```

forge/modal_app.py

```

1  """Modal application — the only compute surface of this repository.
2
3  Local side (tools/cli.py) only submits:
4      modal run forge/modal_app.py --stage ingest --run-id <id> --params '{...}' --size small
5  Inside the container the repository is mounted read-only at /repo and the persistent
6  volume at /vol; stages write exclusively to /vol/runs/<run_id>/<stage>/.
7  """
8
9  from __future__ import annotations
10
11  import contextlib
12  import io
13  import json
14  import os
15  import sys
16  import traceback
17  from pathlib import Path
18
19  import modal
20

```

```

21 _HERE = Path(__file__).resolve()
22 REPO = _HERE.parents[1]
23
24
25 def _load_project() -> dict:
26     for candidate in (REPO / "configs" / "project.json", Path("/repo/configs/project.json")):
27         if candidate.exists():
28             return json.loads(candidate.read_text(encoding="utf-8"))
29     raise FileNotFoundError("configs/project.json not found next to forge/modal_app.py")
30
31
32 PROJECT = _load_project()
33 app = modal.App(PROJECT["app"])
34 volume = modal.Volume.from_name(PROJECT["volume"], create_if_missing=True)
35
36 APT_PACKAGES = [
37     "texlive-xetex",
38     "texlive-lang-chinese",
39     "texlive-latex-base",
40     "texlive-latex-recommended",
41     "texlive-latex-extra",
42     "texlive-science",
43     "texlive-pictures",
44     "texlive-bibtex-extra",
45     "texlive-fonts-recommended",
46     "fonts-noto-cjk",
47     "fonts-noto-core",
48     "fonts-droid-fallback",
49     "fonts-wqy-microhei",
50     "fontconfig",
51     "poppler-utils",
52     "zip",
53     "unzip",
54 ]
55 PIP_PACKAGES = [
56     "numpy==2.2.6",
57     "scipy==1.15.3",
58     "pandas==2.2.3",
59     "pyarrow==20.0.0",
60     "openpyxl==3.1.5",
61     "matplotlib==3.10.3",
62     "scikit-learn==1.6.1",
63     "statsmodels==0.14.4",
64     "ortools==9.12.4544",
65     "highspy==1.10.0",
66     "numba==0.61.2",
67     "cvxpy==1.6.5",
68     "networkx==3.4.2",
69     "pydantic==2.11.4",
70     "pymupdf==1.26.7",
71     "pytest==8.3.5",
72     "pytest-cov==6.1.1",
73     "hypothesis==6.131.9",
74     "ruff==0.11.10",
75     "mypy==1.15.0",
76     "pandas-stubs==2.2.3.250308",
77     "types-openpyxl==3.1.5.20250602",
78 ]
79 SKIP_PARTS = {
80     ".git",
81     "artifacts",
82     "releases",
83     ".forge",
84     "__pycache__",
85     ".mypy_cache",
86     ".ruff_cache",

```

```

87     ".pytest_cache",
88     ".hypothesis",
89     ".DS_Store",
90 }
91
92
93 def _ignore(path: Path) -> bool:
94     return bool(SKIP_PARTS & set(path.parts)) or path.suffix in {".pyc", ".pyo"}
95
96
97 image = (
98     modal.Image.debian_slim(python_version="3.11")
99     .apt_install(*APT_PACKAGES)
100     .pip_install(*PIP_PACKAGES)
101     .env(
102         {
103             "OMP_NUM_THREADS": "1",
104             "OPENBLAS_NUM_THREADS": "1",
105             "MKL_NUM_THREADS": "1",
106             "MPLBACKEND": "Agg",
107             "MPLCONFIGDIR": "/tmp/mpl",
108             "PYTHONHASHSEED": "0",
109             "PYTHONUNBUFFERED": "1",
110             "PYTHONDONTWRITEBYTECODE": "1",
111             "PYTHONPATH": "/repo",
112             "FORGE_REPO": "/repo",
113             "FORGE_VOL": "/vol",
114             "MODAL_IS_REMOTE": "1",
115             "TEXMFVAR": "/tmp/texmf-var",
116             "TEXMFCONFIG": "/tmp/texmf-config",
117             "HOME": "/root",
118         }
119     )
120     .add_local_dir(REPO, "/repo", ignore=_ignore)
121 )
122
123 VOLUME_MOUNT = {"vol": volume}
124
125
126 def _bootstrap() -> None:
127     if "/repo" not in sys.path:
128         sys.path.insert(0, "/repo")
129     os.chdir("/repo")
130     Path("/tmp/mpl").mkdir(exist_ok=True)
131
132
133 def _run(stage: str, run_id: str, params_json: str, force: bool) -> dict:
134     _bootstrap()
135     volume.reload()
136     from forge.runner import execute
137
138     params = json.loads(params_json or "{}")
139     return execute(
140         stage_name=stage,
141         run_id=run_id,
142         params=params,
143         force=force,
144         repo=Path("/repo"),
145         vol=Path("/vol"),
146         commit=volume.commit,
147     )
148
149
150 @app.function(image=image, volumes=VOLUME_MOUNT, cpu=2, memory=4096, timeout=2 * 3600)
151 def run_small(stage: str, run_id: str, params: str, force: bool) -> dict:
152     return _run(stage, run_id, params, force)

```

```

153
154
155 @app.function(image=image, volumes=VOLUME_MOUNT, cpu=8, memory=16384, timeout=6 * 3600)
156 def run_medium(stage: str, run_id: str, params: str, force: bool) -> dict:
157     return _run(stage, run_id, params, force)
158
159
160 @app.function(image=image, volumes=VOLUME_MOUNT, cpu=32, memory=65536, timeout=12 * 3600)
161 def run_large(stage: str, run_id: str, params: str, force: bool) -> dict:
162     return _run(stage, run_id, params, force)
163
164
165 @app.function(image=image, volumes=VOLUME_MOUNT, cpu=4, memory=8192, timeout=3600)
166 def exec_python(code: str, run_id: str) -> str:
167     """Run an ad-hoc Python snippet in the cloud (exploration, status, debugging). Never local."""
168     _bootstrap()
169     volume.reload()
170     buffer = io.StringIO()
171     scope = {
172         "__name__": "__forge_exec__",
173         "RUN_DIR": Path("/vol/runs") / run_id,
174         "VOL": Path("/vol"),
175         "REPO": Path("/repo"),
176         "RUNS": Path("/vol/runs"),
177     }
178     with contextlib.redirect_stdout(buffer), contextlib.redirect_stderr(buffer):
179         try:
180             exec(compile(code, "<forge-exec>", "exec"), scope) # noqa: S102 - explicit cloud REPL
181         except BaseException:
182             traceback.print_exc(file=buffer)
183     volume.commit()
184     return buffer.getvalue()
185
186
187 @app.function(image=image, volumes=VOLUME_MOUNT, cpu=2, memory=4096, timeout=600)
188 def provision() -> dict:
189     """Prove the toolchain: interpreter, packages, TeX engine, fonts, solver availability."""
190     import importlib.metadata
191     import subprocess
192
193     _bootstrap()
194     info: dict = {"modal_task_id": os.getenv("MODAL_TASK_ID"), "cpu_count": os.cpu_count()}
195     info["python"] = sys.version.split()[0]
196     info["packages"] = {p.split("==")[0]: importlib.metadata.version(p.split("==")[0]) for p in PIP_PACKAGES}
197     info["xelatex"] = subprocess.run(["xelatex", "--version"], capture_output=True, text=True).stdout.splitlines()[0]
198     info["bibtex"] = subprocess.run(["bibtex", "--version"], capture_output=True, text=True).stdout.splitlines()[0]
199     info["fc_cjk"] = subprocess.run(["fc-match", "Noto Serif CJK SC"], capture_output=True, text=True).stdout.strip()
200     info["kpse"] = {
201         name: bool(subprocess.run(["kpsewhich", name], capture_output=True, text=True).stdout.strip())
202         for name in (
203             "ctexart.cls",
204             "gbt7714.sty",
205             "siunitx.sty",
206             "algorithm2e.sty",
207             "booktabs.sty",
208             "tikz.sty",
209             "listings.sty",
210             "cleveref.sty",
211             "tcolorbox.sty",
212             "pgfplots.sty",
213         )
214     }
215     from ortools.sat.python import cp_model
216
217     model = cp_model.CpModel()
218     x = model.NewIntVar(0, 10, "x")

```

```

219     model.Add(x >= 3)
220     model.Minimize(x)
221     solver = cp_model.CpSolver()
222     info["cpsat"] = {"status": solver.StatusName(solver.Solve(model)), "x": solver.Value(x)}
223     import highspy
224
225     info["highs"] = highspy.Highs().version()
226     from forge.plotting import cjk_font_family
227
228     info["figure_cjk_font"] = cjk_font_family()
229     return info
230
231
232 @app.local_entrypoint()
233 def main(
234     stage: str = "ingest",
235     run_id: str = "",
236     params: str = "{}",
237     force: bool = False,
238     size: str = "small",
239     spawn: bool = False,
240 ) -> None:
241     """Submit one stage (or ``all`` with params.stages) to the cloud and print its result."""
242     if stage == "provision":
243         print("PROVISION " + json.dumps(provision.remote(), ensure_ascii=False, default=str))
244         return
245     if stage == "exec":
246         payload = json.loads(params or "{}")
247         print(exec_python.remote(payload.get("code", ""), run_id or "adhoc"))
248         return
249     fn = {"small": run_small, "medium": run_medium, "large": run_large}[size]
250     if spawn:
251         call = fn.spawn(stage, run_id, params, force)
252         print(f"FUNCTION_CALL_ID {call.object_id}")
253         return
254     result = fn.remote(stage, run_id, params, force)
255     print("STAGE_RESULT " + json.dumps(result, ensure_ascii=False, default=str))

```

forge/packaging.py

```

1  """Deterministic packaging of support materials (stable member order and timestamps)."""
2
3  from __future__ import annotations
4
5  import zipfile
6  from collections.abc import Iterable
7  from pathlib import Path
8  from typing import Any
9
10 from .hashing import sha256_file
11
12 FIXED_TIME = (2026, 1, 1, 0, 0, 0)
13 SKIP_PARTS = {
14     ".git",
15     "__pycache__",
16     ".DS_Store",
17     ".mypy_cache",
18     ".ruff_cache",
19     ".pytest_cache",
20     ".hypothesis",
21     "artifacts",
22     "releases",
23     ".forge",

```

```

24     "build",
25 }
26
27
28 def code_members(repo: Path, globs: Iterable[str]) -> list[tuple[str, Path]]:
29     found: dict[str, Path] = {}
30     for pattern in globs:
31         for path in sorted(repo.glob(pattern)):
32             if not path.is_file():
33                 continue
34             rel = path.relative_to(repo)
35             if SKIP_PARTS & set(rel.parts) or path.suffix == ".pyc":
36                 continue
37             found[rel.as_posix()] = path
38     return sorted(found.items())
39
40
41 def run_members(run_dir: Path, stages: Iterable[str], *, prefix: str = "runs") -> list[tuple[str, Path]]:
42     members: list[tuple[str, Path]] = []
43     for stage in stages:
44         directory = run_dir / stage
45         if not directory.is_dir():
46             continue
47         for path in sorted(directory.rglob("*")):
48             rel = path.relative_to(directory)
49             if path.is_file() and not (SKIP_PARTS & set(rel.parts)):
50                 members.append((f"{prefix}/{run_dir.name}/{stage}/{rel.as_posix()}", path))
51     return members
52
53
54 def build_zip(members: Iterable[tuple[str, Path]], out: Path, *, comment: str = "") -> dict[str, Any]:
55     ordered = sorted(dict(members).items())
56     out.parent.mkdir(parents=True, exist_ok=True)
57     total = 0
58     with zipfile.ZipFile(out, "w", compression=zipfile.ZIP_DEFLATED, compresslevel=9) as zf:
59         if comment:
60             zf.comment = comment.encode("utf-8")
61         for name, source in ordered:
62             info = zipfile.ZipInfo(name, date_time=FIXED_TIME)
63             info.compress_type = zipfile.ZIP_DEFLATED
64             info.external_attr = 0o644 << 16
65             data = source.read_bytes()
66             total += len(data)
67             zf.writestr(info, data)
68     with zipfile.ZipFile(out) as zf:
69         bad = zf.testzip()
70         count = len(zf.namelist())
71     if bad:
72         raise RuntimeError(f"zip member failed CRC check: {bad}")
73     return {
74         "path": str(out),
75         "members": count,
76         "uncompressed_bytes": total,
77         "bytes": out.stat().st_size,
78         "sha256": sha256_file(out),
79     }

```

forge/pdfqa.py

```
1 """PDF inspection for quality gates (PyMuPDF)."""
2
3 from __future__ import annotations
4
5 import re
6 from collections.abc import Iterable
7 from pathlib import Path
8 from typing import Any
9
10 A4_PT = (595.276, 841.89)
11
12
13 def inspect_pdf(path: Path, *, with_text: bool = True) -> dict[str, Any]:
14     import fitz  # PyMuPDF
15
16     doc = fitz.open(path)
17     pages: list[dict[str, Any]] = []
18     texts: list[str] = []
19     fonts: dict[int, dict[str, Any]] = {}
20     unembedded: list[str] = []
21     for index, page in enumerate(doc, start=1):
22         rect = page.rect
23         pages.append(
24             {
25                 "page": index,
26                 "width_pt": round(rect.width, 2),
27                 "height_pt": round(rect.height, 2),
28                 "a4": abs(rect.width - A4_PT[0]) < 3 and abs(rect.height - A4_PT[1]) < 3,
29             }
30         )
31         if with_text:
32             texts.append(page.get_text("text"))
33             for font in page.get_fonts(full=True):
34                 xref, ftype, basefont = font[0], font[2], font[3]
35                 if xref in fonts:
36                     continue
37                 embedded = ftype == "Type3"
38                 if not embedded and xref:
39                     try:
40                         embedded = len(doc.extract_font(xref)[3]) > 0
41                     except Exception:
42                         embedded = False
43                 fonts[xref] = {"name": basefont, "type": ftype, "embedded": bool(embedded)}
44                 if not embedded:
45                     unembedded.append(basefont)
46     return {
47         "pages": len(doc),
48         "page_boxes": pages,
49         "all_a4": all(p["a4"] for p in pages),
50         "fonts": list(fonts.values()),
51         "unembedded_fonts": sorted(set(unembedded)),
52         "texts": texts,
53         "metadata": doc.metadata,
54     }
55
56
57 def scan_strings(texts: Iterable[str], patterns: Iterable[str], *, context: int = 40) -> list[dict[str, Any]]:
58     compiled = [(p, re.compile(p)) for p in patterns]
59     hits: list[dict[str, Any]] = []
60     for page_no, text in enumerate(texts, start=1):
61         for raw, pattern in compiled:
62             for match in pattern.finditer(text):
63                 start, end = max(0, match.start() - context), min(len(text), match.end() + context)
64                 hits.append({"page": page_no, "pattern": raw, "context": text[start:end].replace("\n", " ")})
```

```
65     return hits
```

forge/plotting.py

```
1  """House style for paper figures: vector PDF output, CJK-capable fonts, accessible palette."""
2
3  from __future__ import annotations
4
5  import logging
6  import warnings
7  from pathlib import Path
8  from typing import Any
9
10 OKABE_ITO = ["#0072B2", "#D55E00", "#009E73", "#E69F00", "#CC79A7", "#56B4E9", "#F0E442", "#000000"]
11 # (family name as matplotlib registers it, font file). TTC files expose only their first face,
12 # which is why the Noto CJK collection is addressed by its JP face name.
13 _CJK_CANDIDATES = (
14     ("WenQuanYi Micro Hei", "/usr/share/fonts/truetype/wqy/wqy-microhei.ttc"),
15     ("Noto Sans CJK JP", "/usr/share/fonts/opentype/noto/NotoSansCJK-Regular.ttc"),
16     ("Droid Sans Fallback", "/usr/share/fonts/truetype/droid/DroidSansFallbackFull.ttf"),
17 )
18 _PROBE_TEXT = "温度 T 水分 C 药材 频段 关键词 abc 0.25"
19 _STATE: dict[str, Any] = {}
20
21
22 class _Catcher(logging.Handler):
23     def __init__(self) -> None:
24         super().__init__()
25         self.records: list[str] = []
26
27     def emit(self, record: logging.LogRecord) -> None:
28         self.records.append(record.getMessage())
29
30
31 def _probe(family: str) -> bool:
32     """True when ``family`` (with DejaVu Sans as Latin fallback) renders the probe text into PDF cleanly."""
33     from matplotlib import pyplot as plt
34
35     catcher = _Catcher()
36     loggers = [
37         logging.getLogger(n) for n in ("matplotlib.font_manager", "matplotlib.backends.backend_pdf", "matplotlib")
38     ]
39     for lg in loggers:
40         lg.addHandler(catcher)
41     try:
42         with warnings.catch_warnings(record=True) as caught:
43             warnings.simplefilter("always")
44             fig, ax = plt.subplots(figsize=(2.5, 1))
45             ax.text(0.05, 0.5, _PROBE_TEXT, fontfamily=[family, "DejaVu Sans"])
46             ax.set_axis_off()
47             probe = Path("/tmp") / f"forge_font_probe_{family.replace(' ', '_')}.pdf"
48             fig.savefig(probe, format="pdf")
49             plt.close(fig)
50             messages = [str(w.message) for w in caught] + catcher.records
51             bad = [m for m in messages if "missing from font" in m or "not found" in m.lower()]
52             return not bad
53     except Exception:
54         return False
55     finally:
56         for lg in loggers:
57             lg.removeHandler(catcher)
58
59
```

```

60 def cjk_font_family() -> str:
61     """Pick a CJK font that renders into PDF without missing glyphs (cached per process)."""
62     if "family" in _STATE:
63         return str(_STATE["family"])
64     import matplotlib
65
66     matplotlib.use("Agg")
67     from matplotlib import font_manager
68
69     chosen = "DejaVu Sans"
70     for family, path in _CJK_CANDIDATES:
71         if not Path(path).exists():
72             continue
73         try:
74             font_manager.fontManager.addfont(path)
75         except Exception:
76             continue
77         if _probe(family):
78             chosen = family
79             break
80     _STATE["family"] = chosen
81     return chosen
82
83
84 def setup(font_size: float = 9.0) -> str:
85     import matplotlib
86
87     matplotlib.use("Agg")
88     from cycler import cycler
89     from matplotlib import pyplot as plt
90
91     family = cjk_font_family()
92     plt.rcParams.update(
93         {
94             "font.family": [family, "DejaVu Sans"],
95             "font.size": font_size,
96             "axes.titlesize": font_size + 1,
97             "axes.labelsize": font_size,
98             "legend.fontsize": font_size - 1,
99             "xtick.labelsize": font_size - 1,
100            "ytick.labelsize": font_size - 1,
101            "axes.prop_cycle": cycler(color=OKABE_ITO),
102            "axes.spines.top": False,
103            "axes.spines.right": False,
104            "axes.grid": True,
105            "grid.alpha": 0.25,
106            "grid.linewidth": 0.6,
107            "lines.linewidth": 1.4,
108            "figure.dpi": 150,
109            "savefig.dpi": 300,
110            "pdf.fonttype": 42,
111            "ps.fonttype": 42,
112            "axes.unicode_minus": False,
113            "figure.constrained_layout.use": True,
114        }
115     )
116     return family
117
118
119 def new_figure(width_in: float = 6.3, height_in: float = 3.4, **kwargs: Any) -> tuple[Any, Any]:
120     from matplotlib import pyplot as plt
121
122     setup()
123     return plt.subplots(figsize=(width_in, height_in), **kwargs)
124
125

```

```

126 def save(fig: Any, pdf_path: Path, *, png_preview: bool = True, dpi: int = 200) -> dict[str, Any]:
127     from matplotlib import pyplot as plt
128
129     pdf_path.parent.mkdir(parents=True, exist_ok=True)
130     fig.savefig(pdf_path, format="pdf")
131     out = {"pdf": str(pdf_path), "bytes": pdf_path.stat().st_size}
132     if png_preview:
133         png = pdf_path.with_suffix(".png")
134         fig.savefig(png, format="png", dpi=dpi)
135         out["png"] = str(png)
136     plt.close(fig)
137     return out

```

forge/runner.py

```

1  """Stage registry and executor. A stage is a function ``fn(ctx) -> metrics``.
2
3  Guarantees: one manifest per (run, stage); dependencies must be completed in the same run
4  (or copied from ``from_run``); a completed stage is skipped unless ``force``; failures are
5  recorded with a traceback and re-raised; the volume is committed after every stage.
6  """
7
8  from __future__ import annotations
9
10 import importlib
11 import json
12 import shutil
13 import time
14 from collections.abc import Callable, Iterable
15 from dataclasses import dataclass
16 from pathlib import Path
17 from typing import Any
18
19 from .context import StageContext
20 from .hashing import hash_tree, tree_digest
21 from .manifest import Manifest
22
23 StageFn = Callable[[StageContext], dict[str, Any] | None]
24 CODE_IGNORE = (
25     ".git",
26     "artifacts",
27     "releases",
28     "__pycache__",
29     ".mypy_cache",
30     ".ruff_cache",
31     ".pytest_cache",
32     ".hypothesis",
33     ".DS_Store",
34     ".forge",
35 )
36
37
38 @dataclass(frozen=True)
39 class StageSpec:
40     name: str
41     fn: StageFn
42     deps: tuple[str, ...]
43     description: str
44
45
46 STAGES: dict[str, StageSpec] = {}
47
48

```

```

49 def stage(name: str, deps: Iterable[str] = (), description: str = "") -> Callable[[StageFn], StageFn]:
50     def decorator(fn: StageFn) -> StageFn:
51         if name in STAGES:
52             raise ValueError(f"duplicate stage registration: {name}")
53         doc = (fn.__doc__ or "").strip().splitlines()
54         STAGES[name] = StageSpec(name, fn, tuple(deps), description or (doc[0] if doc else ""))
55         return fn
56
57     return decorator
58
59
60 def discover(repo: Path) -> None:
61     project = json.loads((repo / "configs" / "project.json").read_text(encoding="utf-8"))
62     importlib.import_module("pipelines.common.stages")
63     importlib.import_module(f"pipelines.{project['package']}.stages")
64
65
66 def list_stages(repo: Path) -> list[dict[str, Any]]:
67     discover(repo)
68     return [{"name": s.name, "deps": list(s.deps), "description": s.description} for s in STAGES.values()]
69
70
71 def _raw_input_prefixes(repo: Path) -> tuple[str, ...]:
72     project = json.loads((repo / "configs" / "project.json").read_text(encoding="utf-8"))
73     return (project["problem_pdf"], project["attachments_dir"].rstrip("/") + "/")
74
75
76 def execute(
77     stage_name: str,
78     run_id: str,
79     params: dict[str, Any],
80     force: bool,
81     repo: Path,
82     vol: Path,
83     commit: Callable[[[]], None] | None = None,
84 ) -> dict[str, Any]:
85     discover(repo)
86     if stage_name == "all":
87         results = [execute(s, run_id, params, force, repo, vol, commit) for s in params.get("stages", [])]
88         return {
89             "stage": "all",
90             "run_id": run_id,
91             "results": results,
92             "status": "completed" if all(r.get("status") in {"completed", "skipped"} for r in results) else "failed",
93         }
94     if stage_name not in STAGES:
95         raise KeyError(f"unknown stage {stage_name!r}; known stages: {sorted(STAGES)}")
96     spec = STAGES[stage_name]
97     run_dir = vol / "runs" / run_id
98     stage_dir = run_dir / stage_name
99     manifest_path = stage_dir / "manifest.json"
100
101     if manifest_path.exists() and not force:
102         previous = Manifest.read(manifest_path)
103         if previous.status == "completed":
104             return {
105                 "stage": stage_name,
106                 "run_id": run_id,
107                 "status": "skipped",
108                 "reason": "already completed in this run (use --force to rerun)",
109                 "outputs_digest": previous.outputs_digest,
110             }
111
112     deps_digest: dict[str, str] = {}
113     for dep in spec.deps:
114         dep_manifest = run_dir / dep / "manifest.json"

```

```

115     if not dep_manifest.exists() and params.get("from_run"):
116         source = vol / "runs" / str(params["from_run"]) / dep
117         if (source / "manifest.json").exists():
118             shutil.copytree(source, run_dir / dep, dirs_exist_ok=True)
119     if not dep_manifest.exists():
120         raise RuntimeError(
121             f"stage '{stage_name}' requires '{dep}', which has not run in run {run_id} "
122             f"(run it first, or pass from_run=<run_id> to reuse its outputs)"
123         )
124     dep_state = Manifest.read(dep_manifest)
125     if dep_state.status != "completed":
126         raise RuntimeError(f"dependency '{dep}' has status {dep_state.status!r} in run {run_id}")
127     deps_digest[dep] = dep_state.outputs_digest
128
129     if stage_dir.exists():
130         shutil.rmtree(stage_dir)
131     ctx = StageContext(run_id, stage_name, params, repo, vol)
132     code_tree = hash_tree(repo, ignore=CODE_IGNORE)
133     prefixes = _raw_input_prefixes(repo)
134     inputs = {k: v for k, v in code_tree.items() if k.startswith(prefixes)}
135     code_ref = {**dict(params.get("code_ref") or {}), "code_digest": tree_digest(code_tree, "files": len(code_tree))}
136     manifest = Manifest.start(run_id, stage_name, params, code_ref, ctx.config.digest, deps_digest, inputs)
137     manifest.write(manifest_path)
138     ctx.log.info("stage.start", deps=list(spec.deps), description=spec.description, size=params.get("size"))
139     t0 = time.monotonic()
140     try:
141         metrics = spec.fn(ctx) or {}
142         ctx.flush_numbers()
143         manifest.complete(stage_dir, metrics)
144         ctx.log.info("stage.completed", duration_s=round(time.monotonic() - t0, 3), metrics=metrics)
145     except BaseException as exc:
146         manifest.fail(exc)
147         ctx.log.error("stage.failed", error=repr(exc))
148         manifest.write(manifest_path)
149         ctx.log.close()
150         if commit:
151             commit()
152         raise
153     manifest.write(manifest_path)
154     ctx.log.close()
155     if commit:
156         commit()
157     return {
158         "stage": stage_name,
159         "run_id": run_id,
160         "status": "completed",
161         "duration_s": manifest.duration_s,
162         "metrics": manifest.metrics,
163         "outputs_digest": manifest.outputs_digest,
164         "modal_task_id": manifest.runtime.get("modal_task_id"),
165     }

```

forge/tex.py

```

1  """XeLaTeX compilation and log analysis (only ever executed inside Modal)."""
2
3  from __future__ import annotations
4
5  import os
6  import re
7  import subprocess
8  from pathlib import Path
9  from typing import Any

```

```

10
11 _UNDEF_REF = re.compile(r"LaTeX Warning: Reference `\[^\]+' on page \d+ undefined")
12 _UNDEF_CIT = re.compile(r"LaTeX Warning: Citation `\[^\]+' on page \d+ undefined")
13 _OVERFULL = re.compile(r"Overfull \\\hbox \\\((\d.)+pt too wide\\")
14 _MISSING = re.compile(r"Missing character: There is no (.*?) in font")
15 _ERROR = re.compile(r"^(.*)$", re.M)
16 _PAGES = re.compile(r"Output written on .*\? \\\((\d+) pages?")
17
18
19 def tex_env() -> dict[str, str]:
20     env = dict(os.environ)
21     env.setdefault("TEXMFVAR", "/tmp/texmf-var")
22     env.setdefault("TEXMFCONFIG", "/tmp/texmf-config")
23     env.setdefault("HOME", "/tmp")
24     env["max_print_line"] = "2000"
25     env["error_line"] = "254"
26     env["half_error_line"] = "238"
27     return env
28
29
30 def summarize_log(log: str) -> dict[str, Any]:
31     overfull = [float(x) for x in _OVERFULL.findall(log)]
32     missing = _MISSING.findall(log)
33     pages = _PAGES.search(log)
34     return {
35         "undefined_references": sorted(set(_UNDEF_REF.findall(log))),
36         "undefined_citations": sorted(set(_UNDEF_CIT.findall(log))),
37         "overfull_count": len(overfull),
38         "overfull_max_pt": max(overfull) if overfull else 0.0,
39         "missing_characters": len(missing),
40         "missing_character_samples": sorted(set(missing))[:10],
41         "errors": _ERROR.findall(log)[:10],
42         "pages_from_log": int(pages.group(1)) if pages else None,
43     }
44
45
46 def aux_page(aux: Path, label: str) -> int | None:
47     """Page number recorded for ``\label{label}`` in the .aux file (hyperref-compatible)."""
48     if not aux.exists():
49         return None
50     pattern = re.compile(r"\\newlabel{ " + re.escape(label) + r"}\{\{[^\}]*\}\{(\d+)\}" )
51     match = pattern.search(aux.read_text(encoding="utf-8", errors="replace"))
52     return int(match.group(1)) if match else None
53
54
55 def compile_tex(
56     workdir: Path,
57     main: str,
58     jobname: str,
59     *,
60     passes: int = 3,
61     bibtex: bool = True,
62     timeout: int = 1200,
63 ) -> dict[str, Any]:
64     env = tex_env()
65     runs: list[dict[str, Any]] = []
66     log_path = workdir / f"{jobname}.log"
67
68     def xelatex(tag: str) -> None:
69         proc = subprocess.run(
70             ["xelatex", "-interaction=nonstopmode", "-halt-on-error", "-file-line-error", f"-jobname={jobname}", main],
71             cwd=workdir,
72             capture_output=True,
73             text=True,
74             env=env,
75             timeout=timeout,

```

```

76         check=False,
77     )
78     runs.append({"pass": tag, "rc": proc.returncode})
79     if proc.returncode != 0:
80         log = log_path.read_text(encoding="utf-8", errors="replace") if log_path.exists() else proc.stdout
81         errors = _ERROR.findall(log)
82         raise RuntimeError(
83             f"xelatex failed on pass {tag} for {main}: "
84             + " | ".join(errors[:5])
85             + "\n--- log tail ---\n"
86             + log[-4000:]
87         )
88
89     xelatex("1")
90     aux = workdir / f"{jobname}.aux"
91     if bibtex and aux.exists():
92         aux_text = aux.read_text(encoding="utf-8", errors="replace")
93         if "\\citation" in aux_text and "\\bibdata" in aux_text:
94             proc = subprocess.run(
95                 ["bibtex", jobname], cwd=workdir, capture_output=True, text=True, env=env, timeout=timeout, check=False
96             )
97             runs.append({"pass": "bibtex", "rc": proc.returncode, "tail": proc.stdout[-1500:]})
98     for i in range(2, passes + 1):
99         xelatex(str(i))
100     log_text = log_path.read_text(encoding="utf-8", errors="replace")
101     return {
102         "pdf": workdir / f"{jobname}.pdf",
103         "log": log_path,
104         "aux": aux,
105         "runs": runs,
106         "summary": summarize_log(log_text),
107     }
108
109
110 def tex_escape(text: str) -> str:
111     replacements = {
112         "\\": r"\textbackslash{}",
113         "&": r"\&",
114         "%": r"\%",
115         "$": r"\$",
116         "#": r"\#",
117         "_": r"\_",
118         "{": r"\{",
119         "}": r"\}",
120         "~": r"\textasciitilde{}",
121         "^": r"\textasciicircum{}",
122     }
123     return "".join(replacements.get(ch, ch) for ch in text)
124
125
126 def tex_path(path: str) -> str:
127     """Typeset a file path so that it can break across lines (hyperref/xurl); falls back to \texttt."""
128     if any(ch in path for ch in "{}%#\\" or not path.isascii()):
129         return f"\\texttt{{{tex_escape(path)}}}" # CJK names need the CJK mono font, not the url font
130     return f"\\nolinkurl{{{path}}}"

```

forge/xlsx.py

```
1 """Workbook I/O: reading attachments, filling result templates, checking result contracts."""
2
3 from __future__ import annotations
4
5 import math
6 from collections.abc import Iterable, Sequence
7 from dataclasses import dataclass
8 from pathlib import Path
9 from typing import Any
10
11 import openpyxl
12 import pandas as pd
13
14 BIG_SHEET_CELLS = 300_000
15
16
17 def read_sheets(path: Path, header: int | None = 0) -> dict[str, pd.DataFrame]:
18     with pd.ExcelFile(path) as xf:
19         return {str(name): xf.parse(name, header=header) for name in xf.sheet_names}
20
21
22 def sheet_headers(path: Path) -> dict[str, list[Any]]:
23     wb = openpyxl.load_workbook(path, read_only=True, data_only=True)
24     out: dict[str, list[Any]] = {}
25     for ws in wb.worksheets:
26         row = list(next(ws.iter_rows(min_row=1, max_row=1, values_only=True), ()))
27         while row and row[-1] is None:
28             row.pop()
29         out[str(ws.title)] = row
30     wb.close()
31     return out
32
33
34 def _clean(value: Any, decimals: int | None) -> Any:
35     if hasattr(value, "item") and not isinstance(value, str | bytes):
36         try:
37             value = value.item()
38         except (ValueError, TypeError):
39             pass
40     if isinstance(value, float):
41         if math.isnan(value) or math.isinf(value):
42             return None
43         return round(value, decimals) if decimals is not None else value
44     return value
45
46
47 def write_result(
48     template: Path,
49     out: Path,
50     sheets: dict[str, tuple[Sequence[Any], Iterable[Sequence[Any]]]],
51     *,
52     decimals: int | None = 4,
53     float_format: str = "0.0000",
54 ) -> dict[str, Any]:
55     """Fill a competition result template.
56
57     ``sheets`` maps sheet name -> (header_row, rows). The template's A1 text is always kept;
58     header cells after A1 are taken from ``header_row[1:]``. Sheets not mentioned keep their
59     template content. Floats are rounded to ``decimals`` and formatted with ``float_format``.
60     Very large sheets fall back to openpyxl's write-only mode (no per-cell number format).
61     """
62     template_wb = openpyxl.load_workbook(template)
63     names = list(template_wb.sheetnames)
64     unknown = [n for n in sheets if n not in names]
```

```

65     if unknown:
66         raise KeyError(f"sheets {unknown} not in template {template.name} (has {names})")
67     a1 = {ws.title: ws.cell(row=1, column=1).value for ws in template_wb.worksheets}
68     materialised = {name: (list(header), [list(r) for r in rows]) for name, (header, rows) in sheets.items()}
69     cells = sum(max(len(h), 1) * (len(rows) + 1) for h, rows in materialised.values())
70     stats: dict[str, Any] = {}
71     out.parent.mkdir(parents=True, exist_ok=True)
72
73     if cells > BIG_SHEET_CELLS:
74         wb = openpyxl.Workbook(write_only=True)
75         for name in names:
76             ws = wb.create_sheet(name)
77             if name in materialised:
78                 header, rows = materialised[name]
79                 full_header = [a1[name], *header[1:]]
80                 ws.append(full_header)
81                 for row in rows:
82                     ws.append([_clean(v, decimals) for v in row])
83                 stats[name] = {"rows": len(rows), "cols": len(full_header)}
84             else:
85                 for tpl_row in template_wb[name].iter_rows(values_only=True):
86                     ws.append(list(tpl_row))
87         wb.save(out)
88     else:
89         wb = template_wb
90         for name, (header, rows) in materialised.items():
91             index = wb.sheetnames.index(name)
92             wb.remove(wb[name])
93             ws = wb.create_sheet(name, index) # fresh sheet: no stale template cells or dimensions
94             full_header = [a1[name], *header[1:]]
95             for j, value in enumerate(full_header, start=1):
96                 ws.cell(row=1, column=j, value=_clean(value, None))
97             for i, row in enumerate(rows, start=2):
98                 for j, value in enumerate(row, start=1):
99                     cleaned = _clean(value, decimals)
100                     cell = ws.cell(row=i, column=j, value=cleaned)
101                     if isinstance(cleaned, float) and j > 1:
102                         cell.number_format = float_format
103             stats[name] = {"rows": len(rows), "cols": len(full_header)}
104         wb.save(out)
105     return {"path": str(out), "sheets": stats, "write_only": cells > BIG_SHEET_CELLS, "cells": cells}
106
107
108 @dataclass(frozen=True)
109 class SheetContract:
110     name: str
111     min_rows: int = 1 # data rows (excluding the header)
112     max_rows: int | None = None
113     header_len: int | None = None # exact number of header cells, if known
114     numeric_from_col: int | None = None # 0-based; every data cell from this column must be numeric
115     allow_blank: bool = False # blank numeric cells permitted (e.g. moving-boundary tables)
116     decimals: int | None = 4 # numeric cells must carry at most this many decimals
117     text_columns: tuple[int, ...] = () # 0-based columns that must be non-empty text
118
119
120 @dataclass(frozen=True)
121 class WorkbookContract:
122     file: str
123     template: str
124     sheets: tuple[SheetContract, ...]
125
126
127 def check_workbook(path: Path, contract: WorkbookContract, template: Path) -> dict[str, Any]:
128     errors: list[str] = []
129     if not path.exists():
130         return {"file": contract.file, "ok": False, "errors": ["file missing"], "sheets": {}}

```

```

131     try:
132         tpl_headers = sheet_headers(template)
133     except Exception as exc:
134         tpl_headers = {}
135         errors.append(f"template unreadable: {exc!r}")
136     wb = openpyxl.load_workbook(path, read_only=True, data_only=True)
137     report: dict[str, Any] = {}
138     names = wb.sheetnames
139     if tpl_headers and names != list(tpl_headers):
140         errors.append(f"sheet names {names} differ from template {list(tpl_headers)}")
141     for sc in contract.sheets:
142         if sc.name not in names:
143             errors.append(f"sheet {sc.name!r} missing")
144             continue
145         ws = wb[sc.name]
146         rows = ws.iter_rows(values_only=True)
147         first = next(rows, None)
148         header: list[Any] = list(first) if first is not None else []
149         while header and header[-1] is None:
150             header.pop()
151         tpl = tpl_headers.get(sc.name)
152         if tpl and header and header[0] != tpl[0]:
153             errors.append(f"{sc.name}: A1 {header[0]!r} differs from template {tpl[0]!r}")
154         if sc.header_len is not None and len(header) != sc.header_len:
155             errors.append(f"{sc.name}: header has {len(header)} cells, expected {sc.header_len}")
156         n = 0
157         bad_numeric = blank = bad_decimals = bad_text = 0
158         for row in rows:
159             values = list(row)
160             if all(v is None or (isinstance(v, str) and not v.strip()) for v in values):
161                 continue
162             n += 1
163             if sc.numeric_from_col is not None:
164                 for v in values[sc.numeric_from_col : len(header) or None]:
165                     if v is None or (isinstance(v, str) and not v.strip()):
166                         blank += 1
167                     elif not isinstance(v, int | float) or isinstance(v, bool):
168                         bad_numeric += 1
169                     elif sc.decimals is not None and isinstance(v, float) and abs(round(v, sc.decimals) - v) > 1e-12:
170                         bad_decimals += 1
171             for j in sc.text_columns:
172                 v = values[j] if j < len(values) else None
173                 if v is None or not str(v).strip():
174                     bad_text += 1
175             if n < sc.min_rows:
176                 errors.append(f"{sc.name}: {n} data rows < {sc.min_rows}")
177             if sc.max_rows is not None and n > sc.max_rows:
178                 errors.append(f"{sc.name}: {n} data rows > {sc.max_rows}")
179             if bad_numeric:
180                 errors.append(f"{sc.name}: {bad_numeric} non-numeric cells in numeric region")
181             if blank and not sc.allow_blank:
182                 errors.append(f"{sc.name}: {blank} blank cells in numeric region")
183             if bad_decimals:
184                 errors.append(f"{sc.name}: {bad_decimals} cells with more than {sc.decimals} decimals")
185             if bad_text:
186                 errors.append(f"{sc.name}: {bad_text} empty cells in required text columns")
187             report[sc.name] = {
188                 "rows": n,
189                 "header": [str(h) for h in header[:6]] + (["..."] if len(header) > 6 else []),
190                 "blank": blank,
191             }
192     wb.close()
193     return {"file": contract.file, "ok": not errors, "errors": errors, "sheets": report}

```

pipelines/__init__.py

pipelines/common/__init__.py

pipelines/common/stages.py

```
1 """Problem-agnostic stages: ingest, lint, test, paper, qa, package, release."""
2
3 from __future__ import annotations
4
5 import importlib
6 import json
7 import os
8 import re
9 import shlex
10 import shutil
11 import subprocess
12 import xml.etree.ElementTree as ET
13 from datetime import UTC, datetime
14 from pathlib import Path
15 from typing import Any
16
17 from forge import packaging, pdfqa, xlsx
18 from forge.context import StageContext
19 from forge.hashing import hash_tree, sha256_file
20 from forge.runner import stage
21 from forge.tex import aux_page, compile_tex, tex_escape, tex_path
22
23 PLANNED_STAGE_FILES = {
24     "lint": ["ruff_check.txt", "ruff_format.txt", "mypy.txt", "lint_report.json", "manifest.json", "events.jsonl"],
25     "test": ["junit.xml", "coverage.json", "pytest.txt", "test_report.json", "manifest.json", "events.jsonl"],
26     "qa": ["qa_report.json", "manifest.json", "events.jsonl"],
27 }
28 MEMBER_DESCRIPTIONS = (
29     (r"~forge/", "云端运行时（阶段执行、清单、契约、排版、打包）"),
30     (r"~pipelines/common/", "通用阶段：数据接入、门禁、论文、质量检查、打包发布"),
31     (r"~pipelines/", "本题建模与求解程序"),
32     (r"~tools/", "本地控制面：提交云端任务、下载、校验散列"),
33     (r"~tests/", "单元 / 性质 / 集成测试"),
34     (r"~configs/", "运行配置与项目元数据"),
35     (r"~manuscript/", "论文 LaTeX 源码"),
36     (r"~docs/", "工程规范、建模标准、决策记录"),
37     (r"/ingest/", "原始数据接入结果（散列、题目文本、规范化数据）"),
38     (r"/validate/", "数据契约校验报告与数据画像"),
39     (r"/results/", "结果工作簿与求解输出"),
40     (r"/figures/", "论文图（矢量 PDF 与预览 PNG）"),
41     (r"/tables/", "论文表（LaTeX 片段与 CSV）"),
42     (r"/lint/", "静态检查报告"),
43     (r"/test/", "测试报告与覆盖率"),
44     (r"/qa/", "论文与结果文件质量门禁报告"),
45     (r"manifest\.json$", "阶段清单（输入输出散列、环境、参数）"),
46     (r"events\.jsonl$", "结构化运行日志"),
47     (r"~AI工具使用详情\.pdf$", "AI 工具使用详情"),
48     (r"~AI_USAGE\.md$", "AI 工具使用记录（源文件）"),
49     (r"~Makefile$", "本地快捷命令（只提交云端任务）"),
50     (r"~pyproject\.toml$", "项目元数据与静态检查配置"),
51     (r"~README\.md$", "仓库说明与复现入口"),
52     (r"~CHANGELOG\.md$", "变更记录"),
```

```

53     (r"~UNLICENSE$", "公有领域声明 (Unlicense) "),
54     (r"~REPRODUCE\.md$", "复现说明"),
55     (r"~MANIFEST\.json$", "支撑材料成员清单与 SHA-256"),
56 )
57
58
59 def _describe(name: str) -> str:
60     for pattern, text in MEMBER_DESCRIPTIONS:
61         if re.search(pattern, name):
62             return text
63     return ""
64
65
66 def _to_parquet(df: Any, target: Path) -> None:
67     frame = df.copy()
68     frame.columns = [str(c).strip() for c in frame.columns]
69     try:
70         frame.to_parquet(target, index=False)
71     except Exception:
72         for col in frame.columns:
73             if frame[col].dtype == object:
74                 frame[col] = frame[col].map(
75                     lambda v: None if v is None or (isinstance(v, float) and v != v) else str(v)
76                 )
77         frame.to_parquet(target, index=False)
78
79
80 @stage("ingest", description="Hash raw inputs, extract the problem text, convert every attachment sheet to parquet")
81 def ingest(ctx: StageContext) -> dict[str, Any]:
82     import fitz
83
84     document = fitz.open(ctx.problem_pdf)
85     text = "\n\n".join(page.get_text("text", sort=True) for page in document)
86     ctx.out("problem_text.txt").write_text(text, encoding="utf-8")
87     inventory: dict[str, Any] = {
88         "problem_pdf": {
89             "path": ctx.project["problem_pdf"],
90             "sha256": sha256_file(ctx.problem_pdf),
91             "pages": len(document),
92         },
93         "files": [],
94     }
95     for path in sorted(ctx.attachments.rglob("*.xlsx")):
96         if path.name.startswith("~$"):
97             continue
98         rel = path.relative_to(ctx.attachments).as_posix()
99         is_template = path.parent == ctx.templates or ctx.templates in path.parents
100         entry: dict[str, Any] = {
101             "path": rel,
102             "sha256": sha256_file(path),
103             "bytes": path.stat().st_size,
104             "template": is_template,
105             "sheets": [],
106         }
107         for name, df in xlsx.read_sheets(path).items():
108             sheet: dict[str, Any] = {"name": name, "rows": len(df), "columns": [str(c) for c in df.columns]}
109             if not is_template:
110                 safe = re.sub(r"[^w.-]+", "_", f"{path.stem}_{name}")
111                 target = ctx.out("data", f"{safe}.parquet")
112                 _to_parquet(df, target)
113                 sheet["parquet"] = target.relative_to(ctx.stage_dir).as_posix()
114                 sheet["dtypes"] = {str(c): str(t) for c, t in df.dtypes.items()}
115                 sheet["head"] = df.head(5).astype(str).values.tolist()
116             entry["sheets"].append(sheet)
117         inventory["files"].append(entry)
118     ctx.log.info("ingest.file", path=rel, template=is_template, sheets=[s["name"] for s in entry["sheets"]])

```

```

119     ctx.write_json("inventory.json", inventory)
120     return {"files": len(inventory["files"]), "problem_chars": len(text)}
121
122
123 @stage("lint", description="Static gates: ruff check, ruff format --check, mypy")
124 def lint(ctx: StageContext) -> dict[str, Any]:
125     returncodes: dict[str, int] = {}
126
127     def run(name: str, cmd: list[str]) -> None:
128         proc = subprocess.run(cmd, cwd=str(ctx.repo), capture_output=True, text=True, check=False)
129         ctx.out(f"{name}.txt").write_text(proc.stdout + ("\n" + proc.stderr if proc.stderr else ""), encoding="utf-8")
130         returncodes[name] = proc.returncode
131         ctx.log.info("lint.tool", tool=name, rc=proc.returncode, tail=(proc.stdout + proc.stderr)[-400:])
132
133     run("ruff_check", ["ruff", "check", "--no-cache", "--output-format", "concise", "."])
134     run("ruff_format", ["ruff", "format", "--check", "--no-cache", "."])
135     run("mypy", ["mypy", "--cache-dir", "/tmp/mypy", *ctx.cfg("gates.mypy_paths", ["forge"])]])
136     required = list(ctx.cfg("gates.lint_required", ["ruff_check", "ruff_format", "mypy"]))
137     passed = all(returncodes.get(name, 1) == 0 for name in required)
138     ctx.write_json("lint_report.json", {"passed": passed, "required": required, "returncodes": returncodes})
139     if not passed:
140         raise RuntimeError(f"lint gate failed: {returncodes}")
141     return {"passed": passed, **returncodes}
142
143
144 @stage("test", description="pytest (unit, property, integration) with JUnit and coverage reports")
145 def test(ctx: StageContext) -> dict[str, Any]:
146     env = {
147         **os.environ,
148         "COVERAGE_FILE": "/tmp/.coverage",
149         "HYPOTHESIS_STORAGE_DIRECTORY": "/tmp/hypothesis",
150         "FORGE_RUN_DIR": str(ctx.run_dir),
151     }
152     cmd = [
153         "python",
154         "-m",
155         "pytest",
156         "tests",
157         "-q",
158         "-p",
159         "no:cacheprovider",
160         "--rootdir",
161         str(ctx.repo),
162         f"--junitxml={ctx.out('junit.xml')}",
163         "--cov=forge",
164         "--cov=pipelines",
165         f"--cov-report=json:{ctx.out('coverage.json')}",
166         "--cov-report=term",
167     ]
168     extra = ctx.param("pytest_args")
169     if extra:
170         cmd += shlex.split(str(extra))
171     proc = subprocess.run(cmd, cwd=str(ctx.repo), capture_output=True, text=True, env=env, check=False)
172     ctx.out("pytest.txt").write_text(proc.stdout + ("\n" + proc.stderr, encoding="utf-8")
173     print(proc.stdout[-5000:], flush=True)
174     summary: dict[str, Any] = {"returncode": proc.returncode}
175     junit = ctx.stage_dir / "junit.xml"
176     if junit.exists():
177         root = ET.parse(junit).getroot()
178         suite = root if root.tag == "testsuite" else root.find("testsuite")
179         if suite is not None:
180             summary.update({k: int(suite.get(k, 0)) for k in ("tests", "failures", "errors", "skipped")})
181     coverage = ctx.stage_dir / "coverage.json"
182     if coverage.exists():
183         summary["coverage_percent"] = round(json.loads(coverage.read_text())["totals"]["percent_covered"], 2)
184     ctx.write_json("test_report.json", summary)

```

```

185     if proc.returncode != 0:
186         raise RuntimeError(f"tests failed: {summary}\n{proc.stdout[-2500:]}")
187     return summary
188
189
190 def _support_members(ctx: StageContext, *, planned: bool) -> list[tuple[str, Path]]:
191     members = packaging.code_members(ctx.repo, ctx.cfg("package.code_globs", []))
192     for name in ctx.cfg("package.stages", []):
193         directory = ctx.run_dir / name
194         if (directory / "manifest.json").exists():
195             members += packaging.run_members(ctx.run_dir, [name])
196         elif planned and name in PLANNED_STAGE_FILES:
197             members += [(f"runs/{ctx.run_id}/{name}/{f}", directory / f) for f in PLANNED_STAGE_FILES[name]]
198     members.append(("AI工具使用详情.pdf", ctx.run_dir / "paper" / "AI工具使用详情.pdf"))
199     members.append(("REPRODUCE.md", ctx.run_dir / "package" / "REPRODUCE.md"))
200     members.append(("MANIFEST.json", ctx.run_dir / "package" / "MANIFEST.json"))
201     return sorted(dict(members).items())
202
203
204 def _load_numbers(run_dir: Path) -> tuple[dict[str, str], list[str]]:
205     numbers: dict[str, str] = {}
206     conflicts: list[str] = []
207     for path in sorted(run_dir.glob("*/numbers.json")):
208         for key, value in json.loads(path.read_text(encoding="utf-8")).items():
209             if key in numbers and numbers[key] != value:
210                 conflicts.append(f"{key}: {numbers[key]!r} vs {value!r} ({path.parent.name})")
211             numbers[key] = value
212     return numbers, conflicts
213
214
215 def _stage_rows(ctx: StageContext) -> list[dict[str, Any]]:
216     rows = []
217     for manifest in sorted(ctx.run_dir.glob("*/manifest.json")):
218         data = json.loads(manifest.read_text(encoding="utf-8"))
219         rows.append(
220             {
221                 "stage": data["stage"],
222                 "status": data["status"],
223                 "duration_s": data.get("duration_s"),
224                 "outputs_digest": data.get("outputs_digest", ""),
225                 "modal_task_id": data.get("runtime", {}).get("modal_task_id"),
226             }
227         )
228     return rows
229
230
231 @stage(
232     "paper",
233     description="Assemble generated inputs (numbers, tables, figures, code, support list) and compile with XeLaTeX",
234 )
235 def paper(ctx: StageContext) -> dict[str, Any]:
236     sources = list(ctx.cfg("paper.sources", []))
237     for source in sources:
238         ctx.dep(source)
239     build = Path("/tmp/build")
240     shutil.rmtree(build, ignore_errors=True)
241     shutil.copytree(
242         ctx.repo / "manuscript", build, ignore=shutil.ignore_patterns("generated", "*.aux", "*.log", "*.pdf")
243     )
244     generated = build / "generated"
245     (generated / "tables").mkdir(parents=True, exist_ok=True)
246     (build / "figures").mkdir(exist_ok=True)
247
248     numbers, conflicts = _load_numbers(ctx.run_dir)
249     for conflict in conflicts:
250         ctx.log.warn("paper.number_conflict", detail=conflict)

```

```

251 lines = ["% generated by forge — do not edit"] + [
252     f"\newcommand{{\val{key}}}{\tex_escape(value)}}" for key, value in sorted(numbers.items())
253 ]
254 (generated / "numbers.tex").write_text("\n".join(lines) + "\n", encoding="utf-8")
255
256 n_tables = n_figures = 0
257 for source in sources:
258     directory = ctx.run_dir / source
259     for table in sorted((directory / "tables").glob("*.tex")) if (directory / "tables").is_dir() else []:
260         shutil.copy2(table, generated / "tables" / table.name)
261         n_tables += 1
262     for figure in sorted((directory / "figures").glob("*")) if (directory / "figures").is_dir() else []:
263         if figure.suffix.lower() in {".pdf", ".png"}:
264             shutil.copy2(figure, build / "figures" / figure.name)
265             n_figures += 1
266
267 code_ref = ctx.params.get("code_ref") or {}
268 meta = [
269     f"\newcommand{{\RunId}}{\tex_escape(ctx.run_id)}",
270     f"\newcommand{{\CodeSha}}{\tex_escape(str(code_ref.get('git_sha', '')))}",
271     f"\newcommand{{\BuildDate}}{\datetime.now(UTC).strftime('%Y-%m-%d')}",
272 ]
273 (generated / "meta.tex").write_text("\n".join(meta) + "\n", encoding="utf-8")
274
275 files = packaging.code_members(ctx.repo, ctx.cfg("paper.code_globs", []))
276 listing = ["% generated by forge — complete source listing"]
277 for rel, src in files:
278     target = build / "code" / rel
279     target.parent.mkdir(parents=True, exist_ok=True)
280     shutil.copy2(src, target)
281     language = {"py": "Python", ".tex": "[LaTeX]TeX"}.get(src.suffix, "")
282     listing.append(f"\needspace{{6\baselineskip}}\subsection*{{\tex_path(rel) }}"
283     listing.append(f"\lstinputlisting[language={language}]{code/{rel}}")
284 (generated / "code_listing.tex").write_text("\n".join(listing) + "\n", encoding="utf-8")
285
286 members = _support_members(ctx, planned=True)
287 rows = [
288     "% generated by forge — support material file list",
289     "\begin{longtable}{{@{}p{0.64\linewidth}p{0.32\linewidth}@{}}",
290     "\toprule 文件 & 说明 \\\midrule \\\endhead \\\bottomrule \\\endfoot",
291 ]
292 rows += [f"\tex_path(name)} & \tex_escape(_describe(name))" for name, _ in members]
293 rows.append("\end{longtable}")
294 (generated / "support_files.tex").write_text("\n".join(rows) + "\n", encoding="utf-8")
295 ctx.write_json("support_members.json", [name for name, _ in members])
296
297 stage_rows = _stage_rows(ctx)
298 repro = [
299     "% generated by forge — reproducibility record",
300     "\begin{longtable}{{@{}llrl@{}}",
301     "\toprule 阶段 & 状态 & 用时/s & 输出摘要 (SHA-256 前 12 位) \\\midrule \\\endhead \\\bottomrule \\\endfoot",
302 ]
303 repro += [
304     f"\texttt{{\tex_escape(r['stage'])}} & {r['status']} & {r['duration_s']} or 0:.1f & "
305     f"\texttt{{r['outputs_digest'][:12]}} \\\\"
306     for r in stage_rows
307 ]
308 repro.append("\end{longtable}")
309 (generated / "repro_table.tex").write_text("\n".join(repro) + "\n", encoding="utf-8")
310
311 result = compile_tex(build, str(ctx.cfg("paper.main", "main.tex")), "main", passes=3, bibtex=True)
312 shutil.copy2(result["pdf"], ctx.out("main.pdf"))
313 shutil.copy2(result["log"], ctx.out("main.log"))
314 if result["aux"].exists():
315     shutil.copy2(result["aux"], ctx.out("main.aux"))
316 body_end = aux_page(result["aux"], "page:body-end")

```

```

317 abstract_end = aux_page(result["aux"], "page:abstract-end")
318 ai_summary = None
319 if (build / "ai_usage.tex").exists():
320     ai = compile_tex(build, "ai_usage.tex", "ai_usage", passes=2, bibtex=False)
321     shutil.copy2(ai["pdf"], ctx.out("AI工具使用详情.pdf"))
322     ai_summary = ai["summary"]
323 info = pdfqa.inspect_pdf(ctx.out("main.pdf"), with_text=False)
324 body_pages = (body_end - abstract_end) if body_end and abstract_end else None
325 report = {
326     "pages": info["pages"],
327     "all_a4": info["all_a4"],
328     "abstract_end_page": abstract_end,
329     "body_end_page": body_end,
330     "body_pages": body_pages,
331     "log": result["summary"],
332     "unembedded_fonts": info["unembedded_fonts"],
333     "fonts": info["fonts"],
334     "numbers": len(numbers),
335     "number_conflicts": conflicts,
336     "tables": n_tables,
337     "figures": n_figures,
338     "code_files": len(files),
339     "support_members": len(members),
340     "ai_usage_log": ai_summary,
341     "stages": stage_rows,
342 }
343 ctx.write_json("paper_report.json", report)
344 summary = result["summary"]
345 return {
346     "pages": info["pages"],
347     "body_pages": body_pages,
348     "undefined_refs": len(summary["undefined_references"]) + len(summary["undefined_citations"]),
349     "overfull_max_pt": summary["overfull_max_pt"],
350     "missing_chars": summary["missing_characters"],
351     "figures": n_figures,
352     "tables": n_tables,
353 }
354
355
356 @stage("qa", deps=("paper",), description="Quality gates over the compiled paper and the result workbooks")
357 def qa(ctx: StageContext) -> dict[str, Any]:
358     for required in ctx.cfg("qa.required_stages", ["lint", "test"]):
359         ctx.dep(required)
360     paper_dir = ctx.dep("paper")
361     report = json.loads((paper_dir / "paper_report.json").read_text(encoding="utf-8"))
362     checks: list[dict[str, Any]] = []
363
364     def check(name: str, ok: bool, detail: Any = "") -> None:
365         checks.append({"name": name, "ok": bool(ok), "detail": detail})
366         ctx.log.info("qa.check", name=name, ok=bool(ok), detail=detail)
367
368     max_body = int(ctx.cfg("paper.max_body_pages", 30))
369     check("abstract_single_page", report["abstract_end_page"] == 1, report["abstract_end_page"])
370     check(
371         "body_pages_within_limit",
372         report["body_pages"] is not None and report["body_pages"] <= max_body,
373         f"{report['body_pages']} <= {max_body}",
374     )
375     log = report["log"]
376     check(
377         "no_undefined_references",
378         not log["undefined_references"] and not log["undefined_citations"],
379         {"refs": log["undefined_references"][:10], "cites": log["undefined_citations"][:10]},
380     )
381     check("no_tex_errors", not log["errors"], log["errors"][:5])
382     check("fonts_embedded", not report["unembedded_fonts"], report["unembedded_fonts"])

```

```

383 check("a4_pages", report["all_a4"])
384 check(
385     "overfull_within_limit",
386     log["overfull_max_pt"] <= float(ctx.cfg("paper.max_overfull_pt", 20.0)),
387     f"max {log['overfull_max_pt']} pt, count {log['overfull_count']}",
388 )
389 check("no_missing_characters", log["missing_characters"] == 0, log["missing_character_samples"])
390 check("no_number_conflicts", not report["number_conflicts"], report["number_conflicts"][:5])
391 check("ai_usage_pdf_present", (paper_dir / "AI工具使用详情.pdf").exists())
392
393 info = pdfqa.inspect_pdf(paper_dir / "main.pdf", with_text=True)
394 deny_hits = pdfqa.scan_strings(info["texts"], ctx.cfg("qa.identity_denylist", []))
395 watch_hits = pdfqa.scan_strings(info["texts"], ctx.cfg("qa.identity_watchlist", []))
396 check("identity_denylist_clean", not deny_hits, deny_hits[:5])
397 check("paper_size_under_20mb", (paper_dir / "main.pdf").stat().st_size <= 20 * 1024 * 1024)
398
399 results_reports: list[dict[str, Any]] = []
400 results_stage = str(ctx.cfg("qa.results_stage", "results"))
401 require_results = bool(ctx.param("require_results", ctx.cfg("qa.require_results", True)))
402 if ctx.has_stage(results_stage):
403     contracts = importlib.import_module(f"pipelines.{ctx.project['package']}.contracts")
404     results_dir = ctx.run_dir / results_stage
405     for contract in contracts.RESULT_CONTRACTS:
406         rep = xlsx.check_workbook(results_dir / contract.file, contract, ctx.templates / contract.template)
407         results_reports.append(rep)
408         check(f"result:{contract.file}", rep["ok"], rep["errors"][:5])
409 else:
410     check("results_stage_present", not require_results, f"stage '{results_stage}' not completed in this run")
411
412 passed = all(c["ok"] for c in checks)
413 ctx.write_json(
414     "qa_report.json",
415     {
416         "passed": passed,
417         "checks": checks,
418         "identity_watchlist_hits": watch_hits,
419         "results": results_reports,
420         "paper": {k: v for k, v in report.items() if k != "fonts"},
421     },
422 )
423 if not passed:
424     raise RuntimeError("QA gate failed: " + " ".join(c["name"] for c in checks if not c["ok"]))
425 return {"passed": passed, "checks": len(checks), "watchlist_hits": len(watch_hits)}
426
427
428 def _reproduce_md(ctx: StageContext) -> str:
429     rows = _stage_rows(ctx)
430     code_ref = ctx.params.get("code_ref") or {}
431     manifest = json.loads((ctx.run_dir / "ingest" / "manifest.json").read_text(encoding="utf-8"))
432     packages = manifest.get("runtime", {}).get("packages", {})
433     lines = [
434         f"# 复现说明 (run `{ctx.run_id}`)",
435         "",
436         "本目录内所有结果均由 Modal 云端按下列阶段生成；本地只提交任务、下载和核对 SHA-256。",
437         "",
438         f"- 代码版本: git `{code_ref.get('git_sha', '')}` (分支 `{code_ref.get('branch', '')}`)",
439         f"- 运行环境: Python {manifest.get('runtime', {}).get('python', '')}, "
440         f"{manifest.get('runtime', {}).get('platform', '')}",
441         "- 依赖版本: " + " ".join(f"{k}=={v}" for k, v in sorted(packages.items())),
442         "",
443         "## 阶段",
444         "",
445         "| 阶段 | 状态 | 用时 (s) | 输出摘要 | Modal 任务 |",
446         "|---|---|---|---|",
447     ]
448     lines += [

```

```

449         f"| {r['stage']} | {r['status']} | {r['duration_s']} or 0:.1f} | {r['outputs_digest'][:16]}` | "
450         f"`{r['modal_task_id']} or ''` | "
451         for r in rows
452     ]
453     lines += [
454         "",
455         "## 复现命令",
456         "",
457         "```bash",
458         "python3 tools/cli.py provision",
459         "python3 tools/cli.py run ingest,validate --new-run",
460         "python3 tools/cli.py run <科学阶段...> --size medium",
461         "python3 tools/cli.py run lint,test,paper,qa,package,release",
462         "python3 tools/cli.py release --version <tag>",
463         "```",
464         "",
465         "每个阶段目录下的 `manifest.json` 记录输入、输出散列、参数与环境; `events.jsonl` 为结构化日志。",
466         "",
467     ]
468     return "\n".join(lines)
469
470
471 @stage("package", deps=("qa",), description="Build the deterministic support-materials zip and its member manifest")
472 def package(ctx: StageContext) -> dict[str, Any]:
473     planned = set(json.loads((ctx.dep("paper") / "support_members.json").read_text(encoding="utf-8")))
474     ctx.out("REPRODUCE.md").write_text(_reproduce_md(ctx), encoding="utf-8")
475     members = [(name, path) for name, path in _support_members(ctx, planned=False) if name != "MANIFEST.json"]
476     missing = [name for name, path in members if not path.exists()]
477     if missing:
478         raise RuntimeError(f"support members missing on disk: {missing[:10]}")
479     actual = {name for name, _ in members} | {"MANIFEST.json"}
480     drift = sorted(actual ^ planned)
481     if drift and not ctx.param("allow_member_drift"):
482         raise RuntimeError(f"support material members differ from the list printed in the paper: {drift[:20]}")
483     manifest = {
484         "run_id": ctx.run_id,
485         "created_at": datetime.now(UTC).isoformat(timespec="seconds"),
486         "members": [
487             {"path": name, "sha256": sha256_file(path), "bytes": path.stat().st_size} for name, path in members
488         ],
489     }
490     manifest_path = ctx.out("MANIFEST.json")
491     manifest_path.write_text(json.dumps(manifest, ensure_ascii=False, indent=2), encoding="utf-8")
492     members.append(("MANIFEST.json", manifest_path))
493     zip_info = packaging.build_zip(members, ctx.out("支撑材料.zip"), comment=f"forge run {ctx.run_id}")
494     limit = int(ctx.cfg("package.max_bytes", 20 * 1024 * 1024))
495     ctx.write_json("package_report.json", {**zip_info, "limit_bytes": limit, "drift": drift})
496     if zip_info["bytes"] > limit:
497         raise RuntimeError(f"support zip {zip_info['bytes']} bytes exceeds limit {limit}")
498     return {"members": zip_info["members"], "bytes": zip_info["bytes"], "drift": len(drift)}
499
500
501 @stage(
502     "release",
503     deps=("package",),
504     description="Assemble deliverables (paper, result workbooks, support zip) with SHA-256 manifest",
505 )
506 def release(ctx: StageContext) -> dict[str, Any]:
507     qa_report = json.loads((ctx.dep("qa") / "qa_report.json").read_text(encoding="utf-8"))
508     if not qa_report["passed"] and not ctx.param("allow_failed_qa"):
509         raise RuntimeError("release refused: QA gate did not pass")
510     paper_dir = ctx.dep("paper")
511     files: dict[str, Path] = {str(ctx.cfg("release.paper_name", "论文.pdf")): paper_dir / "main.pdf"}
512     results_stage = str(ctx.cfg("qa.results_stage", "results"))
513     if ctx.has_stage(results_stage):
514         for name in ctx.project.get("result_files", []):

```

```

515         files[name] = ctx.run_dir / results_stage / name
516     elif not ctx.param("allow_missing_results"):
517         raise RuntimeError("release refused: results stage missing")
518     files["支撑材料.zip"] = ctx.dep("package") / "支撑材料.zip"
519     files["AI工具使用详情.pdf"] = paper_dir / "AI工具使用详情.pdf"
520     manifest: dict[str, Any] = {
521         "run_id": ctx.run_id,
522         "code_ref": ctx.params.get("code_ref"),
523         "files": {},
524         "qa_passed": qa_report["passed"],
525         "created_at": datetime.now(UTC).isoformat(timespec="seconds"),
526     }
527     for name, source in files.items():
528         if not source.exists():
529             raise RuntimeError(f"release file missing: {name} ({source})")
530         target = ctx.out(name)
531         shutil.copy2(source, target)
532         manifest["files"][name] = {"sha256": sha256_file(target), "bytes": target.stat().st_size}
533     notes = [
534         f"# 发布说明 — run {ctx.run_id}",
535         "",
536         f"- 代码: `{(ctx.params.get('code_ref') or {}).get('git_sha', '')}`",
537         f"- QA: {'通过' if qa_report['passed'] else '未通过 (强制发布)'}",
538         "",
539         "| 文件 | 字节 | SHA-256 |",
540         "|---|---|---|",
541     ]
542     notes += [f"| {n} | {m['bytes']} | `{m['sha256']}`" for n, m in manifest["files"].items()]
543     ctx.out("RELEASE_NOTES.md").write_text("\n".join(notes) + "\n", encoding="utf-8")
544     ctx.write_json("release_manifest.json", manifest)
545     return {
546         "files": len(files),
547         "paper_bytes": manifest["files"][str(ctx.cfg("release.paper_name", "论文.pdf"))]["bytes"],
548     }
549
550
551 FMT_IGNORE = (".git", "artifacts", "releases", "__pycache__", ".forge", ".DS_Store", ".mypy_cache", ".ruff_cache")
552
553
554 @stage("fmt", description="ruff --fix and ruff format in the cloud; changed files are exported for a local copy-back")
555 def fmt(ctx: StageContext) -> dict[str, Any]:
556     work = Path("/tmp/fmt")
557     shutil.rmtree(work, ignore_errors=True)
558     shutil.copytree(ctx.repo, work, ignore=shutil.ignore_patterns(*FMT_IGNORE, "*.pyc"))
559     before = hash_tree(work, ignore=FMT_IGNORE)
560     fix = subprocess.run(
561         ["ruff", "check", "--fix", "--no-cache", "--output-format", "concise", "."],
562         cwd=work,
563         capture_output=True,
564         text=True,
565         check=False,
566     )
567     form = subprocess.run(["ruff", "format", "--no-cache", "."], cwd=work, capture_output=True, text=True, check=False)
568     after = hash_tree(work, ignore=FMT_IGNORE)
569     changed = sorted(rel for rel in after if rel not in before or after[rel]["sha256"] != before[rel]["sha256"])
570     for rel in changed:
571         shutil.copy2(work / rel, ctx.out("files", rel))
572     ctx.out("ruff_fix.txt").write_text(fix.stdout + fix.stderr, encoding="utf-8")
573     ctx.out("ruff_format.txt").write_text(form.stdout + form.stderr, encoding="utf-8")
574     ctx.write_json("fmt_report.json", {"changed": changed, "remaining": fix.stdout.strip().splitlines()[-1:]})
575     ctx.log.info("fmt.done", changed=len(changed), remaining=fix.stdout.strip().splitlines()[-1:])
576     return {"changed": len(changed)}

```

pipelines/common/validation.py

```
1 """Shared implementation of the ``validate`` stage (input data contracts + profiling)."""
2
3 from __future__ import annotations
4
5 from typing import Any
6
7 import pandas as pd
8
9 from forge.context import StageContext
10 from forge.contracts import FrameContract, normalise_columns, validate_frame
11
12
13 def run_validation(ctx: StageContext, contracts: dict[str, FrameContract]) -> dict[str, Any]:
14     """Validate every ingested parquet named in ``contracts`` (key = parquet stem)."""
15     data_dir = ctx.dep("ingest") / "data"
16     reports: list[dict[str, Any]] = []
17     profile: dict[str, Any] = {}
18     for stem, contract in contracts.items():
19         path = data_dir / f"{stem}.parquet"
20         if not path.exists():
21             reports.append({"contract": contract.name, "ok": False, "errors": [f"parquet missing: {stem}"], "rows": 0})
22             continue
23         df = normalise_columns(pd.read_parquet(path))
24         report = validate_frame(df, contract)
25         report["parquet"] = path.name
26         reports.append(report)
27         desc = df.describe(include="all").astype(str).to_dict()
28         profile[stem] = {
29             "rows": len(df),
30             "columns": list(df.columns),
31             "describe": desc,
32             "nulls": {c: int(df[c].isna().sum()) for c in df.columns},
33         }
34         ctx.log.info(
35             "validate.frame", contract=contract.name, ok=report["ok"], rows=report["rows"], errors=report["errors"][:5]
36         )
37     ok = all(r["ok"] for r in reports)
38     ctx.write_json("validation_report.json", {"ok": ok, "frames": reports})
39     ctx.write_json("data_profile.json", profile)
40     if not ok:
41         failed = [r["contract"] for r in reports if not r["ok"]]
42         raise RuntimeError(f"input contracts violated: {failed}")
43     return {"ok": ok, "frames": len(reports)}
```

pipelines/d/__init__.py

pipelines/d/contracts.py

```
1 """Data contracts for Problem D (time-frequency conflict detection and resolution)."""
2
3 from __future__ import annotations
4
5 import re
6
7 import pandas as pd
8
9 from forge.contracts import Column, FrameContract
```

```

10 from forge.xlsx import SheetContract, WorkbookContract
11
12 INTERVAL = re.compile(r"^\[(\d+),(\d+)\]$")
13
14
15 def _intervals_well_formed(df: pd.DataFrame) -> list[str]:
16     errors: list[str] = []
17     for col, upper in (("频段区间", 100), ("时间区间", None)):
18         for value in df[col].astype(str):
19             match = INTERVAL.match(value.strip())
20             if not match:
21                 errors.append(f"{col}: malformed interval {value!r}")
22                 continue
23             lo, hi = int(match.group(1)), int(match.group(2))
24             if not lo < hi:
25                 errors.append(f"{col}: empty interval {value!r}")
26             if upper is not None and hi > upper:
27                 errors.append(f"{col}: {value!r} exceeds {upper} bands")
28     return errors[:20]
29
30
31 INPUT_CONTRACTS: dict[str, FrameContract] = {
32     "附件1_Sheet1": FrameContract(
33         name="D.附件1.用频计划",
34         columns=(
35             Column("用频装备编号", "str", regex=r"[ABC]\d{3}", unique=True),
36             Column("频段区间", "str", regex=r"^\[\d+, \d+\]"),
37             Column("时间区间", "str", regex=r"^\[\d+, \d+\]"),
38             Column("间隔时长", "int", min=1),
39             Column("使用次数", "int", min=1),
40         ),
41         min_rows=150,
42         max_rows=150,
43         checks=( _intervals_well_formed, ),
44     ),
45 }
46
47 RESULT_CONTRACTS: list[WorkbookContract] = [
48     WorkbookContract(
49         "result1.xlsx", "result1.xlsx", (SheetContract("Sheet1", min_rows=1, header_len=3, text_columns=(1, 2)),),
50     ),
51     WorkbookContract(
52         "result2.xlsx", "result2.xlsx", (SheetContract("Sheet1", min_rows=1, header_len=4, text_columns=(0,)),),
53     ),
54     WorkbookContract("result3.xlsx", "result3.xlsx", (SheetContract("Sheet1", min_rows=0, header_len=3)),),
55     WorkbookContract(
56         "result4.xlsx", "result4.xlsx", (SheetContract("Sheet1", min_rows=1, header_len=5, text_columns=(0,)),),
57     ),
58 ]

```

pipelines/d/detect.py

```

1 """Conflict detection (Problem 1): pairwise two-pointer method and band-bucketed sweep line.
2
3 Two plans conflict when their band intervals overlap and at least one pair of uses overlaps in time
4 (half-open intervals, MDR-0001). ``detect_pairwise`` is the  $O(n^2 (m_i + m_j))$  reference and
5 ``detect_bandsweep`` the  $O(N \log N + K)$  scalable method; both must return the same pair set.
6 """
7
8 from __future__ import annotations
9
10 from collections import Counter, defaultdict
11 from typing import Any

```

```

12
13 from pipelines.d.plans import CATEGORIES, Plan
14
15
16 def intervals_overlap(a0: int, a1: int, b0: int, b1: int) -> bool:
17     """Half-open intervals ``[a0,a1)`` and ``[b0,b1)`` intersect."""
18     return a0 < b1 and b0 < a1
19
20
21 def freq_overlap(a: Plan, b: Plan) -> bool:
22     return intervals_overlap(a.f, a.f_end, b.f, b.f_end)
23
24
25 def uses_overlap(a: Plan, b: Plan) -> bool:
26     """True when some use of ``a`` overlaps some use of ``b`` in time (two-pointer merge over sorted uses)."""
27     ua, ub = a.uses(), b.uses()
28     i = j = 0
29     while i < len(ua) and j < len(ub):
30         s1, e1 = ua[i]
31         s2, e2 = ub[j]
32         if s1 < e2 and s2 < e1:
33             return True
34         if e1 <= e2:
35             i += 1
36         else:
37             j += 1
38     return False
39
40
41 def overlapping_uses(a: Plan, b: Plan) -> list[tuple[int, int]]:
42     """All (k, l) use pairs of ``a`` and ``b`` that overlap in time (full  $O(m_a m_b)$  enumeration)."""
43     return [
44         (k, l)
45         for k, (s1, e1) in enumerate(a.uses())
46         for l, (s2, e2) in enumerate(b.uses())
47         if intervals_overlap(s1, e1, s2, e2)
48     ]
49
50
51 def plans_conflict(a: Plan, b: Plan) -> bool:
52     return freq_overlap(a, b) and uses_overlap(a, b)
53
54
55 def detect_pairwise(plans: list[Plan]) -> list[tuple[int, int]]:
56     """Reference detector: every pair, band test first, then two-pointer time test. Returns sorted index pairs."""
57     out: list[tuple[int, int]] = []
58     n = len(plans)
59     for i in range(n):
60         a = plans[i]
61         for j in range(i + 1, n):
62             b = plans[j]
63             if freq_overlap(a, b) and uses_overlap(a, b):
64                 out.append((i, j))
65     return out
66
67
68 def detect_bandsweep(plans: list[Plan]) -> list[tuple[int, int]]:
69     """Sweep line per band: bucket every use of every plan into the bands it occupies, sort by start,
70     and report pairs whose uses overlap. Complexity  $O(N \log N + K)$  with  $N$  = total (use, band) items."""
71     buckets: dict[int, list[tuple[int, int, int]]] = defaultdict(list)
72     for idx, p in enumerate(plans):
73         for start, end in p.uses():
74             for band in range(p.f, p.f_end):
75                 buckets[band].append((start, end, idx))
76     found: set[tuple[int, int]] = set()
77     for items in buckets.values():

```

```

78     items.sort()
79     active: list[tuple[int, int]] = [] # (end, idx), pruned lazily
80     for start, end, idx in items:
81         active = [(e, k) for e, k in active if e > start]
82         for _, k in active:
83             found.add((k, idx) if k < idx else (idx, k))
84         active.append((end, idx))
85     return sorted(found)
86
87
88 def pair_key(a: str, b: str) -> str:
89     return "".join(sorted((a, b)))
90
91
92 def conflict_records(plans: list[Plan], pairs: list[tuple[int, int]]) -> list[dict[str, Any]]:
93     """Per conflict pair: ids, categories and the number of overlapping use pairs."""
94     records = []
95     for i, j in pairs:
96         a, b = plans[i], plans[j]
97         uses = overlapping_uses(a, b)
98         records.append(
99             {
100                 "id1": a.pid,
101                 "id2": b.pid,
102                 "pair": pair_key(a.cat, b.cat),
103                 "use_pairs": len(uses),
104                 "first_overlap_time": min(max(a.uses()[k][0], b.uses()[l][0]) for k, l in uses),
105                 "band_overlap": min(a.f_end, b.f_end) - max(a.f, b.f),
106             }
107         )
108     return records
109
110
111 def graph_stats(plans: list[Plan], pairs: list[tuple[int, int]]) -> dict[str, Any]:
112     """Conflict-graph statistics: category-pair counts, degrees, components, involvement."""
113     import networkx as nx
114
115     graph = nx.Graph()
116     graph.add_nodes_from(range(len(plans)))
117     graph.add_edges_from(pairs)
118     degrees = dict(graph.degree())
119     by_pair: Counter[str] = Counter(pair_key(plans[i].cat, plans[j].cat) for i, j in pairs)
120     involved = {c: sum(1 for i, p in enumerate(plans) if p.cat == c and degrees[i] > 0) for c in CATEGORIES}
121     counts = {c: sum(1 for p in plans if p.cat == c) for c in CATEGORIES}
122     components = sorted((len(c) for c in nx.connected_components(graph) if len(c) > 1), reverse=True)
123     degree_hist: Counter[int] = Counter(degrees.values())
124     return {
125         "pairs": len(pairs),
126         "by_pair": {k: int(by_pair.get(k, 0)) for k in ("AA", "AB", "AC", "BB", "BC", "CC")},
127         "plans_involved": int(sum(involved.values())),
128         "plans_involved_by_cat": involved,
129         "plans_by_cat": counts,
130         "max_degree": int(max(degrees.values(), default=0)),
131         "mean_degree": float(sum(degrees.values()) / max(len(plans), 1)),
132         "degree_hist": {str(k): int(v) for k, v in sorted(degree_hist.items())},
133         "components": len(components),
134         "largest_component": int(components[0]) if components else 0,
135         "component_sizes": components,
136         "density": float(len(pairs) / max(len(plans) * (len(plans) - 1) / 2, 1)),
137         "top_degree": sorted(
138             ({ "pid": plans[i].pid, "degree": degrees[i] } for i in range(len(plans))),
139             key=lambda r: (-int(r["degree"]), str(r["pid"])),
140         )[:10],
141     }

```

pipelines/d/highs_util.py

```
1 """HiGHS process-level settings shared by the resolution and packing models."""
2
3 from __future__ import annotations
4
5 _THREADS: list[int] = []
6
7
8 def highs_threads(requested: int) -> int:
9     """Thread count to pass to HiGHS in this process.
10
11     HiGHS initialises its global task executor on the first ``run()`` and a later solve that asks for a
12     different ``threads`` value fails silently (model status "Not Set", objective 0). The first value used
13     in the process is therefore kept for every subsequent solve."""
14     if not _THREADS:
15         _THREADS.append(max(1, int(requested)))
16     return _THREADS[0]
```

pipelines/d/pack.py

```
1 """Problem 3: pack as many additional C-class plans as possible into the free time-frequency cells left by
2 the Problem-2 solution (MDR-0005). Variables x[f,s] (band start, first-use start); AtMostOne per cell.
3 """
4
5 from __future__ import annotations
6
7 import time
8 from dataclasses import replace
9 from typing import Any
10
11 import numpy as np
12
13 from pipelines.d.highs_util import highs_threads
14 from pipelines.d.plans import BANDS, Plan
15
16 C_TEMPLATE = {"w": 3, "d": 2, "g": 8, "n": 12}
17
18
19 def template_span(template: dict[str, int]) -> int:
20     return template["d"] + (template["n"] - 1) * (template["d"] + template["g"])
21
22
23 def template_offsets(template: dict[str, int]) -> np.ndarray:
24     """Time offsets (relative to the first-use start) occupied by one plan of the template."""
25     period = template["d"] + template["g"]
26     return np.array([k * period + j for k in range(template["n"]) for j in range(template["d"])], dtype=np.int64)
27
28
29 def occupancy_grid(plans: list[Plan], horizon: int, bands: int = BANDS) -> np.ndarray:
30     """Boolean (bands × horizon) grid of occupied cells."""
31     grid = np.zeros((bands, horizon), dtype=bool)
32     for p in plans:
33         for start, end in p.uses():
34             grid[p.f : p.f_end, start:end] = True
35     return grid
36
37
38 def candidate_placements(grid: np.ndarray, template: dict[str, int]) -> np.ndarray:
39     """All (f, s) whose template cells are free and end within the horizon; shape (m, 2)."""
40     bands, horizon = grid.shape
41     span = template_span(template)
42     offsets = template_offsets(template)
```

```

43     free = -grid
44     width = template["w"]
45     n_s = horizon - span + 1
46     if n_s <= 0:
47         return np.zeros((0, 2), dtype=np.int64)
48     rows = []
49     for f in range(bands - width + 1):
50         strip = np.all(free[f : f + width], axis=0) # time slots free on all bands of the strip
51         windows = strip[np.arange(n_s)[: , None] + offsets[None, :]] # (n_s, cells per plan)
52         ok = np.all(windows, axis=1)
53         for s in np.flatnonzero(ok):
54             rows.append((f, int(s)))
55     return np.array(rows, dtype=np.int64).reshape(-1, 2)
56
57
58 def cell_rows(candidates: np.ndarray, template: dict[str, int], horizon: int) -> list[list[int]]:
59     """Deduplicated cell rows (lists of candidate indices) with at least two candidates."""
60     offsets = template_offsets(template)
61     cells: dict[int, list[int]] = {}
62     for v, (f, s) in enumerate(candidates.tolist()):
63         for band in range(f, f + template["w"]):
64             base = band * horizon + s
65             for off in offsets.tolist():
66                 cells.setdefault(base + off, []).append(v)
67     rows = {tuple(r) for r in cells.values() if len(r) >= 2}
68     return [list(r) for r in sorted(rows)]
69
70
71 def greedy_pack(candidates: np.ndarray, template: dict[str, int], grid: np.ndarray) -> list[int]:
72     """First-fit greedy (by start time, then band): a feasible lower bound and a CP-SAT hint."""
73     occ = grid.copy()
74     offsets = template_offsets(template)
75     order = np.lexsort((candidates[:, 0], candidates[:, 1]))
76     chosen = []
77     for v in order.tolist():
78         f, s = candidates[v]
79         block = occ[f : f + template["w"]][:, s + offsets]
80         if not block.any():
81             occ[f : f + template["w"], s + offsets] = True
82             chosen.append(v)
83     return chosen
84
85
86 def solve_pack_cpsat(
87     rows: list[list[int]],
88     n_candidates: int,
89     *,
90     hint: list[int] | None,
91     seed: int,
92     workers: int = 8,
93     time_limit: float = 900.0,
94 ) -> dict[str, Any]:
95     from ortools.sat.python import cp_model
96
97     t0 = time.monotonic()
98     cp = cp_model.CpModel()
99     x = [cp.NewBoolVar(f"x{v}") for v in range(n_candidates)]
100     for row in rows:
101         cp.AddAtMostOne(x[v] for v in row)
102     cp.Maximize(sum(x))
103     if hint:
104         hinted = set(hint)
105         for v in range(n_candidates):
106             cp.AddHint(x[v], v in hinted)
107     solver = cp_model.CpSolver()
108     solver.parameters.random_seed = int(seed)

```

```

109 solver.parameters.num_workers = int(workers)
110 solver.parameters.max_time_in_seconds = float(time_limit)
111 status = solver.Solve(cp)
112 chosen = (
113     [v for v in range(n_candidates) if solver.Value(x[v])]
114     if status in (cp_model.OPTIMAL, cp_model.FEASIBLE)
115     else []
116 )
117 return {
118     "solver": "cp-sat",
119     "status": solver.StatusName(status),
120     "value": len(chosen),
121     "bound": float(solver.BestObjectiveBound()) if chosen else None,
122     "chosen": chosen,
123     "seconds": round(time.monotonic() - t0, 3),
124     "n_vars": n_candidates,
125     "n_rows": len(rows),
126 }
127
128
129 def highs_pack(
130     rows: list[list[int]], n_candidates: int, *, integer: bool, time_limit: float, seed: int, threads: int = 8
131 ) -> dict[str, Any]:
132     """LP relaxation upper bound (integer=False) or a MILP cross-check with its dual bound (integer=True)."""
133     import highspy
134
135     t0 = time.monotonic()
136     h = highspy.Highs()
137     h.setOptionValue("output_flag", False)
138     h.setOptionValue("time_limit", float(time_limit))
139     h.setOptionValue("random_seed", int(seed))
140     h.setOptionValue("threads", highs_threads(threads))
141     h.setOptionValue("mip_rel_gap", 0.0)
142     h.setOptionValue("mip_abs_gap", 0.0)
143     n = n_candidates
144     h.addVars(n, np.zeros(n), np.ones(n))
145     if integer:
146         h.changeColsIntegrality(n, np.arange(n, dtype=np.int32), np.array([highspy.HighsVarType.kInteger] * n))
147     starts, index = [], []
148     nnz = 0
149     for row in rows:
150         starts.append(nnz)
151         index.extend(row)
152         nnz += len(row)
153     h.addRows(
154         len(rows),
155         np.full(len(rows), -highspy.kHighsInf),
156         np.ones(len(rows)),
157         nnz,
158         np.array(starts, dtype=np.int32),
159         np.array(index, dtype=np.int32),
160         np.ones(nnz),
161     )
162     h.changeColsCost(n, np.arange(n, dtype=np.int32), -np.ones(n)) # maximise sum x
163     h.run()
164     status = h.modelStatusToString(h.getModelStatus())
165     info = h.getInfo()
166     out: dict[str, Any] = {"status": status, "seconds": round(time.monotonic() - t0, 3), "integer": integer}
167     if integer:
168         out["value"] = -float(info.objective_function_value)
169         out["bound"] = -float(info.mip_dual_bound)
170         sol = np.array(h.getSolution().col_value)
171         out["chosen"] = [int(v) for v in np.flatnonzero(sol > 0.5)]
172     else:
173         out["value"] = -float(info.objective_function_value)
174     return out

```

```

175
176
177 def plan_offsets(plan: Plan) -> np.ndarray:
178     """Time offsets (relative to the first-use start) occupied by ``plan``."""
179     return np.array([k * plan.period + j for k in range(plan.n) for j in range(plan.d)], dtype=np.int64)
180
181
182 def free_positions(grid: np.ndarray, width: int, offsets: np.ndarray, *, first_only: bool = False) -> np.ndarray:
183     """All (f, s) at which a plan of this shape fits in the free cells of ``grid``; shape (m, 2).
184
185     With ``first_only`` the scan stops at the lexicographically smallest (f, s), which is what first-fit
186     needs and keeps the repack linear in the number of plans rather than in the number of positions."""
187     bands, horizon = grid.shape
188     span = int(offsets[-1]) + 1
189     n_s = horizon - span + 1
190     if n_s <= 0 or width > bands:
191         return np.zeros((0, 2), dtype=np.int64)
192     free = ~grid
193     rows = []
194     windows_index = np.arange(n_s)[:, None] + offsets[None, :]
195     for f in range(bands - width + 1):
196         strip = np.all(free[f : f + width], axis=0)
197         ok = np.flatnonzero(np.all(strip[windows_index], axis=1))
198         if first_only:
199             if len(ok):
200                 return np.array([(f, int(ok[0]))], dtype=np.int64)
201             continue
202         rows.extend((f, int(s)) for s in ok)
203     return np.array(rows, dtype=np.int64).reshape(-1, 2)
204
205
206 def repack_existing(plans: list[Plan], horizon: int, bands: int = BANDS) -> tuple[list[Plan], np.ndarray]:
207     """First-fit decreasing re-placement of every plan (MDR-0010, interpretation B).
208
209     Each plan keeps ``w``, ``d``, ``g`` and ``n`` — only its band start and first-use start move, by an
210     unrestricted amount — so the number of cells it occupies is unchanged. Plans are placed largest-first
211     at the lexicographically smallest free (f, s), which compacts them towards low bands and early times
212     and leaves a large contiguous free region. Raises when a plan cannot be placed at all."""
213     grid = np.zeros((bands, horizon), dtype=bool)
214     order = sorted(plans, key=lambda p: (-(p.w * p.d * p.n), -p.w, p.pid))
215     placed: list[Plan] = []
216     for plan in order:
217         offsets = plan_offsets(plan)
218         spots = free_positions(grid, plan.w, offsets, first_only=True)
219         if not len(spots):
220             raise RuntimeError(f"first-fit decreasing could not place {plan.pid} within {bands}x{horizon}")
221         f, s = (int(v) for v in spots[0])
222         grid[f : f + plan.w, s + offsets] = True
223         placed.append(replace(plan, f=f, s=s))
224     return sorted(placed, key=lambda p: p.pid), grid
225
226
227 def plans_from_choice(candidates: np.ndarray, chosen: list[int], template: dict[str, int]) -> list[Plan]:
228     """New C plans (ids N001, N002, ...) in (start, band) order."""
229     picked = sorted((int(candidates[v, 1]), int(candidates[v, 0])) for v in chosen)
230     return [
231         Plan(f"N{k:03d}", "C", f, template["w"], s, template["d"], template["g"], template["n"])
232         for k, (s, f) in enumerate(picked, start=1)
233     ]

```

pipelines/d/plans.py

```
1 """Plan data model for Problem D: parsing attachment rows, use intervals, cells and adjustments.
2
3 A plan occupies the band interval ``[f, f+w)`` during ``n`` uses; use ``k`` (0-based) occupies the
4 time interval ``[s + k(d+g), s + d + k(d+g))``. All intervals are half-open integers (MDR-0001).
5 """
6
7 from __future__ import annotations
8
9 import re
10 from dataclasses import asdict, dataclass, replace
11 from pathlib import Path
12 from typing import Any
13
14 BANDS = 100
15 CATEGORIES = ("A", "B", "C")
16 INTERVAL = re.compile(r"^\s*(\d+)\s*,\s*(\d+)\s*\)\s*$")
17
18
19 def parse_interval(text: Any) -> tuple[int, int]:
20     """Parse ``[lo,hi)`` into ``(lo, hi)``; rejects malformed or empty intervals."""
21     match = INTERVAL.match(str(text))
22     if not match:
23         raise ValueError(f"malformed half-open interval: {text!r}")
24     lo, hi = int(match.group(1)), int(match.group(2))
25     if lo >= hi:
26         raise ValueError(f"empty interval: {text!r}")
27     return lo, hi
28
29
30 def fmt_interval(lo: int, hi: int) -> str:
31     """Format a half-open interval the way the attachments do: ``[lo,hi)``."""
32     return f"[{lo},{hi})"
33
34
35 @dataclass(frozen=True)
36 class Plan:
37     """One frequency-use plan (immutable)."""
38
39     pid: str
40     cat: str
41     f: int # first band occupied
42     w: int # number of bands occupied
43     s: int # start of the first use
44     d: int # duration of every use
45     g: int # gap between consecutive uses
46     n: int # number of uses
47
48     @property
49     def period(self) -> int:
50         return self.d + self.g
51
52     @property
53     def f_end(self) -> int:
54         return self.f + self.w
55
56     @property
57     def end(self) -> int:
58         """Exclusive end time of the last use."""
59         return self.s + self.d + (self.n - 1) * self.period
60
61     def uses(self) -> list[tuple[int, int]]:
62         """Half-open time intervals of the ``n`` uses, in chronological order."""
63         return [(self.s + k * self.period, self.s + k * self.period + self.d) for k in range(self.n)]
64
```

```

65     @property
66     def n_cells(self) -> int:
67         return self.w * self.d * self.n
68
69     def freq_text(self) -> str:
70         return fmt_interval(self.f, self.f_end)
71
72     def time_text(self) -> str:
73         return fmt_interval(self.s, self.s + self.d)
74
75     def shifted(self, df: int = 0, dt: int = 0, gap: int | None = None) -> Plan:
76         """Copy with the band start moved by ``df``, the time start by ``dt`` and/or a new gap."""
77         return replace(self, f=self.f + df, s=self.s + dt, g=self.g if gap is None else gap)
78
79     def record(self) -> dict[str, Any]:
80         return asdict(self)
81
82
83 def plan_from_record(rec: dict[str, Any]) -> Plan:
84     return Plan(
85         str(rec["pid"]),
86         str(rec["cat"]),
87         int(rec["f"]),
88         int(rec["w"]),
89         int(rec["s"]),
90         int(rec["d"]),
91         int(rec["g"]),
92         int(rec["n"]),
93     )
94
95
96 def plan_from_row(pid: Any, freq: Any, time: Any, gap: Any, count: Any) -> Plan:
97     """Build a plan from one attachment row (equipment id, band interval, time interval, gap, count)."""
98     pid = str(pid).strip()
99     f0, f1 = parse_interval(freq)
100     t0, t1 = parse_interval(time)
101     return Plan(pid, pid[0], f0, f1 - f0, t0, t1 - t0, int(gap), int(count))
102
103
104 def load_plans(parquet: Path) -> list[Plan]:
105     """Read the ingested attachment sheet (parquet) into plans, in file order."""
106     import pandas as pd
107
108     df = pd.read_parquet(parquet)
109     df.columns = [str(c).strip() for c in df.columns]
110     return [
111         plan_from_row(r["用频装备编号"], r["频段区间"], r["时间区间"], r["间隔时长"], r["使用次数"])
112         for _, r in df.iterrows()
113     ]
114
115
116 def horizon(plans: list[Plan]) -> int:
117     """Latest exclusive end time over the plans (0 for an empty list)."""
118     return max((p.end for p in plans), key=int, default=0)
119
120
121 def by_category(plans: list[Plan]) -> dict[str, list[Plan]]:
122     out: dict[str, list[Plan]] = {c: [] for c in CATEGORIES}
123     for p in plans:
124         out.setdefault(p.cat, []).append(p)
125     return out

```

pipelines/d/report.py

```
1 """Paper figures and tables for Problem D (stages ``figures`` and ``tables``).
2
3 Figures follow the house style in ``forge.plotting`` (vector PDF + PNG preview, Okabe-Ito palette with a
4 fixed category → hue mapping, one axis per panel, legends whenever two or more series are shown).
5 Tables are booktabs fragments (``tables/*.tex``) with a CSV twin; the paper wraps them in ``table``.
6 """
7
8 from __future__ import annotations
9
10 import csv
11 from collections.abc import Sequence
12 from typing import Any
13
14 from forge import plotting
15 from forge.context import StageContext
16 from forge.runner import stage
17 from forge.tex import tex_escape
18 from pipelines.d.detect import overlapping_uses
19 from pipelines.d.plans import CATEGORIES, Plan, horizon, plan_from_record
20 from pipelines.d.resolve import Limits, enumerate_options
21
22 CAT_COLOR = {"A": "#0072B2", "B": "#D55E00", "C": "#009E73"} # fixed identity → hue (Okabe-Ito slots 1-3)
23 ACTION_COLOR = {"freq": "#0072B2", "time": "#E69F00", "gap": "#CC79A7", "cancel": "#000000", "keep": "#BBBBBB"}
24 ACTION_LABEL = {"freq": "频段平移", "time": "时间平移", "gap": "间隔调整", "cancel": "撤销", "keep": "保持"}
25 GREY = "#C8C8C8"
26 REPORT_DEPS = ("detect", "resolve", "pack", "resolve_interval", "bench", "sensitivity")
27
28
29 def _plans(records: list[dict[str, Any]]) -> list[Plan]:
30     return [plan_from_record(r) for r in records]
31
32
33 def _rects(
34     ax: Any,
35     plans: Sequence[Plan],
36     face: dict[str, str],
37     *,
38     edge: dict[str, str] | None = None,
39     hatch_ids: set[str] = frozenset(),
40     alpha: float = 0.9,
41 ) -> None:
42     from matplotlib.collections import PatchCollection
43     from matplotlib.patches import Rectangle
44
45     edge = edge or {}
46     plain, plain_face, plain_edge, plain_lw = [], [], [], []
47     hatched, hatched_edge = [], []
48     for p in plans:
49         for start, end in p.uses():
50             rect = Rectangle((start, p.f), end - start, p.w)
51             if p.pid in hatch_ids:
52                 hatched.append(rect)
53                 hatched_edge.append(edge.get(p.pid, "#000000"))
54             else:
55                 plain.append(rect)
56                 plain_face.append(face.get(p.pid, GREY))
57                 plain_edge.append(edge.get(p.pid, "none"))
58                 plain_lw.append(0.6 if p.pid in edge else 0.0)
59     if plain:
60         ax.add_collection(
61             PatchCollection(plain, facecolors=plain_face, edgecolors=plain_edge, linewidths=plain_lw, alpha=alpha)
62         )
63     if hatched:
64         ax.add_collection(
```

```

65         PatchCollection(hatched, facecolors="none", edgecolors=hatched_edge, linewidths=0.6, hatch="////")
66     )
67
68
69 def _plane(ax: Any, horizon_t: int, *, xlabel: bool = True) -> None:
70     ax.set_xlim(0, horizon_t)
71     ax.set_ylim(0, 100)
72     ax.set_ylabel("频段编号 ( $\Delta f$ )")
73     if xlabel:
74         ax.set_xlabel("时间 ( $\Delta t$ )")
75     ax.grid(False)
76
77
78 def _legend(ax: Any, entries: list[tuple[str, str, str | None]], **kwargs: Any) -> None:
79     from matplotlib.patches import Patch
80
81     handles = [
82         Patch(
83             facecolor=color if hatch is None else "none",
84             edgecolor="#000000" if hatch else "none",
85             hatch=hatch,
86             label=label,
87         )
88         for label, color, hatch in entries
89     ]
90     ax.legend(handles=handles, frameon=False, **kwargs)
91
92
93 def _conflict_use_ids(plans: list[Plan], conflicts: list[dict[str, Any]]) -> set[tuple[str, int]]:
94     by_id = {p.pid: p for p in plans}
95     out: set[tuple[str, int]] = set()
96     for rec in conflicts:
97         a, b = by_id[rec["id1"]], by_id[rec["id2"]]
98         for k, l in overlapping_uses(a, b):
99             out.add((a.pid, k))
100             out.add((b.pid, l))
101     return out
102
103
104 def fig_q1_gantt(ctx: StageContext, plans: list[Plan], conflicts: list[dict[str, Any]]) -> None:
105     from matplotlib.collections import PatchCollection
106     from matplotlib.patches import Rectangle
107
108     horizon_t = horizon(plans)
109     fig, ax = plotting.new_figure(6.5, 3.4)
110     _rects(ax, plans, {p.pid: CAT_COLOR[p.cat] for p in plans}, alpha=0.75)
111     hot = _conflict_use_ids(plans, conflicts)
112     patches = [
113         Rectangle((start, p.f), end - start, p.w)
114         for p in plans
115         for k, (start, end) in enumerate(p.uses())
116         if (p.pid, k) in hot
117     ]
118     ax.add_collection(PatchCollection(patches, facecolors="none", edgecolors="#000000", linewidths=0.7))
119     _plane(ax, horizon_t)
120     _legend(
121         ax,
122         [
123             ("A 类", CAT_COLOR["A"], None),
124             ("B 类", CAT_COLOR["B"], None),
125             ("C 类", CAT_COLOR["C"], None),
126             ("冲突的使用次 (黑边)", "none", ""),
127         ],
128         loc="upper left",
129         bbox_to_anchor=(1.0, 1.0),
130         ncol=1,

```

```

131     )
132     plotting.save(fig, ctx.out("figures", "fig_q1_gantt.pdf"))
133
134
135 def fig_q1_structure(ctx: StageContext, stats: dict[str, Any]) -> None:
136     import numpy as np
137
138     fig, axes = plotting.new_figure(6.5, 2.7, ncols=2, gridspec_kw={"width_ratios": [1.0, 1.4]})
139     ax = axes[0]
140     matrix = np.zeros((3, 3))
141     for key, value in stats["by_pair"].items():
142         i, j = CATEGORIES.index(key[0]), CATEGORIES.index(key[1])
143         matrix[i, j] = value
144         matrix[j, i] = value
145     ax.imshow(matrix, cmap="Blues", vmin=0)
146     for i in range(3):
147         for j in range(3):
148             ax.text(
149                 j,
150                 i,
151                 f"{int(matrix[i, j])}",
152                 ha="center",
153                 va="center",
154                 color="#000000" if matrix[i, j] < matrix.max() * 0.6 else "#FFFFFF",
155             )
156     ax.set_xticks(range(3), [f"{c} 类" for c in CATEGORIES])
157     ax.set_yticks(range(3), [f"{c} 类" for c in CATEGORIES])
158     ax.set_title("(a) 冲突对数 (按类别对)")
159     ax.grid(False)
160     ax = axes[1]
161     hist = {int(k): v for k, v in stats["degree_hist"].items()}
162     degrees = sorted(hist)
163     ax.bar(degrees, [hist[d] for d in degrees], color="#0072B2", width=0.8)
164     for d in degrees:
165         ax.text(d, hist[d], str(hist[d]), ha="center", va="bottom", fontsize=7)
166     ax.set_xlabel("冲突图中的度 (与之冲突的计划数)")
167     ax.set_ylabel("计划数量")
168     ax.set_title("(b) 度分布")
169     ax.set_xticks(degrees)
170     plotting.save(fig, ctx.out("figures", "fig_q1_structure.pdf"))
171
172
173 def fig_resolution(
174     ctx: StageContext,
175     name: str,
176     plans: list[Plan],
177     decisions: list[dict[str, Any]],
178     conflicts: list[dict[str, Any]],
179     title: str,
180 ) -> None:
181     from matplotlib.patches import FancyArrowPatch
182
183     horizon_t = max(horizon(plans), max((plan_from_record(d["plan"]).end for d in decisions if d["plan"]), default=0))
184     fig, axes = plotting.new_figure(6.5, 5.2, nrows=2, sharex=True)
185     kinds = {d["pid"]: d["kind"] for d in decisions}
186     ax = axes[0]
187     face = {p.pid: (ACTION_COLOR[kinds[p.pid]] if kinds[p.pid] in {"freq", "time", "gap"} else GREY) for p in plans}
188     _rects(ax, [p for p in plans if kinds[p.pid] != "cancel"], face, alpha=0.85)
189     _rects(
190         ax,
191         [p for p in plans if kinds[p.pid] == "cancel"],
192         {},
193         hatch_ids={p.pid for p in plans if kinds[p.pid] == "cancel"},
194     )
195     hot = _conflict_use_ids(plans, conflicts)
196     from matplotlib.collections import PatchCollection

```

```

197 from matplotlib.patches import Rectangle
198
199 ax.add_collection(
200     PatchCollection(
201         [Rectangle((s, p.f), e - s, p.w) for p in plans for k, (s, e) in enumerate(p.uses()) if (p.pid, k) in hot],
202         facecolors="none",
203         edgecolors="#000000",
204         linewidths=0.5,
205     )
206 )
207 _plane(ax, horizon_t, xlabel=False)
208 ax.set_title(f"(a) 消解前: {title}中被调整/撤销的计划以颜色标出, 黑边为冲突使用次")
209 _legend(
210     ax,
211     [
212         ("保持", GREY, None),
213         ("频段平移", ACTION_COLOR["freq"], None),
214         ("时间平移", ACTION_COLOR["time"], None),
215         ("间隔调整", ACTION_COLOR["gap"], None),
216         ("撤销", "none", "////"),
217     ],
218     loc="upper left",
219     bbox_to_anchor=(1.0, 1.0),
220 )
221 ax = axes[1]
222 after = [plan_from_record(d["plan"]) for d in decisions if d["plan"]]
223 _rects(
224     ax, after, {p.pid: (ACTION_COLOR[kinds[p.pid]] if kinds[p.pid] != "keep" else GREY) for p in after}, alpha=0.85
225 )
226 by_id = {p.pid: p for p in plans}
227 for p in after:
228     if kinds[p.pid] in {"freq", "time"}:
229         before = by_id[p.pid]
230         ax.add_patch(
231             FancyArrowPatch(
232                 (before.s, before.f + before.w / 2),
233                 (p.s, p.f + p.w / 2),
234                 arrowstyle="-|>",
235                 mutation_scale=6,
236                 color="#000000",
237                 linewidth=0.5,
238                 alpha=0.6,
239             )
240         )
241 _plane(ax, horizon_t)
242 ax.set_title("(b) 消解后: 无冲突的用频计划 (箭头为平移方向)")
243 plotting.save(fig, ctx.out("figures", f"{name}.pdf"))
244
245
246 def fig_schemes(ctx: StageContext, name: str, alternatives: dict[str, Any]) -> None:
247     import numpy as np
248
249     names = [n for n in ("P", "T", "S", "W") if alternatives.get(n, {}).get("canonical_vector")]
250     fig, axes = plotting.new_figure(6.5, 2.6, ncols=2)
251     ax = axes[0]
252     x = np.arange(len(names))
253     bottom = np.zeros(len(names))
254     for idx, c in enumerate(CATEGORIES):
255         vals = np.array([alternatives[n]["canonical_vector"][3 + idx] for n in names], dtype=float)
256         ax.bar(x, vals, bottom=bottom, color=CAT_COLOR[c], label=f"{c} 类调整", width=0.6)
257         bottom += vals
258     for i in range(len(names)):
259         ax.text(x[i], bottom[i], str(int(bottom[i])), ha="center", va="bottom", fontsize=7)
260     ax.set_xticks(x, [f"方案 {n}" for n in names])
261     ax.set_ylabel("被调整的计划数")
262     ax.set_title("(a) 各目标方案的调整数量")

```

```

263 ax.set_ylim(0, float(bottom.max()) * 1.3)
264 ax.legend(frameon=False, fontsize=7, ncol=3, loc="upper left")
265 ax = axes[1]
266 amps = [alternatives[n]["canonical_vector"][6] / 10.0 for n in names]
267 ax.bar(x, amps, color="#0072B2", width=0.6)
268 for i, a in enumerate(amps):
269     ax.text(x[i], a, f"{a:.1f}", ha="center", va="bottom", fontsize=7)
270 ax.set_xticks(x, [f"方案 {n}" for n in names])
271 ax.set_ylabel("归一化总调整幅度")
272 ax.set_title("(b) 总调整幅度")
273 plotting.save(fig, ctx.out("figures", f"{name}.pdf"))
274
275
276 def fig_q3_layout(
277     ctx: StageContext,
278     existing: list[Plan],
279     new_plans: list[Plan],
280     horizon_t: int,
281     *,
282     name: str = "fig_q3_layout",
283     title: str | None = None,
284 ) -> None:
285     fig, ax = plotting.new_figure(6.5, 3.4)
286     _rects(ax, existing, {p.pid: GREY for p in existing}, alpha=0.9)
287     _rects(ax, new_plans, {p.pid: CAT_COLOR["C"] for p in new_plans}, alpha=0.95)
288     _plane(ax, horizon_t)
289     if title:
290         ax.set_title(title)
291     _legend(
292         ax,
293         [("既有计划", GREY, None), (f"新增 C 类计划 ({len(new_plans)} 个)", CAT_COLOR["C"], None)],
294         loc="upper left",
295         bbox_to_anchor=(1.0, 1.0),
296     )
297     plotting.save(fig, ctx.out("figures", f"{name}.pdf"))
298
299
300 def fig_q4_comparison(
301     ctx: StageContext, q2_table: dict[str, Any], q4_table: dict[str, Any], q4_shifts: dict[str, Any]
302 ) -> None:
303     import numpy as np
304
305     fig, axes = plotting.new_figure(6.5, 2.6, ncols=2)
306     ax = axes[0]
307     x = np.arange(3)
308     width = 0.36
309     q2 = [q2_table[c]["adjust"] for c in CATEGORIES]
310     q4 = [q4_table[c]["adjust"] for c in CATEGORIES]
311     ax.bar(x - width / 2, q2, width, color="#0072B2", label="问题 2")
312     ax.bar(x + width / 2, q4, width, color="#D55E00", label="问题 4")
313     for i in range(3):
314         ax.text(x[i] - width / 2, q2[i], str(q2[i]), ha="center", va="bottom", fontsize=7)
315         ax.text(x[i] + width / 2, q4[i], str(q4[i]), ha="center", va="bottom", fontsize=7)
316     ax.set_xticks(x, [f"{c} 类" for c in CATEGORIES])
317     ax.set_ylabel("被调整的计划数")
318     ax.set_title("(a) 调整数量对照")
319     ax.legend(frameon=False)
320     ax = axes[1]
321     kinds = ["freq", "time", "gap", "cancel"]
322     counts = [q4_shifts[k]["count"] if k != "cancel" else q4_shifts["cancel"] for k in kinds]
323     ax.bar(range(4), counts, color=[ACTION_COLOR[k] for k in kinds], width=0.6)
324     for i, v in enumerate(counts):
325         ax.text(i, v, str(v), ha="center", va="bottom", fontsize=7)
326     ax.set_xticks(range(4), [ACTION_LABEL[k] for k in kinds])
327     ax.set_ylabel("计划数")
328     ax.set_title("(b) 问题 4 的动作构成")

```

```

329     plotting.save(fig, ctx.out("figures", "fig_q4_comparison.pdf"))
330
331
332 def fig_bench_scaling(ctx: StageContext, scaling: list[dict[str, Any]]) -> None:
333     from matplotlib.ticker import NullFormatter
334
335     fig, ax = plotting.new_figure(6.3, 3.0)
336     ns = [r["n"] for r in scaling]
337     limit = max((r["cpsat_seconds"] for r in scaling), default=0.0)
338     series = [
339         ("成对枚举检测", [r["pairwise_seconds"] for r in scaling], "#0072B2", "o"),
340         ("分桶扫描线检测", [r["sweep_seconds"] for r in scaling], "#D55E00", "s"),
341         ("单元格模型构建", [r["build_seconds"] for r in scaling], "#009E73", "^"),
342         (
343             f"CP-SAT 消解 (固定时限 {limit:.0f} s, 到时返回可行解)",
344             [r["cpsat_seconds"] for r in scaling],
345             "#E69F00",
346             "D",
347         ),
348     ]
349     for label, ys, color, marker in series:
350         ax.plot(ns, ys, marker=marker, markersize=4, color=color, label=label)
351     ax.set_xscale("log")
352     ax.set_yscale("log")
353     ax.set_xlabel("计划数量 n (合成实例)")
354     ax.set_ylabel("耗时 (s)")
355     ax.set_xticks(ns, [str(n) for n in ns])
356     ax.xaxis.set_minor_formatter(NullFormatter())
357     ax.legend(frameon=False, fontsize=7, loc="center left")
358     ax.set_xlim(ns[0] * 0.9, ns[-1] * 1.1)
359     plotting.save(fig, ctx.out("figures", "fig_bench_scaling.pdf"))
360
361
362 def fig_sensitivity(ctx: StageContext, cases: list[dict[str, Any]]) -> None:
363     import numpy as np
364
365     rows = [r for r in cases if r["adjust_total"] is not None]
366     labels = [f"({r['fmax']},{r['tmax']})" for r in rows]
367     x = np.arange(len(rows))
368     fig, axes = plotting.new_figure(6.5, 2.7, ncols=2)
369     ax = axes[0]
370     bottom = np.zeros(len(rows))
371     for idx, c in enumerate(CATEGORIES):
372         vals = np.array([r["vector"][3 + idx] for r in rows], dtype=float)
373         ax.bar(x, vals, bottom=bottom, color=CAT_COLOR[c], label=f"{c} 类调整", width=0.65)
374         bottom += vals
375     cancels = np.array([r["cancel_total"] for r in rows], dtype=float)
376     ax.bar(x, cancels, bottom=bottom, color="#000000", label="撤销", width=0.65)
377     for i in range(len(rows)):
378         ax.text(x[i], bottom[i] + cancels[i], str(int(bottom[i] + cancels[i])), ha="center", va="bottom", fontsize=7)
379     ax.set_xticks(x, labels, rotation=45, ha="right")
380     ax.set_xlabel("最大平移幅度 (频段, 时间)")
381     ax.set_ylabel("被调整/撤销的计划数")
382     ax.set_title("(a) 调整与撤销数量")
383     ax.set_ylim(0, float((bottom + cancels).max()) * 1.35)
384     ax.legend(frameon=False, fontsize=7, ncol=2, loc="upper left")
385     ax = axes[1]
386     ax.bar(x, [r["amplitude"] for r in rows], color="#0072B2", width=0.65)
387     for i, r in enumerate(rows):
388         ax.text(x[i], r["amplitude"], f"{r['amplitude']:.1f}", ha="center", va="bottom", fontsize=7)
389     ax.set_xticks(x, labels, rotation=45, ha="right")
390     ax.set_xlabel("最大平移幅度 (频段, 时间)")
391     ax.set_ylabel("归一化总调整幅度")
392     ax.set_title("(b) 总调整幅度")
393     plotting.save(fig, ctx.out("figures", "fig_sensitivity.pdf"))
394

```

```

395
396 @stage("figures", deps=REPORT_DEPS, description="Paper figures (vector PDF + PNG preview)")
397 def figures(ctx: StageContext) -> dict[str, Any]:
398     plans = _plans(ctx.dep_data("detect", "plans.json"))
399     conflicts = ctx.dep_data("detect", "conflicts.json")
400     stats = ctx.dep_data("detect", "stats.json")
401     fig_q1_gantt(ctx, plans, conflicts)
402     fig_q1_structure(ctx, stats)
403     q2_dec = ctx.dep_data("resolve", "decisions.json")
404     fig_resolution(ctx, "fig_q2_resolution", plans, q2_dec, conflicts, "问题 2")
405     fig_schemes(ctx, "fig_q2_schemes", ctx.dep_data("resolve", "alternatives.json"))
406     existing = _plans(ctx.dep_data("resolve", "resolved_plans.json"))
407     new_plans = _plans(ctx.dep_data("pack", "new_plans.json"))
408     pack_report = ctx.dep_data("pack", "pack_report.json")
409     fig_q3_layout(ctx, existing, new_plans, int(pack_report["horizon"]))
410     if "repack" in pack_report.get("alternatives", {}):
411         repack = ctx.dep_data("pack", "repack_plans.json")
412         fig_q3_layout(
413             ctx,
414             _plans(repack["existing"]),
415             _plans(repack["new"]),
416             int(pack_report["horizon"]),
417             name="fig_q3_repack",
418             title="既有计划经首次适配重排后 (MDR-0010 解释 B) ",
419         )
420     q4_dec = ctx.dep_data("resolve_interval", "decisions.json")
421     fig_resolution(ctx, "fig_q4_resolution", plans, q4_dec, conflicts, "问题 4")
422     q2_rep = ctx.dep_data("resolve", "solver_report.json")
423     q4_rep = ctx.dep_data("resolve_interval", "solver_report.json")
424     fig_q4_comparison(ctx, q2_rep["table"], q4_rep["table"], q4_rep["shifts"])
425     fig_bench_scaling(ctx, ctx.dep_data("bench", "scaling.json"))
426     fig_sensitivity(ctx, ctx.dep_data("sensitivity", "cases.json"))
427     files = sorted(p.name for p in (ctx.stage_dir / "figures").glob("*.pdf"))
428     ctx.number("FigureCount", len(files))
429     return {"figures": files}
430
431
432 # ----- tables
433 def _write_table(
434     ctx: StageContext,
435     name: str,
436     header: list[str],
437     rows: list[list[Any]],
438     *,
439     align: str | None = None,
440     long: bool = False,
441 ) -> None:
442     def cell(v: Any) -> str:
443         if isinstance(v, float):
444             return f"{v:.4f}".rstrip("0").rstrip(".") if abs(v - round(v)) > 1e-9 else f"{v:.0f}"
445         return tex_escape(str(v))
446
447     align = align or ("l" + "r" * (len(header) - 1))
448     lines = []
449     if long:
450         lines.append(f"\\begin{{longtable}}{{@{{}}{align}@{{}}}}")
451         lines.append(
452             "\\toprule "
453             + " & ".join(tex_escape(h) for h in header)
454             + " \\\\ \\midrule \\endhead \\bottomrule \\endfoot"
455         )
456         lines += [" & ".join(cell(v) for v in row) + " \\\\" for row in rows]
457         lines.append("\\end{{longtable}}")
458     else:
459         lines.append(f"\\begin{{tabular}}{{@{{}}{align}@{{}}}}")
460         lines.append("\\toprule")

```

```

461         lines.append(" & ".join(tex_escape(h) for h in header) + " \\\\")
462         lines.append("\\midrule")
463         lines += [" & ".join(cell(v) for v in row) + " \\\\" for row in rows]
464         lines.append("\\bottomrule")
465         lines.append("\\end{tabular}")
466     ctx.out("tables", f"{name}.tex").write_text("\n".join(lines) + "\n", encoding="utf-8")
467     with ctx.out("tables", f"{name}.csv").open("w", encoding="utf-8", newline="") as fh:
468         writer = csv.writer(fh)
469         writer.writerow(header)
470         writer.writerows(rows)
471
472
473 def _table1(table: dict[str, dict[str, int]]) -> list[list[Any]]:
474     rows = [[f"{c} 类", table[c]["keep"], table[c]["adjust"], table[c]["cancel"]] for c in CATEGORIES]
475     rows.append(
476         [
477             "合计",
478             sum(table[c]["keep"] for c in CATEGORIES),
479             sum(table[c]["adjust"] for c in CATEGORIES),
480             sum(table[c]["cancel"] for c in CATEGORIES),
481         ]
482     )
483     return rows
484
485
486 def _vector_text(vec: list[int] | None) -> str:
487     return "(" + ", ".join(str(v) for v in vec) + ")" if vec else "-"
488
489
490 def _adjustment_rows(plans: list[Plan], decisions: list[dict[str, Any]], with_gap: bool) -> list[list[Any]]:
491     by_id = {p.pid: p for p in plans}
492     rows = []
493     for dec in sorted(decisions, key=lambda d: d["pid"]):
494         if dec["kind"] == "keep":
495             continue
496         before = by_id[dec["pid"]]
497         after = plan_from_record(dec["plan"]) if dec["plan"] else None
498         row = [
499             dec["pid"],
500             ACTION_LABEL[dec["kind"]],
501             before.freq_text(),
502             after.freq_text() if after else "-",
503             before.time_text(),
504             after.time_text() if after else "-",
505         ]
506         if with_gap:
507             row += [before.g, after.g if after else "-"]
508         rows.append(row)
509     return rows
510
511
512 def _cancelled_rows(
513     plans: list[Plan], decisions: list[dict[str, Any]], conflicts: list[dict[str, Any]], limits: Limits
514 ) -> list[list[Any]]:
515     """Why a plan was cancelled: its conflict degree, higher-priority neighbours and number of legal actions."""
516     by_id = {p.pid: p for p in plans}
517     neighbours: dict[str, list[str]] = {p.pid: [] for p in plans}
518     for rec in conflicts:
519         neighbours[rec["id1"]].append(rec["id2"])
520         neighbours[rec["id2"]].append(rec["id1"])
521     rows = []
522     for dec in sorted(decisions, key=lambda d: d["pid"]):
523         if dec["kind"] != "cancel":
524             continue
525         plan = by_id[dec["pid"]]
526         nbrs = neighbours[plan.pid]

```

```

527         rows.append(
528             [
529                 plan.pid,
530                 plan.freq_text(),
531                 plan.time_text(),
532                 len(nbrs),
533                 sum(1 for n in nbrs if by_id[n].cat in {"A", "B"}),
534                 len(enumerate_options(plan, limits)) - 2, # excluding keep and cancel
535             ]
536         )
537     return rows
538
539
540 def _solver_tables(ctx: StageContext, tag: str, rep: dict[str, Any]) -> None:
541     """Per-level solver evidence (CP-SAT value/bound/status, HiGHS LP and MILP bounds) and model statistics."""
542     solver_rows = []
543     highs_levels = {e["level"]: e for e in rep["highs"]["levels"]}
544     for entry in rep["primary"]["levels"]:
545         h = highs_levels.get(entry["level"], {})
546         solver_rows.append(
547             [
548                 entry["level"],
549                 entry.get("value", "-"),
550                 f"{entry['bound']:.2f}" if "bound" in entry else "-",
551                 entry.get("status", "-"),
552                 f"{entry.get('seconds', 0):.1f}",
553                 f"{h['lp_value']:.2f}" if "lp_value" in h else "-",
554                 f"{h['mip_value']:.0f}" if "mip_value" in h and h["mip_value"] < 1e20 else "-",
555                 f"{h['mip_bound']:.2f}" if "mip_bound" in h and abs(h["mip_bound"]) < 1e20 else "-",
556                 h.get("mip_status", "-"),
557             ]
558         )
559     _write_table(
560         ctx,
561         f"tab_{tag}_solver",
562         [
563             "目标层级",
564             "CP-SAT 值",
565             "CP-SAT 下界",
566             "CP-SAT 状态",
567             "用时/s",
568             "HiGHS LP 下界",
569             "HiGHS MILP 值",
570             "HiGHS MILP 下界",
571             "HiGHS 状态",
572         ],
573         solver_rows,
574         align="lrrlrrrrl",
575     )
576     m = rep["model"]
577     _write_table(
578         ctx,
579         f"tab_{tag}_model",
580         ["项目", "数值"],
581         [
582             ["决策变量数 (计划-动作对)", m["vars"]],
583             ["时频单元数 (含动作扩展)", m["cells"]],
584             ["变量-单元隶属数", m["memberships"]],
585             ["单元格 AtMostOne 约束数 (去重前 / 去重后)", f"{m['rows_raw']} / {m['rows']}"],
586             [
587                 "成对蕴含团约束数 / 强制撤销割数",
588                 f"{m.get('strengthening_groups', 0)} / {m.get('forced_cancel_pairs', 0)}",
589             ],
590             ["可能相互冲突的计划对数", m.get("interacting_pairs", "-")],
591             [
592                 "每计划最多动作数 (A/B/C)",

```

```

593         f"{m['options_per_plan']['A']}/{m['options_per_plan']['B']}/{m['options_per_plan']['C']}",
594     ],
595     ["CP-SAT 词典序总用时/s", f"{rep['primary']['seconds']:.1f}"],
596     ["热启动现任解目标向量 (MDR-0009) ", _vector_text(rep.get("hint", {}).get("vector"))],
597     [
598         "词典序求解起点",
599         {"file": "仓库内现任解", "weighted": "分离权重解".get(rep.get("hint", {}).get("start_from"), "冷启动"),
600     ],
601     ["各级均证明最优", "是" if rep.get("all_optimal") else "否"],
602     ["HiGHS 交叉验证总用时/s", f"{rep['highs']['seconds']:.1f}"],
603     ["HiGHS 证明最优的各级与 CP-SAT 一致", "是" if rep["highs_agree"] else "否"],
604     ["分离权重单目标解与词典序向量一致", "是" if rep["weighted_agree"] else "否"],
605 ],
606 align="lr",
607 )
608
609
610 @stage("tables", deps=REPORT_DEPS, description="Paper tables (booktabs .tex + CSV)")
611 def tables(ctx: StageContext) -> dict[str, Any]:
612     plans = _plans(ctx.dep_data("detect", "plans.json"))
613     stats = ctx.dep_data("detect", "stats.json")
614     conflicts = ctx.dep_data("detect", "conflicts.json")
615
616     # Given data, per category. The table asserts homogeneity instead of assuming it, so that the
617     # paper may quote one (w, d, g, n) per class only when every plan of that class really shares it.
618     data_rows = []
619     for cat in CATEGORIES:
620         group = [p for p in plans if p.cat == cat]
621         if not group:
622             continue
623         shapes = {(p.w, p.d, p.g, p.n) for p in group}
624         if len(shapes) != 1:
625             raise RuntimeError(f"category {cat} is not homogeneous: {sorted(shapes)}")
626         head = group[0]
627         data_rows.append(
628             [
629                 f"{cat} 类",
630                 len(group),
631                 head.w,
632                 head.d,
633                 head.g,
634                 head.n,
635                 head.period,
636                 head.d + (head.n - 1) * head.period,
637                 head.n_cells,
638                 len(group) * head.n_cells,
639             ]
640         )
641         ctx.number(f"DataPlans{cat}", len(group))
642         ctx.number(f"DataWidth{cat}", head.w)
643         ctx.number(f"DataDuration{cat}", head.d)
644         ctx.number(f"DataGap{cat}", head.g)
645         ctx.number(f"DataUses{cat}", head.n)
646         ctx.number(f"DataCells{cat}", head.n_cells)
647     _write_table(
648         ctx,
649         "tab_data",
650         [
651             "类别",
652             "装备数",
653             "频宽 w",
654             "时长 d",
655             "间隔 g",
656             "次数 n",
657             "周期 d+g",
658             "时间跨度",

```

```

659         "单计划单元数",
660         "该类单元合计",
661     ],
662     data_rows,
663     align="lrrrrrrrr",
664 )
665
666 _write_table(
667     ctx,
668     "tab_q1_summary",
669     ["统计量", "数值"],
670     [
671         [
672             "用频计划数 (A/B/C) ",
673             f"{stats['plans_by_cat']['A']}/{stats['plans_by_cat']['B']}/{stats['plans_by_cat']['C']}",
674         ],
675         ["冲突对数", stats["pairs"]],
676         ["冲突的使用次对数", stats["use_pairs"]],
677         [
678             "卷入冲突的计划数 (A/B/C) ",
679             f"{stats['plans_involved']} ({stats['plans_involved_by_cat']['A']}/"
680             f"{stats['plans_involved_by_cat']['B']}/{stats['plans_involved_by_cat']['C']}) ",
681         ],
682         ["冲突图连通分量数 (不少于 2 个节点) ", stats["components"]],
683         ["最大连通分量规模", stats["largest_component"]],
684         ["最大度 / 平均度", f"{stats['max_degree']} / {stats['mean_degree']:.2f}"],
685         ["冲突图密度", f"{stats['density']:.4f}"],
686         [
687             "时频单元总数 / 被占用 / 重叠",
688             f"{stats['cells_total']} / {stats['cells_occupied']} / {stats['cells_overlap']}",
689         ],
690         ["时间范围 (最晚结束时刻, Δt) ", stats["horizon"]],
691     ],
692     align="lr",
693 )
694 matrix = [[f"{a} 类" + [stats["by_pair"][b].join(sorted((a, b)))] for b in CATEGORIES] for a in CATEGORIES]
695 _write_table(ctx, "tab_q1_category", ["类别", "A 类", "B 类", "C 类"], matrix)
696 _write_table(
697     ctx,
698     "tab_q1_top_degree",
699     ["装备编号", "冲突计划数 (度) "],
700     [[r["pid"], r["degree"]] for r in stats["top_degree"]],
701 )
702 mult = stats["use_pair_multiplicity"]
703 _write_table(ctx, "tab_q1_multiplicity", ["冲突使用次对数 k", "冲突对数量"], [[k, v] for k, v in mult.items()])
704 _write_table(
705     ctx,
706     "tab_q1_conflicts",
707     ["序号", "装备 1", "装备 2", "类别对", "冲突使用次对数", "频段重叠宽度"],
708     [
709         [k, r["id1"], r["id2"], r["pair"], r["use_pairs"], r["band_overlap"]]
710         for k, r in enumerate(sorted(conflicts, key=lambda r: (r["id1"], r["id2"])), start=1)
711     ],
712     align="rlllrr",
713     long=True,
714 )
715
716 q2 = ctx.dep_data("resolve", "solver_report.json")
717 q2_alt = ctx.dep_data("resolve", "alternatives.json")
718 q2_dec = ctx.dep_data("resolve", "decisions.json")
719 _write_table(ctx, "tab_q2_table1", ["类别", "保留数量", "调整数量", "撤销数量"], _table1(q2["table"]))
720 _write_table(
721     ctx,
722     "tab_q2_adjustments",
723     ["装备编号", "动作", "原频段", "新频段", "原首次时间", "新首次时间"],
724     _adjustment_rows(plans, q2_dec, False),

```

```

725     align="llllll",
726     long=True,
727 )
728 scheme_rows = []
729 for name, label in (
730     ("P", "P: 类别内嵌七级词典序 (主方案)"),
731     ("T", "T: 总量优先"),
732     ("S", "S: 不含优先级"),
733     ("W", "W: 纯加权和"),
734 ):
735     alt = q2_alt.get(name)
736     if not alt or not alt["canonical_vector"]:
737         continue
738     v = alt["canonical_vector"]
739     scheme_rows.append(
740         [
741             label,
742             f"{v[0]}/{v[1]}/{v[2]}",
743             f"{v[3]}/{v[4]}/{v[5]}",
744             sum(v[3:6]),
745             f"{v[6] / 10:.1f}",
746             "是" if alt["all_optimal"] else "否",
747             "通过" if alt["validator_ok"] else "未通过",
748         ]
749     )
750 _write_table(
751     ctx,
752     "tab_q2_schemes",
753     ["目标方案", "撤销 A/B/C", "调整 A/B/C", "调整合计", "归一化幅度", "各级最优", "独立校验"],
754     scheme_rows,
755     align="lrrrrcc",
756 )
757 _solver_tables(ctx, "q2", q2)
758 _write_table(
759     ctx,
760     "tab_q2_cancelled",
761     ["装备编号", "频段", "首次时间", "冲突邻居数", "其中 A/B 类", "可用平移动作数"],
762     _cancelled_rows(
763         plans, q2_dec, conflicts, Limits(fmax=int(q2["limits"]["fmax"]), tmax=int(q2["limits"]["tmax"]))
764     ),
765     align="lllrrr",
766 )
767
768 pack = ctx.dep_data("pack", "pack_report.json")
769 new_plans = _plans(ctx.dep_data("pack", "new_plans.json"))
770 bounds = pack["bounds"]
771 pack_rows = [
772     ["时间范围 T_end (问题 2 方案最晚结束时刻)", pack["horizon"]],
773     ["既有计划数 / 占用单元 / 自由单元", f"{pack['existing']} / {pack['cells_occupied']} / {pack['free_cells']}"],
774     ["候选放置位置数 (f,s)", pack["candidates"]],
775     ["AtMostOne 约束数", pack["rows"]],
776     ["贪心下界", pack["greedy"]],
777     ["CP-SAT 最优解数量 (状态)", f"{pack['cpsat']['value']} ({pack['cpsat']['status'])"}"],
778     ["CP-SAT 上界", bounds.get("cpsat", "-")],
779     ["HiGHS LP 松弛上界", bounds.get("lp", "-")],
780     [
781         "HiGHS MILP 解 / 上界",
782         f"{int(pack['highs_mip']['value'])} / {bounds.get('highs_mip', '-')} {"if pack.get('highs_mip') else '-'},
783     ],
784     ["自由单元容量界 floor(自由单元数/72)", bounds["free_cells"]],
785     ["逐频段容量界", bounds["per_band"]],
786     ["空区域容量界 (无既有计划)", pack["empty_region_bound"]],
787     ["最终结果 / 最紧上界 / gap", f"{pack['value']} / {pack['upper_bound']} / {pack['gap']}"],
788     ["时频利用率 (前 / 后)", f"{100 * pack['utilisation_before']:.1f}% / {100 * pack['utilisation_after']:.1f}%"],
789 ]
790 if pack.get("alternatives", {}).get("original_horizon"):

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```

791     alt = pack["alternatives"]["original_horizon"]
792     pack_rows.append(
793         [f"备选: T_end 取原始计划最晚结束时刻 {alt['horizon']}", f"{alt['value']} (上界 {alt['upper_bound']}) "]
794     )
795     if pack.get("alternatives", {}).get("repack"):
796         rep = pack["alternatives"]["repack"]
797         pack_rows.append(
798             [
799                 "备选解释 B (既有计划可任意平移, MDR-0010): 下界 / 上界",
800                 f"{rep['value']} / {rep['bounds']['free_cells']}" (重排移动 {rep['existing_moved']} 个既有计划) ",
801             ]
802         )
803     _write_table(ctx, "tab_q3_summary", ["项目", "数值"], pack_rows, align="lr")
804     _write_table(
805         ctx,
806         "tab_q3_plans",
807         ["序号", "频段区间", "首次时间区间", "间隔时长", "使用次数"],
808         [[k, p.freq_text(), p.time_text(), p.g, p.n] for k, p in enumerate(new_plans, start=1)],
809         align="rllrr",
810         long=True,
811     )
812
813     q4 = ctx.dep_data("resolve_interval", "solver_report.json")
814     q4_dec = ctx.dep_data("resolve_interval", "decisions.json")
815     q4_alt = ctx.dep_data("resolve_interval", "alternatives.json")
816     _write_table(ctx, "tab_q4_table1", ["类别", "保留数量", "调整数量", "撤销数量"], _table1(q4["table"]))
817     cmp_rows = [
818         [
819             f"{c} 类",
820             q2["table"][c]["keep"],
821             q4["table"][c]["keep"],
822             q2["table"][c]["adjust"],
823             q4["table"][c]["adjust"],
824             q2["table"][c]["cancel"],
825             q4["table"][c]["cancel"],
826         ]
827         for c in CATEGORIES
828     ]
829     cmp_rows.append(
830         [
831             "合计",
832             sum(q2["table"][c]["keep"] for c in CATEGORIES),
833             sum(q4["table"][c]["keep"] for c in CATEGORIES),
834             sum(q2["table"][c]["adjust"] for c in CATEGORIES),
835             sum(q4["table"][c]["adjust"] for c in CATEGORIES),
836             sum(q2["table"][c]["cancel"] for c in CATEGORIES),
837             sum(q4["table"][c]["cancel"] for c in CATEGORIES),
838         ]
839     )
840     cmp_rows.append(
841         ["归一化幅度", "-", "-", f"{q2['amplitude_normalised']:.1f}", f"{q4['amplitude_normalised']:.1f}", "-", "-"]
842     )
843     _write_table(
844         ctx,
845         "tab_q4_compare",
846         ["类别", "保留(问2)", "保留(问4)", "调整(问2)", "调整(问4)", "撤销(问2)", "撤销(问4)"],
847         cmp_rows,
848     )
849     _write_table(
850         ctx,
851         "tab_q4_adjustments",
852         ["装备编号", "动作", "原频段", "新频段", "原首次时间", "新首次时间", "原间隔", "新间隔"],
853         _adjustment_rows(plans, q4_dec, True),
854         align="llllllrr",
855         long=True,
856     )

```

```

857 scheme_rows4 = []
858 for name in ("P", "T", "S", "W"):
859     alt = q4_alt.get(name)
860     if alt and alt["canonical_vector"]:
861         v = alt["canonical_vector"]
862         scheme_rows4.append(
863             [
864                 name,
865                 f"{v[0]}/{v[1]}/{v[2]}",
866                 f"{v[3]}/{v[4]}/{v[5]}",
867                 sum(v[3:6]),
868                 f"{v[6] / 10:.1f}",
869                 alt["shifts"]["gap"]["count"],
870                 "通过" if alt["validator_ok"] else "未通过",
871             ]
872         )
873 _write_table(
874     ctx,
875     "tab_q4_schemes",
876     ["目标方案", "撤销 A/B/C", "调整 A/B/C", "调整合计", "归一化幅度", "间隔调整数", "独立校验"],
877     scheme_rows4,
878     align="lrrrrrc",
879 )
880 _solver_tables(ctx, "q4", q4)
881 _write_table(
882     ctx,
883     "tab_q4_cancelled",
884     ["装备编号", "频段", "首次时间", "冲突邻居数", "其中 A/B 类", "可用调整动作数"],
885     _cancelled_rows(
886         plans,
887         q4_dec,
888         conflicts,
889         Limits(
890             fmax=int(q4["limits"]["fmax"]),
891             tmax=int(q4["limits"]["tmax"]),
892             gmax=int(q4["limits"]["gmax"]),
893             gap_categories=tuple(q4["limits"]["gap_categories"]),
894         ),
895     ),
896     align="lllrrrr",
897 )
898
899 val_rows = []
900 for stage_name, label in (
901     ("detect", "问题 1 冲突检测"),
902     ("resolve", "问题 2 消解方案"),
903     ("pack", "问题 3 新增计划"),
904     ("resolve_interval", "问题 4 消解方案"),
905 ):
906     rep = ctx.dep_data(stage_name, "validation_report.json")
907     if stage_name == "detect":
908         detail = (
909             f"真值 {rep.get('truth_pairs')} 对, 报告 {rep.get('reported_pairs')} 对, "
910             f"缺失 {len(rep.get('missing', []))}, 多余 {len(rep.get('extra', []))}"
911         )
912     elif stage_name == "pack":
913         detail = f"新增 {rep.get('n_new')}, 剩余冲突 {rep.get('remaining_conflicts')}"
914     else:
915         detail = (
916             f"{rep.get('n_decisions')} 个决策, 存留 {rep.get('n_survivors')}, "
917             f"剩余冲突 {rep.get('remaining_conflicts')}"
918         )
919     val_rows.append([label, rep.get("method", ""), detail, "通过" if rep.get("ok") else "未通过"])
920 _write_table(ctx, "tab_validation", ["对象", "独立校验方法", "校验内容", "结论"], val_rows, align="lllc")
921
922 tiny = ctx.dep_data("bench", "tiny.json")

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```

923 _write_table(
924     ctx,
925     "tab_bench_tiny",
926     ["实例", "计划数", "冲突对", "动作组合数", "穷举最优向量", "CP-SAT 向量", "一致", "穷举用时/s"],
927     [
928         [
929             r["instance"],
930             r["plans"],
931             r["conflicts"],
932             r["combinations"],
933             _vector_text(r["exhaustive_vector"]),
934             _vector_text(r["cpsat_vector"]),
935             "是" if r["agree"] else "否",
936             f"{r['exhaustive_seconds']:.2f}",
937         ]
938         for r in tiny
939     ],
940     align="rrrrllcr",
941     long=True,
942 )
943 scaling = ctx.dep_data("bench", "scaling.json")
944 _write_table(
945     ctx,
946     "tab_bench_scaling",
947     [
948         "n",
949         "时间范围",
950         "冲突对",
951         "成对检测/s",
952         "扫描线/s",
953         "变量数",
954         "约束数",
955         "建模/s",
956         "CP-SAT 状态",
957         "求解/s",
958         "撤销合计",
959         "调整合计",
960     ],
961     [
962         [
963             r["n"],
964             r["horizon"],
965             r["conflicts"],
966             f"{r['pairwise_seconds']:.2f}",
967             f"{r['sweep_seconds']:.3f}",
968             r["vars"],
969             r["rows"],
970             f"{r['build_seconds']:.1f}",
971             r["cpsat_status"],
972             f"{r['cpsat_seconds']:.1f}",
973             sum(r["vector"][0:3]) if r["vector"] else "-",
974             sum(r["vector"][3:6]) if r["vector"] else "-",
975         ]
976         for r in scaling
977     ],
978     align="rrrrrrrrlrrr",
979 )
980 cases = ctx.dep_data("sensitivity", "cases.json")
981 _write_table(
982     ctx,
983     "tab_sensitivity",
984     ["最大频段平移", "最大时间平移", "撤销 A/B/C", "调整 A/B/C", "调整合计", "归一化幅度", "各级最优", "用时/s"],
985     [
986         [
987             r["fmax"],
988             r["tmax"],

```

```

989         f"{r['vector'][0]}/{r['vector'][1]}/{r['vector'][2]}",
990         f"{r['vector'][3]}/{r['vector'][4]}/{r['vector'][5]}",
991         r["adjust_total"],
992         f"{r['amplitude']:.1f}",
993         "是" if r["all_optimal"] else "否",
994         f"{r['seconds']:.1f}",
995     ]
996     for r in cases
997         if r["adjust_total"] is not None
998     ],
999     align="rrrrrrcr",
1000 )
1001 files = sorted(p.name for p in (ctx.stage_dir / "tables").glob("*.tex"))
1002 ctx.number("TableCount", len(files))
1003 return {"tables": files}

```

pipelines/d/resolve.py

```

1  """Conflict resolution (Problems 2 and 4): actions, time-frequency cell model, lexicographic CP-SAT,
2  separated-weight cross-checks and HiGS MILP / LP bounds (MDR-0002, MDR-0003, MDR-0004).
3
4  Model. Plan ``i`` chooses exactly one action ``o`` (Boolean ``y[i,o]``). Two (plan, action) pairs
5  conflict iff they share a (band, time) cell, so ``AtMostOne`` over the variables covering each cell is an
6  exact encoding of "no conflict remains". Objectives are lists of levels; each level is a linear form in
7  the ``y`` variables and is minimised after fixing the optimal values of the previous levels.
8  """
9
10 from __future__ import annotations
11
12 import time
13 from dataclasses import dataclass, field
14 from typing import Any
15
16 from pipelines.d.highs_util import highs_threads
17 from pipelines.d.plans import BANDS, CATEGORIES, Plan
18
19 # amplitude normalisation (MDR-0003): |df|/10 + |dt|/5 + |dg|/10, stored x10 so that it is an integer
20 AMPLITUDE_SCALE = 10
21 FREQ_REF, TIME_REF, GAP_REF = 10, 5, 10
22
23
24 @dataclass(frozen=True)
25 class Option:
26     """One action for a plan: keep, band shift, time shift, gap change or cancel."""
27
28     kind: str
29     delta: int = 0
30
31     def apply(self, plan: Plan) -> Plan | None:
32         if self.kind == "cancel":
33             return None
34         if self.kind == "keep":
35             return plan
36         if self.kind == "freq":
37             return plan.shifted(df=self.delta)
38         if self.kind == "time":
39             return plan.shifted(dt=self.delta)
40         if self.kind == "gap":
41             return plan.shifted(gap=plan.g + self.delta)
42         raise ValueError(f"unknown action {self.kind!r}")
43
44     @property
45     def amplitude(self) -> int:

```

```

46     """Scaled amplitude (x10 of the normalised amplitude)."""
47     if self.kind == "freq":
48         return abs(self.delta) * AMPLITUDE_SCALE // FREQ_REF
49     if self.kind == "time":
50         return abs(self.delta) * AMPLITUDE_SCALE // TIME_REF
51     if self.kind == "gap":
52         return abs(self.delta) * AMPLITUDE_SCALE // GAP_REF
53     return 0
54
55     @property
56     def is_adjust(self) -> bool:
57         return self.kind in {"freq", "time", "gap"}
58
59     def label(self) -> str:
60         return self.kind if self.kind in {"keep", "cancel"} else f"{self.kind}-{self.delta:+d}"
61
62
63 @dataclass(frozen=True)
64 class Limits:
65     fmax: int = 10
66     tmax: int = 5
67     gmax: int = 0
68     gap_categories: tuple[str, ...] = ()
69     allow_cancel: bool = True
70
71
72 def enumerate_options(plan: Plan, limits: Limits) -> list[Option]:
73     """All legal actions of a plan under the limits (band interval inside [0,100), start >= 0, gap >= 1)."""
74     opts = [Option("keep")]
75     for delta in range(-limits.fmax, limits.fmax + 1):
76         if delta and plan.f + delta >= 0 and plan.f + delta + plan.w <= BANDS:
77             opts.append(Option("freq", delta))
78     for delta in range(-limits.tmax, limits.tmax + 1):
79         if delta and plan.s + delta >= 0:
80             opts.append(Option("time", delta))
81     if plan.cat in limits.gap_categories:
82         for delta in range(-limits.gmax, limits.gmax + 1):
83             if delta and plan.g + delta >= 1:
84                 opts.append(Option("gap", delta))
85     if limits.allow_cancel:
86         opts.append(Option("cancel"))
87     return opts
88
89
90 @dataclass
91 class CellModel:
92     """Variables ``y[v]`` (one per plan/action) and the cell rows over which at most one may be chosen."""
93
94     plans: list[Plan]
95     options: list[list[Option]]
96     var_plan: list[int] = field(default_factory=list)
97     var_option: list[int] = field(default_factory=list)
98     plan_vars: list[list[int]] = field(default_factory=list)
99     rows: list[list[int]] = field(default_factory=list)
100     horizon: int = 0
101     n_cells: int = 0
102     n_rows_raw: int = 0
103     memberships: int = 0
104
105     @property
106     def n_vars(self) -> int:
107         return len(self.var_plan)
108
109     def option_of(self, v: int) -> Option:
110         return self.options[self.var_plan[v]][self.var_option[v]]
111

```

```

112     def plan_of(self, v: int) -> Plan:
113         return self.plans[self.var_plan[v]]
114
115
116 def build_cell_model(plans: list[Plan], options: list[list[Option]]) -> CellModel:
117     """Expand every (plan, action) into the cells it occupies; keep rows with >= 2 variables (deduplicated)."""
118     model = CellModel(plans, options)
119     applied: list[Plan | None] = []
120     for i, plan in enumerate(plans):
121         model.plan_vars.append([])
122         for o, opt in enumerate(options[i]):
123             model.plan_vars[i].append(len(model.var_plan))
124             model.var_plan.append(i)
125             model.var_option.append(o)
126             applied.append(opt.apply(plan))
127     horizon = max((p.end for p in applied if p is not None), default=0)
128     model.horizon = horizon
129     cells: dict[int, list[int]] = {}
130     memberships = 0
131     for v, p in enumerate(applied):
132         if p is None:
133             continue
134         for start, end in p.uses():
135             for band in range(p.f, p.f_end):
136                 base = band * horizon
137                 for t in range(start, end):
138                     cells.setdefault(base + t, []).append(v)
139                 memberships += 1
140     model.n_cells = len(cells)
141     model.memberships = memberships
142     raw = [row for row in cells.values() if len(row) >= 2]
143     model.n_rows_raw = len(raw)
144     model.rows = [list(r) for r in sorted({tuple(row) for row in raw})]
145     return model
146
147
148 @dataclass
149 class Strengthening:
150     """Redundant but LP/propagation-strengthening constraints derived from the cell rows (MDR-0004).
151
152     ``groups``: for a variable v and a neighbouring plan j, the options of j that share a cell with v —
153     at most one of {v} together with the group may be chosen (a clique, since j picks exactly one option).
154     ``forced``: plan pairs whose non-cancel options all conflict pairwise → at least one must be cancelled.
155     """
156
157     groups: list[tuple[int, list[int]]] = field(default_factory=list)
158     forced: list[tuple[int, int]] = field(default_factory=list)
159     interacting_pairs: int = 0
160
161
162 def strengthen(model: CellModel) -> Strengthening:
163     conf: list[set[int]] = [set() for _ in range(model.n_vars)]
164     for row in model.rows:
165         for a in row:
166             conf[a].update(row)
167     out = Strengthening()
168     plan_pairs: set[tuple[int, int]] = set()
169     for v in range(model.n_vars):
170         conf[v].discard(v)
171         by_plan: dict[int, list[int]] = {}
172         for u in conf[v]:
173             by_plan.setdefault(model.var_plan[u], []).append(u)
174         for j, us in sorted(by_plan.items()):
175             out.groups.append((v, sorted(us)))
176             i = model.var_plan[v]
177             plan_pairs.add((min(i, j), max(i, j)))

```

```

178 non_cancel = [[v for v in vs if model.option_of(v).kind != "cancel"] for vs in model.plan_vars]
179 cancel_var = {
180     i: next((v for v in vs if model.option_of(v).kind == "cancel"), None) for i, vs in enumerate(model.plan_vars)
181 }
182 for i, j in sorted(plan_pairs):
183     if cancel_var[i] is None or cancel_var[j] is None:
184         continue
185     if all(set(non_cancel[j]) <= conf[v] for v in non_cancel[i]):
186         out.forced.append((i, j))
187 out.interacting_pairs = len(plan_pairs)
188 return out
189
190
191 def _cancel_of(model: CellModel, i: int) -> int:
192     return next(v for v in model.plan_vars[i] if model.option_of(v).kind == "cancel")
193
194
195 Level = tuple[str, dict[int, int]]
196
197
198 def _cancel_vars(model: CellModel, cats: tuple[str, ...]) -> dict[int, int]:
199     return {v: 1 for v in range(model.n_vars) if model.option_of(v).kind == "cancel" and model.plan_of(v).cat in cats}
200
201
202 def _adjust_vars(model: CellModel, cats: tuple[str, ...]) -> dict[int, int]:
203     return {v: 1 for v in range(model.n_vars) if model.option_of(v).is_adjust and model.plan_of(v).cat in cats}
204
205
206 def _amplitude_vars(model: CellModel) -> dict[int, int]:
207     return {v: model.option_of(v).amplitude for v in range(model.n_vars) if model.option_of(v).amplitude}
208
209
210 def _merge(*parts: dict[int, int]) -> dict[int, int]:
211     out: dict[int, int] = {}
212     for part in parts:
213         for v, c in part.items():
214             out[v] = out.get(v, 0) + c
215     return out
216
217
218 def canonical_levels(model: CellModel) -> list[Level]:
219     """The seven reporting levels: cancel A/B/C, adjust A/B/C, scaled amplitude (scheme P, MDR-0003)."""
220     levels: list[Level] = [(f"cancel{c}", _cancel_vars(model, (c,))) for c in CATEGORIES]
221     levels += [(f"adjust{c}", _adjust_vars(model, (c,))) for c in CATEGORIES]
222     levels.append(("amplitude", _amplitude_vars(model)))
223     return levels
224
225
226 def scheme_levels(name: str, model: CellModel) -> list[Level]:
227     """Objective schemes P (primary), T (totals first), S (no priority), W (single weighted sum)."""
228     if name == "P":
229         return canonical_levels(model)
230     if name == "T":
231         levels: list[Level] = [(f"cancel", _cancel_vars(model, CATEGORIES)), (f"adjust", _adjust_vars(model, CATEGORIES))
232     ↪ ]
233         levels += [(f"changed{c}", _merge(_cancel_vars(model, (c,)), _adjust_vars(model, (c,)))) for c in CATEGORIES]
234         levels.append(("amplitude", _amplitude_vars(model)))
235         return levels
236     if name == "S":
237         return [
238             ("cancel", _cancel_vars(model, CATEGORIES)),
239             ("adjust", _adjust_vars(model, CATEGORIES)),
240             ("amplitude", _amplitude_vars(model)),
241         ]
242     if name == "W":
243         weight = {"A": 3, "B": 2, "C": 1}

```

```

243     terms: dict[int, int] = {}
244     for v in range(model.n_vars):
245         opt, plan = model.option_of(v), model.plan_of(v)
246         value = 0
247         if opt.kind == "cancel":
248             value = 100 * weight[plan.cat]
249         elif opt.is_adjust:
250             value = 30 * weight[plan.cat] + opt.amplitude
251         if value:
252             terms[v] = value
253     return [("weighted", terms)]
254     raise KeyError(name)
255
256
257 def level_upper_bounds(model: CellModel, levels: list[Level]) -> list[int]:
258     """Upper bound of each level: each plan contributes at most the largest coefficient among its actions."""
259     bounds = []
260     for _, terms in levels:
261         total = 0
262         for vars_of_plan in model.plan_vars:
263             total += max((terms.get(v, 0) for v in vars_of_plan), default=0)
264         bounds.append(int(total))
265     return bounds
266
267
268 def separated_weights(bounds: list[int]) -> list[int]:
269     """Weights  $W_k = \prod_{j>k} (U_j + 1)$ : a single weighted sum then reproduces the lexicographic optimum."""
270     weights = [1] * len(bounds)
271     for k in range(len(bounds) - 2, -1, -1):
272         weights[k] = weights[k + 1] * (bounds[k + 1] + 1)
273     return weights
274
275
276 def evaluate(model: CellModel, chosen: list[int], levels: list[Level]) -> list[int]:
277     """Value of every level for a solution given as the chosen variable per plan."""
278     chosen_set = set(chosen)
279     return [int(sum(c for v, c in terms.items() if v in chosen_set)) for _, terms in levels]
280
281
282 def decisions_from_choice(model: CellModel, chosen: list[int]) -> list[dict[str, Any]]:
283     out = []
284     for i, plan in enumerate(model.plans):
285         opt = model.option_of(chosen[i])
286         after = opt.apply(plan)
287         out.append(
288             {
289                 "pid": plan.pid,
290                 "cat": plan.cat,
291                 "kind": opt.kind,
292                 "delta": opt.delta,
293                 "label": opt.label(),
294                 "plan": after.record() if after is not None else None,
295             }
296         )
297     return out
298
299
300 def table_from_decisions(decisions: list[dict[str, Any]]) -> dict[str, dict[str, int]]:
301     table = {c: {"keep": 0, "adjust": 0, "cancel": 0} for c in CATEGORIES}
302     for dec in decisions:
303         key = {"keep": "keep", "cancel": "cancel"}.get(dec["kind"], "adjust")
304         table[dec["cat"]][key] += 1
305     return table
306
307
308 def hint_from_decisions(model: CellModel, decisions: list[dict[str, Any]]) -> list[int]:

```

```

309     """Map recorded decisions (`pid`, `kind`, `delta`) onto this model's variables, one per plan.
310
311     Plans without a recorded decision, or whose recorded action does not exist under the current limits,
312     fall back to `keep`; the result is a warm-start *hint* (CP-SAT repairs infeasible hints), never a
313     constraint, so incumbents from a run with different limits can still be reused."""
314     by_pid = {str(d["pid"]): (str(d["kind"]), int(d.get("delta", 0))) for d in decisions}
315     hint: list[int] = []
316     for i, plan in enumerate(model.plans):
317         kind, delta = by_pid.get(plan.pid, ("keep", 0))
318         keep = model.plan_vars[i][0]
319         hint.append(
320             next(
321                 (v for v in model.plan_vars[i] if (model.option_of(v).kind, model.option_of(v).delta) == (kind, delta))
322                 ↵ ,
323                 keep,
324             )
325         )
326     return hint
327
328 def hint_is_feasible(model: CellModel, chosen: list[int]) -> bool:
329     """True when the chosen variables (one per plan) violate no cell row, i.e. no conflict remains."""
330     chosen_set = set(chosen)
331     return all(sum(1 for v in row if v in chosen_set) <= 1 for row in model.rows)
332
333
334 def _cp_base(model: CellModel, strength: Strengthening | None) -> tuple[Any, list[Any]]:
335     from ortools.sat.python import cp_model
336
337     cp = cp_model.CpModel()
338     y = [cp.NewBoolVar(f"y{v}") for v in range(model.n_vars)]
339     for vars_of_plan in model.plan_vars:
340         cp.AddExactlyOne(y[v] for v in vars_of_plan)
341     for row in model.rows:
342         cp.AddAtMostOne(y[v] for v in row)
343     if strength is not None:
344         for v, us in strength.groups:
345             cp.AddAtMostOne([y[v], *(y[u] for u in us)])
346         for i, j in strength.forced:
347             cp.AddBoolOr([y[_cancel_of(model, i)], y[_cancel_of(model, j)]])
348     return cp, y
349
350
351 def _cp_params(solver: Any, seed: int, workers: int, time_limit: float) -> None:
352     solver.parameters.random_seed = int(seed)
353     solver.parameters.num_workers = int(workers)
354     solver.parameters.max_time_in_seconds = float(time_limit)
355     if workers >= 4:
356         solver.parameters.extra_subsolvers.append("core")
357
358
359 def level_value(terms: dict[int, int], chosen: list[int]) -> int:
360     """Value of one level for a solution given as the chosen variable per plan."""
361     picked = set(chosen)
362     return int(sum(c for v, c in terms.items() if v in picked))
363
364
365 def solve_lexicographic_cpsat(
366     model: CellModel,
367     levels: list[Level],
368     *,
369     seed: int,
370     workers: int = 8,
371     time_limit: float | list[float] = 600.0,
372     hint: list[int] | None = None,
373     strength: Strengthening | None = None,

```

```

374 ) -> dict[str, Any]:
375     """Minimise the levels in order with CP-SAT, fixing each level's value before the next.
376
377     Incumbent-preserving search (MDR-0009). A feasible ``hint`` becomes the incumbent ``x*``; before level
378     ``k`` the cut ``L_k <= L_k(x*)`` is added. Because ``x*`` is feasible and satisfies every equality fixed
379     so far, that cut removes no optimal solution, so every level's optimum is unchanged — but the level can
380     never return a value worse than the incumbent. When the solver only matches the incumbent, whichever of
381     the two solutions is lexicographically smaller on the *remaining* levels is carried forward. A level that
382     reaches its time limit continues from its best-found value (recorded with its bound), so the returned
383     vector is always feasible; ``all_optimal`` tells whether every level was proven."""
384     from ortools.sat.python import cp_model
385
386     t0 = time.monotonic()
387     if isinstance(time_limit, int | float):
388         limits = [float(time_limit)] * len(levels)
389     else:
390         limits = [float(x) for x in time_limit]
391     limits += [limits[-1]] * (len(levels) - len(limits))
392     cp, y = _cp_base(model, strength)
393     incumbent: list[int] | None = list(hint) if hint is not None and hint_is_feasible(model, hint) else None
394     if hint is not None:
395         hinted = set(hint)
396         for v in range(model.n_vars):
397             cp.AddHint(y[v], v in hinted)
398     per_level: list[dict[str, Any]] = []
399     values: list[int] = []
400     chosen: list[int] | None = list(incumbent) if incumbent is not None else None
401     all_optimal = True
402     for k, ((name, terms), limit) in enumerate(zip(levels, limits)):
403         expr = cp_model.LinearExpr.WeightedSum([y[v] for v in terms], [c for c in terms.values()])
404         cut = level_value(terms, incumbent) if incumbent is not None else None
405         if cut is not None:
406             cp.Add(expr <= cut) # valid: the incumbent is feasible and meets every equality fixed so far
407             cp.Minimize(expr)
408             solver = cp_model.CpSolver()
409             _cp_params(solver, seed, workers, limit)
410             t1 = time.monotonic()
411             status = solver.Solve(cp)
412             name_status = solver.StatusName(status)
413             entry: dict[str, Any] = {"level": name, "status": name_status, "seconds": round(time.monotonic() - t1, 3)}
414             if status in (cp_model.OPTIMAL, cp_model.FEASIBLE):
415                 value = round(solver.ObjectiveValue())
416                 found = [next(v for v in vars_of_plan if solver.Value(y[v])) for vars_of_plan in model.plan_vars]
417                 entry["bound"] = float(solver.BestObjectiveBound())
418             elif incumbent is not None and cut is not None:
419                 # no solution recovered before the limit: fall back on the incumbent, which attains `cut`
420                 value, found, entry["status"], entry["bound"] = cut, list(incumbent), "INCUMBENT", 0.0
421                 name_status = "INCUMBENT"
422             else:
423                 per_level.append(entry)
424                 all_optimal = False
425                 break
426             if incumbent is not None and cut == value:
427                 # equally good here: keep whichever is lexicographically better on the levels still to come
428                 tail = levels[k + 1 :]
429                 if evaluate(model, incumbent, tail) < evaluate(model, found, tail):
430                     found = list(incumbent)
431             incumbent = chosen = found
432             values.append(value)
433             entry["value"] = value
434             entry["cut"] = cut
435             per_level.append(entry)
436             all_optimal = all_optimal and status == cp_model.OPTIMAL
437             cp.Add(expr == value)
438             cp.ClearHints()
439             picked = set(incumbent)

```

```

440         for v in range(model.n_vars):
441             cp.AddHint(y[v], v in picked)
442     return {
443         "solver": "cp-sat",
444         "levels": per_level,
445         "vector": values,
446         "chosen": chosen,
447         "all_optimal": all_optimal,
448         "seconds": round(time.monotonic() - t0, 3),
449         "n_vars": model.n_vars,
450         "n_rows": len(model.rows),
451     }
452
453
454 def solve_weighted_cpSAT(
455     model: CellModel,
456     levels: list[Level],
457     *,
458     seed: int,
459     workers: int = 8,
460     time_limit: float = 600.0,
461     hint: list[int] | None = None,
462     strength: Strengthening | None = None,
463 ) -> dict[str, Any]:
464     """Single CP-SAT solve of the separated-weight scalarisation (should reproduce the lexicographic vector)."""
465     from ortools.sat.python import cp_model
466
467     t0 = time.monotonic()
468     weights = separated_weights(level_upper_bounds(model, levels))
469     terms: dict[int, int] = {}
470     for w, (_, lv) in zip(weights, levels):
471         for v, c in lv.items():
472             terms[v] = terms.get(v, 0) + w * c
473     cp, y = _cp_base(model, strength)
474     if hint is not None:
475         hinted = set(hint)
476         for v in range(model.n_vars):
477             cp.AddHint(y[v], v in hinted)
478     cp.Minimize(cp_model.LinearExpr.WeightedSum([y[v] for v in terms], list(terms.values())))
479     solver = cp_model.CpSolver()
480     _cp_params(solver, seed, workers, time_limit)
481     status = solver.Solve(cp)
482     chosen = None
483     if status in (cp_model.OPTIMAL, cp_model.FEASIBLE):
484         chosen = [next(v for v in vars_of_plan if solver.Value(y[v])) for vars_of_plan in model.plan_vars]
485     return {
486         "solver": "cp-sat-weighted",
487         "status": solver.StatusName(status),
488         "weights": weights,
489         "objective": float(solver.ObjectiveValue()) if chosen else None,
490         "bound": float(solver.BestObjectiveBound()) if chosen else None,
491         "vector": evaluate(model, chosen, levels) if chosen else None,
492         "chosen": chosen,
493         "seconds": round(time.monotonic() - t0, 3),
494     }
495
496
497 def _highs_base(
498     model: CellModel,
499     *,
500     integer: bool,
501     time_limit: float,
502     seed: int,
503     threads: int,
504     strength: Strengthening | None = None,
505 ) -> Any:

```

```

506 import highspy
507 import numpy as np
508
509 h = highspy.Highs()
510 h.setOptionValue("output_flag", False)
511 h.setOptionValue("log_to_console", False)
512 h.setOptionValue("time_limit", float(time_limit))
513 h.setOptionValue("random_seed", int(seed))
514 h.setOptionValue("threads", highs_threads(threads))
515 h.setOptionValue("mip_rel_gap", 0.0)
516 h.setOptionValue("mip_abs_gap", 0.0)
517 n = model.n_vars
518 h.addVars(n, np.zeros(n), np.ones(n))
519 if integer:
520     h.changeColsIntegrality(n, np.arange(n, dtype=np.int32), np.array([highspy.HighsVarType.kInteger] * n))
521 starts, index, lower, upper = [], [], [], []
522 nnz = 0
523 for vars_of_plan in model.plan_vars:
524     starts.append(nnz)
525     index.extend(vars_of_plan)
526     nnz += len(vars_of_plan)
527     lower.append(1.0)
528     upper.append(1.0)
529 for row in model.rows:
530     starts.append(nnz)
531     index.extend(row)
532     nnz += len(row)
533     lower.append(-highspy.kHighsInf)
534     upper.append(1.0)
535 if strength is not None:
536     for v, us in strength.groups:
537         starts.append(nnz)
538         index.append(v)
539         index.extend(us)
540         nnz += len(us) + 1
541         lower.append(-highspy.kHighsInf)
542         upper.append(1.0)
543     for i, j in strength.forced:
544         starts.append(nnz)
545         index.extend((_cancel_of(model, i), _cancel_of(model, j)))
546         nnz += 2
547         lower.append(1.0)
548         upper.append(highspy.kHighsInf)
549 h.addRows(
550     len(starts),
551     np.array(lower),
552     np.array(upper),
553     nnz,
554     np.array(starts, dtype=np.int32),
555     np.array(index, dtype=np.int32),
556     np.ones(nnz),
557 )
558 return h
559
560
561 def solve_lexicographic_highs(
562     model: CellModel,
563     levels: list[Level],
564     *,
565     seed: int,
566     threads: int = 8,
567     time_limit: float = 600.0,
568     reference: list[int] | None = None,
569     hint: list[int] | None = None,
570     strength: Strengthening | None = None,
571     lp_time_limit: float | None = None,

```

```

572 ) -> dict[str, Any]:
573     """Independent cross-check in HiGHS: the same 0-1 model, level by level.
574
575     For every level the LP relaxation (previous levels fixed to the reference values, i.e. the CP-SAT vector
576     when given) yields a lower bound; the MILP is then run with a time limit. When the MILP proves optimality
577     its value must equal the reference; otherwise its dual bound and incumbent are recorded. The next level is
578     fixed to the reference value (or the MILP optimum when no reference is given)."""
579     import highspy
580     import numpy as np
581
582     t0 = time.monotonic()
583     n = model.n_vars
584     per_level: list[dict[str, Any]] = []
585     values: list[int] = []
586     chosen: list[int] | None = None
587     all_optimal = True
588     agree = True
589     idx_all = np.arange(n, dtype=np.int32)
590     fixed: list[tuple[np.ndarray, np.ndarray, float]] = [] # rows fixing the previous levels
591     lp_limit = float(lp_time_limit) if lp_time_limit is not None else float(time_limit)
592     for k, (name, terms) in enumerate(levels):
593         cost = np.zeros(n)
594         for v, c in terms.items():
595             cost[v] = c
596         entry: dict[str, Any] = {"level": name}
597         # A fresh solver per level and per relaxation: HiGHS' run clock accumulates over successive run()
598         # calls on one object, so re-using it would let the first level's time limit starve the others.
599         lp = _highs_base(model, integer=False, time_limit=lp_limit, seed=seed, threads=threads, strength=strength)
600         mip = _highs_base(model, integer=True, time_limit=time_limit, seed=seed, threads=threads, strength=strength)
601         for h in (lp, mip):
602             for f_idx, f_val, f_value in fixed:
603                 h.addRow(f_value, f_value, len(f_idx), f_idx, f_val)
604             if hint is not None:
605                 hinted = set(hint)
606                 start = highspy.HighsSolution()
607                 start.col_value = [1.0 if v in hinted else 0.0 for v in range(n)]
608                 start.value_valid = True
609                 mip.setSolution(start)
610             for tag, h in (("lp", lp), ("mip", mip)):
611                 h.changeColsCost(n, idx_all, cost)
612                 t1 = time.monotonic()
613                 h.run()
614                 status = h.getModelStatus()
615                 entry[f"{tag}_status"] = h.modelStatusToString(status)
616                 entry[f"{tag}_seconds"] = round(time.monotonic() - t1, 3)
617                 info = h.getInfo()
618                 if status == highspy.HighsModelStatus.kOptimal:
619                     entry[f"{tag}_value"] = float(info.objective_function_value)
620                 elif tag == "mip":
621                     entry["mip_value"] = float(info.objective_function_value)
622                 if tag == "mip":
623                     entry["mip_bound"] = float(info.mip_dual_bound)
624             proven = entry.get("mip_status") == "Optimal"
625             all_optimal = all_optimal and proven
626             if proven:
627                 value = round(entry["mip_value"])
628                 sol = np.array(mip.getSolution().col_value)
629                 chosen = [int(max(vars_of_plan, key=lambda v: sol[v])) for vars_of_plan in model.plan_vars]
630                 if reference is not None and k < len(reference) and value != reference[k]:
631                     agree = False
632                     entry["disagrees_with_reference"] = reference[k]
633             elif reference is not None and k < len(reference):
634                 value = int(reference[k])
635             else:
636                 per_level.append(entry)
637             break

```

```

638         if reference is not None and k < len(reference):
639             value = int(reference[k])
640             entry["fixed_value"] = value
641             values.append(value)
642             per_level.append(entry)
643             idx = np.array(list(terms), dtype=np.int32)
644             val = np.array([float(c) for c in terms.values()])
645             fixed.append((idx, val, float(value)))
646     return {
647         "solver": "highs",
648         "levels": per_level,
649         "vector": values,
650         "chosen": chosen,
651         "all_optimal": all_optimal,
652         "agrees_with_reference": agree,
653         "seconds": round(time.monotonic() - t0, 3),
654         "version": str(highspy.Highs().version()),
655     }

```

pipelines/d/stages.py

```

1  """Problem D stages: validate, detect (Q1), resolve (Q2), pack (Q3), resolve_interval (Q4), bench,
2  sensitivity, results; figures and tables live in ``pipelines.d.report`` (imported here so they register).
3
4  Conventions (docs/ENGINEERING_STANDARD.md): every stage is ``@stage(name, deps=...)`` returning
5  JSON-serialisable metrics; outputs go only to ``ctx.out(...)``; paper numbers via ``ctx.number``;
6  randomness only via ``ctx.seed_everything``; every optimisation result is re-checked by
7  ``pipelines.d.validators`` before it is written.
8  """
9
10 from __future__ import annotations
11
12 import json
13 import math
14 import time
15 from pathlib import Path
16 from typing import Any
17
18 from forge.context import StageContext
19 from forge.runner import stage
20 from forge.xlsx import write_result
21 from pipelines.common.validation import run_validation
22 from pipelines.d import validators
23 from pipelines.d.contracts import INPUT_CONTRACTS
24 from pipelines.d.detect import conflict_records, detect_bandsweep, detect_pairwise, graph_stats
25 from pipelines.d.pack import (
26     C_TEMPLATE,
27     candidate_placements,
28     cell_rows,
29     greedy_pack,
30     highs_pack,
31     occupancy_grid,
32     plans_from_choice,
33     repack_existing,
34     solve_pack_cpsat,
35     template_span,
36 )
37 from pipelines.d.plans import CATEGORIES, Plan, horizon, load_plans, plan_from_record
38 from pipelines.d.resolve import (
39     Limits,
40     build_cell_model,
41     canonical_levels,
42     decisions_from_choice,

```

```

43     enumerate_options,
44     evaluate,
45     hint_from_decisions,
46     hint_is_feasible,
47     scheme_levels,
48     solve_lexicographic_cpsat,
49     solve_lexicographic_highs,
50     solve_weighted_cpsat,
51     strengthen,
52     table_from_decisions,
53 )
54 from pipelines.d.synth import exhaustive_lexicographic, synthetic_instance, tiny_instance
55
56 PARQUET = "附件1__Sheet1.parquet"
57 Q2_LIMITS = Limits(fmax=10, tmax=5)
58 Q4_LIMITS = Limits(fmax=10, tmax=5, gmax=10, gap_categories=("C",))
59 PASS = "通过"
60 FAIL = "未通过"
61 # incumbents of an earlier verified run, used only as warm starts (configs/hints/*.json carry their provenance)
62 Q2_HINT = "configs/hints/q2_decisions.json"
63 Q4_HINT = "configs/hints/q4_decisions.json"
64
65
66 def _vector_text(vec: list[int] | None) -> str:
67     return "(" + ", ".join(str(v) for v in vec) + ")" if vec else "-"
68
69
70 def _load_plans(ctx: StageContext) -> list[Plan]:
71     return load_plans(ctx.dep("ingest") / "data" / PARQUET)
72
73
74 def _records(plans: list[Plan]) -> list[dict[str, Any]]:
75     return [p.record() for p in plans]
76
77
78 def _plans(records: list[dict[str, Any]]) -> list[Plan]:
79     return [plan_from_record(r) for r in records]
80
81
82 @stage("validate", deps=("ingest",), description="Validate the 150 frequency-use plans against their contract")
83 def validate(ctx: StageContext) -> dict[str, Any]:
84     return run_validation(ctx, INPUT_CONTRACTS)
85
86
87 # ----- Q1
88 @stage("detect", deps=("ingest", "validate"), description="Q1: exact conflict detection with two algorithms")
89 def detect(ctx: StageContext) -> dict[str, Any]:
90     plans = _load_plans(ctx)
91     t0 = time.perf_counter()
92     pairs = detect_pairwise(plans)
93     t_pairwise = time.perf_counter() - t0
94     t0 = time.perf_counter()
95     sweep = detect_bandsweep(plans)
96     t_sweep = time.perf_counter() - t0
97     if pairs != sweep:
98         raise RuntimeError(f"detectors disagree: pairwise {len(pairs)} vs sweep {len(sweep)}")
99     records = conflict_records(plans, pairs)
100     stats = graph_stats(plans, pairs)
101     use_pairs = sum(r["use_pairs"] for r in records)
102     horizon_t = horizon(plans)
103     grid = occupancy_grid(plans, horizon_t)
104     import numpy as np
105
106     counts = np.zeros(grid.shape, dtype=int)
107     for p in plans:
108         for start, end in p.uses():

```

```

109         counts[p.f : p.f_end, start:end] += 1
110     validation = validators.validate_detection(plans, [(r["id1"], r["id2"]) for r in records])
111     if not validation["ok"]:
112         raise RuntimeError(f"independent detection check failed: {validation}")
113     multiplicity = {
114         str(k): int(sum(1 for r in records if r["use_pairs"] == k)) for k in sorted({r["use_pairs"] for r in records})
115     }
116     stats.update(
117         {
118             "use_pairs": int(use_pairs),
119             "use_pair_multiplicity": multiplicity,
120             "horizon": int(horizon_t),
121             "cells_total": int(counts.size),
122             "cells_occupied": int((counts >= 1).sum()),
123             "cells_overlap": int((counts >= 2).sum()),
124             "cells_max_multiplicity": int(counts.max()),
125             "cells_by_cat": {c: int(sum(p.n_cells for p in plans if p.cat == c)) for c in CATEGORIES},
126             "seconds_pairwise": t_pairwise,
127             "seconds_sweep": t_sweep,
128             "detectors_agree": True,
129             "pair_multiplicity_max": max((r["use_pairs"] for r in records), default=0),
130         }
131     )
132     ctx.write_json("plans.json", _records(plans))
133     ctx.write_json("conflicts.json", records)
134     ctx.write_json("stats.json", stats)
135     ctx.write_json("validation_report.json", validation)
136     ctx.number("QonePlans", len(plans))
137     ctx.number("QoneConflictPairs", stats["pairs"])
138     for key, value in stats["by_pair"].items():
139         ctx.number(f"QoneConflictPairs{key}", value)
140     ctx.number("QonePlansInConflict", stats["plans_involved"])
141     for c in CATEGORIES:
142         ctx.number(f"QonePlansInConflict{c}", stats["plans_involved_by_cat"][c])
143     ctx.number("QoneComponents", stats["components"])
144     ctx.number("QoneLargestComponent", stats["largest_component"])
145     ctx.number("QoneMaxDegree", stats["max_degree"])
146     ctx.number("QoneMeanDegree", stats["mean_degree"], ".2f")
147     ctx.number("QoneDensity", stats["density"], ".4f")
148     ctx.number("QoneUsePairs", use_pairs)
149     ctx.number("QoneHorizon", horizon_t)
150     ctx.number("QoneCellsTotal", stats["cells_total"])
151     ctx.number("QoneCellsOccupied", stats["cells_occupied"])
152     ctx.number("QoneCellsOverlap", stats["cells_overlap"])
153     ctx.number("QoneOccupancyPercent", 100.0 * stats["cells_occupied"] / stats["cells_total"], ".1f")
154     ctx.number("QonePairwiseMs", 1000 * t_pairwise, ".0f")
155     ctx.number("QoneSweepMs", 1000 * t_sweep, ".0f")
156     ctx.number("QoneValidator", PASS)
157     ctx.number("QoneTopDegreeId", stats["top_degree"][0]["pid"] if stats["top_degree"] else "-")
158     ctx.log.info("detect.done", pairs=stats["pairs"], by_pair=stats["by_pair"], seconds=t_pairwise)
159     return {"pairs": stats["pairs"], "by_pair": stats["by_pair"], "use_pairs": use_pairs, "validator_ok": True}
160
161
162 # ----- Q2 / Q4
163 def _shift_stats(decisions: list[dict[str, Any]]) -> dict[str, Any]:
164     freq = [abs(d["delta"]) for d in decisions if d["kind"] == "freq"]
165     tim = [abs(d["delta"]) for d in decisions if d["kind"] == "time"]
166     gap = [abs(d["delta"]) for d in decisions if d["kind"] == "gap"]
167
168     def stat(xs: list[int]) -> dict[str, Any]:
169         return {"count": len(xs), "mean": (sum(xs) / len(xs)) if xs else 0.0, "max": max(xs, default=0), "sum": sum(xs)}
170
171     return {
172         "freq": stat(freq),
173         "time": stat(tim),

```

```

174         "gap": stat(gap),
175         "cancel": sum(1 for d in decisions if d["kind"] == "cancel"),
176     }
177
178
179 def _read_decisions(path: Path) -> list[dict[str, Any]]:
180     """Decisions from a stage output (a list) or an in-repo incumbent file ({'provenance', 'decisions'})."""
181     data = json.loads(path.read_text(encoding="utf-8"))
182     return list(data.get("decisions", [])) if isinstance(data, dict) else list(data)
183
184
185 def _load_hint(ctx: StageContext, model: Any, source: str | None) -> tuple[list[int] | None, dict[str, Any]]:
186     """Warm start (MDR-0004 addendum): ``source`` is a path relative to the run directory or to the repository.
187
188     Returns the hint (one variable per plan) and a small provenance record; a missing file means no hint."""
189     if not source:
190         return None, {"source": None}
191     for path in (ctx.run_dir / source, ctx.repo / source):
192         if path.exists():
193             decisions = _read_decisions(path)
194             hint = hint_from_decisions(model, decisions)
195             info = {
196                 "source": source,
197                 "path": str(path),
198                 "feasible": hint_is_feasible(model, hint),
199                 "vector": evaluate(model, hint, canonical_levels(model)),
200             }
201             ctx.log.info("hint.loaded", **info)
202             return hint, info
203     ctx.log.warn("hint.missing", source=source)
204     return None, {"source": source, "path": None}
205
206
207 def _run_resolution(
208     ctx: StageContext, plans: list[Plan], limits: Limits, prefix: str, *, default_hint: str | None = None
209 ) -> dict[str, Any]:
210     """Shared Q2/Q4 pipeline: model, lexicographic CP-SAT, cross-checks, alternatives, validation, outputs."""
211     workers = int(ctx.param("workers", 8))
212     level_limits = ctx.param("level_time_limits", None) or float(ctx.param("time_limit", 360))
213     highs_limit = float(ctx.param("highs_time_limit", 120))
214     weighted_limit = float(ctx.param("weighted_time_limit", 180))
215     alt_limit = float(ctx.param("alt_time_limit", 60))
216     seed = ctx.seed_everything(prefix)
217     options = [enumerate_options(p, limits) for p in plans]
218     t0 = time.perf_counter()
219     model = build_cell_model(plans, options)
220     strength = strengthen(model) if bool(ctx.param("strengthen", True)) else None
221     build_seconds = time.perf_counter() - t0
222     levels = canonical_levels(model)
223     ctx.log.info(
224         "model.built",
225         vars=model.n_vars,
226         rows=len(model.rows),
227         rows_raw=model.n_rows_raw,
228         groups=len(strength.groups) if strength else 0,
229         forced=len(strength.forced) if strength else 0,
230         seconds=build_seconds,
231     )
232
233     hint, hint_info = _load_hint(ctx, model, ctx.param("hint_from", default_hint))
234     # The separated-weight scalarisation runs first (MDR-0009): warm-started from the *file* incumbent it is an
235     # independent second search, and its solution seeds the lexicographic pass, so the reported vector can
236     # never be worse than either route. (Previously it was hinted by the lexicographic solution — circular.)
237     weighted = solve_weighted_cpsat(
238         model, levels, seed=seed, workers=workers, time_limit=weighted_limit, hint=hint, strength=strength
239     )

```

```

240 start = hint
241 if weighted["chosen"] is not None and hint_is_feasible(model, weighted["chosen"]):
242     if hint is None or list(weighted["vector"]) < evaluate(model, hint, levels):
243         start = weighted["chosen"]
244 hint_info["start_from"] = "file" if start is hint else "weighted"
245 hint_info["start_vector"] = evaluate(model, start, levels) if start is not None else None
246 primary = solve_lexicographic_cpsat(
247     model, levels, seed=seed, workers=workers, time_limit=level_limits, strength=strength, hint=start
248 )
249 ctx.log.info("cpsat.primary", vector=primary["vector"], levels=primary["levels"], seconds=primary["seconds"])
250 if primary["chosen"] is None or len(primary["vector"]) != len(levels):
251     raise RuntimeError(f"CP-SAT found no complete solution: {primary['levels']}")
252 if bool(ctx.param("require_optimal", False)) and not primary["all_optimal"]:
253     raise RuntimeError(f"CP-SAT did not prove optimality on every level: {primary['levels']}")
254 deterministic: bool | None = None
255 if bool(ctx.param("repeat", False)):
256     repeat = solve_lexicographic_cpsat(
257         model, levels, seed=seed, workers=workers, time_limit=level_limits, strength=strength
258     )
259     deterministic = repeat["chosen"] == primary["chosen"] and repeat["vector"] == primary["vector"]
260 weighted_agree = weighted["vector"] == primary["vector"]
261 ctx.log.info(
262     "cpsat.weighted",
263     vector=weighted["vector"],
264     status=weighted["status"],
265     agree=weighted_agree,
266     start_from=hint_info["start_from"],
267 )
268 highs = solve_lexicographic_highs(
269     model,
270     levels,
271     seed=seed,
272     threads=workers,
273     time_limit=highs_limit,
274     reference=primary["vector"],
275     hint=primary["chosen"],
276     strength=strength,
277     lp_time_limit=float(ctx.param("lp_time_limit", highs_limit)),
278 )
279 highs_agree = highs["agrees_with_reference"]
280 ctx.log.info("highs.done", levels=highs["levels"], all_optimal=highs["all_optimal"], seconds=highs["seconds"])
281
282 decisions = decisions_from_choice(model, primary["chosen"])
283 table = table_from_decisions(decisions)
284 validation = validators.validate_resolution(
285     plans,
286     decisions,
287     fmax=limits.fmax,
288     tmax=limits.tmax,
289     gmax=limits.gmax,
290     gap_categories=limits.gap_categories,
291     reported_table=table,
292     reported_vector=primary["vector"],
293 )
294 if not validation["ok"]:
295     raise RuntimeError(f"independent validation failed: {validation['errors']}")
296 survivors = [plan_from_record(d["plan"]) for d in decisions if d["plan"] is not None]
297
298 alternatives: dict[str, Any] = {}
299 for name in ("P", "T", "S", "W"):
300     if name == "P":
301         res, chosen = primary, primary["chosen"]
302     else:
303         res = solve_lexicographic_cpsat(
304             model,
305             scheme_levels(name, model),

```

```

306         seed=seed,
307         workers=workers,
308         time_limit=alt_limit,
309         hint=primary["chosen"],
310         strength=strength,
311     )
312     chosen = res["chosen"]
313     alt_decisions = decisions_from_choice(model, chosen) if chosen else []
314     alt_valid = (
315         validators.validate_resolution(
316             plans,
317             alt_decisions,
318             fmax=limits.fmax,
319             tmax=limits.tmax,
320             gmax=limits.gmax,
321             gap_categories=limits.gap_categories,
322         )
323         if chosen
324         else {"ok": False}
325     )
326     alternatives[name] = {
327         "scheme_vector": res["vector"],
328         "canonical_vector": evaluate(model, chosen, levels) if chosen else None,
329         "table": table_from_decisions(alt_decisions) if chosen else None,
330         "all_optimal": res["all_optimal"],
331         "levels": res["levels"],
332         "seconds": res["seconds"],
333         "validator_ok": bool(alt_valid["ok"]),
334         "shifts": _shift_stats(alt_decisions),
335     }
336     ctx.log.info(
337         "alternative", scheme=name, vector=alternatives[name]["canonical_vector"], optimal=res["all_optimal"]
338     )
339
340 shifts = _shift_stats(decisions)
341 report = {
342     "limits": limits.__dict__,
343     "model": {
344         "vars": model.n_vars,
345         "rows": len(model.rows),
346         "rows_raw": model.n_rows_raw,
347         "cells": model.n_cells,
348         "memberships": model.memberships,
349         "horizon": model.horizon,
350         "options_per_plan": {
351             c: max(len(options[i]) for i, p in enumerate(plans) if p.cat == c) for c in CATEGORIES
352         },
353         "build_seconds": build_seconds,
354         "strengthening_groups": len(strength.groups) if strength else 0,
355         "forced_cancel_pairs": len(strength.forced) if strength else 0,
356         "interacting_pairs": strength.interacting_pairs if strength else None,
357     },
358     "primary": {k: v for k, v in primary.items() if k != "chosen"},
359     "hint": hint_info,
360     "all_optimal": primary["all_optimal"],
361     "repeat_deterministic": deterministic,
362     "weighted": {k: v for k, v in weighted.items() if k != "chosen"},
363     "weighted_agree": weighted_agree,
364     "highs": {k: v for k, v in highs.items() if k != "chosen"},
365     "highs_agree": highs_agree,
366     "vector": primary["vector"],
367     "table": table,
368     "shifts": shifts,
369     "amplitude_normalised": primary["vector"][-1] / 10.0,
370     "horizon_after": horizon(survivors),
371 }

```

```

372 ctx.write_json("decisions.json", decisions)
373 ctx.write_json("resolved_plans.json", _records(survivors))
374 ctx.write_json("solver_report.json", report)
375 ctx.write_json("alternatives.json", alternatives)
376 ctx.write_json("validation_report.json", validation)
377
378 for c in CATEGORIES:
379     ctx.number(f"{prefix}Keep{c}", table[c]["keep"])
380     ctx.number(f"{prefix}Adjust{c}", table[c]["adjust"])
381     ctx.number(f"{prefix}Cancel{c}", table[c]["cancel"])
382     ctx.number(f"{prefix}KeepTotal", sum(table[c]["keep"] for c in CATEGORIES))
383     ctx.number(f"{prefix}AdjustTotal", sum(table[c]["adjust"] for c in CATEGORIES))
384     ctx.number(f"{prefix}CancelTotal", sum(table[c]["cancel"] for c in CATEGORIES))
385     ctx.number(f"{prefix}FreqShifted", shifts["freq"]["count"])
386     ctx.number(f"{prefix}TimeShifted", shifts["time"]["count"])
387     ctx.number(f"{prefix}GapChanged", shifts["gap"]["count"])
388     ctx.number(f"{prefix}MeanAbsFreqShift", shifts["freq"]["mean"], ".2f")
389     ctx.number(f"{prefix}MaxAbsFreqShift", shifts["freq"]["max"])
390     ctx.number(f"{prefix}SumAbsFreqShift", shifts["freq"]["sum"])
391     ctx.number(f"{prefix}MeanAbsTimeShift", shifts["time"]["mean"], ".2f")
392     ctx.number(f"{prefix}MaxAbsTimeShift", shifts["time"]["max"])
393     ctx.number(f"{prefix}SumAbsTimeShift", shifts["time"]["sum"])
394     ctx.number(f"{prefix}MeanAbsGapChange", shifts["gap"]["mean"], ".2f")
395     ctx.number(f"{prefix}MaxAbsGapChange", shifts["gap"]["max"])
396     ctx.number(f"{prefix}Amplitude", primary["vector"][-1] / 10.0, ".1f")
397     ctx.number(f"{prefix}AmplitudeScaled", primary["vector"][-1])
398     ctx.number(f"{prefix}Vars", model.n_vars)
399     ctx.number(f"{prefix}Rows", len(model.rows))
400     ctx.number(f"{prefix}RowsRaw", model.n_rows_raw)
401     ctx.number(f"{prefix}Memberships", model.memberships)
402     ctx.number(f"{prefix}Cells", model.n_cells)
403     ctx.number(f"{prefix}CpsatSeconds", primary["seconds"], ".1f")
404     ctx.number(f"{prefix}HighsSeconds", highs["seconds"], ".1f")
405     ctx.number(f"{prefix}WeightedSeconds", weighted["seconds"], ".1f")
406     ctx.number(f"{prefix}HighsAgree", PASS if highs_agree else FAIL)
407     ctx.number(f"{prefix}HighsAllOptimal", PASS if highs["all_optimal"] else FAIL)
408     ctx.number(f"{prefix}WeightedAgree", PASS if weighted_agree else FAIL)
409     ctx.number(f"{prefix}AllOptimal", PASS if primary["all_optimal"] else FAIL)
410     ctx.number(f"{prefix}Deterministic", "未检验" if deterministic is None else (PASS if deterministic else FAIL))
411     ctx.number(f"{prefix}ForcedCancelPairs", len(strength.forced) if strength else 0)
412     ctx.number(f"{prefix}InteractingPairs", strength.interacting_pairs if strength else 0)
413     ctx.number(f"{prefix}Groups", len(strength.groups) if strength else 0)
414     for entry in primary["levels"]:
415         key = entry["level"].capitalize()
416         ctx.number(f"{prefix}Value{key}", entry.get("value", "-"))
417         ctx.number(f"{prefix}Bound{key}", math.ceil(entry.get("bound", 0) - 1e-6) if "bound" in entry else "-")
418         ctx.number(f"{prefix}Status{key}", entry.get("status", "-"))
419     ctx.number(f"{prefix}Validator", PASS if validation["ok"] else FAIL)
420     ctx.number(f"{prefix}HorizonAfter", report["horizon_after"])
421     ctx.number(f"{prefix}HintUsed", "是" if hint is not None else "否")
422     ctx.number(f"{prefix}HintFeasible", PASS if hint_info.get("feasible") else FAIL)
423     ctx.number(f"{prefix}HintVector", _vector_text(hint_info.get("vector")))
424     ctx.number(f"{prefix}Vector", _vector_text(primary["vector"]))
425     ctx.number(f"{prefix}Levels", len(levels))
426     for entry in highs["levels"]:
427         if "lp_value" in entry:
428             ctx.number(f"{prefix}LpBound{entry['level'].capitalize()}", entry["lp_value"], ".2f")
429     for name, alt in alternatives.items():
430         vec = alt["canonical_vector"]
431         if vec is not None:
432             ctx.number(f"{prefix}Scheme{name}AdjustTotal", sum(vec[3:6]))
433             ctx.number(f"{prefix}Scheme{name}CancelTotal", sum(vec[0:3]))
434             ctx.number(f"{prefix}Scheme{name}Amplitude", vec[6] / 10.0, ".1f")
435             ctx.number(f"{prefix}Scheme{name}AdjustA", vec[3])
436     return {
437         "vector": primary["vector"],

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438         "table": table,
439         "all_optimal": primary["all_optimal"],
440         "deterministic": deterministic,
441         "weighted_agree": weighted_agree,
442         "highs_agree": highs_agree,
443         "validator_ok": validation["ok"],
444         "cpsat_seconds": primary["seconds"],
445         "highs_seconds": highs["seconds"],
446         "horizon_after": report["horizon_after"],
447     }
448
449
450 @stage("resolve", deps=("detect",), description="Q2: lexicographic conflict resolution (CP-SAT, HiGHS cross-check)")
451 def resolve(ctx: StageContext) -> dict[str, Any]:
452     plans = _plans(ctx.dep_data("detect", "plans.json"))
453     limits = Limits(fmax=int(ctx.param("fmax", Q2_LIMITS.fmax)), tmax=int(ctx.param("tmax", Q2_LIMITS.tmax)))
454     return _run_resolution(ctx, plans, limits, "Qtwo", default_hint=Q2_HINT)
455
456
457 @stage(
458     "resolve_interval",
459     deps=("detect",),
460     description="Q4: resolution with C-class gap adjustments (|dg| <= 10) as an additional single-parameter action",
461 )
462 def resolve_interval(ctx: StageContext) -> dict[str, Any]:
463     plans = _plans(ctx.dep_data("detect", "plans.json"))
464     limits = Limits(
465         fmax=int(ctx.param("fmax", Q4_LIMITS.fmax)),
466         tmax=int(ctx.param("tmax", Q4_LIMITS.tmax)),
467         gmax=int(ctx.param("gmax", Q4_LIMITS.gmax)),
468         gap_categories=("C",),
469     )
470     return _run_resolution(ctx, plans, limits, "Qfour", default_hint=Q4_HINT)
471
472
473 @stage(
474     "compare",
475     deps=("resolve", "resolve_interval"),
476     description="Q4 vs Q2: lexicographic comparison of the two resolutions",
477 )
478 def compare(ctx: StageContext) -> dict[str, Any]:
479     q2 = ctx.dep_data("resolve", "solver_report.json")
480     q4 = ctx.dep_data("resolve_interval", "solver_report.json")
481     q2_vec, q4_vec = list(q2["vector"]), list(q4["vector"])
482     not_worse = q4_vec <= q2_vec # Q4's action set is a superset of Q2's (MDR-0006)
483     comparison = {
484         "q2_vector": q2_vec,
485         "q4_vector": q4_vec,
486         "lexicographically_not_worse": not_worse,
487         "cancel_total_delta": sum(q4_vec[0:3]) - sum(q2_vec[0:3]),
488         "adjust_total_delta": sum(q4_vec[3:6]) - sum(q2_vec[3:6]),
489         "amplitude_delta": (q4_vec[6] - q2_vec[6]) / 10.0,
490         "q2_table": q2["table"],
491         "q4_table": q4["table"],
492         "q2_horizon_after": q2["horizon_after"],
493         "q4_horizon_after": q4["horizon_after"],
494         "q4_gap_changed": q4["shifts"]["gap"]["count"],
495     }
496     ctx.write_json("comparison.json", comparison)
497     ctx.number("QfourLexNotWorse", PASS if not_worse else FAIL)
498     ctx.number("QfourCancelDelta", comparison["cancel_total_delta"])
499     ctx.number("QfourAdjustDelta", comparison["adjust_total_delta"])
500     ctx.number("QfourAmplitudeDelta", comparison["amplitude_delta"], ".1f")
501     ctx.number("QfourHorizonDelta", q4["horizon_after"] - q2["horizon_after"])
502     return {"lexicographically_not_worse": not_worse, "q2": q2_vec, "q4": q4_vec}
503

```

```

504 # ----- Q3
505
506 def _pack_once(
507     ctx: StageContext,
508     existing: list[Plan],
509     horizon_t: int,
510     *,
511     tag: str,
512     time_limit: float,
513     mip_limit: float,
514     lp_limit: float,
515 ) -> dict[str, Any]:
516     import numpy as np
517
518     workers = int(ctx.param("workers", 8))
519     seed = ctx.seed(f"pack-{tag}")
520     grid = occupancy_grid(existing, horizon_t)
521     free_cells = int((~grid).sum())
522     free_by_band = (-grid).sum(axis=1)
523     cells_per_plan = C_TEMPLATE["w"] * C_TEMPLATE["d"] * C_TEMPLATE["n"]
524     t0 = time.perf_counter()
525     cands = candidate_placements(grid, C_TEMPLATE)
526     rows = cell_rows(cands, C_TEMPLATE, horizon_t)
527     build_seconds = time.perf_counter() - t0
528     greedy = greedy_pack(cands, C_TEMPLATE, grid)
529     ctx.log.info(
530         "pack.model", tag=tag, candidates=len(cands), rows=len(rows), greedy=len(greedy), seconds=build_seconds
531     )
532     cp = solve_pack_cpsat(rows, len(cands), hint=greedy, seed=seed, workers=workers, time_limit=time_limit)
533     ctx.log.info(
534         "pack.cpsat", tag=tag, status=cp["status"], value=cp["value"], bound=cp["bound"], seconds=cp["seconds"]
535     )
536     lp = (
537         highs_pack(rows, len(cands), integer=False, time_limit=lp_limit, seed=seed, threads=workers)
538         if lp_limit > 0
539         else None
540     )
541     mip = (
542         highs_pack(rows, len(cands), integer=True, time_limit=mip_limit, seed=seed, threads=workers)
543         if mip_limit > 0
544         else None
545     )
546     bounds: dict[str, float] = {
547         "free_cells": math.floor(free_cells / cells_per_plan),
548         "per_band": math.floor(
549             sum(int(b) // (C_TEMPLATE["d"] * C_TEMPLATE["n"]) for b in free_by_band) / C_TEMPLATE["w"]
550         ),
551         "cpsat": math.floor(cp["bound"] + 1e-6) if cp["bound"] is not None else math.inf,
552     }
553     if lp is not None and lp["status"] == "Optimal":
554         bounds["lp"] = math.floor(lp["value"] + 1e-6)
555     if mip is not None and mip.get("bound") is not None and math.isfinite(mip["bound"]):
556         bounds["highs_mip"] = math.floor(mip["bound"] + 1e-6)
557     upper = int(min(bounds.values()))
558     best_name, best = "cpsat", cp
559     if mip is not None and mip.get("chosen") and len(mip["chosen"]) > cp["value"]:
560         best_name, best = "highs_mip", mip
561     new_plans = plans_from_choice(cands, best["chosen"], C_TEMPLATE)
562     validation = validators.validate_packing(existing, new_plans, horizon=horizon_t, template=C_TEMPLATE)
563     if not validation["ok"]:
564         raise RuntimeError(f"independent packing validation failed: {validation['errors']}")
565     value = len(new_plans)
566     return {
567         "tag": tag,
568         "horizon": int(horizon_t),
569         "existing": len(existing),

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570     "cells_total": int(grid.size),
571     "cells_occupied": int(grid.sum()),
572     "free_cells": free_cells,
573     "candidates": len(cands),
574     "rows": len(rows),
575     "build_seconds": build_seconds,
576     "greedy": len(greedy),
577     "cpsat": {k: v for k, v in cp.items() if k != "chosen"},
578     "lp": lp,
579     "highs_mip": {k: v for k, v in (mip or {}).items() if k != "chosen"} if mip else None,
580     "bounds": bounds,
581     "upper_bound": upper,
582     "value": value,
583     "gap": upper - value,
584     "optimal": upper == value,
585     "source": best_name,
586     "utilisation_before": float(grid.sum() / grid.size),
587     "utilisation_after": float((grid.sum() + value * cells_per_plan) / grid.size),
588     "new_plans": _records(new_plans),
589     "validation": validation,
590     "free_by_band": [int(b) for b in free_by_band],
591     "span": template_span(C_TEMPLATE),
592     "seed": seed,
593     "np_version": str(np.__version__),
594 }
595
596
597 @stage("pack", deps=("detect", "resolve"), description="Q3: maximum number of additional C-class plans (set packing)")
598 def pack(ctx: StageContext) -> dict[str, Any]:
599     existing = _plans(ctx.dep_data("resolve", "resolved_plans.json"))
600     original = _plans(ctx.dep_data("detect", "plans.json"))
601     horizon_t = horizon(existing)
602     result = _pack_once(
603         ctx,
604         existing,
605         horizon_t,
606         tag="main",
607         time_limit=float(ctx.param("time_limit", 900)),
608         mip_limit=float(ctx.param("mip_time_limit", 300)),
609         lp_limit=float(ctx.param("lp_time_limit", 300)),
610     )
611     ctx.write_json("new_plans.json", result["new_plans"])
612     ctx.write_json("validation_report.json", result["validation"])
613     alternatives: dict[str, Any] = {}
614     alt_horizon = horizon(original)
615     if bool(ctx.param("alt_horizon", True)) and alt_horizon != horizon_t:
616         alt = _pack_once(
617             ctx,
618             existing,
619             alt_horizon,
620             tag="original-horizon",
621             time_limit=float(ctx.param("alt_time_limit", 240)),
622             mip_limit=0,
623             lp_limit=float(ctx.param("lp_time_limit", 600)),
624         )
625         alternatives["original_horizon"] = {k: v for k, v in alt.items() if k not in {"new_plans", "free_by_band"}}
626     # MDR-0010 interpretation B: the existing plans may themselves be shifted by an unrestricted amount.
627     # A first-fit-decreasing re-placement is one feasible layout, so the number it admits is a valid lower
628     # bound on that interpretation's optimum; the free-cell capacity bound is a valid upper bound for both
629     # interpretations, because shifting never changes how many cells a plan occupies.
630     if bool(ctx.param("repack", True)):
631         repacked, _ = repack_existing(existing, horizon_t)
632         moved = sum(1 for a, b in zip(sorted(existing, key=lambda p: p.pid), repacked) if (a.f, a.s) != (b.f, b.s))
633         rep = _pack_once(
634             ctx,
635             repacked,

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636         horizon_t,
637         tag="repack",
638         time_limit=float(ctx.param("repack_time_limit", 300)),
639         mip_limit=0,
640         lp_limit=float(ctx.param("repack_lp_time_limit", 120)),
641     )
642     params_kept = all(
643         (a.w, a.d, a.g, a.n) == (b.w, b.d, b.g, b.n)
644         for a, b in zip(sorted(existing, key=lambda p: p.pid), repacked)
645     )
646     contained = all(0 <= p.f and p.f_end <= 100 and p.s >= 0 and p.end <= horizon_t for p in repacked)
647     if not params_kept or not contained or len(repacked) != len(existing):
648         raise RuntimeError("repack changed a plan parameter, dropped a plan, or left the resource region")
649     alternatives["repack"] = {
650         k: v for k, v in rep.items() if k not in {"new_plans", "free_by_band", "validation"}
651     } | {"existing_moved": moved, "params_kept": params_kept}
652     ctx.write_json("repack_plans.json", {"existing": _records(repacked), "new": rep["new_plans"]})
653     ctx.log.info("pack.repack", moved=moved, new_plans=rep["value"], bound=rep["bounds"]["free_cells"])
654     ctx.number("QthreeAltRepackNewPlans", rep["value"])
655     ctx.number("QthreeAltRepackBound", rep["bounds"]["free_cells"])
656     ctx.number("QthreeAltRepackMoved", moved)
657     ctx.number("QthreeAltRepackStatus", rep["cpsat"]["status"])
658     ctx.number("QthreeAltRepackGreedy", rep["greedy"])
659     ctx.number("QthreeAltRepackValidator", PASS)
660     # capacity of the empty region (what "no extra resource" could hold at most without any existing plan)
661     empty_bound = math.floor(100 * horizon_t / (C_TEMPLATE["w"] * C_TEMPLATE["d"] * C_TEMPLATE["n"]))
662     report = {k: v for k, v in result.items() if k not in {"new_plans"}}
663     report["alternatives"] = alternatives
664     report["empty_region_bound"] = empty_bound
665     ctx.write_json("pack_report.json", report)
666     ctx.number("QthreeNewPlans", result["value"])
667     ctx.number("QthreeUpperBound", result["upper_bound"])
668     ctx.number("QthreeGap", result["gap"])
669     ctx.number("QthreeOptimal", PASS if result["optimal"] else FAIL)
670     ctx.number("QthreeHorizon", horizon_t)
671     ctx.number("QthreeFreeCells", result["free_cells"])
672     ctx.number("QthreeCellsTotal", result["cells_total"])
673     ctx.number("QthreeCandidates", result["candidates"])
674     ctx.number("QthreeRows", result["rows"])
675     ctx.number("QthreeGreedy", result["greedy"])
676     ctx.number("QthreeCpsatStatus", result["cpsat"]["status"])
677     ctx.number("QthreeCpsatSeconds", result["cpsat"]["seconds"], ".1f")
678     ctx.number("QthreeCpsatValue", result["cpsat"]["value"])
679     ctx.number("QthreeCpsatBound", result["bounds"]["cpsat"] if math.isfinite(result["bounds"]["cpsat"]) else "--")
680     ctx.number("QthreeFreeCellBound", result["bounds"]["free_cells"])
681     ctx.number("QthreePerBandBound", result["bounds"]["per_band"])
682     ctx.number("QthreeEmptyRegionBound", empty_bound)
683     if "lp" in result["bounds"]:
684         ctx.number("QthreeLpBound", result["bounds"]["lp"])
685         ctx.number("QthreeLpValue", result["lp"]["value"], ".2f")
686     if result["highs_mip"]:
687         ctx.number("QthreeHighsStatus", result["highs_mip"]["status"])
688         ctx.number("QthreeHighsValue", int(result["highs_mip"].get("value", 0)))
689         if "highs_mip" in result["bounds"]:
690             ctx.number("QthreeHighsBound", result["bounds"]["highs_mip"])
691     ctx.number("QthreeUtilBefore", 100 * result["utilisation_before"], ".1f")
692     ctx.number("QthreeUtilAfter", 100 * result["utilisation_after"], ".1f")
693     ctx.number("QthreeValidator", PASS)
694     ctx.number("QthreeSpan", result["span"])
695     if "original_horizon" in alternatives:
696         ctx.number("QthreeAltHorizon", alt_horizon)
697         ctx.number("QthreeAltNewPlans", alternatives["original_horizon"]["value"])
698         ctx.number("QthreeAltUpperBound", alternatives["original_horizon"]["upper_bound"])
699     return {
700         "new_plans": result["value"],
701         "upper_bound": result["upper_bound"],

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702     "gap": result["gap"],
703     "status": result["cpsat"]["status"],
704     "validator_ok": True,
705 }
706
707
708 # ----- bench
709 @stage("bench", deps=("detect",), description="Exhaustive small-instance comparison and scaling benchmark")
710 def bench(ctx: StageContext) -> dict[str, Any]:
711     workers = int(ctx.param("workers", 8))
712     seed = ctx.seed_everything("bench")
713     tiny_limits = Limits(fmax=2, tmax=1)
714     tiny_rows = []
715     agree = 0
716     deterministic = 0
717     for k in range(int(ctx.param("n_tiny", 24))):
718         n_plans = 4 + k % 3
719         plans = tiny_instance(seed + k, n_plans=n_plans)
720         options = [enumerate_options(p, tiny_limits) for p in plans]
721         model = build_cell_model(plans, options)
722         t0 = time.perf_counter()
723         truth = exhaustive_lexicographic(plans, tiny_limits, canonical_levels, model)
724         t_exh = time.perf_counter() - t0
725         res = solve_lexicographic_cpsat(model, canonical_levels(model), seed=seed, workers=workers, time_limit=120)
726         again = solve_lexicographic_cpsat(model, canonical_levels(model), seed=seed, workers=workers, time_limit=120)
727         same = res["vector"] == truth
728         agree += int(same)
729         deterministic += int(again["chosen"] == res["chosen"] and again["vector"] == res["vector"])
730         combos = 1
731         for o in options:
732             combos *= len(o)
733         tiny_rows.append(
734             {
735                 "instance": k + 1,
736                 "plans": n_plans,
737                 "conflicts": len(detect_pairwise(plans)),
738                 "combinations": combos,
739                 "exhaustive_vector": truth,
740                 "cpsat_vector": res["vector"],
741                 "agree": same,
742                 "exhaustive_seconds": t_exh,
743                 "cpsat_seconds": res["seconds"],
744             }
745         )
746     ctx.log.info("bench.tiny", agree=agree, total=len(tiny_rows))
747     ctx.write_json("tiny.json", tiny_rows)
748
749     scaling = []
750     sizes = [int(v) for v in ctx.param("sizes", [150, 300, 600])]
751     scale_limit = float(ctx.param("scale_time_limit", 120))
752     for n in sizes:
753         plans = synthetic_instance(n, seed + n)
754         t0 = time.perf_counter()
755         pairs = detect_pairwise(plans)
756         t_pair = time.perf_counter() - t0
757         t0 = time.perf_counter()
758         sweep = detect_bandsweep(plans)
759         t_sweep = time.perf_counter() - t0
760         options = [enumerate_options(p, Q2_LIMITS) for p in plans]
761         t0 = time.perf_counter()
762         model = build_cell_model(plans, options)
763         t_build = time.perf_counter() - t0
764         levels = canonical_levels(model)
765         res = solve_weighted_cpsat(model, levels, seed=seed, workers=workers, time_limit=scale_limit)
766         row = {
767             "n": n,

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768         "horizon": horizon(plans),
769         "conflicts": len(pairs),
770         "detectors_agree": pairs == sweep,
771         "pairwise_seconds": t_pair,
772         "sweep_seconds": t_sweep,
773         "vars": model.n_vars,
774         "rows": len(model.rows),
775         "memberships": model.memberships,
776         "build_seconds": t_build,
777         "cpsat_status": res["status"],
778         "cpsat_seconds": res["seconds"],
779         "vector": res["vector"],
780     }
781     scaling.append(row)
782     ctx.log.info("bench.scale", **row)
783     ctx.write_json("scaling.json", scaling)
784     ctx.number("BenchTinyInstances", len(tiny_rows))
785     ctx.number("BenchTinyAgree", agree)
786     ctx.number("BenchTinyAllAgree", PASS if agree == len(tiny_rows) else FAIL)
787     ctx.number("BenchTinyDeterministic", deterministic)
788     ctx.number("BenchTinyAllDeterministic", PASS if deterministic == len(tiny_rows) else FAIL)
789     ctx.number("BenchTinyMaxCombinations", max(r["combinations"] for r in tiny_rows))
790     ctx.number("BenchScaleMaxN", max(sizes))
791     ctx.number("BenchScaleMinN", min(sizes))
792     last = scaling[-1]
793     ctx.number("BenchScaleMaxPairwiseSeconds", last["pairwise_seconds"], ".2f")
794     ctx.number("BenchScaleMaxSweepSeconds", last["sweep_seconds"], ".2f")
795     ctx.number("BenchScaleMaxCpsatSeconds", last["cpsat_seconds"], ".1f")
796     ctx.number("BenchScaleMaxStatus", last["cpsat_status"])
797     ctx.number("BenchScaleAllOptimal", PASS if all(r["cpsat_status"] == "OPTIMAL" for r in scaling) else FAIL)
798     ctx.number("BenchScaleDetectorsAgree", PASS if all(r["detectors_agree"] for r in scaling) else FAIL)
799     return {
800         "tiny_agree": agree,
801         "tiny_deterministic": deterministic,
802         "tiny_total": len(tiny_rows),
803         "scaling": [(r["n"], r["cpsat_status"], round(r["cpsat_seconds"], 1)) for r in scaling],
804     }
805
806
807 # ----- sensitivity
808 @stage(
809     "sensitivity",
810     deps=("detect", "resolve"),
811     description="Q2 sensitivity to the maximum shift amplitudes (base case taken from `resolve`, others warm-started)",
812 )
813 def sensitivity(ctx: StageContext) -> dict[str, Any]:
814     workers = int(ctx.param("workers", 8))
815     seed = ctx.seed_everything("sensitivity")
816     plans = _plans(ctx.dep_data("detect", "plans.json"))
817     base_report = ctx.dep_data("resolve", "solver_report.json")
818     base_decisions = ctx.dep_data("resolve", "decisions.json")
819     base_limits = (int(base_report["limits"]["fmax"]), int(base_report["limits"]["tmax"]))
820     cases = ctx.param("cases", [[5, 2], [5, 5], [10, 2], [10, 5], [15, 5], [20, 10]])
821     level_limits = ctx.param("level_time_limits", None) or float(ctx.param("time_limit", 60))
822     rows = []
823     for fmax, tmax in cases:
824         limits = Limits(fmax=int(fmax), tmax=int(tmax))
825         options = [enumerate_options(p, limits) for p in plans]
826         model = build_cell_model(plans, options)
827         if (limits.fmax, limits.tmax) == base_limits:
828             # identical model: reuse the primary Q2 solution instead of re-solving with a shorter limit
829             decisions = base_decisions
830             vec = list(base_report["vector"])
831             all_optimal, seconds, source = (
832                 bool(base_report["all_optimal"]),
833                 float(base_report["primary"]["seconds"]),

```

```

834         "resolve",
835     )
836 else:
837     hint = hint_from_decisions(model, base_decisions)
838     res = solve_lexicographic_cpsat(
839         model,
840         canonical_levels(model),
841         seed=seed,
842         workers=workers,
843         time_limit=level_limits,
844         hint=hint,
845         strength=strengthen(model),
846     )
847     decisions = decisions_from_choice(model, res["chosen"]) if res["chosen"] else []
848     vec, all_optimal, seconds, source = res["vector"], res["all_optimal"], res["seconds"], "cp-sat"
849     valid = (
850         validators.validate_resolution(plans, decisions, fmax=limits.fmax, tmax=limits.tmax)
851         if decisions
852         else {"ok": False}
853     )
854     rows.append(
855         {
856             "fmax": int(fmax),
857             "tmax": int(tmax),
858             "vars": model.n_vars,
859             "vector": vec,
860             "cancel_total": sum(vec[0:3]) if len(vec) == 7 else None,
861             "adjust_total": sum(vec[3:6]) if len(vec) == 7 else None,
862             "amplitude": vec[6] / 10.0 if len(vec) == 7 else None,
863             "table": table_from_decisions(decisions) if decisions else None,
864             "all_optimal": all_optimal,
865             "seconds": seconds,
866             "source": source,
867             "validator_ok": bool(valid["ok"]),
868             "shifts": _shift_stats(decisions),
869         }
870     )
871     ctx.log.info("sensitivity.case", fmax=fmax, tmax=tmax, vector=vec, seconds=seconds, source=source)
872     ctx.write_json("cases.json", rows)
873     base = next((r for r in rows if (r["fmax"], r["tmax"]) == base_limits), rows[0])
874     # lexicographic monotonicity: a larger action set can never give a worse vector than the base solution
875     monotone = all(
876         list(r["vector"]) <= list(base["vector"])
877         for r in rows
878         if len(r["vector"]) == 7 and r["fmax"] >= base_limits[0] and r["tmax"] >= base_limits[1]
879     ) and all(
880         list(r["vector"]) >= list(base["vector"])
881         for r in rows
882         if len(r["vector"]) == 7 and r["fmax"] <= base_limits[0] and r["tmax"] <= base_limits[1]
883     )
884     ctx.number("SensCases", len(rows))
885     ctx.number("SensBaseAdjustTotal", base["adjust_total"])
886     ctx.number("SensBaseCancelTotal", base["cancel_total"])
887     ctx.number("SensMinAdjustTotal", min(r["adjust_total"] for r in rows if r["adjust_total"] is not None))
888     ctx.number("SensMaxAdjustTotal", max(r["adjust_total"] for r in rows if r["adjust_total"] is not None))
889     ctx.number("SensMaxCancelTotal", max(r["cancel_total"] for r in rows if r["cancel_total"] is not None))
890     ctx.number("SensMinCancelTotal", min(r["cancel_total"] for r in rows if r["cancel_total"] is not None))
891     ctx.number("SensAllOptimal", PASS if all(r["all_optimal"] for r in rows) else FAIL)
892     ctx.number("SensAllValid", PASS if all(r["validator_ok"] for r in rows) else FAIL)
893     ctx.number("SensMonotone", PASS if monotone else FAIL)
894     return {"cases": [(r["fmax"], r["tmax"], r["vector"]) for r in rows], "monotone": monotone}
895
896
897 # ----- results
898 @stage(
899     "results",

```

```

900     deps=("detect", "resolve", "pack", "resolve_interval"),
901     description="Fill result1-4.xlsx from the organisers' templates after every validator passed",
902 )
903 def results(ctx: StageContext) -> dict[str, Any]:
904     for name in ("detect", "resolve", "pack", "resolve_interval"):
905         report = ctx.dep_data(name, "validation_report.json")
906         if not report.get("ok"):
907             raise RuntimeError(f"validator of stage {name} did not pass; refusing to write result workbooks")
908     templates = ctx.templates
909     written: dict[str, Any] = {}
910
911     conflicts = ctx.dep_data("detect", "conflicts.json")
912     rows1 = [
913         [k, r["id1"], r["id2"]] for k, r in enumerate(sorted(conflicts, key=lambda r: (r["id1"], r["id2"])), start=1)
914     ]
915     written["result1.xlsx"] = write_result(
916         templates / "result1.xlsx", ctx.out("result1.xlsx"), {"Sheet1": ([ "序号", "冲突装备1", "冲突设备2"], rows1)}
917     )
918
919     def adjustment_rows(decisions: list[dict[str, Any]], with_gap: bool) -> list[list[Any]]:
920         rows: list[list[Any]] = []
921         for dec in sorted(decisions, key=lambda d: d["pid"]):
922             if dec["kind"] == "keep":
923                 continue
924             plan = plan_from_record(dec["plan"]) if dec["plan"] else None
925             freq = plan.freq_text() if plan and dec["kind"] == "freq" else None
926             tim = plan.time_text() if plan and dec["kind"] == "time" else None
927             gap = plan.g if plan and dec["kind"] == "gap" else None
928             cancel = "是" if dec["kind"] == "cancel" else None
929             rows.append([dec["pid"], freq, tim, gap, cancel] if with_gap else [dec["pid"], freq, tim, cancel])
930         return rows
931
932     rows2 = adjustment_rows(ctx.dep_data("resolve", "decisions.json"), with_gap=False)
933     written["result2.xlsx"] = write_result(
934         templates / "result2.xlsx",
935         ctx.out("result2.xlsx"),
936         {"Sheet1": ([ "用频装备编号", "调整后频段区间", "调整后时间区间", "是否撤销用频计划"], rows2)},
937     )
938     new_plans = _plans(ctx.dep_data("pack", "new_plans.json"))
939     rows3 = [[k, p.freq_text(), p.time_text()] for k, p in enumerate(new_plans, start=1)]
940     written["result3.xlsx"] = write_result(
941         templates / "result3.xlsx",
942         ctx.out("result3.xlsx"),
943         {"Sheet1": ([ "新增用频装备序号", "调整后频段区间", "调整后时间区间"], rows3)},
944     )
945     rows4 = adjustment_rows(ctx.dep_data("resolve_interval", "decisions.json"), with_gap=True)
946     written["result4.xlsx"] = write_result(
947         templates / "result4.xlsx",
948         ctx.out("result4.xlsx"),
949         {"Sheet1": ([ "用频装备编号", "调整后频段范围", "调整后时间区间", "调整后间隔时长", "是否撤销用频计划"], rows4)},
950     )
951     ctx.write_json("results_report.json", written)
952     ctx.number("ResultOneRows", len(rows1))
953     ctx.number("ResultTwoRows", len(rows2))
954     ctx.number("ResultThreeRows", len(rows3))
955     ctx.number("ResultFourRows", len(rows4))
956     return {name: info["sheets"]["Sheet1"]["rows"] for name, info in written.items()}
957
958
959 from pipelines.d import report as _report # noqa: E402,F401 (registers figures/tables stages)

```

pipelines/d/synth.py

```
1 """Synthetic instances and an exhaustive reference solver for benchmarking (MDR-0007)."""
2
3 from __future__ import annotations
4
5 import random
6 from itertools import combinations
7 from typing import Any
8
9 from pipelines.d.detect import plans_conflict
10 from pipelines.d.plans import BANDS, Plan
11 from pipelines.d.resolve import CellModel, Level, Limits, Option, enumerate_options
12
13 CATEGORY_PARAMS = {"A": (10, 5, 60, 3), "B": (15, 3, 40, 4), "C": (3, 2, 8, 12)} # (w, d, g, n)
14 CATEGORY_SHARE = {"A": 20, "B": 40, "C": 90}
15
16
17 def synthetic_instance(n: int, seed: int, *, base_n: int = 150, base_start_max: int = 531) -> list[Plan]:
18     """`n` plans with the attachment's category mix and parameters; the time range scales with `n`
19     so that the occupancy density stays comparable to the real instance."""
20     rng = random.Random(seed)
21     start_max = max(1, round(base_start_max * n / base_n))
22     plans: list[Plan] = []
23     counts = {c: round(n * share / base_n) for c, share in CATEGORY_SHARE.items()}
24     counts["C"] += n - sum(counts.values())
25     for cat, count in counts.items():
26         w, d, g, m = CATEGORY_PARAMS[cat]
27         for k in range(count):
28             plans.append(
29                 Plan(f"{cat}-{k + 1:03d}", cat, rng.randint(0, BANDS - w), w, rng.randint(0, start_max), d, g, m)
30             )
31     return plans
32
33
34 def tiny_instance(seed: int, *, n_plans: int = 5, bands: int = 12) -> list[Plan]:
35     """A crowded miniature instance for exhaustive comparison."""
36     rng = random.Random(seed)
37     plans = []
38     for k in range(n_plans):
39         cat = rng.choice("ABC")
40         w = rng.randint(2, 3)
41         plans.append(
42             Plan(
43                 f"{cat}-{k + 1:03d}",
44                 cat,
45                 rng.randint(0, bands - w),
46                 w,
47                 rng.randint(0, 6),
48                 rng.randint(1, 2),
49                 rng.randint(1, 3),
50                 rng.randint(2, 3),
51             )
52         )
53     return plans
54
55
56 def exhaustive_lexicographic(plans: list[Plan], limits: Limits, levels_fn: Any, model: CellModel) -> list[int] | None:
57     """Depth-first enumeration of every action combination; returns the lexicographically smallest level vector.
58
59     ``levels_fn(model)`` gives the levels; conflicts are decided with ``plans_conflict`` on the applied plans,
60     i.e. by interval arithmetic rather than by the cell rows, so the comparison with CP-SAT is meaningful."""
61     levels: list[Level] = levels_fn(model)
62     n = len(plans)
63     applied: list[list[Plan | None]] = [[opt.apply(p) for opt in model.options[i]] for i, p in enumerate(plans)]
64     conflict: dict[tuple[int, int, int, int], bool] = {}
```

```

65     for i, j in combinations(range(n), 2):
66         for oi, pi in enumerate(applied[i]):
67             for oj, pj in enumerate(applied[j]):
68                 conflict[(i, oi, j, oj)] = pi is not None and pj is not None and plans_conflict(pi, pj)
69     best: list[int] | None = None
70     choice: list[int] = []
71
72     def rec(i: int) -> None:
73         nonlocal best
74         if i == n:
75             vec = [sum(terms.get(model.plan_vars[k][choice[k]], 0) for k in range(n)) for _, terms in levels]
76             if best is None or vec < best:
77                 best = vec
78             return
79         for o in range(len(model.options[i])):
80             if all(not conflict[(k, choice[k], i, o)] for k in range(i)):
81                 choice.append(o)
82                 rec(i + 1)
83                 choice.pop()
84
85     rec(0)
86     return best
87
88
89 def options_for(plans: list[Plan], limits: Limits) -> list[list[Option]]:
90     return [enumerate_options(p, limits) for p in plans]

```

pipelines/d/validators.py

```

1  """Independent validators for Problem D (deliberately separate from detect.py / resolve.py / pack.py).
2
3  Every check here re-derives what a plan occupies from first principles — the plan is expanded into an
4  explicit set of (band, time) cells and conflicts are found by set intersection — so that agreement with
5  the interval-arithmetic detectors and the CP-SAT/HiGHS solutions is genuine, not tautological.
6  """
7
8  from __future__ import annotations
9
10 from itertools import combinations
11 from typing import Any
12
13 from pipelines.d.plans import BANDS, CATEGORIES, Plan
14
15 KINDS = ("keep", "freq", "time", "gap", "cancel")
16
17
18 def cell_set(plan: Plan) -> set[tuple[int, int]]:
19     """Every (band, time) cell the plan occupies, from the definition of its uses."""
20     cells: set[tuple[int, int]] = set()
21     for k in range(plan.n):
22         start = plan.s + k * (plan.d + plan.g)
23         for band in range(plan.f, plan.f + plan.w):
24             for t in range(start, start + plan.d):
25                 cells.add((band, t))
26     return cells
27
28
29 def conflicts_by_cells(plans: list[Plan]) -> list[tuple[str, str]]:
30     """All conflicting id pairs (id1 < id2), found by intersecting cell sets."""
31     cells = {p.pid: cell_set(p) for p in plans}
32     out = []
33     for a, b in combinations(plans, 2):
34         if cells[a.pid] & cells[b.pid]:

```

```

35         out.append((min(a.pid, b.pid), max(a.pid, b.pid)))
36     return sorted(out)
37
38
39 def plan_bounds_errors(plan: Plan) -> list[str]:
40     errors = []
41     if plan.f < 0 or plan.f + plan.w > BANDS:
42         errors.append(f"{plan.pid}: bands {plan.f}..{plan.f + plan.w} outside [0,{BANDS}]")
43     if plan.s < 0:
44         errors.append(f"{plan.pid}: start {plan.s} < 0")
45     if plan.g < 1 or plan.n < 1 or plan.d < 1 or plan.w < 1:
46         errors.append(f"{plan.pid}: non-positive parameter (w={plan.w}, d={plan.d}, g={plan.g}, n={plan.n})")
47     return errors
48
49
50 def validate_detection(plans: list[Plan], reported: list[tuple[str, str]]) -> dict[str, Any]:
51     """Compare a reported conflict list with the cell-set ground truth."""
52     truth = conflicts_by_cells(plans)
53     rep = sorted((min(a, b), max(a, b)) for a, b in reported)
54     missing = sorted(set(truth) - set(rep))
55     extra = sorted(set(rep) - set(truth))
56     duplicates = len(rep) - len(set(rep))
57     ok = not missing and not extra and duplicates == 0
58     return {
59         "ok": ok,
60         "method": "cell-set intersection",
61         "truth_pairs": len(truth),
62         "reported_pairs": len(rep),
63         "missing": missing[:20],
64         "extra": extra[:20],
65         "duplicates": duplicates,
66     }
67
68
69 def _changed_fields(before: Plan, after: Plan) -> list[str]:
70     return [name for name in ("f", "w", "s", "d", "g", "n") if getattr(before, name) != getattr(after, name)]
71
72
73 def validate_resolution(
74     original: list[Plan],
75     decisions: list[dict[str, Any]],
76     *,
77     fmax: int,
78     tmax: int,
79     gmax: int = 0,
80     gap_categories: tuple[str, ...] = (),
81     reported_table: dict[str, dict[str, int]] | None = None,
82     reported_vector: list[int] | None = None,
83 ) -> dict[str, Any]:
84     """Check a resolution: one legal action per plan, one changed parameter, amplitude limits, bounds,
85     zero conflicts among surviving plans, and (optionally) the reported Table-1 statistics / objective."""
86     errors: list[str] = []
87     by_id = {p.pid: p for p in original}
88     seen: set[str] = set()
89     survivors: list[Plan] = []
90     table = {c: {"keep": 0, "adjust": 0, "cancel": 0} for c in CATEGORIES}
91     vector = {"cancel": dict.fromkeys(CATEGORIES, 0), "adjust": dict.fromkeys(CATEGORIES, 0), "amplitude": 0}
92     for dec in decisions:
93         pid = str(dec["pid"])
94         kind = str(dec["kind"])
95         if pid not in by_id:
96             errors.append(f"unknown plan {pid}")
97             continue
98         if pid in seen:
99             errors.append(f"{pid}: more than one decision")
100             continue

```

```

101     seen.add(pid)
102     before = by_id[pid]
103     if kind not in KINDS:
104         errors.append(f"{pid}: unknown action {kind!r}")
105         continue
106     if kind == "cancel":
107         table[before.cat]["cancel"] += 1
108         vector["cancel"][before.cat] += 1
109         continue
110     after = Plan(**dec["plan"]) if dec.get("plan") else before
111     if after.pid != pid or after.cat != before.cat:
112         errors.append(f"{pid}: identity changed in adjusted plan")
113     changed = _changed_fields(before, after)
114     expected = {"keep": [], "freq": ["f"], "time": ["s"], "gap": ["g"]}[kind]
115     if changed != expected:
116         errors.append(f"{pid}: action {kind} but changed fields {changed}")
117     errors += plan_bounds_errors(after)
118     if kind == "freq" and abs(after.f - before.f) > fmax:
119         errors.append(f"{pid}: band shift {after.f - before.f} exceeds {fmax}")
120     if kind == "time" and abs(after.s - before.s) > tmax:
121         errors.append(f"{pid}: time shift {after.s - before.s} exceeds {tmax}")
122     if kind == "gap":
123         if before.cat not in gap_categories:
124             errors.append(f"{pid}: gap change not allowed for category {before.cat}")
125         if abs(after.g - before.g) > gmax:
126             errors.append(f"{pid}: gap change {after.g - before.g} exceeds {gmax}")
127     survivors.append(after)
128     if kind == "keep":
129         table[before.cat]["keep"] += 1
130     else:
131         table[before.cat]["adjust"] += 1
132         vector["adjust"][before.cat] += 1
133         vector["amplitude"] += {
134             "freq": abs(after.f - before.f),
135             "time": 2 * abs(after.s - before.s),
136             "gap": abs(after.g - before.g),
137         }[kind]
138     undecided = sorted(set(by_id) - seen)
139     if undecided:
140         errors.append(f"{len(undecided)} plans without decision, e.g. {undecided[:5]}")
141     remaining = conflicts_by_cells(survivors)
142     if remaining:
143         errors.append(f"{len(remaining)} conflicts remain, e.g. {remaining[:5]}")
144     if reported_table is not None and reported_table != table:
145         errors.append(f"reported table {reported_table} differs from recomputed {table}")
146     recomputed_vector = [vector["cancel"][c] for c in CATEGORIES] + [vector["adjust"][c] for c in CATEGORIES]
147     recomputed_vector.append(int(vector["amplitude"]))
148     if reported_vector is not None and list(reported_vector) != recomputed_vector:
149         errors.append(f"reported objective {reported_vector} differs from recomputed {recomputed_vector}")
150     return {
151         "ok": not errors,
152         "method": "independent per-plan legality checks + cell-set conflict scan",
153         "errors": errors[:30],
154         "n_decisions": len(decisions),
155         "n_survivors": len(survivors),
156         "remaining_conflicts": len(remaining),
157         "table": table,
158         "vector": recomputed_vector,
159         "amplitude_normalised": vector["amplitude"] / 10.0,
160         "horizon": max((p.end for p in survivors), default=0),
161     }
162
163
164 def validate_packing(
165     existing: list[Plan], new_plans: list[Plan], *, horizon: int, template: dict[str, int]
166 ) -> dict[str, Any]:

```

```

167     """Check new plans: template parameters, bounds, inside the horizon, no conflicts with anything."""
168     errors: list[str] = []
169     for p in new_plans:
170         if (p.w, p.d, p.g, p.n) != (template["w"], template["d"], template["g"], template["n"]):
171             errors.append(f"{p.pid}: parameters {(p.w, p.d, p.g, p.n)} differ from template")
172         if p.cat != "C":
173             errors.append(f"{p.pid}: category {p.cat} is not C")
174         errors += plan_bounds_errors(p)
175         if p.end > horizon:
176             errors.append(f"{p.pid}: ends at {p.end} > horizon {horizon}")
177     ids = [p.pid for p in new_plans]
178     if len(ids) != len(set(ids)) or set(ids) & {p.pid for p in existing}:
179         errors.append("duplicate or clashing plan ids")
180     remaining = conflicts_by_cells(existing + new_plans)
181     if remaining:
182         errors.append(f"{len(remaining)} conflicts, e.g. {remaining[:5]}")
183     return {
184         "ok": not errors,
185         "method": "parameter/bound checks + cell-set conflict scan over existing and new plans",
186         "errors": errors[:30],
187         "n_new": len(new_plans),
188         "remaining_conflicts": len(remaining),
189         "cells_new": sum(p.n_cells for p in new_plans),
190     }

```

pyproject.toml

```

1 [project]
2 name = "cumcm2026-d"
3 version = "0.1.0"
4 description = "时频冲突检测与消解 — cloud-only reproducible modelling study (CUMCM 2026 Problem D)"
5 requires-python = ">=3.11"
6 license = "Unlicense"
7
8 [tool.ruff]
9 line-length = 120
10 target-version = "py311"
11 extend-exclude = ["artifacts", "releases", "manuscript"]
12
13 [tool.ruff.lint]
14 select = ["E", "F", "W", "I", "B", "UP", "NPY", "RUF", "S102"]
15 ignore = ["RUF001", "RUF002", "RUF003", "E741", "B905"]
16
17 [tool.ruff.lint.per-file-ignores]
18 "tests/**" = ["S101", "B011"]
19
20 [tool.mypy]
21 python_version = "3.11"
22 ignore_missing_imports = true
23 check_untyped_defs = true
24 warn_unused_ignores = false
25 no_implicit_optional = true
26 exclude = "(artifacts|releases|manuscript)/"
27
28 [tool.pytest.ini_options]
29 testpaths = ["tests"]
30 addopts = "-p no:cacheprovider -q"
31 markers = ["integration: needs the mounted repository inputs (runs in Modal)"]
32
33 [tool.coverage.run]
34 source = ["forge", "pipelines"]

```

tests/__init__.py

tests/conftest.py

```
1 """Shared test configuration: deterministic hypothesis profile, no on-disk example database."""
2
3 from __future__ import annotations
4
5 import os
6 from pathlib import Path
7
8 import pytest
9 from hypothesis import HealthCheck, settings
10
11 settings.register_profile(
12     "forge", database=None, deadline=None, max_examples=60, suppress_health_check=[HealthCheck.too_slow]
13 )
14 settings.load_profile("forge")
15
16 REPO = Path(os.environ.get("FORGE_REPO", Path(__file__).resolve().parents[1]))
17
18
19 @pytest.fixture(scope="session")
20 def repo() -> Path:
21     return REPO
22
23
24 @pytest.fixture
25 def tmp_repo(tmp_path: Path) -> Path:
26     """A minimal repository layout (configs only) for runtime tests that must not touch real data."""
27     (tmp_path / "configs").mkdir()
28     (tmp_path / "configs" / "default.toml").write_text("[run]\nseed = 7\n[paper]\nnsources = []\n", encoding="utf-8")
29     (tmp_path / "configs" / "project.json").write_text(
30         '{"letter": "T", "package": "t", "problem_pdf": "T.pdf", "attachments_dir": "att", '
31         '"templates_dir": "att/tpl", "result_files": []}',
32         encoding="utf-8",
33     )
34     (tmp_path / "att" / "tpl").mkdir(parents=True)
35     return tmp_path
```

tests/integration/__init__.py

tests/integration/test_ingest_validate.py

```
1 """End-to-end ingest + validate against the real attachments (runs inside Modal where /repo exists)."""
2
3 from __future__ import annotations
4
5 import json
6 import os
7 from pathlib import Path
8
9 import pytest
10
11 from forge import runner
```

```

12
13 REPO = Path(os.environ.get("FORGE_REPO", "/repo"))
14 pytestmark = pytest.mark.integration
15
16
17 @pytest.mark.skipif(not (REPO / "configs" / "project.json").exists(), reason="repository inputs not mounted")
18 def test_ingest_then_validate_on_real_inputs(tmp_path: Path) -> None:
19     vol = tmp_path / "vol"
20     ingest = runner.execute("ingest", "itest", {"code_ref": {"git_sha": "test"}}, False, REPO, vol)
21     assert ingest["status"] == "completed" and ingest["metrics"]["files"] >= 1
22     inventory = json.loads((vol / "runs" / "itest" / "ingest" / "inventory.json").read_text(encoding="utf-8"))
23     assert any(not f["template"] for f in inventory["files"])
24     validate = runner.execute("validate", "itest", {}, False, REPO, vol)
25     assert validate["status"] == "completed" and validate["metrics"]["ok"]

```

tests/unit/__init__.py

tests/unit/test_config.py

```

1 from __future__ import annotations
2
3 from pathlib import Path
4
5 import pytest
6
7 from forge.config import Config, deep_merge, load_config
8
9
10 def test_deep_merge_is_recursive_and_non_destructive() -> None:
11     base = {"a": {"x": 1, "y": 2}, "b": 1}
12     merged = deep_merge(base, {"a": {"y": 3}, "c": 4})
13     assert merged == {"a": {"x": 1, "y": 3}, "b": 1, "c": 4}
14     assert base == {"a": {"x": 1, "y": 2}, "b": 1}
15
16
17 def test_config_dotted_access_and_digest() -> None:
18     cfg = Config({"paper": {"sources": ["q1"], "max_body_pages": 30}})
19     assert cfg.get("paper.max_body_pages") == 30
20     assert cfg.get("paper.missing", "d") == "d"
21     assert cfg["paper.sources"] == ["q1"]
22     with pytest.raises(KeyError):
23         cfg["nope.nope"]
24     assert cfg.digest == Config({"paper": {"max_body_pages": 30, "sources": ["q1"]}}).digest
25
26
27 def test_load_config_with_profile_and_overrides(tmp_repo: Path) -> None:
28     (tmp_repo / "configs" / "fast.toml").write_text("[run]\nseed = 99\n", encoding="utf-8")
29     cfg = load_config(tmp_repo, profile="fast", overrides={"paper": {"sources": ["x"]}})
30     assert cfg.get("run.seed") == 99
31     assert cfg.get("paper.sources") == ["x"]

```

tests/unit/test_contracts.py

```
1 from __future__ import annotations
2
3 import numpy as np
4 import pandas as pd
5 from hypothesis import given
6 from hypothesis import strategies as st
7
8 from forge.contracts import Column, FrameContract, validate_frame
9
10 CONTRACT = FrameContract(
11     name="t",
12     columns=(
13         Column("t", "int", min=0, unique=True, monotonic="increasing", step=60),
14         Column("v", "float", min=0, max=1),
15     ),
16     min_rows=2,
17 )
18
19
20 def test_valid_frame_passes() -> None:
21     df = pd.DataFrame({"t": [0, 60, 120], "v": [0.1, 0.2, 0.3]})
22     report = validate_frame(df, CONTRACT)
23     assert report["ok"], report["errors"]
24
25
26 def test_violations_are_reported_precisely() -> None:
27     df = pd.DataFrame({"t": [0, 60, 60], "v": [0.1, 1.5, np.nan], "extra": [1, 2, 3]})
28     report = validate_frame(df, CONTRACT)
29     joined = " | ".join(report["errors"])
30     assert not report["ok"]
31     assert "unexpected columns" in joined
32     assert "duplicated" in joined
33     assert "max 1.5 > 1" in joined
34     assert "null" in joined
35     assert "not increasing" in joined
36
37
38 def test_column_names_are_stripped() -> None:
39     df = pd.DataFrame({"t ": [0, 60], " v": [0.5, 0.6]})
40     assert validate_frame(df, CONTRACT)["ok"]
41
42
43 @given(st.lists(st.floats(min_value=0, max_value=1, allow_nan=False), min_size=2, max_size=40))
44 def test_property_ranges_within_bounds_always_pass(values: list[float]) -> None:
45     df = pd.DataFrame({"t": np.arange(len(values)) * 60, "v": values})
46     assert validate_frame(df, CONTRACT)["ok"]
47
48
49 def test_regex_and_custom_checks() -> None:
50     contract = FrameContract(
51         "ids", (Column("id", "str", regex=r"[ABC]\d{3}"),), checks=(lambda df: ["custom"] if len(df) > 1 else [],)
52     )
53     assert validate_frame(pd.DataFrame({"id": ["A001"]}), contract)["ok"]
54     report = validate_frame(pd.DataFrame({"id": ["A001", "Z9"]}), contract)
55     assert any("violate" in e for e in report["errors"]) and "custom" in report["errors"]
```

tests/unit/test_d_hints.py

```
1 """Problem D: warm-start hints mapped from recorded decisions, and the HiGHS per-level cross-check."""
2
3 from __future__ import annotations
4
5 from pipelines.d.plans import Plan
6 from pipelines.d.resolve import (
7     Limits,
8     build_cell_model,
9     canonical_levels,
10    decisions_from_choice,
11    enumerate_options,
12    evaluate,
13    hint_from_decisions,
14    hint_is_feasible,
15    solve_lexicographic_cpsat,
16    solve_lexicographic_highs,
17 )
18 from pipelines.d.synth import tiny_instance
19
20
21 def _model(plans: list[Plan], limits: Limits): # type: ignore[no-untyped-def]
22     return build_cell_model(plans, [enumerate_options(p, limits) for p in plans])
23
24
25 def test_hint_round_trips_and_falls_back_to_keep() -> None:
26     plans = tiny_instance(31, n_plans=5)
27     model = _model(plans, Limits(fmax=2, tmax=1))
28     res = solve_lexicographic_cpsat(model, canonical_levels(model), seed=1, workers=2, time_limit=60)
29     decisions = decisions_from_choice(model, res["chosen"])
30     hint = hint_from_decisions(model, decisions)
31     assert hint == res["chosen"]
32     assert hint_is_feasible(model, hint)
33     assert evaluate(model, hint, canonical_levels(model)) == res["vector"]
34     # unknown plans are ignored, missing plans and actions outside the limits fall back to keep
35     partial = [{"pid": "Z999", "kind": "cancel", "delta": 0}, {"pid": plans[0].pid, "kind": "freq", "delta": 9}]
36     fallback = hint_from_decisions(model, partial)
37     assert fallback == [vars_of_plan[0] for vars_of_plan in model.plan_vars]
38     assert all(model.option_of(v).kind == "keep" for v in fallback)
39
40
41 def test_hint_infeasibility_is_detected() -> None:
42     a = Plan("A001", "A", 0, 10, 0, 5, 60, 3)
43     b = Plan("B001", "B", 0, 10, 0, 5, 60, 3)
44     model = _model([a, b], Limits(fmax=10, tmax=5))
45     keep_both = [vars_of_plan[0] for vars_of_plan in model.plan_vars]
46     assert not hint_is_feasible(model, keep_both)
47     cancel_b = [model.plan_vars[0][0], next(v for v in model.plan_vars[1] if model.option_of(v).kind == "cancel")]
48     assert hint_is_feasible(model, cancel_b)
49
50
51 def test_highs_levels_are_independent_of_earlier_time_limits() -> None:
52     # every level gets a fresh solver, so a tiny LP limit on one level cannot starve the following ones
53     plans = tiny_instance(32, n_plans=6)
54     model = _model(plans, Limits(fmax=2, tmax=1))
55     levels = canonical_levels(model)
56     res = solve_lexicographic_cpsat(model, levels, seed=1, workers=2, time_limit=60)
57     highs = solve_lexicographic_highs(
58         model, levels, seed=1, threads=2, time_limit=60, reference=res["vector"], hint=res["chosen"], lp_time_limit=60
59     )
60     assert highs["agrees_with_reference"] and highs["all_optimal"] and highs["vector"] == res["vector"]
61     assert all(e.get("lp_status") == "Optimal" for e in highs["levels"])
```

tests/unit/test_d_pack.py

```
1 """Problem D: packing model on analytically solvable regions."""
2
3 from __future__ import annotations
4
5 import numpy as np
6 import pytest
7
8 from pipelines.d.pack import (
9     C_TEMPLATE,
10     candidate_placements,
11     cell_rows,
12     greedy_pack,
13     highs_pack,
14     occupancy_grid,
15     plans_from_choice,
16     repack_existing,
17     solve_pack_cpsat,
18     template_span,
19 )
20 from pipelines.d.plans import Plan
21 from pipelines.d.validators import conflicts_by_cells, validate_packing
22
23
24 def test_empty_strip_packs_five_phases() -> None:
25     # 3 free bands over exactly one span: five plans with starts 0,2,4,6,8 tile the strip (period 10, duration 2)
26     horizon = template_span(C_TEMPLATE) # 112
27     grid = np.ones((100, horizon), dtype=bool)
28     grid[0:3, :] = False
29     cands = candidate_placements(grid, C_TEMPLATE)
30     assert cands.tolist() == [[0, 0]] # only s = 0 fits inside the horizon
31     grid = np.ones((100, horizon + 8), dtype=bool)
32     grid[0:3, :] = False
33     cands = candidate_placements(grid, C_TEMPLATE)
34     assert len(cands) == 9
35     rows = cell_rows(cands, C_TEMPLATE, horizon + 8)
36     res = solve_pack_cpsat(rows, len(cands), hint=None, seed=0, workers=1, time_limit=30)
37     assert res["status"] == "OPTIMAL" and res["value"] == 5
38     lp = highs_pack(rows, len(cands), integer=False, time_limit=30, seed=0, threads=1)
39     assert lp["value"] >= 5 - 1e-6
40     new_plans = plans_from_choice(cands, res["chosen"], C_TEMPLATE)
41     report = validate_packing([], new_plans, horizon=horizon + 8, template=C_TEMPLATE)
42     assert report["ok"], report["errors"]
43
44
45 def test_existing_plan_blocks_candidates_and_greedy_is_feasible() -> None:
46     existing = [Plan("C001", "C", 0, 3, 0, 2, 8, 12)]
47     horizon = 130
48     grid = occupancy_grid(existing, horizon)
49     assert grid.sum() == 72
50     cands = candidate_placements(grid, C_TEMPLATE)
51     assert not any((f < 3 and s % 10 in (0, 1, 9)) for f, s in cands.tolist() if s <= 110)
52     greedy = greedy_pack(cands, C_TEMPLATE, grid)
53     new_plans = plans_from_choice(cands, greedy, C_TEMPLATE)
54     report = validate_packing(existing, new_plans, horizon=horizon, template=C_TEMPLATE)
55     assert report["ok"] and len(new_plans) == len(greedy) > 0
56
57
58 def test_repack_keeps_parameters_compacts_and_stays_conflict_free() -> None:
59     """MDR-0010 interpretation B: first-fit decreasing may move a plan anywhere but must not change it."""
60     scattered = [
61         Plan("A001", "A", 80, 10, 30, 5, 60, 3),
62         Plan("B001", "B", 40, 15, 7, 3, 40, 4),
63         Plan("B002", "B", 60, 15, 90, 3, 40, 4),
64         Plan("C001", "C", 20, 3, 11, 2, 8, 12),
```

```

65     Plan("C002", "C", 97, 3, 40, 2, 8, 12),
66 ]
67 horizon = 300
68 repacked, grid = repack_existing(scattered, horizon)
69 assert [p.pid for p in repacked] == sorted(p.pid for p in scattered)
70 before = {p.pid: p for p in scattered}
71 for p in repacked:
72     q = before[p.pid]
73     assert (p.w, p.d, p.g, p.n, p.cat) == (q.w, q.d, q.g, q.n, q.cat) # only f and s may move
74     assert 0 <= p.f and p.f_end <= 100 and p.s >= 0 and p.end <= horizon
75     assert not conflicts_by_cells(repacked) # the layout itself is conflict-free
76     assert grid.sum() == sum(p.n_cells for p in scattered) # no cell is double-counted
77     assert max(p.f_end for p in repacked) <= max(p.f_end for p in scattered) # compacted towards low bands
78     # the repacked layout admits at least as many new C plans as the scattered one
79     scattered_grid = occupancy_grid(scattered, horizon)
80     assert len(candidate_placements(grid, C_TEMPLATE)) >= len(candidate_placements(scattered_grid, C_TEMPLATE))
81
82
83 def test_repack_fills_a_strip_exactly_and_refuses_an_impossible_region() -> None:
84     # period 10, duration 2: five phases (s = 0,2,4,6,8) tile a 3-band strip, but only once every offset
85     # still ends inside the horizon, i.e. from span + 8 = 120 onwards
86     horizon = template_span(C_TEMPLATE) + 8
87     strip = [Plan(f"C{k:03d}", "C", 0, 3, 2 * k, 2, 8, 12) for k in range(5)]
88     repacked, grid = repack_existing(strip, horizon)
89     assert not conflicts_by_cells(repacked)
90     assert sorted(p.s for p in repacked) == [0, 2, 4, 6, 8] # first-fit reproduces the five phases
91     assert grid[0:3].all() and not grid[3:].any() # and they tile exactly the first three bands
92     with pytest.raises(RuntimeError, match="could not place"):
93         repack_existing([*strip, Plan("C999", "C", 0, 3, 0, 2, 8, 12)], horizon, bands=3)

```

tests/unit/test_d_plans_detect.py

```

1  """Problem D: plan semantics, the two detectors and the independent cell-set validator."""
2
3  from __future__ import annotations
4
5  from hypothesis import given
6  from hypothesis import strategies as st
7
8  from pipelines.d.detect import detect_bandsweep, detect_pairwise, overlapping_uses, plans_conflict
9  from pipelines.d.plans import Plan, fmt_interval, parse_interval, plan_from_row
10 from pipelines.d.synth import synthetic_instance
11 from pipelines.d.validators import cell_set, conflicts_by_cells, validate_detection
12
13
14 def test_problem_statement_example_a001() -> None:
15     plan = plan_from_row("A001", "[80,90)", "[35,40)", 60, 3)
16     assert plan.cat == "A" and plan.w == 10 and plan.d == 5 and plan.period == 65
17     assert plan.uses() == [(35, 40), (100, 105), (165, 170)]
18     assert plan.end == 170 and plan.n_cells == 150
19     assert parse_interval("[3, 7) ") == (3, 7) and fmt_interval(3, 7) == "[3,7)"
20
21
22 def test_half_open_intervals_do_not_touch() -> None:
23     a = Plan("A001", "A", 0, 10, 0, 5, 60, 3)
24     b = Plan("B001", "B", 10, 15, 0, 3, 40, 4) # bands [10,25) touch [0,10) without overlap
25     c = Plan("C001", "C", 5, 3, 5, 2, 8, 1) # single use [5,7) touches [0,5)
26     d = Plan("C002", "C", 5, 3, 4, 2, 8, 1) # single use [4,6) overlaps [0,5)
27     e = Plan("C003", "C", 5, 3, 5, 2, 8, 12) # 7th use [65,67) overlaps A's second use [65,70)
28     assert not plans_conflict(a, b) and not plans_conflict(a, c) and plans_conflict(a, d)
29     assert overlapping_uses(a, d) == [(0, 0)] and overlapping_uses(a, e) == [(1, 6)]
30
31

```

```

32 def test_detectors_agree_with_cell_sets_on_synthetic_instances() -> None:
33     for seed in (1, 2, 3):
34         plans = synthetic_instance(60, seed)
35         pairs = detect_pairwise(plans)
36         assert pairs == detect_bandsweep(plans)
37         truth = conflicts_by_cells(plans)
38         assert [(plans[i].pid, plans[j].pid) for i, j in pairs] == truth
39         report = validate_detection(plans, truth)
40         assert report["ok"] and report["truth_pairs"] == len(pairs)
41
42
43 def test_validate_detection_flags_missing_and_extra() -> None:
44     plans = synthetic_instance(40, 7)
45     truth = conflicts_by_cells(plans)
46     assert truth, "instance must contain conflicts"
47     report = validate_detection(plans, truth[1:] + [("A001", "A002")])
48     assert not report["ok"] and report["missing"] == truth[:1]
49     assert ("A001", "A002") in report["extra"] or truth[0] == ("A001", "A002")
50
51
52 plan_strategy = st.builds(
53     Plan,
54     pid=st.just("X"),
55     cat=st.just("C"),
56     f=st.integers(0, 12),
57     w=st.integers(1, 4),
58     s=st.integers(0, 15),
59     d=st.integers(1, 3),
60     g=st.integers(1, 6),
61     n=st.integers(1, 4),
62 )
63
64
65 @given(plan_strategy, plan_strategy)
66 def test_property_interval_arithmetic_matches_cell_sets(a: Plan, b: Plan) -> None:
67     assert plans_conflict(a, b) == bool(cell_set(a) & cell_set(b))
68     assert len(cell_set(a)) == a.n_cells

```

tests/unit/test_d_report.py

```

1  """Problem D: report helpers (tables) on hand-made inputs."""
2
3  from __future__ import annotations
4
5  from pipelines.d.plans import Plan
6  from pipelines.d.report import _cancelled_rows, _table1, _vector_text
7  from pipelines.d.resolve import Limits
8
9
10 def test_cancelled_rows_explain_each_cancelled_plan() -> None:
11     a = Plan("A001", "A", 0, 10, 0, 5, 60, 3)
12     c1 = Plan("C001", "C", 5, 3, 3, 2, 8, 12)
13     c2 = Plan("C002", "C", 95, 3, 3, 2, 8, 12)
14     decisions = [
15         {"pid": "A001", "kind": "keep", "delta": 0, "plan": a.record()},
16         {"pid": "C001", "kind": "cancel", "delta": 0, "plan": None},
17         {"pid": "C002", "kind": "cancel", "delta": 0, "plan": None},
18     ]
19     conflicts = [{"id1": "A001", "id2": "C001"}]
20     rows = _cancelled_rows([a, c1, c2], decisions, conflicts, Limits(fmax=10, tmax=5))
21     assert [r[0] for r in rows] == ["C001", "C002"]
22     assert rows[0][3] == 1 and rows[0][4] == 1 # one neighbour, which is an A plan
23     assert rows[1][3] == 0 and rows[1][4] == 0

```

```

24     assert rows[0][5] == 15 + 8 # band start 5: shifts -5..10 minus 0 (15); time start 3: shifts -3..5 minus 0 (8)
25     assert rows[1][5] == 12 + 8 # band start 95: only -10..2 (12 shifts) keep the plan inside [0,100)
26
27
28 def test_table1_and_vector_text() -> None:
29     table = {
30         "A": {"keep": 1, "adjust": 2, "cancel": 0},
31         "B": {"keep": 0, "adjust": 0, "cancel": 1},
32         "C": {"keep": 3, "adjust": 4, "cancel": 5},
33     }
34     rows = _table1(table)
35     assert rows[-1] == ["合计", 4, 6, 6]
36     assert _vector_text([0, 1, 2]) == "(0, 1, 2)" and _vector_text(None) == "-"

```

tests/unit/test_d_resolve.py

```

1  """Problem D: actions, cell model, lexicographic CP-SAT vs exhaustive enumeration, HiGHS agreement, validator."""
2
3  from __future__ import annotations
4
5  import pytest
6
7  from pipelines.d.plans import Plan
8  from pipelines.d.resolve import (
9      Limits,
10     Option,
11     build_cell_model,
12     canonical_levels,
13     decisions_from_choice,
14     enumerate_options,
15     evaluate,
16     hint_is_feasible,
17     level_upper_bounds,
18     level_value,
19     scheme_levels,
20     separated_weights,
21     solve_lexicographic_cpsat,
22     solve_lexicographic_highs,
23     solve_weighted_cpsat,
24     strengthen,
25     table_from_decisions,
26 )
27 from pipelines.d.synth import exhaustive_lexicographic, tiny_instance
28 from pipelines.d.validators import validate_resolution
29
30
31 def test_option_enumeration_respects_bounds_and_single_parameter() -> None:
32     plan = Plan("C001", "C", 95, 3, 2, 2, 8, 12)
33     opts = enumerate_options(plan, Limits(fmax=10, tmax=5))
34     freq = sorted(o.delta for o in opts if o.kind == "freq")
35     time = sorted(o.delta for o in opts if o.kind == "time")
36     assert freq == [*range(-10, 0), 1, 2] # band interval must stay inside [0,100)
37     assert time == [-2, -1, 1, 2, 3, 4, 5] # start must stay >= 0
38     assert [o.kind for o in opts].count("keep") == 1 and [o.kind for o in opts].count("cancel") == 1
39     assert all(o.kind != "gap" for o in opts)
40     with_gap = enumerate_options(plan, Limits(fmax=10, tmax=5, gmax=10, gap_categories=("C",)))
41     gaps = sorted(o.delta for o in with_gap if o.kind == "gap")
42     assert gaps == [d for d in range(-7, 11) if d != 0] # g' = 8 + delta >= 1
43     assert Option("time", -3).amplitude == 6 and Option("freq", 4).amplitude == 4 and Option("gap", 5).amplitude == 5
44     assert Option("freq", 2).apply(plan) == plan.shifted(df=2)
45     assert Option("cancel").apply(plan) is None
46
47

```

```

48 def test_separated_weights_dominate_lower_levels() -> None:
49     bounds = [20, 40, 90, 20, 40, 90, 1500]
50     weights = separated_weights(bounds)
51     for k in range(len(bounds) - 1):
52         assert weights[k] > sum(w * b for w, b in zip(weights[k + 1 :], bounds[k + 1 :]))
53
54
55 @pytest.mark.parametrize("seed", [11, 12, 13, 14])
56 def test_cpsat_lexicographic_matches_exhaustive_and_validator(seed: int) -> None:
57     plans = tiny_instance(seed, n_plans=5)
58     limits = Limits(fmax=2, tmax=1)
59     options = [enumerate_options(p, limits) for p in plans]
60     model = build_cell_model(plans, options)
61     levels = canonical_levels(model)
62     truth = exhaustive_lexicographic(plans, limits, canonical_levels, model)
63     res = solve_lexicographic_cpsat(model, levels, seed=1, workers=2, time_limit=60)
64     assert res["all_optimal"] and res["vector"] == truth
65     decisions = decisions_from_choice(model, res["chosen"])
66     report = validate_resolution(
67         plans, decisions, fmax=2, tmax=1, reported_table=table_from_decisions(decisions), reported_vector=res["vector"]
68     )
69     assert report["ok"], report["errors"]
70     weighted = solve_weighted_cpsat(model, levels, seed=1, workers=2, time_limit=60)
71     assert weighted["vector"] == truth
72     highs = solve_lexicographic_highs(model, levels, seed=1, threads=2, time_limit=60)
73     assert highs["all_optimal"] and highs["vector"] == truth
74     assert evaluate(model, highs["chosen"], levels) == truth
75     assert level_upper_bounds(model, levels)[-1] >= truth[-1]
76
77
78 def test_schemes_are_well_formed_and_forced_cancellation_is_detected() -> None:
79     # two identical plans can never coexist: the optimum must cancel one of them (no shift can separate them
80     # because they are also identical after any common shift? no - shifting one of them resolves it)
81     a = Plan("A001", "A", 0, 10, 0, 5, 60, 3)
82     b = Plan("B001", "B", 0, 10, 0, 5, 60, 3)
83     plans = [a, b]
84     model = build_cell_model(plans, [enumerate_options(p, Limits(fmax=0, tmax=0)) for p in plans])
85     for name in ("P", "T", "S", "W"):
86         assert scheme_levels(name, model)
87     res = solve_lexicographic_cpsat(model, canonical_levels(model), seed=0, workers=1, time_limit=30)
88     assert res["vector"] == [0, 1, 0, 0, 0, 0, 0] # the B plan is cancelled, never the A plan
89     model2 = build_cell_model(plans, [enumerate_options(p, Limits(fmax=10, tmax=5)) for p in plans])
90     res2 = solve_lexicographic_cpsat(model2, canonical_levels(model2), seed=0, workers=1, time_limit=30)
91     assert res2["vector"][:6] == [0, 0, 0, 0, 1, 0] # with shifts allowed: adjust the B plan instead
92
93
94 def test_validator_rejects_illegal_decisions() -> None:
95     a = Plan("A001", "A", 0, 10, 0, 5, 60, 3)
96     b = Plan("C001", "C", 5, 3, 3, 2, 8, 12)
97     bad = [
98         {"pid": "A001", "kind": "keep", "delta": 0, "plan": a.record()},
99         {"pid": "C001", "kind": "freq", "delta": 12, "plan": b.shifted(df=12).record()},
100     ]
101     report = validate_resolution([a, b], bad, fmax=10, tmax=5)
102     assert not report["ok"] and any("exceeds" in e for e in report["errors"])
103     two_params = [
104         {"pid": "A001", "kind": "keep", "delta": 0, "plan": a.record()},
105         {"pid": "C001", "kind": "time", "delta": 1, "plan": b.shifted(df=1, dt=1).record()},
106     ]
107     report = validate_resolution([a, b], two_params, fmax=10, tmax=5)
108     assert not report["ok"] and any("changed fields" in e for e in report["errors"])
109     unresolved = [
110         {"pid": "A001", "kind": "keep", "delta": 0, "plan": a.record()},
111         {"pid": "C001", "kind": "keep", "delta": 0, "plan": b.record()},
112     ]
113     report = validate_resolution([a, b], unresolved, fmax=10, tmax=5)

```

```

114     assert not report["ok"] and report["remaining_conflicts"] == 1
115
116
117 def test_strengthen_detects_forced_cancellation_and_keeps_optimum() -> None:
118     a = Plan("A001", "A", 0, 10, 0, 5, 60, 3)
119     b = Plan("B001", "B", 0, 10, 0, 5, 60, 3)
120     model = build_cell_model([a, b], [enumerate_options(p, Limits(fmax=0, tmax=0)) for p in (a, b)])
121     strength = strengthen(model)
122     assert strength.forced == [(0, 1)] and strength.interacting_pairs == 1 and strength.groups
123     for seed in (21, 22):
124         plans = tiny_instance(seed, n_plans=5)
125         limits = Limits(fmax=2, tmax=1)
126         model = build_cell_model(plans, [enumerate_options(p, limits) for p in plans])
127         truth = exhaustive_lexicographic(plans, limits, canonical_levels, model)
128         res = solve_lexicographic_cpsat(
129             model, canonical_levels(model), seed=1, workers=2, time_limit=60, strength=strengthen(model)
130         )
131         assert res["all_optimal"] and res["vector"] == truth
132         highs = solve_lexicographic_highs(
133             model,
134             canonical_levels(model),
135             seed=1,
136             threads=2,
137             time_limit=60,
138             reference=truth,
139             strength=strengthen(model),
140         )
141         assert highs["agrees_with_reference"] and highs["vector"] == truth
142
143
144 @pytest.mark.parametrize("seed", [31, 32, 33])
145 def test_incumbent_cut_preserves_optimum_and_never_worsens_the_hint(seed: int) -> None:
146     """MDR-0009: a feasible hint bounds every level from above without removing any optimum.
147
148     The lexicographic vector must equal the exhaustive optimum whether or not a hint is supplied, and it
149     must never be lexicographically worse than the hint it started from."""
150     plans = tiny_instance(seed, n_plans=5)
151     limits = Limits(fmax=2, tmax=1)
152     model = build_cell_model(plans, [enumerate_options(p, limits) for p in plans])
153     levels = canonical_levels(model)
154     truth = exhaustive_lexicographic(plans, limits, canonical_levels, model)
155     cold = solve_lexicographic_cpsat(model, levels, seed=1, workers=2, time_limit=60)
156     assert cold["all_optimal"] and cold["vector"] == truth
157     # "cancel everything" is always feasible and (except in degenerate instances) far from optimal
158     cancel_all = [next(v for v in vs if model.option_of(v).kind == "cancel") for vs in model.plan_vars]
159     assert hint_is_feasible(model, cancel_all)
160     warm = solve_lexicographic_cpsat(model, levels, seed=1, workers=2, time_limit=60, hint=cancel_all)
161     assert warm["all_optimal"] and warm["vector"] == truth # the cut preserved every level's optimum
162     assert warm["vector"] <= evaluate(model, cancel_all, levels) # and never returned a worse vector
163     cuts = [lv["cut"] for lv in warm["levels"]]
164     assert cuts[0] == evaluate(model, cancel_all, levels)[0] # first cut comes straight from the hint
165     assert all(c is not None and v <= c for c, v in zip(cuts, warm["vector"])) # every level respects its cut
166
167
168 def test_lexicographic_never_worse_than_a_good_hint() -> None:
169     """A one-second budget is far too short for the search, but the incumbent must still be returned."""
170     plans = tiny_instance(41, n_plans=6)
171     limits = Limits(fmax=2, tmax=1)
172     model = build_cell_model(plans, [enumerate_options(p, limits) for p in plans])
173     levels = canonical_levels(model)
174     truth = exhaustive_lexicographic(plans, limits, canonical_levels, model)
175     optimal = solve_lexicographic_cpsat(model, levels, seed=1, workers=2, time_limit=60)
176     starved = solve_lexicographic_cpsat(
177         model, levels, seed=1, workers=1, time_limit=0.001, hint=optimal["chosen"], strength=strengthen(model)
178     )
179     assert starved["vector"] == truth

```

```

180     assert all(lv["status"] in {"OPTIMAL", "FEASIBLE", "INCUMBENT"} for lv in starved["levels"])
181
182
183 def test_level_value_matches_evaluate() -> None:
184     plans = tiny_instance(51, n_plans=4)
185     model = build_cell_model(plans, [enumerate_options(p, Limits(fmax=1, tmax=1)) for p in plans])
186     levels = canonical_levels(model)
187     chosen = [vs[0] for vs in model.plan_vars]
188     assert [level_value(terms, chosen) for _, terms in levels] == evaluate(model, chosen, levels)

```

tests/unit/test_hashing.py

```

1 from __future__ import annotations
2
3 from pathlib import Path
4
5 from hypothesis import given
6 from hypothesis import strategies as st
7
8 from forge.hashing import hash_tree, sha256_bytes, sha256_json, tree_digest
9
10
11 @given(st.binary())
12 def test_sha256_bytes_is_deterministic(data: bytes) -> None:
13     assert sha256_bytes(data) == sha256_bytes(data)
14     assert len(sha256_bytes(data)) == 64
15
16
17 @given(
18     st.dictionaries(
19         st.text(min_size=1, max_size=8), st.integers() | st.floats(allow_nan=False) | st.text(max_size=5), max_size=6
20     )
21 )
22 def test_sha256_json_ignores_key_order(mapping: dict) -> None:
23     reversed_mapping = dict(reversed(list(mapping.items())))
24     assert sha256_json(mapping) == sha256_json(reversed_mapping)
25
26
27 def test_hash_tree_ignores_records_and_is_sorted(tmp_path: Path) -> None:
28     (tmp_path / "b.txt").write_text("b")
29     (tmp_path / "a.txt").write_text("a")
30     (tmp_path / "manifest.json").write_text("{}")
31     (tmp_path / "__pycache__").mkdir()
32     (tmp_path / "__pycache__" / "x.pyc").write_bytes(b"\x00")
33     tree = hash_tree(tmp_path, ignore=("manifest.json", "__pycache__"))
34     assert list(tree) == ["a.txt", "b.txt"]
35     digest = tree_digest(tree)
36     (tmp_path / "manifest.json").write_text('{"changed": true}')
37     assert tree_digest(hash_tree(tmp_path, ignore=("manifest.json", "__pycache__"))) == digest

```

tests/unit/test_manifest.py

```

1 from __future__ import annotations
2
3 from pathlib import Path
4
5 from forge.manifest import Manifest
6
7
8 def test_manifest_lifecycle_and_verification(tmp_path: Path) -> None:

```

```

9     stage_dir = tmp_path / "stage"
10    stage_dir.mkdir()
11    (stage_dir / "out.json").write_text("{}")
12    (stage_dir / "events.jsonl").write_text("{}\n")
13    manifest = Manifest.start("run1", "stage", {"p": 1}, {"git_sha": "abc"}, "cfg", {"ingest": "d"}, {})
14    assert manifest.status == "running" and manifest.runtime["python"]
15    manifest.complete(stage_dir, {"m": 1.5})
16    assert manifest.status == "completed"
17    assert list(manifest.outputs) == ["out.json"]
18    manifest.write(stage_dir / "manifest.json")
19    loaded = Manifest.read(stage_dir / "manifest.json")
20    assert loaded.outputs_digest == manifest.outputs_digest
21    assert loaded.verify_outputs(stage_dir) == []
22    (stage_dir / "out.json").write_text('{ "x": 1 }')
23    assert loaded.verify_outputs(stage_dir) == ["sha256 mismatch: out.json"]
24
25
26 def test_manifest_failure_records_traceback() -> None:
27     manifest = Manifest.start("run1", "stage", {}, {}, "cfg", {}, {})
28     try:
29         raise ValueError("boom")
30     except ValueError as exc:
31         manifest.fail(exc)
32     assert manifest.status == "failed"
33     assert manifest.error and manifest.error["type"] == "ValueError"
34     assert manifest.duration_s is not None

```

tests/unit/test_packaging_tex.py

```

1 from __future__ import annotations
2
3 from pathlib import Path
4
5 from forge.packaging import build_zip, code_members, run_members
6 from forge.tex import aux_page, summarize_log, tex_escape
7
8
9 def test_zip_is_deterministic(tmp_path: Path) -> None:
10     src = tmp_path / "src"
11     src.mkdir()
12     (src / "b.py").write_text("print(2)")
13     (src / "a.py").write_text("print(1)")
14     (src / "__pycache__").mkdir()
15     (src / "__pycache__" / "a.pyc").write_bytes(b"x")
16     members = code_members(src, ["**/*.py"])
17     assert [m[0] for m in members] == ["a.py", "b.py"]
18     one = build_zip(members, tmp_path / "one.zip")
19     two = build_zip(list(reversed(members)), tmp_path / "two.zip")
20     assert one["sha256"] == two["sha256"] and one["members"] == 2
21
22
23 def test_run_members_prefix(tmp_path: Path) -> None:
24     run = tmp_path / "runs" / "r1"
25     (run / "q1").mkdir(parents=True)
26     (run / "q1" / "out.json").write_text("{}")
27     assert run_members(run, ["q1", "missing"]) == [(("runs/r1/q1/out.json", run / "q1" / "out.json"))]
28
29
30 def test_log_summary_and_aux_page(tmp_path: Path) -> None:
31     log = (
32         "LaTeX Warning: Reference `fig:x' on page 3 undefined on input line 10.\n"
33         "Overfull \\hbox (12.5pt too wide) in paragraph\n"
34         "Missing character: There is no  in font Latin Modern!\n"

```

```

35         "!\n\n"
36         "Output written on main.pdf (7 pages).\n"
37     )
38     summary = summarize_log(log)
39     assert summary["undefined_references"] == ["fig:x"]
40     assert summary["overfull_max_pt"] == 12.5 and summary["missing_characters"] == 1
41     assert summary["errors"] == ["Undefined control sequence."] and summary["pages_from_log"] == 7
42     aux = tmp_path / "main.aux"
43     aux.write_text("\n\\newlabel{page:body-end}{27}{page:body-end}{\\n\\newlabel{sec:a}{2.1}{5}{}}\n")
44     assert aux_page(aux, "page:body-end") == 27 and aux_page(aux, "sec:a") == 5 and aux_page(aux, "nope") is None
45
46
47 def test_tex_escape() -> None:
48     assert tex_escape("a_b & 100% #1") == r"a_b \& 100\% \#1"

```

tests/unit/test_runner.py

```

1  from __future__ import annotations
2
3  import json
4  from pathlib import Path
5
6  import pytest
7
8  from forge import runner
9  from forge.context import StageContext
10
11
12 @pytest.fixture(autouse=True)
13 def isolated_registry(monkeypatch: pytest.MonkeyPatch) -> None:
14     monkeypatch.setattr(runner, "STAGES", {})
15     monkeypatch.setattr(runner, "discover", lambda repo: None)
16
17
18 def test_execute_records_manifest_skips_reruns_and_enforces_deps(tmp_repo: Path, tmp_path: Path) -> None:
19     vol = tmp_path / "vol"
20
21     @runner.stage("first", description="first stage")
22     def first(ctx: StageContext) -> dict:
23         ctx.out("a.txt").write_text("hello")
24         ctx.number("Answer", 42)
25         return {"n": 1}
26
27     @runner.stage("second", deps=("first",))
28     def second(ctx: StageContext) -> dict:
29         assert (ctx.dep("first") / "a.txt").read_text() == "hello"
30         return {"ok": True}
31
32     with pytest.raises(RuntimeError):
33         runner.execute("second", "r1", {}, False, tmp_repo, vol)
34     result = runner.execute("first", "r1", {"code_ref": {"git_sha": "x"}}, False, tmp_repo, vol)
35     assert result["status"] == "completed" and result["metrics"] == {"n": 1}
36     manifest = json.loads((vol / "runs" / "r1" / "first" / "manifest.json").read_text())
37     assert (
38         manifest["status"] == "completed" and "a.txt" in manifest["outputs"] and "numbers.json" in manifest["outputs"]
39     )
40     assert json.loads((vol / "runs" / "r1" / "first" / "numbers.json").read_text()) == {"Answer": "42"}
41     assert runner.execute("first", "r1", {}, False, tmp_repo, vol)["status"] == "skipped"
42     assert runner.execute("second", "r1", {}, False, tmp_repo, vol)["status"] == "completed"
43     combined = runner.execute("all", "r1", {"stages": ["first", "second"]}, False, tmp_repo, vol)
44     assert combined["status"] == "completed" and [r["status"] for r in combined["results"]] == ["skipped", "skipped"]
45
46

```

```

47 def test_failed_stage_writes_failed_manifest(tmp_repo: Path, tmp_path: Path) -> None:
48     vol = tmp_path / "vol"
49
50     @runner.stage("bad")
51     def bad(ctx: StageContext) -> dict:
52         raise ValueError("boom")
53
54     with pytest.raises(ValueError):
55         runner.execute("bad", "r1", {}, False, tmp_repo, vol)
56     manifest = json.loads((vol / "runs" / "r1" / "bad" / "manifest.json").read_text())
57     assert manifest["status"] == "failed" and manifest["error"]["message"] == "boom"
58
59
60 def test_from_run_copies_dependency_outputs(tmp_repo: Path, tmp_path: Path) -> None:
61     vol = tmp_path / "vol"
62
63     @runner.stage("base")
64     def base(ctx: StageContext) -> dict:
65         ctx.out("x.txt").write_text("x")
66         return {}
67
68     @runner.stage("child", deps=("base",))
69     def child(ctx: StageContext) -> dict:
70         return {"seen": (ctx.dep("base") / "x.txt").read_text()}
71
72     runner.execute("base", "old", {}, False, tmp_repo, vol)
73     result = runner.execute("child", "new", {"from_run": "old"}, False, tmp_repo, vol)
74     assert result["metrics"] == {"seen": "x"}

```

tests/unit/test_xlsx.py

```

1 from __future__ import annotations
2
3 from pathlib import Path
4
5 import openpyxl
6 import pytest
7
8 from forge.xlsx import SheetContract, WorkbookContract, check_workbook, read_sheets, sheet_headers, write_result
9
10
11 def _template(path: Path) -> Path:
12     wb = openpyxl.Workbook()
13     ws = wb.active
14     ws.title = "温度"
15     ws.append(["时间\\到药材中心的距离", 0, 0.1, "...", 2])
16     ws.append([1, None, None, None, None])
17     ws2 = wb.create_sheet("水分浓度")
18     ws2.append(["时间\\到药材中心的距离", 0, 0.1, "...", 2])
19     wb.save(path)
20     return path
21
22
23 def test_write_result_keeps_all_rounds_and_passes_contract(tmp_path: Path) -> None:
24     template = _template(tmp_path / "tpl.xlsx")
25     header = ["ignored", 0.0, 0.1, 0.2]
26     rows = [[t, 28.123456, 28.2, 28.3] for t in range(1, 6)]
27     info = write_result(template, tmp_path / "out.xlsx", {"温度": (header, rows), "水分浓度": (header, rows)})
28     assert info["sheets"]["温度"]["rows"] == 5
29     headers = sheet_headers(tmp_path / "out.xlsx")
30     assert headers["温度"][0] == "时间\\到药材中心的距离"
31     assert headers["温度"][1:] == [0.0, 0.1, 0.2]
32     frames = read_sheets(tmp_path / "out.xlsx")

```

```

33     assert float(frames["温度"].iloc[0, 1]) == pytest.approx(28.1235)
34     contract = WorkbookContract(
35         "out.xlsx",
36         "tpl.xlsx",
37         (
38             SheetContract("温度", min_rows=5, max_rows=5, header_len=4, numeric_from_col=0),
39             SheetContract("水分浓度", min_rows=5, header_len=4, numeric_from_col=0),
40         ),
41     )
42     report = check_workbook(tmp_path / "out.xlsx", contract, template)
43     assert report["ok"], report["errors"]
44
45
46 def test_check_workbook_flags_blank_and_text_cells(tmp_path: Path) -> None:
47     template = _template(tmp_path / "tpl.xlsx")
48     rows = [[1, 1.0, None, "x"], [2, 2.0, 2.0, 2.0]]
49     write_result(template, tmp_path / "out.xlsx", {"温度": (["h", 0, 0.1, 0.2], rows)})
50     contract = WorkbookContract("out.xlsx", "tpl.xlsx", (SheetContract("温度", min_rows=2, numeric_from_col=0),))
51     report = check_workbook(tmp_path / "out.xlsx", contract, template)
52     joined = " ".join(report["errors"])
53     assert not report["ok"] and "blank" in joined and "non-numeric" in joined
54
55
56 def test_write_result_rejects_unknown_sheet(tmp_path: Path) -> None:
57     template = _template(tmp_path / "tpl.xlsx")
58     with pytest.raises(KeyError):
59         write_result(template, tmp_path / "out.xlsx", {"nope": (["h"], [])})
60
61
62 def test_write_only_path_for_large_sheets(tmp_path: Path, monkeypatch: pytest.MonkeyPatch) -> None:
63     import forge.xlsx as mod
64
65     monkeypatch.setattr(mod, "BIG_SHEET_CELLS", 10)
66     template = _template(tmp_path / "tpl.xlsx")
67     rows = [[t, 1.0, 2.0] for t in range(1, 8)]
68     info = write_result(template, tmp_path / "big.xlsx", {"温度": (["h", 0, 0.1], rows)})
69     assert info["write_only"]
70     headers = sheet_headers(tmp_path / "big.xlsx")
71     assert list(headers) == ["温度", "水分浓度"]
72     assert headers["温度"][0] == "时间\\到药材中心的距离"

```

tools/_init_.py

tools/cli.py

```

1  #!/usr/bin/env python3
2  """Forge control plane (local). Never computes: submits cloud jobs, downloads, verifies hashes.
3
4  Examples
5      python3 tools/cli.py run ingest                                # new run id, small container
6      python3 tools/cli.py run validate --run-id <id>              # continue the same run
7      python3 tools/cli.py run q1,q2 --size medium                 # sequential stages in one container
8      python3 tools/cli.py run q3 --size large --spawn             # detached; poll with `wait`/`status`
9      python3 tools/cli.py exec --code 'print(sorted(p.name for p in RUN_DIR.iterdir()))'
10     python3 tools/cli.py status                                    # manifests of the last run
11     python3 tools/cli.py download --stage results                 # into artifacts/, verified against manifests
12     python3 tools/cli.py release --version v1.0.0                 # deliverables into releases/v1.0.0/
13
14

```

```

15 from __future__ import annotations
16
17 import argparse
18 import hashlib
19 import json
20 import os
21 import shutil
22 import subprocess
23 import sys
24 from datetime import datetime
25 from pathlib import Path
26
27 REPO = Path(__file__).resolve().parents[1]
28 PROJECT = json.loads((REPO / "configs" / "project.json").read_text(encoding="utf-8"))
29 APP_FILE = REPO / "forge" / "modal_app.py"
30 STATE_DIR = REPO / ".forge"
31 LAST_RUN = STATE_DIR / "last_run"
32 MODAL = shutil.which("modal") or str(Path.home() / ".local" / "bin" / "modal")
33
34
35 def _sh(cmd: list[str], **kw) -> subprocess.CompletedProcess:
36     print("$ " + " ".join(cmd), flush=True)
37     return subprocess.run(cmd, cwd=REPO, check=False, **kw)
38
39
40 def git_ref() -> dict:
41     def out(*args: str) -> str:
42         try:
43             return subprocess.run(["git", *args], cwd=REPO, capture_output=True, text=True, check=False).stdout.strip()
44         except OSError:
45             return ""
46
47     sha = out("rev-parse", "HEAD") or "nogit"
48     return {
49         "git_sha": sha,
50         "branch": out("rev-parse", "--abbrev-ref", "HEAD"),
51         "dirty": bool(out("status", "--porcelain")),
52     }
53
54
55 def new_run_id() -> str:
56     return datetime.now().strftime("%Y%m%d-%H%M%S") + "-" + git_ref()["git_sha"][:7]
57
58
59 def remember(run_id: str) -> None:
60     STATE_DIR.mkdir(exist_ok=True)
61     LAST_RUN.write_text(run_id, encoding="utf-8")
62
63
64 def resolve_run_id(explicit: str | None, create: bool) -> str:
65     if explicit:
66         remember(explicit)
67         return explicit
68     if create or not LAST_RUN.exists():
69         run_id = new_run_id()
70         remember(run_id)
71         return run_id
72     return LAST_RUN.read_text(encoding="utf-8").strip()
73
74
75 def parse_params(items: list[str] | None) -> dict:
76     params: dict = {}
77     for item in items or []:
78         key, _, raw = item.partition("=")
79         try:
80             params[key] = json.loads(raw)

```

```

81         except json.JSONDecodeError:
82             params[key] = raw
83         return params
84
85
86 def modal_run(args: list[str], detach: bool = False) -> int:
87     cmd = [MODAL, "run"] + ([ "--detach" ] if detach else []) + [str(APP_FILE)] + args
88     return _sh(cmd).returncode
89
90
91 def cmd_run(ns: argparse.Namespace) -> int:
92     stages = [s for s in ns.stage.split(",") if s]
93     run_id = resolve_run_id(ns.run_id, create=ns.new_run)
94     params = parse_params(ns.param)
95     params.update(
96         {"code_ref": git_ref(), "size": ns.size, "submitted_at": datetime.now().isoformat(timespec="seconds")}
97     )
98     if ns.from_run:
99         params["from_run"] = ns.from_run
100     if ns.profile:
101         params["profile"] = ns.profile
102     stage = stages[0] if len(stages) == 1 else "all"
103     if stage == "all":
104         params["stages"] = stages
105     args = ["--stage", stage, "--run-id", run_id, "--params", json.dumps(params, ensure_ascii=False), "--size", ns.size
106         ↵ ]
107     if ns.force:
108         args.append("--force")
109     if ns.spawn:
110         args.append("--spawn")
111     print(f"run_id={run_id} stages={stages} size={ns.size} spawn={ns.spawn}")
112     return modal_run(args, detach=ns.spawn)
113
114
115 def cmd_exec(ns: argparse.Namespace) -> int:
116     code = Path(ns.file).read_text(encoding="utf-8") if ns.file else ns.code
117     if not code:
118         print("provide --code or --file", file=sys.stderr)
119         return 2
120     run_id = resolve_run_id(ns.run_id, create=False)
121     return modal_run(
122         ["--stage", "exec", "--run-id", run_id, "--params", json.dumps({"code": code}, ensure_ascii=False)]
123     )
124
125
126 def cmd_provision(_: argparse.Namespace) -> int:
127     return modal_run(["--stage", "provision"])
128
129
130 STATUS_SNIPPET = r"""
131 import json
132 from pathlib import Path
133 run = RUN_DIR
134 if not run.exists():
135     print("run directory missing:", run)
136 else:
137     rows = []
138     for m in sorted(run.glob("*/manifest.json")):
139         d = json.loads(m.read_text())
140         err = (d.get("error") or {}).get("message", "")[:80]
141         digest = d.get("outputs_digest", "")[:12]
142         rows.append((d["stage"], d["status"], d.get("duration_s"), digest, len(d.get("outputs", {})), err))
143     print(f'{"stage":14s} {"status":10s} {"seconds":>9s} {"digest":12s} {"files":>5s} error')
144     for r in rows:
145         print(f'{r[0]:14s} {r[1]:10s} {str(r[2]):>9s} {r[3]:12s} {r[4]:5d} {r[5]}')
146     print("run_dir:", run)

```

```

146 """
147
148
149 def cmd_status(ns: argparse.Namespace) -> int:
150     run_id = resolve_run_id(ns.run_id, create=False)
151     return modal_run(["--stage", "exec", "--run-id", run_id, "--params", json.dumps({"code": STATUS_SNIPPET})])
152
153
154 def cmd_runs(_: argparse.Namespace) -> int:
155     return _sh([MODAL, "volume", "ls", PROJECT["volume"], "runs"]).returncode
156
157
158 def cmd_logs(ns: argparse.Namespace) -> int:
159     run_id = resolve_run_id(ns.run_id, create=False)
160     code = (
161         "from pathlib import Path\n"
162         f"p = RUN_DIR / {ns.stage!r} / 'events.jsonl'\n"
163         "print(p.read_text()[-%d:] if p.exists() else 'no events for stage')\n" % ns.tail
164     )
165     return modal_run(["--stage", "exec", "--run-id", run_id, "--params", json.dumps({"code": code})])
166
167
168 def sha256(path: Path) -> str:
169     h = hashlib.sha256()
170     with path.open("rb") as fh:
171         for chunk in iter(lambda: fh.read(1 << 20), b""):
172             h.update(chunk)
173     return h.hexdigest()
174
175
176 def verify_dir(root: Path) -> int:
177     """Verify every downloaded stage directory against its manifest (transport check only)."""
178     problems = 0
179     manifests = sorted(root.rglob("manifest.json"))
180     if not manifests:
181         print(f"no manifests under {root}")
182         return 1
183     for manifest in manifests:
184         data = json.loads(manifest.read_text(encoding="utf-8"))
185         stage_dir = manifest.parent
186         bad = []
187         for rel, meta in data.get("outputs", {}).items():
188             file = stage_dir / rel
189             if not file.exists():
190                 bad.append(f"missing {rel}")
191             elif sha256(file) != meta["sha256"]:
192                 bad.append(f"sha256 mismatch {rel}")
193         status = "OK " if not bad else "BAD"
194         print(
195             f"[{status}] {data.get('stage'):14s} {data.get('status'):10s} "
196             f"files={len(data.get('outputs', {}))} {stage_dir}"
197         )
198         for b in bad:
199             print("    " + b)
200         problems += len(bad)
201     return 0 if problems == 0 else 1
202
203
204 def volume_get(remote: str, local_parent: Path) -> int:
205     """Download a remote directory into ``local_parent/<basename>`` (replacing any previous copy)."""
206     target = local_parent / Path(remote).name
207     shutil.rmtree(target, ignore_errors=True)
208     local_parent.mkdir(parents=True, exist_ok=True)
209     return _sh([MODAL, "volume", "get", "--force", PROJECT["volume"], remote, str(local_parent) + "/"]).returncode
210
211

```

```

212 def cmd_download(ns: argparse.Namespace) -> int:
213     run_id = resolve_run_id(ns.run_id, create=False)
214     if ns.stage:
215         remote, parent = f"runs/{run_id}/{ns.stage}", REPO / ns.dest / "runs" / run_id
216     else:
217         remote, parent = f"runs/{run_id}", REPO / ns.dest / "runs"
218     rc = volume_get(remote, parent)
219     if rc != 0:
220         return rc
221     return verify_dir(parent / Path(remote).name)
222
223
224 def cmd_verify(ns: argparse.Namespace) -> int:
225     return verify_dir(Path(ns.path))
226
227
228 def cmd_release(ns: argparse.Namespace) -> int:
229     run_id = resolve_run_id(ns.run_id, create=False)
230     dest = REPO / "releases" / ns.version
231     if dest.exists() and not ns.force:
232         print(f"{dest} exists; pass --force to overwrite", file=sys.stderr)
233         return 2
234     staging = REPO / "artifacts" / "release_staging" / run_id
235     rc = volume_get(f"runs/{run_id}/release", staging)
236     if rc != 0:
237         return rc
238     shutil.rmtree(dest, ignore_errors=True)
239     shutil.copytree(staging / "release", dest)
240     manifest = json.loads((dest / "release_manifest.json").read_text(encoding="utf-8"))
241     bad = 0
242     for name, meta in manifest["files"].items():
243         file = dest / name
244         ok = file.exists() and sha256(file) == meta["sha256"]
245         bad += 0 if ok else 1
246         print(f"[{'OK' if ok else 'BAD'}] {name} {meta['bytes']} bytes sha256={meta['sha256'][:16]}")
247     print(f"release {ns.version} from run {run_id}: {'verified' if not bad else f'{bad} problems'} -> {dest}")
248     return 0 if not bad else 1
249
250
251 def cmd_wait(ns: argparse.Namespace) -> int:
252     interpreter = (
253         Path(MODAL).read_text(encoding="utf-8", errors="ignore").splitlines()[0].lstrip("#!").strip()
254         if Path(MODAL).exists()
255         else sys.executable
256     )
257     code = (
258         "import json, modal, sys\n"
259         f"call = modal.FunctionCall.from_id({ns.call_id!r})\n"
260         "print('STAGE_RESULT ' + json.dumps(call.get(), ensure_ascii=False, default=str))\n"
261     )
262     return _sh([interpreter, "-c", code]).returncode
263
264
265 def cmd_fmt(ns: argparse.Namespace) -> int:
266     """Format in the cloud, then copy the exported files back into the working tree."""
267     run_id = resolve_run_id(ns.run_id, create=False)
268     params = {"code_ref": git_ref(), "size": "small"}
269     rc = modal_run(["--stage", "fmt", "--run-id", run_id, "--params", json.dumps(params), "--force"])
270     if rc != 0:
271         return rc
272     dest = REPO / "artifacts" / "runs" / run_id / "fmt"
273     rc = volume_get(f"runs/{run_id}/fmt", dest.parent)
274     if rc != 0:
275         return rc
276     files = dest / "files"
277     changed = sorted(p for p in files.rglob("*") if p.is_file()) if files.exists() else []

```

```

278     for src in changed:
279         rel = src.relative_to(files)
280         target = REPO / rel
281         target.parent.mkdir(parents=True, exist_ok=True)
282         shutil.copy2(src, target)
283         print(f"updated {rel}")
284     report = json.loads((dest / "fmt_report.json").read_text(encoding="utf-8"))
285     print(f"{len(changed)} files formatted; remaining ruff findings: {report.get('remaining')}")
286     return 0
287
288
289 def cmd_new_run(_: argparse.Namespace) -> int:
290     run_id = new_run_id()
291     remember(run_id)
292     print(run_id)
293     return 0
294
295
296 def build_parser() -> argparse.ArgumentParser:
297     p = argparse.ArgumentParser(prog="forge", description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter
298     ↪ )
299     sub = p.add_subparsers(dest="command", required=True)
300
301     r = sub.add_parser("run", help="run one stage or a comma-separated sequence in one container")
302     r.add_argument("stage")
303     r.add_argument("--run-id")
304     r.add_argument("--new-run", action="store_true", help="start a fresh run id")
305     r.add_argument("--size", default="small", choices=["small", "medium", "large"])
306     r.add_argument("--force", action="store_true", help="re-run even if the stage already completed")
307     r.add_argument("--spawn", action="store_true", help="detach: return immediately, poll with status/wait")
308     r.add_argument("--from-run", help="copy missing dependency outputs from this earlier run")
309     r.add_argument("--profile", help="overlay configs/<profile>.toml")
310     r.add_argument("--param", action="append", help="key=value (JSON-parsed) forwarded to the stage")
311     r.set_defaults(fn=cmd_run)
312
313     e = sub.add_parser("exec", help="run an ad-hoc Python snippet in the cloud against the run directory")
314     e.add_argument("--code")
315     e.add_argument("--file")
316     e.add_argument("--run-id")
317     e.set_defaults(fn=cmd_exec)
318
319     sub.add_parser("provision", help="verify the cloud toolchain").set_defaults(fn=cmd_provision)
320
321     s = sub.add_parser("status", help="print manifests of a run")
322     s.add_argument("--run-id")
323     s.set_defaults(fn=cmd_status)
324
325     lg = sub.add_parser("logs", help="tail the event log of a stage")
326     lg.add_argument("stage")
327     lg.add_argument("--run-id")
328     lg.add_argument("--tail", type=int, default=6000)
329     lg.set_defaults(fn=cmd_logs)
330
331     sub.add_parser("runs", help="list runs on the volume").set_defaults(fn=cmd_runs)
332
333     d = sub.add_parser("download", help="download a run (or one stage) into artifacts/ and verify hashes")
334     d.add_argument("--run-id")
335     d.add_argument("--stage")
336     d.add_argument("--dest", default="artifacts")
337     d.set_defaults(fn=cmd_download)
338
339     v = sub.add_parser("verify", help="verify a downloaded directory against its manifests")
340     v.add_argument("path")
341     v.set_defaults(fn=cmd_verify)
342
343     rel = sub.add_parser("release", help="download the release stage into releases/<version>/ and verify")

```

```

343     rel.add_argument("--version", required=True)
344     rel.add_argument("--run-id")
345     rel.add_argument("--force", action="store_true")
346     rel.set_defaults(fn=cmd_release)
347
348     w = sub.add_parser("wait", help="block on a spawned function call id and print its result")
349     w.add_argument("call_id")
350     w.set_defaults(fn=cmd_wait)
351
352     f = sub.add_parser("fmt", help="ruff --fix + format in the cloud and copy the results back")
353     f.add_argument("--run-id")
354     f.set_defaults(fn=cmd_fmt)
355
356     sub.add_parser("new-run", help="mint a run id").set_defaults(fn=cmd_new_run)
357     return p
358
359
360 def main(argv: list[str] | None = None) -> int:
361     os.chdir(REPO)
362     ns = build_parser().parse_args(argv)
363     return int(ns.fn(ns))
364
365
366 if __name__ == "__main__":
367     sys.exit(main())

```